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Danos et al.(10) **Pub. No.: US 2025/0276097 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **FULLY-HUMAN POST-TRANSLATIONALLY
MODIFIED ANTIBODY THERAPEUTICS***C07K 16/40* (2006.01)*C12N 7/00* (2006.01)*C12N 15/86* (2006.01)(71) Applicant: **REGENXBIO Inc.**, Rockville, MD
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2750/14143 (2013.01)(72) Inventors: **Olivier Danos**, New York, NY (US);
Zhuchun Wu, North Potomac, MD
(US); **Franz Michael Gerner**,
Myersville, MD (US); **Sherri Van**
Everen, Menlo Park, CA (US)(21) Appl. No.: **19/213,663**(22) Filed: **May 20, 2025****Related U.S. Application Data**(60) Division of application No. 16/706,065, filed on Dec.
6, 2019, which is a continuation of application No.
PCT/US2018/056346, filed on Oct. 17, 2018.(60) Provisional application No. 62/609,750, filed on Dec.
22, 2017, provisional application No. 62/574,106,
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62/700,124, filed on Jul. 18, 2018.**Publication Classification**(51) **Int. Cl.***A61K 48/00* (2006.01)*A61K 9/00* (2006.01)*C07K 16/18* (2006.01)*C07K 16/22* (2006.01)*C07K 16/24* (2006.01)*C07K 16/28* (2006.01)

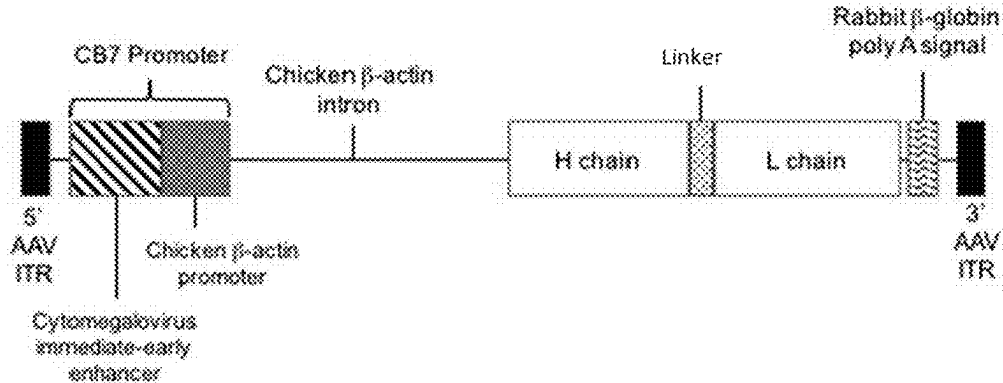
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ABSTRACT

Provided are methods and compositions for the delivery of fully human post-translationally modified therapeutic monoclonal antibodies and antigen-binding fragments thereof. The fully human post-translationally modified therapeutic monoclonal antibodies may be preferably delivered by gene therapy methods, particularly as a recombinant adeno-associated virus (rAAV) vector to the appropriate tissue. Methods of manufacture of the AAV vectors, pharmaceutical compositions and methods of treatment are also provided. In addition, provided are methods of producing therapeutic antibodies that are "biobetters" as fully human post-translationally modified. These fully human post-translationally modified therapeutic antibodies may be administered to a subject in need of treatment with the therapeutic antibody.

Specification includes a Sequence Listing.

FIG. 1



FIGS. 2A-2F

FIG. 2A ADUCANUMAB Fab IgG1 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Aducanumab Heavy Chain (SEQ ID NO: 1)

XVQLVESGGG VVQPGKSLRL SCAASGFAFS SYGMHWVRQA PGKGLEWVAV IWFDGTKKKYY 60
TDSVKGREFTI SRDNSKNTLY LQMNTLRAED TAVYYCARDR GIGARRGEYY MDVWGKGTTV 120
TVSSASTRGE SVFPLAPSSK STSGGTAALG CLVKDYFPEP VTVSGNSGAL TSGVHTFPAV 180
LQSSGLYSLS SVVTVPSSSL GTQTYICNVN HKPSNTKVDK RVEPKSCD +/- KTHI (or KTHL) +/-
CPPCPA +/-PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Aducanumab Light Chain (SEQ ID NO: 2)

DIQMTQSPSS LSASVGDRVT ITCRASQIS SYLNWYQRP GKAPKLLIYA ASSLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EDFATYFCQQ SYSTPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSRN RGEC 214

Heavy Chain Legend:

CDRs are underscored (predicted for Aducanumab)

EDTAVY = Y-sulfation site (SEQ ID NO: 248)

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

DKTHICPPCPAPELLCG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Aducanumab)

EDFATY = Y-sulfation site (SEQ ID NO: 250)

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 2A-2F (CON'T)

FIG. 2B CRENEZUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Crenezumab Heavy Chain (SEQ ID NO: 3)

EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYGMSWVRQA PGKGLELVA S INSNGGSTYY 60
PDSVKGRFTI SRDNAKNSLY LQMSNLRAED TAVYYCASGD YWG Q STTVTV SSASTKGPSV 120
FPLAPCSRST SESTAALGCL VKDYFPEPVT VSWSGALTS GVHTFPAVLQ SSGLYSLSSV 180
VTVPSSSLGT KTY ICVDHK PSNTKVDKRV ESKY +/- GPPCPCPA +/- PEFLGGPSVE

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Crenezumab Light Chain (SEQ ID NO: 4)

DIVMTQSPLS LPVTGPGEPA ISCRSSQSLV YSNGDTYLHW YLQKPGQSPQ LLIYKVSNRF 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCSQSTHVP WTFG QYKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVCL LNNFYPREAK VQWKVDNALQ SCNSQESVTEQDSKDYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TK STNRGEC 219

Heavy Chain Legend:

CDRs are underscored (predicted for Crenezumab)

EDTAVYY = Y-sulfation site (SEQ ID NO: 248)

Q = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

GPPCPPCPAPPEFLG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for Crenezumab)

EDVGVYY = Y-sulfation site (SEQ ID NO: 251)

Q = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 2A-2F (CON'T)

FIG. 2C GANTENERUMAB Fab IgG1 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Gantenerumab Heavy Chain (SEQ ID NO: 5)

QVELVESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA I GTRITYY 60ADSVKGRFTI SRDNS LQMNSLRAED TAVY CARGK GNTHKPYGYV RYFDVWG 120LVTVSSASTK GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTV SG ALTSGVHTFP 180AVLQSSGLYS LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KIHT (or KIHL) CPPCPA PELLSGPSVFL 250

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Gantenerumab Light Chain (SEQ ID NO: 6)

DIVLTQSPAT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGOAPRLLIY GASSRAIGVP 60

ARFSGSGSGT DFTLTSSLE PEPFATY CL QIYNMPITFG KVEIKRT VAAPSVFIFF 120PSDEQLKSGT ASVVCLLNNE YPREAKVQWK VDNALQ QS QESVTEQDSK DSTYSLSTL 180TLISKADYKH KVIACEVTHQ GLSSPVTK NRGEC 215

Heavy Chain Legend:

CDRs are underscored (predicted for Gantenerumab)

 EDTAVY = Y-sulfation site (SEQ ID NO: 248) = Asparaginal (N) glycosylation site(S/T) = Non-consensus Asparaginal (N) glycosylation site Q = Glutamine (Q) glycosylation site DTATCPPCPAPELLSS = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Gantenerumab)

 EDFATY = Y-sulfation site (SEQ ID NO: 250) Q = Glutamine (Q) glycosylation site(S/T) = Non-consensus Asparaginal (N) glycosylation site

FIGS. 2A-2F (CON'T)

FIG. 2D aTAU FAB CONSTRUCT:

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

aTAU Heavy Chain (SEQ ID NO: 53)

EVKVVESGGG LVQPGGSMKL SCVVSGETFS NYWVNWVRQA PGKGLEWVAQ IRLKSDNYAT
HYEESVKGRF TISRDDSKSS VYLQMNRLRA EDSGIYYCTN WEDYWGQGT VTVSSASTKG
PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSNISGA LTSGVHTFPA VLQSSGLYSL
SSVVTVPSSS LGTKTYICNV DHKPSNTRVD KRVESKY +/- GPPCPPCPA +/- PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

aTAU Light Chain (SEQ ID NO: 54)

DIVLTQSPDS LAVSLGERAT ISCRASQSVS TSRYSYIHWHY QOKPGQPPKL LIKYASNLES
GVPSRFSGSG SGTDFILNIH PLEPEDFATY KCHHSWEIPL TFGQGTKLEI KRTVAAPSVF
IFPPSDEQLK SGTASVVCLL NNFYPREAKV QWKVDNALQS GMSQESVTEQ DSKDSTYSL
STLTLSKADY EKHKVYACEV THQGLSSPVT KSIIRGEC

Heavy Chain Legend:

CDRs are underscored (predicted for aTAU)

EDTAVYY and WNEDYW = Y-sulfation site (SEQ ID NOS 248 and 252)

QGT = Glutamine (Q) glycosylation site

(S/T)IN = Non-consensus Asparaginal (N) glycosylation site

GPPCPPCPAPEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for aTAU)

EDFATYY = Y-sulfation site (SEQ ID NO: 250)

QGT = Glutamine (Q) glycosylation site

(S/T)IN = Non-consensus Asparaginal (N) glycosylation site

FIGS. 2A-2F (CON'T)

FIG. 2E ERENUMAB FAB CONSTRUCT:

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Erenumab Heavy Chain (SEQ ID NO: 55)

QVQLVESGGG VVQPGRLRL SCAASGFTFS SFGMHVVRQA PGKGLEWVAV ISFDGSIKYS 60
VDSVKGRFTI SRDNTLTLE LQMNSLRAD TAVYTCARDR LNYYDSSGYY HYKYYGMAVW 120
GCGTIVTVSS ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPTVS ~~W~~SGALTSGV 180
HTFPAVLQSS GLYSLSSVVT VPSSENGFTQT YTCNVDHKPS NTKVDKTVR KCCVE ~~+/~~CEPCFA ~~+/~~PPVAG
235

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Erenumab Light Chain (SEQ ID NO: 56)

QSVLTQPPSV SAAPGQKVTI SCSSGSSNIG NNYVSWYQQL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT STTLGITGLQ TQDEADYYCG TWDSRLSAVV FGGGTKLTVL GOPKANPTVT 120
LFPSSSEELQ ANKATLVCLI SDFYPGAVTV AWKADGSEVK AGVETTKPSK ~~Q~~SNKYAASS 180
YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS 216

Heavy Chain Legend:

CDRs are underscored (reported for Erenumab)

EDTAVYI = Y-sulfation site (SEQ ID NO: 248)

GGT = Glutamine (Q) glycosylation site

(S/T)N = Nonconsensus Asparaginal (N) glycosylation site

VECPPCPAPPVAG = Hinge region (SEQ ID NO: 253)

Light Chain Legend:

CDRs are underscored (reported for Erenumab)

QDEADYY = Y-sulfation site (SEQ ID NO: 256)

(S/T)N = Nonconsensus Asparaginal (N) glycosylation site

FIGS. 2A-2F (CON'T)

FIG. 2F BAN2401 Fab IgG1 CONSTRUCT:

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

BAN2401 Heavy Chain (SEQ ID NO: 57)

EVQLVESGGG LVQPGGSLRL SCSASGFTFS SFGMHVVRQA PGKGLEWVAY ISSGSSTIYY 60
GDTVKGRTI SRDPAKNSLF LQMSSLRAED EDTAVYYCAREG GYYYGRSYYT MDYWG QTV 120
TVSSASTKGP SVFPLAPSSK STSGGTAALG CLVKDYFPEP VTV SWNSGAL TSGVHTFPAV 180
LQSSGLYSLS SVVTPSSSL GTQTYICNVN HKPSNTKVDK KVEPKSCD DKTHLCPPCPAPELLGG 165
DKTHLCPPCPAPELLGG DKTHLCPPCPAPELLGG

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

BAN2401 Light Chain (SEQ ID NO: 58)

DVVMTQSPLS LPVTPGAPAS ISCRSSQSIV HSGNNTYLEW YLQKPGQSPK LLIYKVSNRF 60
SGVPDRFSGS GSGTDFTLRI SRVEAEDVGI EDVGIYCQSHVP PTFGPGTKLE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPRKAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TK SWNRGEC 219

Heavy Chain Legend:

CDRs are underscored

EDTAVYY = Y-sulfation site (SEQ ID NO: 248)Q = Glutamine (Q) glycosylation site(S/T)XN = Non-consensus Asparaginal (N) glycosylation siteDKTHLCPPCPAPELLGG = Hinge Region (SEQ ID NO: 249)

Or

DKTHLCPPCPAPELLGG = Hinge Region (SEQ ID NO: 254)

Light Chain Legend:

CDRs are underscored

EDVGIY = Y-sulfation site (SEQ ID NO: 255)(S/T)XN = Non-consensus Asparaginal (N) glycosylation site

FIGS. 3A-3E

FIG. 3A DUPILUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMLLLLLIALSLALVTNS

Dupilumab Heavy Chain (SEQ ID NO: 7)

EVQLVESGGG LEQPGGSLRL SCAGSGFTFR DYAMTWVRQA PGKGLEWVSS ISGSGGNTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED IAVYKCAKDR LSITIRPRYY GLDVWGQST 120
VTVSSASTKG PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTVPSST LGTKTYTCNV DHKPSNTKVD KRVESKY +/- GPPCPFPCEA 240
PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMLLLLLIALSLALVTNS

Dupilumab Light Chain (SEQ ID NO: 8)

DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL YSIGYNYLDW YLQKSGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEQVGF YFCMQALQTP YTFGSKLE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

Heavy Chain Legend:

CDRs are underscored (predicted for Dupilumab)

KQTAFT = Y-sulfation site (SEQ ID NO: 248)

QQT = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginyl (N) glycosylation site

GPPCPFPCEAFLQG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for Dupilumab)

EDVGFY = Y-sulfation site (SEQ ID NO: 257)

QQT = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginyl (N) glycosylation site

FIGS. 3A-3E (CON'T)

FIG. 3B IXEKIZUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Ixekizumab Heavy Chain (SEQ ID NO: 9)

QVQLVQSGAE VKKPGSSVKV SCKASGYSET DYHIHWVRQA PGQGLEWMGV INPMYGTIDY 60
 NORFKGRVTI TADESTSTAY MELSSLRSED TAVYYCARYD YFTGTGVYWG QGLLVTVSSA 120
 STKGPSVEPL APCSRSTSES TAALGCLVKD YFPEPVTV SW NSGALTSGVH TFPAVLQSSG 180
 LYSLSVVTV PSSSLGTKTY TCNVDHKPSN TKVDKRVESK Y +/- GPPCPPCPA +/- PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Ixekizumab Light Chain (SEQ ID NO: 10)

DIVMTQTPLS LSVTPGQPAS ISCRSSRSIV HSRGNTYLHW YLQKPGQSPQ LLIYKVSNRF 60
 IGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YFCQSSTHLP FTFG QSKLE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
 SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKS ENRGEC 219

Heavy Chain Legend:

CDRs are underscored (predicted for

Ixekizumab)

EDTAVYY = Y-sulfation site (SEQ ID NO: 248)

QGL = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

GPPCPPCPAPEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for

Ixekizumab)

EDVGVY = Y-sulfation site (SEQ ID NO: 251)

QSK = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

FIGS. 3A-3E (CON'T)

FIG. 3C SECUKINUMAB IgG1 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Secukinumab Heavy Chain (SEQ ID NO: 11)

EVQLVESGGG LVQPGGSLRL SCAASGFTFS NYWMNWVRQA PGKGLEWVAA INQDGSEKYY 60
VGSVKGRETI SRDNAKNSLY LQMNSLRVED TAVYYCVRDY YDILTDYTH YWYFDLWGRG 120
TLVTSSAST KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVMNS GALTSGVHTF 180
PAVLQSSGLY SLSSVTVPS SSLGTQTYIC NVNHKPSNTK VDKRVEPKSC D +/- KTHI (or KTHL)
+/- CPPCPA +/- PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGF

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Secukinumab Light Chain (SEQ ID NO: 12)

EIVLTQSPGT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGIP 60
DRFSGSGSGT DFTLTISRLE PEPFAYFCQ QYGSSPCTFG GTRLEIKRT VAAPSVFIFP 120
PSDEQLKSGT ASVVCLLNNF YPREAKVQWK VDNALQSGNS QESVTEQDSK DSTYLSSTL 180
TLKADYEKH KVIACEVTHQ GLSSPVTKSF NRGEC 215

Heavy Chain Legend:

CDRs are underscored (predicted for Secukinumab)

EDTAVY and YDILTDYTH = Y-sulfation site (SEQ ID NOS 248 and 258)

GTR = Non-consensus Asparaginal (N) glycosylation site

DKTHICPPCPAPELLG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Secukinumab)

EPFAYFY = Y-sulfation site (SEQ ID NO: 259)

GTR = Glutamine (Q) glycosylation site

GTR = Non-consensus Asparaginal (N) glycosylation site

FIG. 3D USTEKINUMAB IgG1 Construct

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Ustekinumab Heavy Chain (SEQ ID NO: 13)

EVQLVQSGAE VKKPGESLKI SCKGSGYSFT TYWLQWVRQM PGKGLDWIGI MSPVDSDIRY 60
SPSFQGVITM SVDKSITTAY LQWNSLKASD TAMYYCARRR PGQGYFDFWG GTRLVTVSSS 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVM NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCD +/- KTHI (or KTHL) +/- CPPCPA
+/- PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGF

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Ustekinumab Light Chain (SEQ ID NO: 14)

DIQMTQSPSS LSASVGDRVT ITCRASQGIS SWLAWYQQK EKAPKSLIYA ASSLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EPFAYFCQ YNIYPYTFG GTRLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQ LSSPVTKSF NRGEC 214

Heavy Chain Legend:

CDRs are underscored (reported for Ustekinumab)

GTR = Glutamine (Q) glycosylation site

GTR = Non-consensus Asparaginal (N) glycosylation site

DKTHICPPCPAPELLG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (reported for Ustekinumab)

EPFAYFY = Y-sulfation site (SEQ ID NO: 250)

GTR = Glutamine (Q) glycosylation site

GTR = Non-consensus Asparaginal (N) glycosylation site

FIGS. 3A-3E (CON'T)

FIG. 3E MEPOLIZUMAB IgG1 Fab Construct

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Mepolizumab Heavy Chain (SEQ ID NO: 15)

QVTLRESGPA LVKPTQTTLT TCTVSGFSLT SYSVHWVRQP PGKGLEWLGW IWASGGTDYN 60
 SALMSRLSIS KDTERRQVVL TMTNMDPVDT ATYYCARDPP SSLLRLDYWG RGTPVTVSSA 120
 STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSN NSGALTSGVH TFPVAVLQSSG 180
 LYSLSVVTV PSSSLGTQTY ICNVNHHKPSN TKVDKRVEPK SCD +/- KTHT (or KTBL) +/- CPPCPA
 +/-PELLGQPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGF

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Mepolizumab Light Chain (SEQ ID NO: 16)

DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSGNQKNYLA WXQOKPGQPP KLLIYGASTR 60
 ESGVPDRESG SGSGTDFTLT ISSLAEDVA VYYQNVHSE PFTFGGGTKL EIKRTVAAPS 120
 VFIFPPSDEQ LKSGTASVVC LLNNEYPREA KVQWKVDNAL QSGNSQESVT EQDSKDSTYS 180
 LSSTLTLSKA DYEKHKVYAC EVTHQGLSSP VTKSPNRGEC 220

Heavy Chain Legend:

CDRs are underscored (reported for Mepolizumab)

DTATYY = Y-sulfation site (SEQ ID NO: 260)

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

DKTHTCPPCPAPELGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (reported for Mepolizumab)

EDVAVYY = Y-sulfation site (SEQ ID NO: 261)

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 4A-4B

FIG. 4A VEDOLIZUMAB IgG1 Fab CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Vedolizumab Heavy Chain (SEQ ID NO: 17)

QVQLVQSGAE VKKPGASVKV SCKGSGYTFT SYWMHWVRQA PGQRLEWIGE IDPSESNTNY 60
 NQKFKGRVIL TVDISASTAY MELSSLRSED TAVYYCARGG YDQWDYAIQY WGQSTLTVTS 120
 SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTY SWSGALTSG VHTFPAVLQS 180
 SGLYSLSSVV TVPSSSLGTQ TYICNVNHPK SNTKVDKKVE PKSCD +/- KTHL (or KTHL) +/-
 CPPCPA // PEIACAPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Vedolizumab Light Chain (SEQ ID NO: 18)

DVVMTQSPFS LPVTPGEPAS ISCRSSQSLA KSYGNTYLSW YLQKPGQSPQ LLIYGISNRF 60
 SGVPDRFSGS GSGTIDFTLKI SRVEAEDGVV FYCLQGTROP YTFGQKVE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSSTYSL 180
 SSTLTLSKAD YEKHKVYACE VTHQGLSPV TKSTPRGEC 219

Heavy Chain Legend:

CDRs are underscored (predicted for Vedolizumab)

EDTAVYY and YDQWDYAIQY = Y-sulfation site (SEQ ID NOS 248 and 262)

Q = Glutamine (Q) glycosylation site

(S/T/N) = Non-consensus Asparaginal (N) glycosylation site

PKSCD +/- KTHL (or KTHL) +/- = Hinge region (SEQ ID NO: 263)

Light Chain Legend:

CDRs are underscored (predicted for Vedolizumab)

YLQKPGQSPQ = Y-sulfation site (SEQ ID NO: 251)

Q = Glutamine (Q) glycosylation site

(S/T/N) = Non-consensus Asparaginal (N) glycosylation site

FIG. 4B NATALIZUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Natalizumab Heavy Chain (SEQ ID NO: 19):

QVQLVQSGAE VKKPGASVKV SCKASGFENIK DTYIHWVRQA PGQRLEWMGR IDPANGYTKY 60
 DPKEQGRVIT TADTSASTAY MELSSLRSED TAVYYCAREG YYGNYGVYAM DYWGQSTLVT 120
 VSSASTKGPS VFPLAPCSRS TSESTAALGC LVKDYFPEPV TVSWWSGALT SGVHTFPAVL 180
 QSSGLYSLSS VVTVPSSSLG TKTYTCNVVDH KPSNTKVDKR VESKY +/- GPPCPFCPA // PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Natalizumab Light Chain (SEQ ID NO: 20):

DIQMTQSPSS LSASVGDRVT ITCKTSQDIN KYMAWYQOTP GKAPRLLIHY TSALQPGIPS 60
 RFSGSGSGRD YTFTISSLPQ EDIATYYCLQ YDNLWTFGQKVEIKRTVA APSVFIFPPS 120
 DEQLKSGTAS VVCLLNNFYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSSTLTL 180
 SKADYEKHKV YACEVTHQGL SSPVTKSTPR GEC 213

Heavy Chain Legend:

CDRs are underscored.

EDTAVYY = Y-sulfation site (SEQ ID NO: 248).

Q = Glutamine (Q) glycosylation site.

(S/T/N) = Non-consensus Asparaginal (N) glycosylation site.

GPPCPFCPA // PEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored.

EDIATYY = Y-sulfation site (SEQ ID NO: 264).

Q = Glutamine (Q) glycosylation site

(S/T/N) = Non-consensus Asparaginal (N) glycosylation site.

FIGS. 5A-5C

FIG. 5A ALIROCUMAB Fab IgG1 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMOLLLIALSLALVTNS

Alirocumab Heavy Chain (SEQ ID NO: 21)

EVQLVESGGG LVQPGGSLRL SCAASGFF~~EN~~ NYAMNWVROA PGKGLDWVST ISGSGG~~TTY~~ 60
ADSVKGRFTI SRDSSKHTLY LQMNLSRAED LAYVYCAKDS NWGNFDLWGR GTLVTVSSAS 120
TRGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTV~~SW~~ SGALTSGVHT FPAVLQSSGL 180
YSLSSVTVTP SSSLGTQTYI CNVNHKPSNT KVDKKVEPKS CD +/- ~~KTHT~~ (or KTHL) +/- CPFCPA
- PEILGPSVFT,

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKOTLNFDLLKLAGDVESNFGP

Leader (SEO ID NO: 161)

MYRMOILLIALSLALVTNS

Alirocumab Light Chain (SEQ ID NO: 22)

DIVMTQSPDS LAVSLGERAT INCKSSQSVL YRSHNRNFLG WYQKPGQPP NLLIYWASTR 60
 ESGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCQYYTT PYTFGQK KL EIKRTVAAPS 120
 VFIFPPSDEQ LKSGTASVVC LLNNFYPREA KVQWKVDNAL QSGNSQESVT EQDSKSTYS 180
 LSSTLTLSKA DYEKKHVIAC EVTHOGLSSP VTKSTNRGEC 220

Heavy Chain Legend:

CDRs are underscored (predicted for Alirocumab)

S/T/A = Non-consensus Asparaginal (N) glycosylation site

E/D/V/Y = Y-sulfation site (SEQ ID NO: 248)

DKTHICPPCPAPELGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Alirocumab)

SYYY = Y-sulfation site (SEQ ID NO: 261)

STN = Non-consensus Asparaginal (N) glycosylation site

G = Glutamine (Q) glycosylation site

FIG. 5B EVOLUCOMAB Fab IgG2 CONSTRUCT

Leader (SEO ID NO: 161)

MYRMOLLIALLSLALVTNS

Evolucomab Heavy Chain (SEQ ID NO: 23)

EVQLVQSGAE VKKPGASVKV SCKASGYTLT SYGISWVRQA PGQGLEWMGW VSFYNGNTNY 60
 AQKLQGRGTM TTDPTSTAY MELRSLRSDD TAVYTCARGY GMDVWGQGT VTVSSASTKG 120
 PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTQSNKSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSK EGTQTYTCNV DHKPSNTKVD KIVERKCCVE +/- CPPCPA +/- PPVAG 231

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKOTLNFDLLKLAGDVESNP GP

Leader (SEQ ID NO: 161)

MYRMOLLLIALSLALVTNS

Evolucomab Light Chain (SEQ ID NO: 24)

ESALSTQPASV SGSPGQSITI SCTGTSSDVG GYNSVSWYQQ HPGKAPKLMI YEVSNRPSGV 60
SNRFGSKSGC NTASLTISGL QAEDDADYYC NSYTSTSMVF GGGTKLTVLG QPKAAPSVTL 120
FPPSSEELQA NKATLVCLIS DFYPGAVTVA WKADSSPVKA GVETTTPSKQ SNNKYAASSY 180
LSLTPEQWKS HRSYSQVTH EGSTVEKTVA PTECS 215

Heavy Chain Legend:

CDRs are underscored (predicted for Evolucomab)
DRIATYY = Y-sulfation site (SEQ ID NO: 265)
QQT = Glutamine (Q) glycosylation site
TS/TXX = Nonconsensus Asparaginyl (N) glycosylation site

VECPFCFAPFVAG = Hinge region (SEQ ID NO: 253)

Light Chain Legend:

CDRs are underscored (predicted for
Evolucomab)
EDEADYY = Y-sulfation site (SEQ ID NO:
266)

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FIGS. 5A-5C (CON'T)

FIG. 5C EVINACUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLIALSLALVTNS

Evinacumab Heavy Chain (SEQ ID NO: 25)

EVQLVESGGG VIQPGGSLRL SCAASGFTFD DYAMNWVROG PGKGLEWVSA ISGDGGSTYY 60
 ADSVKGRFTI SRDN~~SN~~SLY LQMNSLRAED ~~TAFFY~~CAKDL RNTIFGVVIP DAFDIWG~~Q~~ 120
 MVTVSSASTK GPSVFPLAPC SRSTSESTAA LGCLVKDYFP EPVTV~~SW~~SG ALTSGVHTFP 180
 AVLQSSGLYS LSSVTVFPSS SLGTKITYTCN VDHKPSNTKV DKRVESKYGP P +/- CPDCEA +/-
 PEFLC~~Q~~PSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLIALSLALVTNS

Evinacumab Light Chain (SEQ ID NO: 26)

DIQMTQSPST LSASVGDRVT ITCRASQSI~~R~~ SWLAWYQQKP GKAPKLLIYK ASSLESGVPS 60
 RFSGSGSGTE FTLTISSLPQ ~~DDFATYYCQ~~Q YNSYSYTFG~~Q~~ ~~Q~~KLEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNFEY PREAKVQWKV DNALQ~~SG~~ESQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTK~~SR~~R RGEK 214

Heavy Chain Legend:

CDRs are underscored (predicted for Evinacumab)

~~EDTAFFY~~ = Y-sulfation site (SEQ ID NO: 267)

~~Q~~ = Glutamine (Q) glycosylation site

~~(S/T)N~~ = Nonconsensus Asparaginal (N) glycosylation site

~~CPDCEAPEFLC~~ = Hinge region (SEQ ID NO: 268)

Light Chain Legend:

CDRs are underscored (predicted for Evinacumab)

~~DDFATYY~~ = Y-sulfation site (SEQ ID NO: 269)

~~(S/T)N~~ = Nonconsensus Asparaginal (N) glycosylation site

~~Q~~ = Glutamine (Q) glycosylation site

FIG. 5D E06 Single Chain Construct

Leader (SEQ ID NO: 244):

MYRMQLLLIALSLALVTNS

E06 Light Chain (SEQ ID NO: 60)

DIVMTQSPSS LSVSAGKKVT ISCTASESLY SSKHKVHYLA WYQ~~KK~~FEQSP KLLIYGASNR 60
 YIGVPDRFTG SGSGTDFTLT ISSVQVEDLT HYYCAQFY~~SY~~ PLTFGAGTKL EIK 113

Linker (SEQ ID NO: 245)

GGGGSGGGSGGGGS

E06 Heavy Chain (SEQ ID NO: 59)

EVKLVESGGG LVQPGGSLRL SCATSGFTFS DFYMEWVRQA PGKRLEWIAA ~~SR~~KANDYIT 60
~~EY~~ADSVKGRF IVSRDTSQSI LYLQMNALRA ~~EDTAFFY~~CAR DYYGSSYWF DVWGAGTIVT 120
 VSS

Legend

Light Chain:

~~YQKKE~~ = Y-sulfation site (SEQ ID NO: 270)

Heavy Chain:

~~SR~~ = Non-consensus Asparaginal (N) glycosylation site

~~DYITTEY~~, ~~EDTAFFY~~ = Y-sulfation site (SEQ ID NOS 271 and 272)

FIG. 6

FIG. 6 DENOSUMAB Fab IgG1 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Denosumab Heavy Chain (SEQ ID NO: 27)

EVQLLES^{GGG} LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSG ITGSGGSTYY 60
 ADSVEGRFTI SRD^{SKNT}LY LQMNSLRAED TAVTYCAKDP GTTVIMSWFD PWG^{GG}LVTV 120
 SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFFPEPVT V^{SW}SGALTS GVHTFPAVLQ 180
 SSGLYSLSSV VTVFSSSLGT QTYICNVNHK PSNTKVDKKV EPKSCD +/- ^{KTHT (or KTHL) +/-}
 CFFCPA +/- FELLCP^{PSVFL}

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Denosumab Light Chain (SEQ ID NO: 28)

EIVLTQSPGT LSLPGERAT LSCRASQSVR GRYLAWYQQK PGQAPRLLIY GASSRATGIP 60
 DRFSGSGSGT DFTLTISRLE PEDFAVEYCQ QYGSSPRTFG ^{GG}KVEIKRT VAAPSVFIFF 120
 PSDEQLKSGT ASVVCLLNNE YPREAKVQWK VDNALQ^{SGNS} QESVTEQDSK DSTYLSSTL 180
 TLSKADYEKH KVIACEVTHQ GLSSPVT^{SR}RGEC 215

Heavy Chain Legend:

CDRs are underscored (predicted for Denosumab)

^{EDFAVYY} = Y-sulfation site (SEQ ID NO: 248)

^{GG} = Glutamine (Q) glycosylation site

^{(S/T)X} = Non-consensus Asparaginal (N) glycosylation site

^{DRTHICFFCPAPELLCG} = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Denosumab)

^{EDFAVYY} = Y-sulfation site (SEQ ID NO: 273)

^{GG} = Glutamine (Q) glycosylation site

^{(S/T)X} = Non-consensus Asparaginal (N) glycosylation site

FIGS. 7A-7B

FIG. 7A NIVOLUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Nivolumab Heavy Chain (SEQ ID NO: 29)

QVQLVESGGG VVQPGRSRLR DCKASGITFS NSGMHWVRQA PGKGLEWVAV IWYDGSKRYY 60
 ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATND DYWGQCTLVLT VSSASTKGPS 120
 VFPLAPCSRS TSESTAALGC LVKDYFPEPV TVSNNSGALT SGVHTFPAVL QSSGLYSLS 180
 VVTVPSSSLG TKTYTCNVDH KPSNTKVDKR VE SKY +/- GPPCPPCPA +/- PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Nivolumab Light Chain (SEQ ID NO: 30)

EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKE GQAPRLLIYD ASNRATGIPA 60
 RFGSGSGGTD FTLTISSLEP EDFAVYYCOQ SSNWPRTEFG QKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

Heavy Chain Legend:

CDRs are underscored (predicted for Nivolumab)

EDFAVYY = Y-sulfation site (SEQ ID NO: 248)

QQT = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

GPPCPPCPAPEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for Nivolumab)

EDFAVYY = Y-sulfation site (SEQ ID NO: 259)

QQT = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

FIG. 7B PEMBROLIZUMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Pembrolizumab Heavy Chain (SEQ ID NO: 31)

QVQLVQSGVE VKKPGASVKV SCKASGYIFT NYMYWVRQA PGQGLEWMGG INPSNGGTNF 60
 NEKFKNRVTL TDSSTTTAY MELKSLQFDD TAVYYCARRD YRFDMGFDYW GGGTIVTVSS 120
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS NNSGALTSGV HTFPAVLQSS 180
 GLYLSVSVVT VPSSSLGTKT YTCNVDHKPS NTKVDKRVES KY +/- GPPCPPCPA +/- PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Pembrolizumab Light Chain (SEQ ID NO: 32)

EIVLTQSPAT LSLSPGERAT LSCRASKGVS TSGYSYLHWY QQKPGQAPRL LIYLAAYLES 60
 GVPARFSGSG SGTIDFTLTIS SLEPEDFAVY YQHSRDLP TFGGGTKVEI KRTVAAPSVF 120
 IFPPSDEQLK SGTASVCLL NNFYPREAKV QWKVDNALQSGNSQESVTEQ DSKDSTYLS 180
 STLTLKADY EKHKVYACEV THQGLSSPVT KSFNRGEC 218

Heavy Chain Legend:

CDRs are underscored (predicted for Pembrolizumab)

EDFAVYY = Y-sulfation site (SEQ ID NO: 265)

QQT = Glutamine (Q) glycosylation site

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

GPPCPPCPAPEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored (predicted for Pembrolizumab)

EDFAVYY = Y-sulfation site (SEQ ID NO: 259)

(S/T)X = Non-consensus Asparaginal (N) glycosylation site

FIGS. 8A-8H

FIG. 8A RANIBIZUMAB IgG1 CONSTRUCT

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Ranibizumab Heavy Chain (SEQ ID NO: 33):

EVQLVESGGG LVQPGGSLRL SCAASGYDET HYGMINWVRQA PGKGLEWVGW INTYTGEPTY 60
 AADFKRRFTF SLDTSKSTAY LQMNSLRAED TAVYYCAKYP YYYGTSHWYF DVWGQGLVT 120
 VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWMSGALT SGVHTFPAVL 180
 QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCD +/- KTHI (or KTHL) +/-
 CFFCPA +/- FELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Ranibizumab Light Chain (SEQ ID NO: 34):

DIQLTQSPSS LSASVGDRVT ITCSASQDIS NYLNWYQQKPK GKAPKVLIIYF TSSLHSGVPS 60
 RFSGSGSGTD FTLTISSLQP EDFAIYYCQQ YSTVPTFTGQ GIKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKRW RGEK 214

Heavy Chain Legend:

CDRs are underscored

EDFAIYY = Y-sulfation site (SEQ ID NO: 248)

Q = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

DRTHICFFCPAPELGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored

EDFAIYY = Y-sulfation site (SEQ ID NO: 250)

Q = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 8A-8H (CON'T)

FIG. 8B BEVACIZUMAB IgG1 Fab CONSTRUCT

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Bevacizumab Heavy Chain (SEQ ID NO: 35):

EVQLVESGGG LVQPGGSLRL SCAASGYTFT NYGMNWVRQA PGKGLEWVGW INTYTGEPTY 60
 AAEFKRRRTF SLDTSKSTAY LQMNSLRAED TAVYYCARYP HYYGSSHWYF DVWGKLLVT 120
 VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYPPEPV TVSWNSGALT SGVHTFPAVL 180
 QSSGLYSLSV VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCD +/- KTHI (KTHL) +/- CPPCPA
 +/- PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Bevacizumab Light Chain (SEQ ID NO: 36):

DIQMTQSPSS LSASVGDRVT ITCSASQDIS NYLNWYQQK E GAPKVLIIYF TSSLHSGVPS 60
 RFGSGSGTD FTLTISSLQP EDAFYFCQQ YSTVPWTFG G KVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

Heavy Chain Legend:

CDRs are underscored

EEYAVYY = Y-sulfation site (SEQ ID NO: 249)

QQT = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

DKTHICEPCPAPELLGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored

EDYAVYY = Y-sulfation site (SEQ ID NO: 274)

QQT = Glutamine (Q) glycosylation site (SEQ ID NO: 275)

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 8A-8H (CON'T)

FIG. 8C LAMPALIZUMAB IgG1 Fab CONSTRUCT

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Lampalizumab Heavy Chain (SEQ ID NO: 37):

EVQLVQSGPE LKKPGASVKV SCKASGYFT NYGMNWVROA PGQGLEWMGW INTYTGETTY 60
 ADDFKGRFVF SLDTSVSTAY LQISLKAED TAVYYCEREG GVNNWC~~Q~~SL VTVSSASTKG 120
 PSVFPLAPSS KSTISGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSSS LGTQTYICNV NHKPSNIKVD KKVEPKSCD +/- KTHI (or KTHL) +/- CPPCPA +/-
 PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Lampalizumab Light Chain (SEQ ID NO: 38):

DIQVTQSPSS LSASVGDRVT ITCITSTDID DDMNWYQKP GKVPKLLISG GNTLRPGVPS 60
 RFGSGSGTD FTLTISSLQP EDVATYYCLQ SDSLPTIFG~~Q~~ GKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYERHK VYACEVTHQ LSSPVTK~~SEN~~ RGEC 214

Heavy Chain Legend:

CDRs are underscored (predicted for Lampalizumab)
~~EV~~YAD~~Q~~ and ~~ED~~VATYY = Y-sulfation site (SEQ ID NOS 276 and 248)
~~Q~~ = Glutamine (Q) glycosylation site
~~(S/T)~~Y = Non-consensus Asparaginal (N) glycosylation site
~~DKTHICPPCPAPELLGG~~ = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Lampalizumab)
~~ED~~VATYY = Y-sulfation site (SEQ ID NO: 274)
~~Q~~ = Glutamine (Q) glycosylation site
~~(S/T)~~Y = Non-consensus Asparaginal (N) glycosylation site

FIG. 8D BROLUCIZUMAB SINGLE CHAIN CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLIALLSLALVTNS

Brolucizumab Light Chain (SEQ ID NO: 40):

EIVMTQSPST LSASVGDRVI ITCQASEIIH SWLAWYQKP GKAPKLLIYL ASTLASGVPS 60
 RFGSGSGAE FTLTISSLQP ~~DD~~FATYYCQN VYLASTNGAN FG~~Q~~KLTVL G 111

Linker (SEQ ID NO: 246):

GGGGSGGGSGGGSGGGGS

Brolucizumab Heavy Chain (SEQ ID NO: 39):

EVQLVESGGG LVQPGGSLRL SCTASGFSLT DYWMTWVRQ APGKGLEWVG FIDPDDDPYY 60
 ATWAKGRFTI SRDN~~SKNTLY~~ LQMNSLRAED TAVYYCAGGD HNSGWGLDIW GGG~~Q~~LTVSS 120

Heavy Chain Legend:

CDRs are underscored (predicted for Brolucizumab)
~~DD~~FATYY = Y-sulfation site (SEQ ID NO: 269)
~~Q~~ = Glutamine (Q) glycosylation site
~~(S/T)~~Y = Non-consensus Asparaginal (N) glycosylation site

Light Chain Legend:

CDRs are underscored (predicted for Brolucizumab)
~~ED~~VATYY and ~~SK~~TDPYY and ~~DD~~DPYY = Y-sulfation site (SEQ ID NOS 274, 277 and 278)
~~Q~~ = Glutamine (Q) glycosylation site
~~(S/T)~~Y = Non-consensus Asparaginal (N) glycosylation site

FIGS. 8A-8H (CON'T)

FIG. 8E BELIMUMAB IgG1 Fab CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Belimumab Heavy Chain (SEQ ID NO: 41):

QVQLQQSGAE VKRPGSSVRV SCKASGCTEN NNAINWVRQA PGQGLEWMGG IIPMEGTAKY 60
 SQRFQGRVAI TADESTGTAS MELSSLRSED TAVYFCARSR DLLLEPHHAL SEWGR GTMVT 120
 VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSSSGALT SGVHTFPAVL 180
 QSSGLYSLSV VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCD +/- KTHT (or KTHL) +/-
 CPPCPA +/-PELLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Belimumab Light Chain (SEQ ID NO: 42):

SSELTQDPAV SVALGQTVRV TCQGDLSRSY YASWYQOKPG QAPVLVIYVK NNRPSGIPDR 60
 FSGSSSGDTA SLTITGAQAE DEADYFCSSR DSSGHHWVFG GGTELTVLGQ PKAAPSVTLF 120
 PPSSEELQAN KATLVCLISD FYPGAVTVAW KADSSPVKAG VETTPSKQS NNKYAASSYL 180
 SLTPEQWKSH RSYSCQVTHE GSTVEKTVAP TECS

Heavy Chain Legend:

CDRs are underscored (predicted for Belimumab)

SYAVY = Y-sulfation site (SEQ ID NO: 248)

STAN = Non-consensus Asparaginal (N) glycosylation site

STHTEPCFAPELLSS = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Belimumab)

DEADY = Y-sulfation site (SEQ ID NO: 266)

STAN = Non-consensus Asparaginal (N) glycosylation site

FIG 8F. ECULIZUMAB FAB CONSTRUCT

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Eculizumab Heavy Chain (SEQ ID NO: 43)

QVQLVQSGAE VKKPGASVRV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE ILPGSGSTEV 60
 LRFKDRVTM TRDTSTSTVY MELSSLRSED TAVYFCARYF FGSSPNWYFD VWGQGLVTV 120
 SSASTKGPSV FPLAPCSRST SESTAAALGCL VKDYFPEPVT VSSSGALTS GVHTFPAVLQ 180
 SSGLYSLSSV VIVPSSNFGT QTYTCNVDHK PSNTKVDKTV ERKCCVE +/- CPPCPA +/- PFVAG 238

Furin-F2A Linker (SEQ ID NO: 242)

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161)

MYRMQLLLLLIALSLALVTNS

Eculizumab Light Chain (SEQ ID NO: 44)

DIQMTQSPSS LSASVGDRVT ITCGASENIY GALNWIYQOKP GKAPKLLIYG ATNLADGVPS 60
 RFGSGSGSGTD FTLTISSLQP EDIATYFCQN VLNTPITFGQ EAKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNFEY PREAKVQWKV DNALQSSNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

Heavy Chain Legend:

CDRs are underscored (predicted for Eculizumab)

QGL = Glutamine (Q) glycosylation site

EDIATY = Y-sulfation site (SEQ ID NO: 248)

STAN = Non-consensus Asparaginal (N) glycosylation site

VECPFCFAPFVAG = Hinge region (SEQ ID NO: 253)

Light Chain Legend:

CDRs are underscored (predicted for Eculizumab)

QGL = Glutamine (Q) glycosylation site

EDIATY = Y-sulfation site (SEQ ID NO: 250)

STAN = Non-consensus Asparaginal (N) glycosylation site

FIGS. 8A-8H (CON'T)

FIG. 8G ANDECALIXIMAB Fab IgG4 CONSTRUCT

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Andecaliximab Heavy Chain (SEQ ID NO: 45):

QVQLQESGPG LVKPSSETLSL TCTVSGFSLL SYGVHWVRQP PGKGLEWLGW IWTGGITINYN 60
 SALMSRFTIS KDDSKNTVYL KMNSLKTEDI AIYYCARYYY GMDYWGCTLL VTVSSASTKG 120
 PSVFPLAPCS RSTSESTAAL GCLVRDYFPE PVTVSQNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVFSSS LGTKTYTCNV DHKPSNTKVD KRVESKY +/- GPFCPPCPA +/- PEFLGGPSVFL

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Andecaliximab Light Chain (SEQ ID NO: 46):

DIQMTQSPSS LSASVGDRVT ITCKASQDVR NTVAWYQQKP GKAPKLLIYS SSYRNTGVPD 60
 RFSGSGSGTD FTLTISSLQA EDVAVYTCQQ HYITPYTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYERHK VYACEVTHQG LSSPVTKSEN RGEK 214

Heavy Chain Legend:

CDRs are underscored.

EDVAVY = Y-sulfation site (SEQ ID NO: 279).

QST = Glutamine (Q) glycosylation site.

(S/T)XX = Non-consensus Asparaginal (N) glycosylation site.

GPFCPPCPA +/- PEFLGG = Hinge region (SEQ ID NO: 231)

Light Chain Legend:

CDRs are underscored.

EDVAVY = Y-sulfation site (SEQ ID NO: 261).

QST = Glutamine (Q) glycosylation site

(S/T)XX = Non-consensus Asparaginal (N) glycosylation site

FIG. 8H LANADELUMAB CONSTRUCT

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Lanadelumab Heavy Chain (SEQ ID NO: 47):

EVQLLESGGG LVQPGGSLRL SCAASGFTFS HYIMMWVRQA PGKGLEWVSG IYSSGGITVY 60
 ADSVKGRETI SRDNSKNTLY LQMNSLRAED TAVTYCAYRR IGVPRRDEFD IWGGIMVTV 120
 SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
 SSGLYSLSSV VIVPSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCD +/- KTHI (or KTHL) +/-
 CPPCPA
 +/- PEFLGGPSVFL 242

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Lanadelumab Light Chain (SEQ ID NO: 48):

DIQMTQSPST LSASVGDRVT ITCRASQSIG SWLAWYQQKP GKAPKLLIYK ASTLESQVPS 60
 RFSGSGSGTE FTLTISSLQP EDVAVYTCQQ YNTYWTFGG GTKVEIKRTVA APSVFIFPPS 120
 DEQLKSGTAS VVCLLNNFYF REAKVQWKVD NALQSGNSQ SVTEQDSKDS TYSLSSTLT 180
 SKADYERHKV YACEVTHQGL SSPVTKSEN RGEK 213

Heavy Chain Legend:

CDRs are underscored.

EDVAVY = Y-sulfation site (SEQ ID NO: 248).

QST = Glutamine (Q) glycosylation site.

(S/T)XX = Nonconsensus Asparaginal (N) glycosylation site

DKTHICPPCPAPEFLGG = Hinge region (SEQ ID NO: 249).

Light Chain Legend:

CDRs are underscored.

EDVAVY = Y-sulfation site (SEQ ID NO: 269)

QST = Glutamine (Q) glycosylation site.

(S/T)XX = Nonconsensus Asparaginal (N) glycosylation site.

FIGS. 9A-9B

FIG. 9A ADALIMUMAB Fab Construct

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Adalimumab Heavy Chain (SEQ ID NO: 49):

EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYAMHWVRQA PGKGLEWVSA ITWNSGHIDY 60
 ADSVEGRFTI SRDNAKNSLY LQMNSLRAED IAVYYCAKVS YLSTASSLDY WGQGLVTVS 120
 SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWISGALTSG VHTFPAVLQS 180
 SGLYSLSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKKVE PKSCD +/- KTHT (KTHL) +/- CPFCPA
 +/-PELLGGPSVF L

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQTLNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLIALLSLALVTNS

Adalimumab Light Chain (SEQ ID NO: 247)

DIQMTQSPSS LSASVGDRVT ITCRASQIR NYLAWYQQKPK GKAPKLLIYA ASTLQSGVPS 60
 RFGSGSGGTD FTLTISSLQP EDVATYYCQR YNRAPYTFGQ GTKVEIKRTV AAPSVFIFFP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSRN RGEC 214

Heavy Chain Legend:

CDRs are underscored (predicted for Adalimumab)
FTFDYA and EDVATYY = Y-sulfation site (SEQ ID NOS 280 and 248)
QST = Glutamine (Q) glycosylation site
(S/T)N = Non-consensus Asparaginal (N) glycosylation site
DRHTCPFCPAPELLGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Adalimumab)
EDVATYY = Y-sulfation site (SEQ ID NO: 274)
QST = Glutamine (Q) glycosylation site
(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIGS. 9A-9B (CONT'D)

FIG. 9B INFLIXIMAB Fab Construct

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Infliximab Heavy Chain (SEQ ID NO: 51):

EVKLEESGGG LVQPGGSMKL SCVASGFIFS NHWMNWVRQS PEKGLEWVAE IRSKSINSAT 60
 HYAESVKGRF TISRDDSKSA VYLQMTDLRT EDTGVTCSP NYYGSTYDYW GILTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTV WMSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCD +/- KTHL (KTHL) +/- CPPCPA
 +/- PELLGQPSVF L

Furin-F2A Linker (SEQ ID NO: 242):

RKRRAPVKQILNFDLLKLAGDVESNPGP

Leader (SEQ ID NO: 161):

MYRMQLLLLLIALSLALVTNS

Infliximab Light Chain (SEQ ID NO: 52)

DILLTQSPAI LSVSPGERVS FSCRASQFVG SSIHWYQQRT NGSPRELLIKY ASESMSGIPS 60
 RFGSGSGGTD FTLSINTVES EDIADYFCQQ SHSWPFTFGS GTNLEVKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

Heavy Chain Legend:

CDRs are underscored (predicted for Infliximab)

EDTGVTCSP = Y-sulfation site (SEQ ID NO: 281)

Q = Glutamine (Q) glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

DKYTHCPPEAPEELGG = Hinge region (SEQ ID NO: 249)

Light Chain Legend:

CDRs are underscored (predicted for Infliximab)

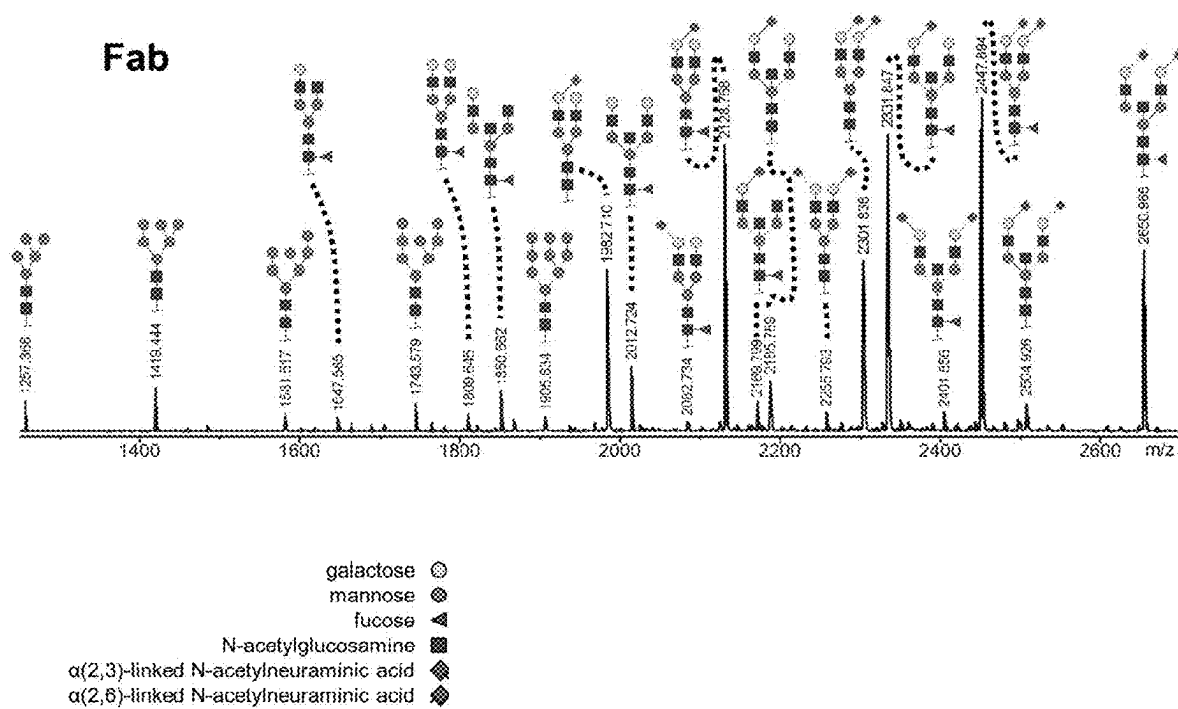
EDIADYFCQQ = Y-sulfation site (SEQ ID NO: 282)

Q = Glutamine (Q) glycosylation site

NY(S/T) = Consensus glycosylation site

(S/T)N = Non-consensus Asparaginal (N) glycosylation site

FIG. 10



FIGS. 11A-11B

FIG. 11A CLUSTAL O(1.2.4) ALIGNMENT OF THERAPEUTIC Fab HEAVY CHAIN

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Aducanumab   XVQLVESGGGVVQPGRSLRLSCAASGFAFSSY--GMHWVRQAPGKGLEWVAIVWFDG---TKK--YY 60
Crenezumab   EVQLVESGGGLVQPGGSLRLSCAASGFTFSSY--GMSWVRQAPGKGLELVASINSNG---GST--YY 60
Gantenerumab QVELVESGGGLVQPGGSLRLSCAASGFTFSSY--AMSWVRQAPGKGLEWVSAINASG---TRT--YY 60
Dupilumab    EVQLVESGGGLEQPGGSLRLSCAGSGFTFRDY--AMTWVRQAPGKGLEWVSSISGSG---GNT--YY 60
Ixekezumab   QVQLVQSGAEVKKPGSSVKVSKASGYSTFDY--HIHWVRQAPGQGLEWMGVINPMY---GTT--DY 60
Secukinumab  EVQLVESGGGLVQPGGSLRLSCAASGFTFSNY--WMNWVRQAPGKGLEWVAAINQDG---SEK--YY 60
Ustekinumab  EVQLVQSGAEVKKPGESLKISCKGSGYSFTTY--WLGWVRQMPGKGLDWIGIMSPVD----SDI--RY 60
Mepolizumab QVTLRESGPALVKPTQTTLTICTVSGFSLTSY--SVHWVRQPPGKGLEWLGVWIASG---G-T-DY 59
Vedolizumab  QVQLVQSGAEVKKPGASVKVSKASGYTFTSY--WMHWVRQAPGQRLEWIGIDPSE---SNTNY-- 60
Natalizumab  QVQLVQSGAEVKKPGASVKVSKASGFNIKDT-YIHWVRQAPGQRLEWMGRIDPAN---GYTKYD 61
Alirocumab   EVQLVESGGGLVQPGGSLRLSCAASGFTFNNTY--AMNWVRQAPGKGLDWVSTISGSG---GTTNY-- 60
Evolucumab   EVQLVQSGAEVKKPGASVKVSKASGYTFTSY--GISWVRQAPGQGLEWMGVVSFYN----GNTNY-- 60
Evinacumab   EVQLVESGGGVIQPGGSLRLSCAASGFTFDDY--AMNWVRQGPKGKLEWVSAISGDG---GST--YY 60
Denosumab    EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYA--MSWVRQAPGKGLEWVSGITGSG---GST--YY 60
Nivolumab    QVQLVESGGGVVQPGRSLRLDCKASGITFSNS--GMHWVRQAPGKGLEWVAIVWYDG---SKRY-- 60
Pembrolizumab QVQLVQSGVEVKKPGASVKVSKASGYTFTNY--YMWVRQAPGQGLEWMGGINPSN---GGTNF-- 60
Ranibizumab  EVQLVESGGGLVQPGGSLRLSCAASGYDFTY--GMNWVRQAPGKGLEWVGWINTYT---GEP--TY 60
E06.scFv     EVKLVESGGGLVQPGGSLRLSCATSGFTFSDY--YMEWVRQAPGKRLEWIAASRNKANDYTT--EY 62
BAN2401      EVQLVESGGGLVQPGGSLRLSCASGFTFSSF--GMHWVRQAPGKGLEWVAIVSSGS---STI--YY 60
Bevacizumab  EVQLVESGGGLVQPGGSLRLSCAASGYTFTNY--GMNWVRQAPGKGLEWVGWINTYT---GEP--TY 60
Erenumab     QVQLVESGGGVVQPGRSLRLSCAASGFTFSSF--GMHWVRQAPGKGLEWVAIVISF--D---GSIKYS 60
aTAU         EVKVVESGGGLVQPGGSMKLSCVSGFTFSNY--WVNWVRQAPGKGLEWVAQIRLKSNDYATHYE 63
Brolucizumab EVQLVESGGGLVQPGGSLRLSCTASGFTLDYYMTWVRQAPGKGLEWVGFIID----PDD--PYY 60
Lampalizumab EVQLVQSGPELKKPGASVKVSKASGYTFTNY--GMNWVRQAPGQGLEWMGWINTYT---GETTY-- 60
Andecaliximab QVQLQESGPGLVKPSSETLSLTCTVSGFSLLSY--GVHWVRQPPGKGLEWLGVWITGG---TT--NYN 60
Belimumab    QVQLQQSGAEVKKPGSSVRVSKASGTFNNN--AINWVRQAPGQGLEWMGGIIPMF---GTAKYS 61
Eculizumab   QVQLVQSGAEVKKPGASVKVSKASGYIFSNTY--WIQWVRQAPGQGLEWMGEILPGS---GSTEYT 61
Lanadelumab  EVQLLESGGGLVQPGGSLRLSCAASGFTFSHY--IMMWVRQAPGKGLEWVSGIYSSG---GIT--VY 60
Adalimumab   EVQLVESGGGLVQPGRSLRLSCAASGFTFDDY--AMHWVRQAPGKGLEWVSAITWNS---GHIDY-- 60
Infliximab   EVKLEESGGGLVQPGGSMKLSCVASGFIISNH--WMNWVRQSPEKGLEWVAEIRSKSINSATHY-- 62
      * * : ** : : * :: : * ** : : **** **: * : . : :

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"Aducanumab   TDSVKGRFTISRDN SKNTLYLQMN TLR AEDTAVYYCARDR--GIGARRGPY--YMDVWGKG 117
"Crenezumab   PDSVKGRFTISRDN AKNSLYLQMN SLRAEDTAVYYCASG-----DYWGQG 105
"Gantenerumab ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARGKGNTHKPY--GYVRYFDVWGQG 119
"Dupilumab    ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAKDRLSITIR--PRYYGLDVWGQG 118
"Ixekezumab   NQRFKGRVTITADESTSTAYMELSSLRSED TAVYYCARYDYFT-----GTGV----YWGQG 112
"Secukinumab  VGSVKGRFTISRDN AKNSLYLQMN SLRVEDTAVYYCVRDYDILTDYYIHYWYFDLWGGR 120
"Ustekinumab  SPSFQGGVTMSVDKSIT TAYLQWNSLKASDTAMYYCARRRPG-----QGYFDFWGQG 112
"Mepolizumab  NSALMSRLSISKDTSRNQVVL TMTNMDPVD TATYYCARDPFSSLL-----RLDYWGGR 112
"Vedolizumab  NQKFKGRVTLTVDISASTAYMELSSLRSED TAVYYCARGGYDG-----W--DYAIDYWGQG 114
"Natalizumab  --PKFQGRVTITADTSASTAYMELSSLRSED TAVYYCAREGYYG-----NYGVYAMDYWGQG 116
"Alirocumab   ADSVKGRFIIISRDS SKHTLYLQMN SLRAEDTAVYYCAKDSN-----WGNFDLWGGR 111
"Evolucumab   AQKLQGRGTM TDPSTSTAYMELSLRSDDTAVYYCARG-----YGMVDVWGQG 108
"Evinacumab   ADSVKGRFTISRDN SKNSLYLQMN SLRAEDTAFFYCAKDLRNTIFGV--VIPDAFDIWGQG 119
"Denosumab    ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAKDPGTTV-----IMSWFDPWGQG 115
"Nivolumab    ADSVKGRFTISRDN SKNTLFLQMN SLRAEDTAVYYCATND-----DYWGQG 106
"Pembrolizumab NEKFKNRVTLTDSSTTTAYMELKSLQFDDTAVYYCARRDYRF-----DMG-----FDYWGQG 113
"Ranibizumab  AADFKKRRFTFSLDTSKSTAYLQMN SLRAEDTAVYYCAKYP--YY---YGTSHWYFDVWGQG 116

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FIGS. 11A-11B (CON'T)

"E06.scFv	ADSVKGRFIVSRDTSQSILYLQMNALRAEDTAIYYCARD-----YYGSSSYWYFDVWGAG	116
"BAN2401	GDTVKGRTTISRDNAKNSLFLQMSSLRAEDTAVYYCAREGGYY---YGRSYYTM DYWGQG	117
"Bevacizumab	AADFKRRTFSLDTSKSTAYLQMNLSLAEDTAVYYCAKYPH-Y---YGSSHWFYFDVWGQG	116
"Erenumab	VDSVKGRTTISRDNSKNTLFLQMNSLRAEDTAVYYCARDRLNYYDS[6] KYYGMAVWGQG	123
"aTAU	-ESVKGRFTISRDDSKSSVLYLQMNNLRAEDSGIYYCT----NWE-----DYWGQG	108
"Brolucizumab	ATWAKGRFTISRDNKSNTLYLQMSNLRAEDTAVYYCAGDG-----HNSGWGLDIWGQG	113
"Lampalizumab	AddFKGREVFSLDTSVSTAYLQLISSLKAEDTAVYYCEREG-----GVNNWGQG	108
"Andecaliximab	S-ALMSRFTISKDDSKNVTYVLKMNSLKTEDTAIYYCAR-----YYYGM DYWGQG	108
"Belimumab	--QNFGQRVAITADESTGTASME LSSLRSED TAVYYCARSDLL---LFP HHALSPWGRG	116
"Eculizumab	--ENFKDRVTMTTRDTSTSTVYMELSSLRSED TAVYYCARY--FFG-----SSPNWYFDVWGQG	115
"Lanadelumab	ADSVKGRFTISRDNKSNTLYLQMNLSLAEDTAVYYCAYRRIGVPR-----RDEFDWGQG	115
"Adalimumab	ADSVEGRFTISRDNAKNSLYLQMNLSLAEDTAVYYCAKVSYLS-----TASSLDYW GQG	114
"Infliximab	AESVKGRFTISRDDSKSAVYLQMTDLRTEDTGVIYYCSR-----NY Y--GSTY----DYWGQG	113
	: : : * : : : : *** : *	** :

	N	
"Aducanumab	TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	177
"Crenezumab	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	165
"Gantenerumab	TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	179
"Dupilumab	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	178
"Ixekezumab	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	172
"Secukinumab	TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	180
"Ustekinumab	TLVTVSSSSSTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	172
"Mepolizumab	TPVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	172
"Vedolizumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	174
"Natalizumab	TLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	176
"Alirocumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	171
"Evolucomab	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	168
"Evinacumab	TMVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	179
"Denosumab"	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	175
"Nivolumab	TLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	166
"Pembrolizumab	TLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	173
"Ranibizumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	176
"E06.scFv	TTVTVSS 123	
"BAN2401	TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	177
"Bevacizumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	176
"Erenumab	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	183
"aTAU	TTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	168
"Brolucizumab	TEVTVSS 120	
"Lampalizumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	168
"Andecaliximab	TLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	168
"Belimumab	TMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	176
"Eculizumab	TLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	175
"Lanadelumab	TMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	175
"Adalimumab	TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	174
"Infliximab	TELTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF	173
* *****		

FIGS. 11A-11B (CON'T)

"Aducanumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	232
"Crenezumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKY-----GPP	217
"Gantenerumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	234
"Dupilumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKY-----GPP	230
"Ixekezumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKYG---PP	224
"Secukinumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	235
"Ustekinumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	227
"Mepolizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	227
"Vedolizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	229
"Natalizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKYG---PP	228
"Alirocumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	226
"Evolucomab	PAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKRVESKSCDKTHT	220
"Evinacumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKY-----GPP	231
"Denosumab"	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	230
"Nivolumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKY---GPP	218
"Pembrolizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKYG---PP	225
"Ranibizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	226
"BAN2401	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	228
"Bevacizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	231
"Erenumab	PAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKRVESKSCDKTHT	235
"aTAU	PAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKY---GPP	220
"Lampalizumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	218
"Andecaliximab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKY---GPP	220
"Belimumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	231
"Eculizumab	PAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKRVESKSCDKTHT	227
"Lanadelumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	230
"Adalimumab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	229
"Infliximab	PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVESKSCDKTHT	228

*****. :*: ** * :***** *

FIG. 11B CLUSTAL O(1.2.4) ALIGNMENT THERAPEUTIC Fab LIGHT CHAIN

"Aducanumab	DIQMTQSPSSLSASVGDRVTITCRASQSI-----SSYLWYQQKPGKAPKLLIYAASSL	54
"Crenezumab	DIVMTQSPFLSLPVTGPGEPAISCRSSQSLVYSN-GDTYHLWYLQKPGQSPQLLIYKVSNR	59
"Gantenerumab	DIVLTQSPATLSLSPGERATLSCRASQSVS-----SSYLAWYQQKPGQAPRLIIYGASSR	55
"Dupilumab	DIVMTQSPFLSLPVTGPGEPAISCRSSQSLLYSI-GYNYLDWYLQKSGQSPQLLIYLGSNR	59
"Ixekezumab	DIVMTQTPFLSLSVTPGQPASISCRSSRSLVHSR-GNTYHLWYLQKPGQSPQLLIYKVSNR	59
"Secukinumab	EIVLTQSPGTLSLSPGERATLSCRASQSVS-----SSYLAWYQQKPGQAPRLIIYGASSR	55
"Ustekinumab	DIQMTQSPSSLSASVGDRVTITCRASQGI-----SSWLAWYQQKPEKAPKSLIYAASSL	54
"Mepolizumab	DIVMTQSPDSLAVSLGERATINCKSSQSLNLSGNQKNYLAWXQQKPGQPPKLLIYGASTR	60
"Vedolizumab	DVVMTQSPFLSLPVTGPGEPAISCRSSQSLAKSY-GNTYLSWYLQKPGQSPQLLIYGISNR	59
"Natalizumab	DIQMTQSPSSLSASVGDRVTITCKTSQDI-----NKMAYWYQQTPGKAPRLIIHYTSAL	54
"Alirocumab	DIVMTQSPDSLAVSLGERATINCKSSQSVLYRSNNRNLGWYQQKPGQPPNLLIYWASTR	60
"Evolucomab	ESALT-QPASVSGSPGQSITISCTGTSS---DVGGYNSVSWYQQHPGKAPKLMIEVSNR	56
"Evinacumab	DIQMTQSPSTLSASVGDRVTITCRASQSI-----RSWLAWYQQKPGKAPKLLIYKASSL	54
"Denosumab	EIVLTQSPGTLSLSPGERATLSCRASQSVR-----GRYLAWYQQKPGQAPRLIIYGASSR	55
"Nivolumab	EIVLTQSPATLSLSPGERATLSCRASQSVS-----SYLAWYQQKPGQAPRLIIYDASNR	54
"Pembrolizumab	EIVLTQSPATLSLSPGERATLSCRASKGVSTSG--YSLHWYQQKPGQAPRLIIYLASYL	58
"Ranibizumab	DIQMTQSPSSLSASVGDRVTITCSAQDI---SN---YLNWYQQKPGKAPKVLIIYFTSSL	54
"EO6.scFv	DIVMTQSPSSLSVAGKKVTISCTASESLYSSKHKVHYLAWYQKKPEQSPKLLIYGASNR	60
"BAN2401	DVVMTQSPFLSLPVTGPGEPAISCRSSQSIIVH--SNGNTYLEWYLQKPGQSPKLLIYKVSNR	59

FIGS. 11A-11B (CON'T)

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"Bevacizumab  DIQMTQSPSSLSASVGDRVITTCASQDI-----SNYLNWYQQKPGKAPKVLIIYFTSSL 54
"aIAU          DIVLTQSPDSLAVSLGERATISCRASQSVSTS--RYSYIHWYQQKPGQPPKLLIKYASNL 58
"Erenumab      QSVLTQ--PPSVSAAPGQKVITISCSGSSSNIG----NNYVSWYQQLPGTAPKLLIYDNNKR 55
"Brolucizumab  EIVMTQSPSTLSASVGDRVITTCQASEI-I-----SWLAWYQQKPGKAPKLLIYLASTL 54
"Lampalizumab  DIQVTQSPSSLSASVGDRVITTCITSTDI-----DDDMNWYQQKPGKVPKLLISGGNTL 54
"Andecaliximab -IQMTQSPSSLSASVGDRVITTCRASQDV-----RNTVAWYQQKPGKAPKLLIYSSSYR 53
"Belimumab     SSELT-QDPAVSVALGQTVRVTCQGDSL---R---SYYASWYQQKPGQAPVLVIYGNKNNR 53
"Eculizumab    DIQMTQSPSSLSASVGDRVITTCGASENI-----YGALNWYQQKPGKAPKLLIYGAINL 54
"Lanadelumab   DIQMTQSPSTLSASVGDRVITTCRASQSI-----SSWLAWYQQKPGKAPKLLIYKASTL 54
"Adalimumab    DIQMTQSPSSLSASVGDRVITTCRASQGI-----KNYLAWYQQKPGKAPKLLIYAASTL 54
"Infliximab     DILLTQSPAILSVSPGERVSFSCRASQFV-----SSIHWYQQRTNGSPRLLIKYASES 54
               :*      ::      : * :      :.*      *      *      : *      :*      .

"Aducanumab    QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQSYS---TPLTFGGGTKVEIK-RTVA 111
"Crenezumab    FSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQSTH---VPWTFGGGTKVEIK-RTVA 116
"Gantenerumab  ATGVPARFSGSGSGTDFTLTISLLEPEDFATYYCLQIYN----MPITFGGQGTKEIK-RTVA 112
"Dupilumab     ASGVPDFRFSGSGSGTDFTLKISRVEAEDVGFYYCMQALQ----TPYTFGGGTKLEIK-RTVA 116
"Ixekizumab    FIGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQSTH---LPFTFGGQGTKEIK-RTVA 116
"Secukinumab   ATGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQYGS---SPCTFGGQGTKEIK-RTVA 112
"Ustekinumab   QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYNI---YPYTFGGGTKLEIK-RTVA 111
"Nepolizumab   ESGVPDRFSGSGSGTDFTLTISSLQAEADVAVYYCQNVHS---FPPTFGGQGTKEIK-RTVA 117
"Vedoluzumab   FSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCLQGTH---QPYTFGGGTKVEIK-RTVA 116
"Natalizumab   QPGIPSRFSGSGSGRDYFTLTISLQPEDFATYYCLQYD---NLWTFGGGTKVEIK-RTVA 110
"Alirocumab    ESGVPDRFSGSGSGTDFTLTISSLQAEADVAVYYCQQYYT---TPYTFGGGTKLEIK-RTVA 117
"Evolucomab    PSGVSNRFSGSKSGNTASLTISGLQAEDEADYYCNSYTS---TSMVFGGGTKLTVLQPKA 114
"Evinacumab    ESGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQQYNS----YSYTFGGGTKLEIK-RTVA 111
"Denosumab     ATGIPDRFSGSGSGTDFTLTISRLEPEDFAVFYCCQYGS----SPRTFGGQGTKEIK-RTVA 112
"Nivolumab      ATGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCQQSSN-----WPRTFGGQGTKEIK-RTVA 111
"Pembrolizumab ESGVPARFSGSGSGTDFTLTISLLEPEDFAVYYCQHSRD----LPLTFGGGTKVEIK-RTVA 115
"Ranibizumab   HSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYST---VPWTFGGGTKVEIK-RTVA 111
"E06.scFv      YIGVPDRFTGSGSGTDFTLTISSVQVEDLTHYYCAQFYS---YPLTFGAGTKLEIK 113
"BAN2401       FSGVPDRFSGSGSGTDFTLTISRVEAEDVGIYYCFQGSN-----VPPTFGPGTKLEIK-RTVA 116
"Bevacizumab   HSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYST----VPWTFGGGTKVEIK-RTVA 111
"aIAU          ESGVPSRFSGSGSGTDFTLTNIHLEPEDFATYYCHRSWEI-----PLTFGGGTKLEIK-RTVA 115
"Erenumab      [4]PPDRFSGSKSGTSTTLGITGLQTGDEADYYCG-TWDSRLSAVVFGGGTKLTVLQPKA 115
"Brolucizumab  ASGVPSRFSGSGSGAEFTLTISLQPDDEFATYYCQNVYLASTNGANFGGQGTKEIK-RTVA 111
"Lampalizumab  RPGVPSRFSGSGSGTDFTLTISLQPEDFATYYCLQSDS---LPYTFGGGTKVEIK-RTVA 111
"Adecaliximab  NTGVPDFRFSGSGSGTDFTLTISLQAEADVAVYYCQQHYI-----TPYTFGGGTKVEIK-RTVA 110
"Belimumab     PSGIPDRFSGSSSGNTASLTITGAQAEDEADYYCSSRDS--SGNHWVFGGGTKEIK-RTVA 113
"Eculizumab    ADGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQNVLN-----TPLTFGGGTKVEIK-RTVA 111
"Lanadelumab   ESGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQQYNT---Y-WTFGGGTKVEIK-RTVA 110
"Adalimumab    QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRYNR---APYTFGGGTKVEIK-RTVA 111
"Infliximab    MSGIPSRFSGSGSGTDFTLTISLQPEDFATYYCQSHSWP---FTFGSGTNLEIK-RTVA 111
               *:      *****      *      :.:.*      :      :.*      .;*      .**      **.:      :      :      *

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FIGS. 11A-11B (CON'T)

		S/N	
"Aducanumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Crenezumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	176	
"Gantenerumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	172	
"Dupilumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	176	
"Izekizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	176	
"Secukinumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	172	
"Ustekinumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Mepolizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	177	
"Vedolizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	176	
"Natalizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	170	
"Alirocumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	177	
"Evolucumab	APSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGV-E--TTTPSKQS	171	
"Evinacumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Denosumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	172	
"Nivolumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Pembrolizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	175	
"Ranibizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"BAN2401	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	176	
"Bevacizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"aTAU	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	175	
"Erenumab	NPTVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADGSPVKAGV-E---TTKPSKQS	172	
"Lampalizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Adecaximab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Belimumab	APSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGV-ETTTTPSKQSN	172	
"Eculizumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Lanadelumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	170	
"Adalimumab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
"Infliximab	APSVFIFPPSDEQLKSGTASVCLLNFFYPREAKVQWKVDNALQSGNSQESVTEQDSKDS	171	
	**** :****.*:*:..*:*:****.:**** ..* **.*.: ... *:*		

FIGS. 11A-11B (CON'T)

	N	
"Aducanumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Crenezumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	219
"Gantenerumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	215
"Dupilumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	219
"Ixekezumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	219
"Secukinumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	215
"Ustekinumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Mepolizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	220
"Vedolizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	219
"Natalizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	213
"Alirocumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	220
"Evolucumab	NNKYAASSYLSLTPEQWKSHRSYSCQVTHEGST---VEKTVAPTECS	215
"Evinacumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Denosumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	215
"Nivolumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Pembrolizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	218
"Ranibizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"BAN2401	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	219
"Bevacizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"aTAU	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC	218
"Erenumab	NNKYAASSYLSLTPEQWKSHRSYSCQVTHEGST---VEKTVAPTECS	216
"Lampalizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Adecalliximab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Belimumab	---KYAASSYLSLTPEQWKSHRSYSCQVTHEGST---VEKTVAPTECS	214
"Eculizumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Lanadelumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	213
"Adalimumab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214
"Infliximab	---TYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC-	214

. *: ** *: *: : : : . *: *: *: *: *: : * *: . **

FIG. 12

	VP1 ₁₋₇₃₆ →													
AAV1	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV2	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV3-3	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV4-4	-MTDGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	59									
AV5	MSFVDHPPDWLEE-VGEGLREFLGL	EAGPPKPKPNQQHQDQARGLVLP	PGYN	NYLGP	GNGLD	59								
AAV6	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV7	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV8	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
hu31	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
hu32	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
AAV9	MAADGYLPDWLEDNLSEGIREWWDLKP	GAPKPKANQQKQDDGRGLVLP	PGYKYLGP	FNGLD	60									
SUBS	-STVDHP-----ETVG--V-QFLK-QA-P-K-PAERKK-DG-----N-----F-----													
	MF		L	D	E	V	P	QS						
				G			Q	R						
AAV1	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	RYNHADA	AEFQERLQ	EDTSFGG	NLGRAVFQ	120								
AAV2	KGEPVNEADAAALEHDKAYDRQLDS	GDNPYLKYNHADA	AEFQERLK	EDTSFGG	NLGRAVFQ	120								
AAV3-3	KGEPVNEADAAALEHDKAYDQQLKAGDNPYL	KYNHADA	AEFQERLQ	EDTSFGG	NLGRAVFQ	120								
AAV4-4	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	KYNHADA	AEFQQRLQ	GDTSFGG	NLGRAVFQ	119								
AV5	RGEPVNRADAEVAREHDISYNEQLEAGDNPYL	KYNHADA	AEFQEKI	ADDSFGG	NLGRAVFQ	119								
AAV6	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	RYNHADA	AEFQERLQ	EDTSFGG	NLGRAVFQ	120								
AAV7	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	RYNHADA	AEFQERLQ	EDTSFGG	NLGRAVFQ	120								
AAV8	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	RYNHADA	AEFQERLQ	EDTSFGG	NLGRAVFQ	120								
hu31	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	KYNHADA	AEFQERLK	EDTSFGG	NLGRAVFQ	120								
hu32	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	KYNHADA	AEFQERLK	EDTSFGG	NLGRAVFQ	120								
AAV9	KGEPVNAADAAALEHDKAYDQQLKAGDNPYL	KYNHADA	AEFQERLK	EDTSFGG	NLGRAVFQ	120								
SUBS	R-----E--EV-R-----IS-NE--DS-----R-----QK-QD-----K-----													
	R			R	E			AG						
				Q										
	VP2 ₁₃₈ → ---HVRI---													
AAV1	AKKRVLEPLGLVEEGAKTAPGKKRPVEQSPQ	-EPDSSSGIGKTGQQPAKKRLNFGQTGDS	179											
AAV2	AKKRVLEPLGLVEEPVKTAPGKKRPVEHSFV	-EPDSSSGTGKAGQQPARKRLNFGQTGDA	179											
AAV3-3	AKKRILEPLGLVEEAAKTAPGKKGA	VDQSPQ-EPDSSSGVGKSGKQPARKRLNFGQTGDS	179											
AAV4-4	AKKRVLEPLGLVEQAGETAPGKKRPLIESPQ	-QPDSSSTGIGKKKGQPARKKLVEFEDETGA	178											
AV5	AKKRVLEPFGLVEEGAKTAPTGKRIDDHFP	-----KRKKARTEEDSKPSTSSDA	168											
AAV6	AKKRVLEPFGLVEEGAKTAPGKKRPVEQSPQ	-EPDSSSGIGKTGQQPAKKRLNFGQTGDS	179											
AAV7	AKKRVLEPLGLVEEGAKTAPAKKRPVEPSPQRS	PDSSSTGIGKKKGQPARKRLNFGQTGDS	180											
AAV8	AKKRVLEPLGLVEEGAKTAPGKKRPVEPSPQRS	PDSSSTGIGKKKGQPARKRLNFGQTGDS	180											
hu31	AKKRILEPLGLVEEAAKTAPGKKRPVEQSPQ	-EPDSSAGIGKSGSQPAKKKLNFGQTGDT	179											
hu32	AKKRILEPLGLVEEAAKTAPGKKRPVEQSPQ	-EPDSSAGIGKSGSQPAKKKLNFGQTGDT	179											
AAV9	AKKRILEPLGLVEEAAKTAPGKKRPVEQSPQ	-EPDSSAGIGKSGAQPARKRLNFGQTGDT	179											
SUBS	---V---F---QGGE---TG-GIDHFP-V-S---S-T---KKQARTREKSVPEDETGA													
	I	PV	A	ALIP	Q	T	V	T	K	E	D	K	STSS	S
				E					A	S				
									R	A				

FIG. 12 (CON'T)

--HVR2--
 VP3₂₀₃→

AAV1	ESVFD--PQPLGEPPATPAAVGPPTTMA	SGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	238
AAV2	DSVFD--PQPLGQPPAAPSGLGTNTMATG	SGGAPMADNNEGADGVGNSSGNWHCDSTWLGDR	238
AAV3-3	ESVFD--PQPLGEPPAAP	TSLGSTNTMASGGGAPMADNNEGADGVGNSSGNWHCDSTWLGDR	238
AAV4-4	GDGP-----PEGSTSGAMS----	DDSEMRAAAGGAAVEGGQGADGVGNASGDWHCDSTWSEGH	232
AV5	EASPSGSQQLQIPAQPASSL	GADTMSAGGGGGLGDNNGQADGVGNASGDWHCDSTWLGDR	228
AAV6	ESVFD--PQPLGEPPATPAAVGPPTTMA	SGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	238
AAV7	ESVFD--PQPLGEPPAAPSSVGS	GTVAAGGGAPMADNNEGADGVGNASGNWHCDSTWLGDR	239
AAV8	ESVFD--PQPLGEPPAAPSGVGPNTMA	AGGGAPMADNNEGADGVGSSSGNWHCDSTWLGDR	239
hu31	ESVFD--PQPIGEPPAAPSGVGS	LTMASGGGAPVADNNEGADGVGSSSGNWHCDSTWLGDR	238
hu32	ESVFD--PQPIGEPPAAPSGVGS	LTMASGGGAPVADNNEGADGVGSSSGNWHCDSTWLGDR	238
AAV9	ESVFD--PQPIGEPPAAPSGVGS	LTMASGGGAPVADNNEGADGVGSSSGNWHCDSTWLGDR	238
SUBS	GDG-S-S-QLQQTSGTMA	SLDPNEVRAAA--GAMGEGGQ-----NA--D-----T-MEGH	
	DA	E S AQPATA AG ST S LV	S
		I --- DT A	
		TD	
		S	

HVR3

AAV1	VITTSTRTWALPTYNNHLYKQIS--SASTGASNDN	HYFGYSTPWGYFDFNRFHCHFS	SPRDW	297
AAV2	VITTSTRTWALPTYNNHLYKQIS---	SQSGASNDNHYFGYSTPWGYFDFNRFHCHFS	SPRDW	296
AAV3-3	VITTSTRTWALPTYNNHLYKQIS---	SQSGASNDNHYFGYSTPWGYFDFNRFHCHFS	SPRDW	296
AAV4-4	VTTTSTRTWVLPPTYNNHLYKRLG-----	ESLQSN	TYNGFSTPWGYFDFNRFHCHFS	SPRDW
AV5	VVTKSTRTWVLP	SYNNHQYREIKS--GSVDGSNANAYFGYSTPWGYFDFNRFHSHW	SPRDW	287
AAV6	VITTSTRTWALPTYNNHLYKQISSAST--	GASNDNHYFGYSTPWGYFDFNRFHCHFS	SPRDW	297
AAV7	VITTSTRTWALPTYNNHLYKQISS--	ETAGSTNDNTYFGYSTPWGYFDFNRFHCHFS	SPRDW	298
AAV8	VITTSTRTWALPTYNNHLYKQISNGTSGG	ATNDNTYFGYSTPWGYFDFNRFHCHFS	SPRDW	299
hu31	VITTSTRTWALPTYNNHLYKQISNSTSGG	SSNDNAYFGYSTPWGYFDFNRFHCHFS	SPRDW	298
hu32	VITTSTRTWALPTYNNHLYKQISNSTSGG	SSNDNAYFGYSTPWGYFDFNRFHCHFS	SPRDW	298
AAV9	VITTSTRTWALPTYNNHLYKQISNSTSGG	SSNDNAYFGYSTPWGYFDFNRFHCHFS	SPRDW	298
SUBS	-T-K-----V--S-----Q-RR	LGSGSQSDATQA-T-----S-W-----		
	V	E K AATTEGL S H		
		G V		
		E A		

AAV1	QRLINNNWGF	RPKRLNFKLFNIQVKEVTTNDGVT	TTIANNLTSTVQVFS	SDSEYQLPYVLGS	357
AAV2	QRLINNNWGF	RPKRLNFKLFNIQVKEVTQNDG	TTTIANNLTSTVQVFTD	SEYQLPYVLGS	356
AAV3-3	QRLINNNWGF	RPKLSFKLFNIQVRGVTQNDG	TTTIANNLTSTVQVFTD	SEYQLPYVLGS	356
AAV4-4	QRLINNNWGM	RPKAMRVKIFNIQVKEVTT	SNGETTVANNLTSTVQIF	ADSSYEPLPYVMDA	347
AV5	QRLINNNWGF	RPKSLRVKIFNIQVKEVTVQD	STTTIANNLTSTVQVFTD	DDYQLPYVVG	347
AAV6	QRLINNNWGF	RPKRLNFKLFNIQVKEVTTNDGVT	TTIANNLTSTVQVFS	SDSEYQLPYVLGS	357
AAV7	QRLINNNWGF	RPKRLSFKLFNIQVKEVTTNDGVT	TTIANNLTSTIQVFS	SDSEYQLPYVLGS	358
AAV8	QRLINNNWGF	RPKRLSFKLFNIQVKEVTQNE	GTKTIANNLTSTIQVFTD	SEYQLPYVLGS	359
hu31	QRLINNNWGF	RPKRLNFKLFNIQVKEVT	DNGVKTIANNLTSTVQVFTD	SDYQLPYVLGS	358
hu32	QRLINNNWGF	RPKRLNFKLFNIQVKEVT	DNGVKTIANNLTSTVQVFTD	SDYQLPYVLGS	358
AAV9	QRLINNNWGF	RPKRLNFKLFNIQVKEVT	DNGVKTIANNLTSTVQVFTD	SDYQLPYVLGS	358
SUBS	-----M--RAMRV-I-----	VQDSTT-----I-I-S-DE-E-----MDA			
		K S	QSE E	A S	
		S			

HVR4

(b)(6) (b)(7)(C), (b)(7)(D) NVR-5 (b)(6) (b)(7)(C), (b)(7)(D)

-----HVR6-----HVR7-----HVR8-----

AAV1	QPKNWLP	GP	CYRQQRV	SKTKIDN-----	NNSNFTWTG	ASKYNLNGRES	IINPGTAMASHK	528
AAV2	QSRNWLP	GP	CYRQQRV	SKTSADN-----	NNSEYSWTG	ATKYHLNGRDS	LVNPGPAMASHK	527
AAV3-3	QARNWLP	GP	CYRQQR	LSKTANDN-----	NNSNFPWTA	ASKYHLNGRDS	LVNPGPAMASHK	528
AAV4-4	FKKNWLP	GPS	SIKQOGF	SKTANQNYKIPATG	SDSLIKYETH	STLDGRWSALT	PFPMATAG	527
AV5	TYKNWFP	GP	MGRTQ	GWNLGSGVN-----	RASVSAFAT	TNRMELEGAS	YQVPPQFNGMTNN	514
AAV6	QPKNWLP	GP	CYRQQRV	SKTKIDN-----	NNSNFTWTG	ASKYNLNGRES	IINPGTAMASHK	528
AAV7	QAKNWLP	GP	CFRQQRV	SKTLDQN-----	NNSNFAWTG	ATKYHLNGRNS	LVNPGVAMATHK	530
AAV8	QAKNWLP	GP	CYRQQRV	STTTGQN-----	NNSNFAWTG	ATKYHLNGRNS	LANPGIAMATHK	530
hu31	QGRNYIP	GPS	SYRQQRV	STTTVTQN-----	NNSEFAWPG	ASSWALNGRNS	LMNPGPAMASHK	528
hu32	QGRNYIP	GPS	SYRQQRV	STTTVTQN-----	NNSEFAWPG	ASSWALNGRNS	LMNPGPAMASHK	528
AAV9	QGRNYIP	GPS	SYRQQRV	STTTVTQN-----	NNSEFAWPG	ASSWALNGRNS	LMNPGPAMASHK	528
SUBS	FAK-WL---	CIKT-G	WNLGSGV-----	-----	TG-DS	SLIKYETHST-D-ASYQVP-Q	TFPGMTAG	
	TP	F	MG	F K AND	RA	NYTFAT	TNRME E D ALT VN NN	
	K		F	L KA		V P TAG KYN	W I I I	
	Y			LD		S	H E A	
	S			T				

FIG. 12 (CON'T)

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-----HVR9-----
AAV1  DDEDKFFPMSCVMIFGKESA----CASNTALD--NVMITDEEEIKATNPVATERFGTVAVNLFQ 585
AAV2  DDEEKFFPQSGVLIFGKQGS----EXTNVDIE--KVMITDEEEIRTTNPVATEQYGSVSTNLQ 584
AAV3-3 DDEEKFFPMHGNLIFGKQGT---TASNAELD--NVMITDEEEIRTTNPVATEQYGTVANNLQ 585
AAV4-4 PADSKFS--NSQLIFAGPKQN---GNTATVPG--TLIFTSEEEIAATNATDIDMWGNLPGGDQ 583
AV5    LQGSNTYALENTMIFNSQFANPGTTATYLEGNMLITSESETQPVNRVAYNVGGQMATNNQ 574
AAV6  DDKDKFFPMSCVMIFGKESA----CASNTALD--NVMITDEEEIKATNPVATERFGTVAVNLFQ 585
AAV7  DDEDRFFPSSGVLIFGKTGA---TN-KTTLE--NVLMTNEEEIRPTNPVATEEYGVVSSNLQ 586
AAV8  DDEERFFPSSGVLIFGKQNA---ARDNADYS--DVMLTSEEEIKTTNPVATEEYGVVADNLQ 587
hu31  EGEDRFFPLSGSLIFGKQGT---GRDNVDAD--KVMITNEEEIKTTNPVATESYGVVATNNQ 585
hu32  EGEDRFFPLSGSLIFGKQGT---GRDNVDAD--KVMITNEEEIKTTNPVATESYGVVATNNQ 585
AAV9  EGEDRFFPLSGSLIFGKQGT---GRDNVDAD--KVMITNEEEIKTTNPVATESYGVVATNNQ 585
SUBS  LQGSNTYAMENTMFANPKQN---TNTATVPG--TLIF-S-S-TQPV-ATDYDMW-NLPGGD-
      PADEK S QHQLI  SESA  EASKAALE-NMLM D  RA  R  NVF TMSN L
      DDK      NN V      TPS  AK  KTY      L      A      QG I V N
              S I      N      EI      E S S F
              N      Y      R      D

---HVR10---
AAV1  SSSTDPATGCVHAMGALPGMVWQDRDVYLQGP IWAKIPHTDGHFHPSPLMGGFGLKNPPP 645
AAV2  RGNRQAATADVNTQGVLPGMVWQDRDVYLQGP IWAKIPHTDGHFHPSPLMGGFGLKHPPP 644
AAV3-3 SSNTAPTGTGVNHQCALPGMVWQDRDVYLQGP IWAKIPHTDGHFHPSPLMGGFGLKHPPP 645
AAV4-4 SNSNLPTVDRLTALGAVPGMVWQNRDIYQGP IWAKIPHTDGHFHPSPLMGGFGLKHPPP 643
AV5    SSTTAPATGTYNLQEIIVPGSVWMERDVYLQGP IWAKIPETGAHFHPSPAMGGFGLKHPPP 634
AAV6  SSSTDPATGCVHVMGALPGMVWQDRDVYLQGP IWAKIPHTDGHFHPSPLMGGFGLKHPPP 645
AAV7  AANTAQAQTQVNNQCALPGMVWQNRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGLKHPPP 646
AAV8  QQNTAPQIGTVNSQCALPGMVWQNRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGLKHPPP 647
hu31  SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPPP 645
hu32  SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPPP 645
AAV9  SAQAQAQTGWVQNQGILPGMVWQDRDVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPPP 645
SUBS  RNSNLPTVDRLTALAV--S--ME--I-----E-GAH-----AI-----L-N---
      ASNTA AIADYHTM V      N
      QGTRD  QT NH
      Q      V L
              V
              S

---HVR11---
AAV1  QILIKNTPVPANPPAEFSATKFAFASFITQYSTGQVSVEIEWELQKENS KRWNPEVQYTSNY 705
AAV2  QILIKNTPVPANPSTTFSAAKFAFASFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 704
AAV3-3 QIMIKNTPVPANPPTTFSPAFAFASFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 705
AAV4-4 QIFIKNTPVPANPATTFSSPTVNSFITQYSTGQVSVQIDWEIQKERSKRWNPEVQYTSNY 703
AV5    MMLIKNTPVPGNI--TSFSQVPVSSFITQYSTGQVTVEMEWELKKENS KRWNPEIQYTNNY 693
AAV6  QILIKNTPVPANPPAEFSATKFAFASFITQYSTGQVSVEIEWELQKENS KRWNPEVQYTSNY 705
AAV7  QILIKNTPVPANPPEVFTPAKFAFASFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNF 706
AAV8  QILIKNTPVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 707
hu31  QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 705
hu32  QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 705
AAV9  QILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNY 705
SUBS  MMM-----G-IAAE--SDVPVS-----QMD--IK--R-----V-----
      F      SET TAA FA
              S PT

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FIG. 12 (CON'T)

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          V  QS
          S
-----EVR12-----
AAV1    AKSANVDFTVDNNGLYTEPRPIGTRYLTRPL 736
AAV2    NKSUNVDFTVDINGVYSEPRPIGTRYLTRNL 735
AAV3-3  NKSUNVDFTVDINGVYSEPRPIGTRYLTRNL 736
AAV4-4  GQQNSLLWAPDAACKYTEPRAIGTRYLTHHL 734
AV5     NDPQFVDFAPDSTGEYRTTRPIGTRYLTRPL 724
AAV6    AKSANVDFTVDNNGLYTEPRPIGTRYLTRPL 736
AAV7    EKQTGVDFAVDSQGVYSEPRPIGTRYLTRNL 737
AAV8    YKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL 738
hu31    YKSNNVEFAVSTEGVYSEPRPIGTRYLTRNL 736
hu32    YKSNNVEFAVNTEGVYSEPRPIGTRYLTRNL 736
AAV9    YKSNNVEFAVNTEGVYSEPRPIGTRYLTRNL 736
SUBS    GQQVSLWTPDAA-K-RTT-A-----HP--
        NDPQF D   SSN E T           H
        A  TG     NQ L
        E  A      T

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FULLY-HUMAN POST-TRANSLATIONALLY MODIFIED ANTIBODY THERAPEUTICS

0. SEQUENCE LISTING

[0001] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Oct. 17, 2018, is named 26115_105004_SL.txt and is 400,185 bytes in size.

1. INTRODUCTION

[0002] Compositions and methods are described for the delivery of a fully human post-translationally modified (HuPTM) therapeutic monoclonal antibody (“mAb”) or the HuPTM antigen-binding fragment of a therapeutic mAb—e.g., a fully human-glycosylated (HuGly) Fab of the therapeutic mAb—to a human subject diagnosed with a disease or condition indicated for treatment with the therapeutic mAb.

2. BACKGROUND OF THE INVENTION

[0003] Therapeutic mAbs have been shown to be effective in treating a number of diseases and conditions. However, because these agents are effective for only a short period of time, repeated injections for long durations are often required, thereby creating considerable treatment burden for patients.

3. SUMMARY OF THE INVENTION

[0004] Compositions and methods are described for the delivery of a HuPTM mAb or a HuPTM antigen-binding fragment of a therapeutic mAb (for example, a fully human-glycosylated Fab (HuGlyFab) of a therapeutic mAb) to a patient (human subject) diagnosed with a disease or condition indicated for treatment with the therapeutic mAb. Such antigen-binding fragments of therapeutic mAbs include a Fab, F(ab')₂, or scFv (single-chain variable fragment) (collectively referred to herein as “antigen-binding fragment”). “HuPTM Fab” as used herein may include other antigen binding fragments of a mAb. In an alternative embodiment, full-length mAbs can be used. Delivery may be advantageously accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a therapeutic mAb or its antigen-binding fragment (or a hyperglycosylated derivative of either) to a patient (human subject) diagnosed with a condition indicated for treatment with the therapeutic mAb—to create a permanent depot in a tissue or organ of the patient that continuously supplies the HuPTM mAb or antigen-binding fragment of the therapeutic mAb, i.e., a human-glycosylated transgene product, to a target tissue where the mAb or antigen-binding fragment thereof exerts its therapeutic effect.

[0005] The HuPTM mAb or HuPTM antigen-binding fragment encoded by the transgene can include, but is not limited to, a full-length or an antigen-binding fragment of a therapeutic antibody that binds to:

[0006] Nervous System Targets, including Amyloid beta (A β or Abeta) peptides derived from the amyloid precursor protein (APP) implicated in Alzheimer’s disease, including but not limited to, aducanumab, cren-
ezumab, gantenerumab, and BAN2401, indicated for treating Alzheimer’s disease (see FIGS. 2A-2C and 2F); Tau protein implicated in tauopathies, including

Alzheimer’s disease, progressive supranuclear palsy, frontotemporal dementia, chronic traumatic encephalopathy, Pick’s Complex, primary age-related tauopathy, including but not limited to “aTAU” (see FIG. 2D) for treating tauopathies; and CGRP receptor implicated in migraines and cluster headaches including but not limited to erenumab (AIMOVIG™) (see FIG. 2E), eptinezumab, fremanezumab, and galcanezumab for treating migraines and cluster headaches;

[0007] Interleukins or interleukin receptors, including but not limited to, IL4R, such as dupilumab (see FIG. 3A), indicated for treating atopic dermatitis; IL17A such as ixekizumab (TALTZ®) or secukinumab (COSENTYX®) (see FIGS. 3B and 3C) indicated for treating plaque psoriasis, psoriatic arthritis, and ankylosing spondylitis; IL-5, such as mepolizumab (NUCALA®) (see FIG. 3D), indicated for treating asthma; and IL12/IL23 such as ustekinumab (STELARA®) (see FIG. 3E) indicated for treating psoriasis and Crohn’s disease;

[0008] Integrin, including but not limited to, vedolizumab (ENTYVIO®), indicated for treating ulcerative colitis and Crohn’s disease (see FIG. 4A) and natalizumab (anti-integrin alpha 4) for treating multiple sclerosis and Crohn’s disease (see FIG. 4B);

[0009] Hypercholesterolemia and Cardiovascular Disease Targets, such as PCSK9, including but not limited to, alirocumab (PRALUENT®) and evolocumab (REPATHA®), indicated for treating HeFH and HoFH (see FIGS. 5A and 5B); or ANGPTL3, including but not limited to, evinacumab (see FIG. 5C), indicated for the treatment of HoFH and severe forms of dyslipidemia and proinflammatory/proatherogenic phospholipids including but not limited to E06-scFv for the treatment of cardiovascular disease, including atherosclerosis (see FIG. 5D);

[0010] RANKL, including but not limited to, denosumab (XGEVA® and PROLIA®), indicated for treating osteoporosis, increasing bone mass in breast and prostate cancer patients, and preventing skeletal-related events due to bone metastasis (see FIG. 6);

[0011] PD-1, or PD-L1 or PD-L2, (these antibodies sometimes referred to herein as PD-1 blockers), including but not limited to, nivolumab (OPDIVO®) and pembrolizumab (KEYTRUDA®), indicated for treating metastatic melanoma, lymphomas, and non-small cell lung carcinomas (see FIGS. 7A and 7B);

[0012] BLyS (B-lymphocyte stimulator, also known as B-cell activating factor (BAFF)), including but not limited to, belimumab (BENLYSTA®), indicated for the treatment of systemic lupus erythematosus (SLE) (see FIG. 8E);

[0013] Ocular Targets, including but not limited to, VEGF (vascular endothelial growth factor), including but not limited to, ranibizumab (LUCENTIS®), bevacizumab (AVASTIN®), and brolucizumab indicated for treating neovascular age-related macular degeneration (e.g., “wet AMD”) (see FIGS. 8A, 8B and 8D); factor D, including but not limited to lampalizumab, for treating dry AMD (see FIG. 8C); and matrix metalloproteinase 9 (MMP9), including but not limited to andecaliximab, for treating dry AMD (FIG. 8G);

[0014] TNF-alpha, including but not limited, to adalimumab (HUMIRA®) and infliximab (REMICADE®)

indicated for treating rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, plaque psoriasis, and ulcerative colitis (FIG. 9A for adalimumab and FIG. 9B for infliximab); and

[0015] Plasma Protein targets, such as human complement proteins including but not limited to anti-C5 and C5a complement proteins, such as eculizumab (SO-LIRIS®) for the treatment of patients with paroxysmal nocturnal hemoglobinuria (PNH) to reduce hemolysis, or the treatment of atypical hemolytic uremic syndrome (aHUS) to inhibit complement-mediated thrombotic microangiopathy (FIG. 8F); and plasma kallikrein, including but not limited to lanadelumab for treating hereditary angioedema (see FIG. 8H);

[0016] or such mAbs or antigen-binding fragments engineered to contain additional glycosylation sites on the Fab domain (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety for its description of derivatives of antibodies that are hyperglycosylated on the Fab domain of the full-length antibody).

[0017] The recombinant vector used for delivering the transgene includes non-replicating recombinant adeno-associated virus vectors ("rAAV"). However, other viral vectors may be used, including but not limited to lentiviral vectors; vaccinia viral vectors, or non-viral expression vectors referred to as "naked DNA" constructs. Expression of the transgene can be controlled by constitutive or tissue-specific expression control elements.

[0018] Gene therapy constructs are designed such that both the heavy and light chains are expressed. The coding sequences for the heavy and light chains can be engineered in a single construct in which the heavy and light chains are separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed. In certain embodiments, the coding sequences encode for a Fab or F(ab')₂ or an scFv. In other embodiments, the constructs express an scFv in which the heavy and light chain variable domains are connected via a flexible, non-cleavable linker. In certain embodiments, the construct expresses, from the N-terminus, NH₂-V_L-linker-V_H-COOH or NH₂-V_H-linker-V_L-COOH.

[0019] Therapeutic antibodies delivered by gene therapy have several advantages over injected or infused therapeutic antibodies that dissipate over time resulting in peak and trough levels. Sustained expression of the transgene product antibody, as opposed to injecting an antibody repeatedly, allows for a more consistent level of antibody to be present at the site of action, and is less risky and more convenient for patients, since fewer injections need to be made. Furthermore, antibodies expressed from transgenes are post-translationally modified in a different manner than those that are directly injected because of the different microenvironment present during and after translation. Without being bound by any particular theory, this results in antibodies that have different diffusion, bioactivity, distribution, affinity, pharmacokinetic, and immunogenicity characteristics, such that the antibodies delivered to the site of action are "bio-betters" in comparison with directly injected antibodies.

[0020] In addition, antibodies expressed from transgenes in vivo are not likely to contain degradation products associated with antibodies produced by recombinant technologies, such as protein aggregation and protein oxidation. Aggregation is an issue associated with protein production and storage due to high protein concentration, surface inter-

action with manufacturing equipment and containers, and purification with certain buffer systems. These conditions, which promote aggregation, do not exist in transgene expression in gene therapy. Oxidation, such as methionine, tryptophan, and histidine oxidation, is also associated with protein production and storage, and is caused by stressed cell culture conditions, metal and air contact, and impurities in buffers and excipients. The proteins expressed from transgenes in vivo may also oxidize in a stressed condition. However, humans, and many other organisms, are equipped with an antioxidation defense system, which not only reduces the oxidation stress, but sometimes also repairs and/or reverses the oxidation. Thus, proteins produced in vivo are not likely to be in an oxidized form. Both aggregation and oxidation could affect the potency, pharmacokinetics (clearance), and immunogenicity.

[0021] Pharmaceutical compositions suitable for administration to human subjects comprise a suspension of the recombinant vector in a formulation buffer comprising a physiologically compatible aqueous buffer, a surfactant and optional excipients.

[0022] The invention is based, in part, on the following principles:

[0023] (i) The mAb therapeutics currently on the market are of the immunoglobulin G (IgG) isotypes, such as IgG1, IgG2, and IgG4, which in general have pharmacokinetic (PK) characteristics, such as slow clearance, long half-life, and limited tissue distribution. After intravenous administration, typical mAb serum PK profiles are biphasic with a rapid distribution phase and a slower elimination phase; thus, repeat administration is required to maintain doses required to treat chronic conditions. Moreover, the distribution of mAbs is generally limited to the vascular and interstitial spaces due to their large size and hydrophilicity. The extent of mAb partitioning from circulation into most tissues generally ranges from about 5-15%, except for brain where it is much lower. (See, e.g., Kamath, 2016, Drug Discovery Today: Technologies 21-22:75-83, which is incorporated by reference herein in its entirety). Continuous production of HuPTMmAbs or HuPTM Fabs in situ avoids repeat administrations and allows the use of Fabs, which would otherwise have too short a systemic half-life to achieve efficacy; and the methods of administration described allow direct access to target tissues, such as the brain, where the delivery of higher doses to such tissues can be achieved.

[0024] (ii) The Fab region of a number of therapeutic mAbs possesses glycosylation sites. For example, see FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H and 9A-9B which identify and highlight in blue and green, respectively, consensus and non-consensus asparaginal ("N") glycosylation sites as well as glutamine ("Q") residues that are glycosylation sites in the Fab region of certain therapeutic mAbs. (See, e.g., Valliere-Douglass et al., 2009, J. Biol. Chem. 284: 32493-32506, and Valliere-Douglass et al., 2010, J. Biol. Chem. 285:16012-16022, each of which is incorporated by reference in its entirety for the identification of N-linked glycosylation sites in antibodies). In addition, O-glycosylation comprises the addition of N-acetyl-galactosamine to serine or threonine residues by the enzyme. It has been demonstrated that amino acid residues present in the hinge region of antibodies

can be O-glycosylated. The possibility of O-glycosylation confers another advantage to the therapeutic antibodies provided herein, as compared to, e.g., antigen-binding fragments produced in *E. coli*, again because the *E. coli* naturally does not contain machinery equivalent to that used in human O-glycosylation. (Instead, O-glycosylation in *E. coli* has been demonstrated only when the bacteria is modified to contain specific O-glycosylation machinery. See, e.g., Farid-Moayer et al., 2007, *J. Bacteriol.* 189:8088-8098.) Moreover, the Fab amino acid sequence may be modified to engineer hyperglycosylated variants (e.g., see amino acid substitutions that can be made to engineer hyperglycosylated Fab regions of therapeutic antibodies shown in FIGS. 11A and 11B; and Courtois et al., 2016, *mAbs* 8:99-112 which is incorporated by reference herein in its entirety for its description of derivatives of antibodies that are hyperglycosylated on the Fab domain of the full-length antibody).

[0025] (iii) In addition to the glycosylation sites, the Fab regions can contain tyrosine (“Y”) sulfation sites in or near the CDRs; see FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H and 9A-9B which identify tyrosine-O-sulfation sites in the Fab region of certain therapeutic mAbs, as highlighted in yellow. (See, e.g., Yang et al., 2015, *Molecules* 20:2138-2164 (particularly at 2154), which is incorporated by reference in its entirety for the analysis of amino acids surrounding tyrosine residues subjected to protein tyrosine sulfation). The “rules” can be summarized as follows: Y residues with E or D within +5 to −5 position of Y, and where position −1 of Y is a neutral or acidic charged amino acid—but not a basic amino acid, e.g., R, K, or H that abolishes sulfation.

[0026] (iv) The glycosylation of Fab regions, such as those shown in FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H and 9A-9B by human cells will result in the addition of glycans that can improve stability, half-life and reduce unwanted aggregation and/or immunogenicity of the transgene product. (See, e.g., Bovenkamp et al., 2016, *J. Immunol.* 196:1435-1441 for a review of the emerging importance of Fab glycosylation; and FIG. 10 which identifies glycans that can be attached to HuGlyFab (adapted from Bondt et al., 2014, *Mol. & Cell. Proteomics* 13.1:3029-2029)). The Fab and Fc portions of antibodies have been shown to have distinct glycosylation patterns, with Fab glycans being high in galactosylation, sialylation, and bisection (e.g., with bisecting GlcNAc) but low in fucosylation with respect to Fc glycans. (E.g., see Bondt et al., 2014, *Mol. & Cell. Proteomics* 13.11: 3029-3039, incorporated by reference herein in its entirety for its disclosure of Fab-associated N-glycans).

[0027] (v) Significantly, glycans that are added to HuGlyFab of the invention are highly processed complex-type N-glycans that contain 2,6-sialic acid. Such glycans are not present in (a) therapeutic mAbs produced in *E. coli* (which are not glycosylated at all); (b) in therapeutic antibodies produced in CHO cells that do not have the 2,6-sialyltransferase required to add 2,6-sialic acid during glycosylation; or (c) in therapeutic antibodies produced in either CHO or murine cell lines that add N-Glycolylneuraminic acid (“Neu5Gc” or “NeuGc”) which is not natural to humans (and poten-

tially immunogenic), instead of N-Acetylneuraminic acid (“Neu5Ac”) the predominant human sialic acid. See, e.g., Dumont et al., 2015, *Crit. Rev. Biotechnol.* 36 (6): 1110-1122; Huang et al., 2006, *Anal. Biochem.* 349:197-207 (NeuGc is the predominant sialic acid in murine cell lines such as SP2/0 and NS0); and Song et al., 2014, *Anal. Chem.* 86:5661-5666, each of which is incorporated by reference herein in its entirety.

[0028] (vi) The human glycosylation pattern of the HuGlyFab of the invention should reduce immunogenicity of the transgene product and improve efficacy. Importantly, when the antigen-binding fragments, used in accordance with the methods described herein are expressed in human target cells, the need for in vitro production in prokaryotic host cells (e.g., *E. coli*) or eukaryotic host cells (e.g., CHO cells or murine NS0 or SP2/0 cells) is circumvented. Instead, as a result of the methods described herein (e.g., use of human target cells to express the antigen-binding fragments), N-glycosylation sites of the antigen-binding fragments are advantageously decorated with glycans relevant to and beneficial to treatment of humans. Such an advantage is unattainable when CHO cells, murine cells, or *E. coli* are utilized in antibody/antigen-binding fragment production, because, e.g., (a) CHO cells lack components needed for addition of certain glycans (e.g., 2,6 sialic acid and bisecting GlcNAc); (b) CHO cells and murine cells (NS0 and SP2/0 cells) add Neu5Gc as sialic acid not typical to humans instead of Neu5Ac; (c) CHO cells can also produce an immunogenic glycan, the α -Gal antigen, which reacts with anti- α -Gal antibodies present in most individuals, which at high concentrations can trigger anaphylaxis (see, e.g., Bosques, 2010, *Nat Biotech* 28:1153-1156); and (d) *E. coli* does not naturally contain components needed for N-glycosylation.

[0029] (vii) Tyrosine-sulfation of Fab regions, such as those shown in FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H and 9A-9B—a robust post-translational process in many human cells—should result in transgene products with increased avidity for their molecular targets. Indeed, tyrosine-sulfation of the Fab of antibodies has been shown to dramatically increase avidity for antigen and activity. (See, e.g., Loos et al., 2015, *PNAS* 112:12675-12680, and Choc et al., 2003, *Cell* 114:161-170). Such post-translational modifications are not present on therapeutic antibodies made in *E. coli* (a host that does not possess the enzymes required for tyrosine-sulfation), and at best are under-represented in therapeutic mAbs made in CHO cells. CHO cells are not secretory cells and have a limited capacity for post-translational tyrosine-sulfation. (See, e.g., Mikkelsen & Ezban, 1991, *Biochemistry* 30:1533-1537, especially discussion at p. 1537).

[0030] For the foregoing reasons, the production of HuPTM mAb or HuPTM Fab should result in a “biobetter” molecule for the treatment of disease accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a full-length or HuPTM Fab of a therapeutic mAb to a patient (human subject) diagnosed with a disease indication for that mAb, to create a permanent depot in the subject that continuously supplies the human-glycosylated, sulfated transgene product produced by the subject’s transduced cells. The cDNA construct for the

HuPTMmAb or HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced human cells.

[0031] As an alternative, or an additional treatment to gene therapy, the full-length or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and the glycoprotein can be administered to patients.

[0032] Combination therapies involving delivery of the full-length or HuPTM Fab to the patient accompanied by administration of other available treatments are encompassed by the methods of the invention. The additional treatments may be administered before, concurrently or subsequent to the gene therapy treatment. Such additional treatments can include but are not limited to co-therapy with the therapeutic mAb.

[0033] Also provided are methods of manufacturing the viral vectors, particularly the AAV based viral vectors. In specific embodiments, provided are methods of producing recombinant AAVs comprising culturing a host cell containing an artificial genome comprising a cis expression cassette flanked by AAV ITRs, wherein the cis expression cassette comprises a transgene encoding a therapeutic antibody operably linked to expression control elements that will control expression of the transgene in human cells; a trans expression cassette lacking AAV ITRs, wherein the trans expression cassette encodes an AAV rep and capsid protein operably linked to expression control elements that drive expression of the AAV rep and capsid proteins in the host cell in culture and supply the rep and cap proteins in trans; sufficient adenovirus helper functions to permit replication and packaging of the artificial genome by the AAV capsid proteins; and recovering recombinant AAV encapsidating the artificial genome from the cell culture.

3.1 Illustrative Embodiments

Compositions of Matter

[0034] 1. A pharmaceutical composition for treating Alzheimer's disease, migraines, cluster headaches, or tauopathies including chronic traumatic encephalopathy, progressive supranuclear palsy, and frontotemporal dementia in a human subject in need thereof, comprising an adeno-associated virus (AAV) vector having:

[0035] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 capsid (SEQ ID NO: 79) or AAVrh10 capsid (SEQ ID NO: 80); and

[0036] (b) an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-amyloid beta, anti-Tau, or anti-CGRPR mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human CNS cells;

[0037] wherein said AAV vector is formulated for intrathecal administration to the CNS of said subject.

[0038] 2. The pharmaceutical composition of paragraph 1, wherein the anti-amyloid β mAb is aducanumab, crenelumab, gantenerumab, or BAN2401 and the anti-Tau mAb is aTAU and the anti-CGRPR is erenumab, eptinezumab, fremanezumab, or galcanezumab.

[0039] 3. The pharmaceutical composition of paragraphs 1 or 2, wherein the antigen-binding fragment is a Fab, a $F(ab')_2$, or a single chain variable domain (scFv).

[0040] 4. The pharmaceutical composition of any of paragraphs 1 to 3, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 1 and a light chain with an amino acid sequence of SEQ ID NO: 2; or a heavy chain with an amino acid sequence of SEQ ID NO: 3 and a light chain with an amino acid sequence of SEQ ID NO: 4; or a heavy chain with an amino acid sequence of SEQ ID NO: 5 and a light chain with an amino acid sequence of SEQ ID NO: 6; or a heavy chain with an amino acid sequence of SEQ ID NO: 53 and a light chain with an amino acid sequence of SEQ ID NO: 54; a heavy chain with an amino acid sequence of SEQ ID NO: 55 and a light chain with an amino acid sequence of SEQ ID NO: 56; or a heavy chain with an amino acid sequence of SEQ ID NO: 57 and a light chain with an amino acid sequence of SEQ ID NO: 58.

[0041] 5. The pharmaceutical composition of paragraph 4, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 101 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 102 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 103 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 104 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 105 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 106 encoding the light chain; or a heavy chain with a nucleotide sequence of SEQ ID NO: 153 and a light chain with a nucleotide sequence of SEQ ID NO: 154; a heavy chain with a nucleotide sequence of SEQ ID NO: 155 and a light chain with a nucleotide sequence of SEQ ID NO: 156; or a heavy chain with a nucleotide sequence of SEQ ID NO: 157 and a light chain with a nucleotide sequence of SEQ ID NO: 158.

[0042] 6. The pharmaceutical composition of any of paragraphs 1 to 4, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0043] 7. The pharmaceutical composition of any of paragraphs 1 to 6, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human CNS cells.

[0044] 8. The pharmaceutical composition of paragraph 7, wherein said signal sequence is MYRMQLLLLLIALSLA-LVTNS (SEQ ID NO: 161) or a signal sequence from Table 1.

[0045] 9. The pharmaceutical composition of any of paragraphs 1 to 8, wherein the AAV capsid is AAV9.

[0046] 10. A pharmaceutical composition for treating atopic dermatitis in a human subject in need thereof, comprising an AAV vector comprising:

[0047] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0048] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-LAR mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

- [0049] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.
- [0050] 11. The pharmaceutical composition of paragraph 10 wherein the anti-ILAR mAb is dupilumab.
- [0051] 12. The pharmaceutical composition of paragraphs 10 or 11, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0052] 13. The pharmaceutical composition of any of paragraphs 10 to 12, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 7 and a light chain with an amino acid sequence of SEQ ID NO: 8.
- [0053] 14. The pharmaceutical composition of paragraph 13, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 107 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 108 encoding the light chain.
- [0054] 15. The pharmaceutical composition of any of paragraphs 10 to 13, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0055] 16. The pharmaceutical composition of any of paragraphs 10 to 15, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.
- [0056] 17. The pharmaceutical composition of paragraph 16, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.
- [0057] 18. The pharmaceutical composition of any of paragraphs 10 to 17, wherein the AAV capsid is AAV8.
- [0058] 19. A pharmaceutical composition for treating psoriasis, psoriatic arthritis, ankylosing spondylitis, or Crohn's disease in a human subject in need thereof, comprising an AAV vector comprising:
- [0059] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or an AAV9 capsid (SEQ ID NO: 79); and
- [0060] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-IL17A mAb or anti-IL12/IL23 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;
- [0061] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.
- [0062] 20. The pharmaceutical composition of paragraph 19 wherein the anti-IL17A or anti-IL12/IL23 mAb is ixekizumab, secukinumab or ustekinumab.
- [0063] 21. The pharmaceutical composition of paragraphs 19 or 20, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0064] 22. The pharmaceutical composition of any of paragraphs 19 to 21, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 9 and a light chain with an amino acid sequence of SEQ ID NO: 10; or a heavy chain with an amino acid sequence of SEQ ID NO: 11 and a light chain with an amino acid sequence of SEQ ID NO: 12; or a heavy chain with an amino acid sequence of SEQ ID NO: 13 and a light chain with an amino acid sequence of SEQ ID NO: 14.
- [0065] 23. The pharmaceutical composition of paragraph 22, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 109 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 110 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 111 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 112 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 113 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 114 encoding the light chain.
- [0066] 24. The pharmaceutical composition of any of paragraphs 19 to 22, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0067] 25. The pharmaceutical composition of any of paragraphs 19 to 24, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.
- [0068] 26. The pharmaceutical composition of paragraph 25, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.
- [0069] 27. The pharmaceutical composition of any of paragraphs 19 to 26, wherein the AAV capsid is AAV8.
- [0070] 28. A pharmaceutical composition for treating asthma in a human subject in need thereof, comprising an AAV vector comprising:
- [0071] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or an AAV9 capsid (SEQ ID NO: 79); and
- [0072] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-IL-5 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;
- [0073] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.
- [0074] 29. The pharmaceutical composition of paragraph 28 wherein the anti-IL-5 mAb is mepolizumab.
- [0075] 30. The pharmaceutical composition of paragraphs 28 or 29, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0076] 31. The pharmaceutical composition of any of paragraphs 28 to 30, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 15 and a light chain with an amino acid sequence of SEQ ID NO: 16.
- [0077] 32. The pharmaceutical composition of paragraph 31, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 115 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 116 encoding the light chain.
- [0078] 33. The pharmaceutical composition of any of paragraphs 28 to 31, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0079] 34. The pharmaceutical composition of any of paragraphs 28 to 33, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light

chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0080] 35. The pharmaceutical composition of paragraph 34, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0081] 36. The pharmaceutical composition of any of paragraphs 28 to 35, wherein the AAV capsid is AAV8.

[0082] 37. A pharmaceutical composition for treating multiple sclerosis, ulcerative colitis or Crohn's disease in a human subject in need thereof, comprising an AAV vector comprising:

[0083] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78), an AAV9 capsid (SEQ ID NO: 79), or an AAVrh10 capsid (SEQ ID NO: 80); and

[0084] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-integrin mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells or human CNS cells;

[0085] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject or for the intrathecal administration to the CNS of said subject.

[0086] 38. The pharmaceutical composition of paragraph 37, wherein the anti-integrin mAb is vedolizumab or natalizumab.

[0087] 39. The pharmaceutical composition of paragraphs 37 or 38, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0088] 40. The pharmaceutical composition of any of paragraphs 37 to 39, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 17 and a light chain with an amino acid sequence of SEQ ID NO: 18; or a heavy chain with an amino acid sequence of SEQ ID NO: 19 and a light chain with an amino acid sequence of SEQ ID NO: 20.

[0089] 41. The pharmaceutical composition of paragraph 40, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 117 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 118 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 119 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 120 encoding the light chain.

[0090] 42. The pharmaceutical composition of any of paragraphs 37 to 41, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0091] 43. The pharmaceutical composition of any of paragraphs 37 to 42, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0092] 44. The pharmaceutical composition of paragraph 43, wherein said signal sequence is selected from the signal sequences in Table 1, 2 or 3.

[0093] 45. The pharmaceutical composition of any of paragraphs 37 to 44, wherein the AAV capsid is AAV8.

[0094] 46. A pharmaceutical composition for treating HeFH, HoFH, dyslipidemia, cardiovascular disease including atherosclerotic cardiovascular disease (ACD), athero-

sclerotic plaque formation, abnormally high levels of non-HDL cholesterol and LDL, aortic stenosis, hepatic stenosis, or hypercholesterolemia in a human subject in need thereof, comprising an AAV vector comprising:

[0095] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or an AAV9 capsid (SEQ ID NO: 79); and

[0096] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-PCSK9, anti-ANGPTL3, or anti-OxPL mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0097] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.

[0098] 47. The pharmaceutical composition of paragraph 46, wherein the anti-PCSK9 or anti-ANGPTL3 mAb is alirocumab, evolocumab or evinacumab or the anti-OxPL is E06.

[0099] 48. The pharmaceutical composition of paragraphs 46 or 47, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0100] 49. The pharmaceutical composition of any of paragraphs 46 to 48, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 21 and a light chain with an amino acid sequence of SEQ ID NO: 22; or a heavy chain with an amino acid sequence of SEQ ID NO: 23 and a light chain with an amino acid sequence of SEQ ID NO: 24; a heavy chain with an amino acid sequence of SEQ ID NO: 25 and a light chain with an amino acid sequence of SEQ ID NO: 26; or a heavy chain with an amino acid sequence of SEQ ID NO: 59 and a light chain with an amino acid sequence of SEQ ID NO: 60.

[0101] 50. The pharmaceutical composition of paragraph 49, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 121 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 122 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 123 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 124 encoding the light chain; a nucleotide sequence of SEQ ID NO: 125 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 126 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 159 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 160 encoding the light chain.

[0102] 51. The pharmaceutical composition of any of paragraphs 44 to 50, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0103] 52. The pharmaceutical composition of any of paragraphs 44 to 51, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0104] 53. The pharmaceutical composition of paragraph 52, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0105] 54. The pharmaceutical composition of any of paragraphs 44 to 53, wherein the AAV capsid is AAV8.

[0106] 55. A pharmaceutical composition for treating osteoporosis in a human subject in need thereof, comprising an AAV vector comprising:

[0107] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0108] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-RANKL mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0109] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.

[0110] 56. The pharmaceutical composition of paragraph 55, wherein the anti-RANKL mAb is denosumab.

[0111] 57. The pharmaceutical composition of paragraphs 55 or 56, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0112] 58. The pharmaceutical composition of any of paragraphs 55 to 57, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 27 and a light chain with an amino acid sequence of SEQ ID NO:28.

[0113] 59. The pharmaceutical composition of paragraph 58, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 127 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 128 encoding the light chain.

[0114] 60. The pharmaceutical composition of any of paragraphs 55 to 59, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0115] 61. The pharmaceutical composition of any of paragraphs 55 to 60, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0116] 62. The pharmaceutical composition of paragraph 61, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0117] 63. The pharmaceutical composition of any of paragraphs 55 to 62, wherein the AAV capsid is AAV8.

[0118] 64. A pharmaceutical composition for treating metastatic melanoma, lymphoma or non-small cell lung carcinoma in a human subject in need thereof, comprising an AAV vector comprising:

[0119] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0120] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding a PD-1 blocker mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0121] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.

[0122] 65. The pharmaceutical composition of paragraph 64, wherein the PD-1 blocker mAb is nivolumab or pembrolizumab.

[0123] 66. The pharmaceutical composition of paragraphs 64 or 65, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0124] 67. The pharmaceutical composition of any of paragraphs 64 to 66, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 29 and a light chain with an amino acid sequence of SEQ ID NO: 30; or a heavy chain with an amino acid sequence of SEQ ID NO: 31 and a light chain with an amino acid sequence of SEQ ID NO: 32.

[0125] 68. The pharmaceutical composition of paragraph 67, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 129 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 130 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 131 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 132 encoding the light chain.

[0126] 69. The pharmaceutical composition of any of paragraphs 64 to 68, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0127] 70. The pharmaceutical composition of any of paragraphs 64 to 69, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0128] 71. The pharmaceutical composition of paragraph 70, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0129] 72. The pharmaceutical composition of any of paragraphs 64 to 71, wherein the AAV capsid is AAV8.

[0130] 73. A pharmaceutical composition for treating systemic lupus erythematosus (SLE) in a human subject in need thereof, comprising an AAV vector comprising:

[0131] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or an AAV9 capsid (SEQ ID NO: 79); and

[0132] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-BLyS mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0133] wherein said AAV vector is formulated for intravenous administration to the liver or muscle of said subject.

[0134] 74. The pharmaceutical composition of paragraph 73, wherein the anti-BLyS mAb is belimumab.

[0135] 75. The pharmaceutical composition of paragraphs 73 or 74, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0136] 76. The pharmaceutical composition of any of paragraphs 73 to 75, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 41 and a light chain with an amino acid sequence of SEQ ID NO:42.

[0137] 77. The pharmaceutical composition of paragraph 76, wherein the transgene comprises a nucleotide sequence

of SEQ ID NO: 141 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 142 encoding the light chain.

[0138] 78. The pharmaceutical composition of any of paragraphs 73 to 77, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0139] 79. The pharmaceutical composition of any of paragraphs 73 to 78, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0140] 80. The pharmaceutical composition of paragraph 79, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0141] 81. The pharmaceutical composition of any of paragraphs 73 to 80, wherein the AAV capsid is AAV8.

[0142] 82. A pharmaceutical composition for treating ocular disorders, including age-related macular degeneration, in a human subject in need thereof, comprising an AAV vector comprising:

[0143] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0144] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-MMP9, anti-VEGF or anti-Id mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human retinal cells;

[0145] wherein said AAV vector is formulated for sub-retinal, intravitreal or suprachoroidal administration to the eye of said subject.

[0146] 83. The pharmaceutical composition of paragraph 82, wherein the anti-VEGF mAb is ranibizumab, bevacizumab, or brodalumab, said anti-Fd mAb is lampalizumab or said anti-MMP9 mAb is andecaliximab.

[0147] 84. The pharmaceutical composition of paragraphs 82 or 83, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0148] 85. The pharmaceutical composition of any of paragraphs 82 to 84, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 33 and a light chain with an amino acid sequence of SEQ ID NO: 34, or a heavy chain with an amino acid sequence of SEQ ID NO: 35 and a light chain with an amino acid sequence of SEQ ID NO: 36; or a heavy chain with an amino acid sequence of SEQ ID NO: 37 and a light chain with an amino acid sequence of SEQ ID NO: 38; or a heavy chain with an amino acid sequence of SEQ ID NO: 39 and a light chain with an amino acid sequence of SEQ ID NO: 40; or a heavy chain with an amino acid sequence of SEQ ID NO: 45 and a light chain with an amino acid sequence of SEQ ID NO: 46.

[0149] 86. The pharmaceutical composition of paragraph 85, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 133 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 134 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 135 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 136 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 137 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 138 encoding the light

chain; or a nucleotide sequence of SEQ ID NO: 139 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 140 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 145 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 146 encoding the light chain.

[0150] 87. The pharmaceutical composition of any of paragraphs 82 to 85, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0151] 88. The pharmaceutical composition of any of paragraphs 82 to 87, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human retinal cells.

[0152] 89. The pharmaceutical composition of paragraph 88, wherein said signal sequence is selected from the signal sequences in Table 1.

[0153] 90. The pharmaceutical composition of any of paragraphs 82 to 89, wherein the AAV capsid is AAV8.

[0154] 91. A pharmaceutical composition for treating rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, plaque psoriasis, or ulcerative colitis, in a human subject in need thereof, comprising an AAV vector comprising:

[0155] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0156] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-TNF antibody, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0157] wherein said AAV vector is formulated for intravenous administration to said subject.

[0158] 92. The pharmaceutical composition of paragraph 91, wherein the anti-TNF-alpha mAb is adalimumab or infliximab.

[0159] 93. The pharmaceutical composition of paragraphs 91 or 92, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0160] 94. The pharmaceutical composition of any of paragraphs 91 to 93, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 49 and a light chain with an amino acid sequence of SEQ ID NO: 50; or a heavy chain with an amino acid sequence of SEQ ID NO: 51 and a light chain with an amino acid sequence of SEQ ID NO: 52.

[0161] 95. The pharmaceutical composition of paragraph 94, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 149 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 150 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 151 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 152 encoding the light chain.

[0162] 96. The pharmaceutical composition of any of paragraphs 91 to 94, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0163] 97. The pharmaceutical composition of any of paragraphs 91 to 96, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light

chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells or human muscle cells.

[0164] 98. The pharmaceutical composition of paragraph 97, wherein said signal sequence is selected from the signal sequences in Table 2 or 3.

[0165] 99. The pharmaceutical composition of any of paragraphs 91 to 98, wherein the AAV capsid is AAV8.

[0166] 100. A pharmaceutical composition for treating paroxysmal nocturnal hemoglobinuria (PNH) or atypical hemolytic uremic syndrome (aHUS), in a human subject in need thereof, comprising an AAV vector comprising:

[0167] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0168] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-C5 or C5a complement protein mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells; wherein said AAV vector is formulated for intravenous administration to said subject.

[0169] 101. The pharmaceutical composition of paragraph 100, wherein the anti-C5 or C5a complement protein mAb is eculizumab.

[0170] 102. The pharmaceutical composition of paragraphs 100 or 101, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0171] 103. The pharmaceutical composition of any of paragraphs 100 to 102, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 43 and a light chain with an amino acid sequence of SEQ ID NO: 44.

[0172] 104. The pharmaceutical composition of paragraph 103, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 143 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 144 encoding the light chain.

[0173] 105. The pharmaceutical composition of any of paragraphs 101 to 104, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0174] 106. The pharmaceutical composition of any of paragraphs 100 to 105, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells.

[0175] 107. The pharmaceutical composition of paragraph 106, wherein said signal sequence is selected from the signal sequences in Table 3.

[0176] 108. The pharmaceutical composition of any of paragraphs 101 to 107, wherein the AAV capsid is AAV8.

[0177] 109. A pharmaceutical composition for treating hereditary angiodema, in a human subject in need thereof, comprising an AAV vector comprising:

[0178] (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78) or AAV9 capsid (SEQ ID NO: 79); and

[0179] (b) an artificial genome comprising an expression cassette flanked by AAV ITRs wherein the expression cassette comprises a transgene encoding an anti-plasma kallikrein mAb, or an antigen-binding fragment

thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells;

[0180] wherein said AAV vector is formulated for intravenous administration to said subject.

[0181] 110. The pharmaceutical composition of paragraph 109, wherein the anti-plasma kallikrein mAb is lanadelumab.

[0182] 111. The pharmaceutical composition of paragraphs 109 or 111, wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0183] 112. The pharmaceutical composition of any of paragraphs 109 to 111, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 47 and a light chain with an amino acid sequence of SEQ ID NO: 48.

[0184] 113. The pharmaceutical composition of paragraph 112, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 147 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 148 encoding the light chain.

[0185] 114. The pharmaceutical composition of any of paragraphs 110 to 113, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0186] 115. The pharmaceutical composition of any of paragraphs 109 to 114, wherein the transgene encodes a signal sequence at the N-terminus of the heavy chain and the light chain of said antigen-binding fragment that directs secretion and post translational modification in said human liver cells.

[0187] 116. The pharmaceutical composition of paragraph 115, wherein said signal sequence is selected from the signal sequences in Table 3.

[0188] 117. The pharmaceutical composition of any of paragraphs 110 to 116, wherein the AAV capsid is AAV8.

Method of Treatment

[0189] 118. A method of treating Alzheimer's disease, migraines, cluster headaches, or tauopathies including chronic traumatic encephalopathy, progressive supranuclear palsy, and frontotemporal dementia in a human subject in need thereof, comprising delivering to the cerebrospinal fluid (CSF) of said human subject, a therapeutically effective amount of an anti-amyloid beta, anti-Tau, or anti-CGRPR mAb or antigen-binding fragment thereof, produced by human central nervous system (CNS) cells.

[0190] 119. A method of treating Alzheimer's disease, migraines, cluster headaches, or tauopathies including chronic traumatic encephalopathy, progressive supranuclear palsy, and frontotemporal dementia in a human subject in need thereof, comprising:

[0191] administering to the cisterna *magna* of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-amyloid beta, anti-Tau, or anti-CGRPR mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human CNS cells, so that a depot is formed that releases a human post-translationally modified (HuPTM) form of said mAb or antigen-binding fragment thereof.

[0192] 120. The method of paragraphs 118 or 119 wherein the anti-amyloid beta mAb is aducanumab, crenezumab, gantenerumab, or BAN2401 or wherein the anti-Tau mAb is

aTAU or wherein the anti-CGRPR is erenumab, eptinezumab, fremanezumab, or galcanezumab.

[0193] 121. The method of any of paragraphs 118 to 120 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0194] 122. The method of any of paragraphs 118 to 121, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 1 and a light chain with an amino acid sequence of SEQ ID NO:2; or a heavy chain with an amino acid sequence of SEQ ID NO: 3 and a light chain with an amino acid sequence of SEQ ID NO:4; or a heavy chain with an amino acid sequence of SEQ ID NO: 5 and a light chain with an amino acid sequence of SEQ ID NO:6; or a heavy chain with an amino acid sequence of SEQ ID NO: 53 and a light chain with an amino acid sequence of SEQ ID NO:54; a heavy chain with an amino acid sequence of SEQ ID NO: 55 and a light chain with an amino acid sequence of SEQ ID NO:56; or a heavy chain with an amino acid sequence of SEQ ID NO: 57 and a light chain with an amino acid sequence of SEQ ID NO:58.

[0195] 123. The method of claim 122, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 101 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 102 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 103 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 104 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 105 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 106 encoding the light chain; or a heavy chain with an amino acid sequence of SEQ ID NO: 153 and a light chain with an amino acid sequence of SEQ ID NO: 154; a heavy chain with an amino acid sequence of SEQ ID NO: 155 and a light chain with an amino acid sequence of SEQ ID NO:156 or a heavy chain with an amino acid sequence of SEQ ID NO: 157 and a light chain with an amino acid sequence of SEQ ID NO:158.

[0196] 124. The method of any of paragraphs 118 to 122, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0197] 125. The method of any of paragraphs 118 to 124 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0198] 126. The method of any of paragraphs 118 to 125 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc and/or α-Gal.

[0199] 127. The method of any of paragraphs 118 to 126 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0200] 128. The method of any of paragraphs 119 to 127 wherein the recombinant expression vector is AAV9 or AAVrh10.

[0201] 129. The method of any of paragraphs 119 to 128 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human CNS cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0202] 130. A method of treating psoriasis, psoriatic arthritis, ankylosing spondylitis, or Crohn's disease in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective

amount of an anti-IL17A or anti-IL12/IL23 mAb or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0203] 131. A method of treating psoriasis, psoriatic arthritis, ankylosing spondylitis, or Crohn's disease in a human subject in need thereof, comprising:

[0204] administering to the liver or muscle of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-IL17A or anti-IL12/IL23 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

[0205] 132. The method of paragraph 130 or 131 wherein the anti-IL17A or anti-IL12/IL23 mAb is ixekizumab, secukinumab, or ustekinumab.

[0206] 133. The method of any of paragraphs 130 to 132 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0207] 134. The method of any of paragraphs 130 to 133, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 9 and a light chain with an amino acid sequence of SEQ ID NO:10; or a heavy chain with an amino acid sequence of SEQ ID NO: 11 and a light chain with an amino acid sequence of SEQ ID NO: 12; or a heavy chain with an amino acid sequence of SEQ ID NO: 13 and a light chain with an amino acid sequence of SEQ ID NO: 14.

[0208] 135. The method of claim 134, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 109 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 110 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 111 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 112 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 113 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 114 encoding the light chain.

[0209] 136. The method of any of paragraphs 132 to 134, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0210] 137. The method of any of paragraphs 132 to 136 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0211] 138. The method of any of paragraphs 132 to 137 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.

[0212] 139. The method of any of paragraphs 132 to 138 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0213] 140. The method of any of paragraphs 133 to 139 wherein the recombinant expression vector is AAV8 or AAV9.

[0214] 141. The method of any of paragraphs 133 to 140 in which production of said HuPTM form of the mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0215] 142. A method of treating multiple sclerosis, ulcerative colitis or Crohn's disease in a human subject in need

thereof, comprising delivering to the CSF or circulation of said human subject, a therapeutically effective amount of an anti-integrin mAb or antigen-binding fragment thereof, produced by human CNS cells, human liver cells or human muscle cells.

[0216] 143. A method of treating multiple sclerosis, ulcerative colitis or Crohn's disease in a human subject in need thereof, comprising:

[0217] administering to the CNS, liver or muscle of said human subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-integrin mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human CNS cells, human liver cells or in human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment.

[0218] 144. The method of paragraphs 142 or 143 wherein the anti-integrin mAb is natalizumab or vedolizumab.

[0219] 145. The method of any of paragraphs 142 to 144 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0220] 146. The method of any of paragraphs 142 to 145, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 17 and a light chain with an amino acid sequence of SEQ ID NO: 18; or a heavy chain with an amino acid sequence of SEQ ID NO: 19 and a light chain with an amino acid sequence of SEQ ID NO: 20.

[0221] 147. The method of claim 146, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 117 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 118 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 119 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 120 encoding the light chain.

[0222] 148. The method of any of paragraphs 142 to 145, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0223] 149. The method of any of paragraphs 142 to 148 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0224] 150. The method of any of paragraphs 142 to 149 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0225] 151. The method of any of paragraphs 142 to 150 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0226] 152. The method of any of paragraphs 143 to 151 wherein the recombinant expression vector is AAV8, AAV9, or AAVrh10.

[0227] 153. The method of any of paragraphs 143 to 152 in which production of the HuPTM form of the mAb or antigen-binding fragment thereof is confirmed by transducing human CNS cells, human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0228] 154. A method of treating atopic dermatitis in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective

amount of an anti-ILAR mAb or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0229] 155. A method of treating atopic dermatitis in a human subject in need thereof, comprising: administering to the liver or muscle of said human subject, a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-ILAR mAb, or antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells, so that a depot is formed that released a HuPTM form of said mAb or antigen-binding fragment thereof.

[0230] 156. The method of paragraphs 154 or 155 wherein the anti-IL-4R mAb is dupilumab.

[0231] 157. The method of any of paragraphs 154 to 156 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0232] 158. The method of any of paragraphs 154 to 157, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 7 and a light chain with an amino acid sequence of SEQ ID NO: 8.

[0233] 159. The method of claim 158, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 107 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 108 encoding the light chain.

[0234] 160. The method of any of paragraphs 154 to 158, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0235] 161. The method of any of paragraphs 154 to 160 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0236] 162. The method of any of paragraphs 154 to 161 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0237] 163. The method of any of paragraphs 154 to 162 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0238] 164. The method of any of paragraphs 155 to 163 wherein the recombinant expression vector is AAV8 or AAV9.

[0239] 165. The method of any of paragraphs 155 to 164 in which production of the HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0240] 166. A method of treating asthma in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-IL-5 mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0241] 167. A method of treating asthma in a human subject in need thereof, comprising:

[0242] administering to the liver or muscle of said human subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-IL-5 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle

cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

[0243] 168. The method of paragraphs 166 or 167 wherein the anti-IL-5 mAb is mepolizumab.

[0244] 169. The method of any of paragraphs 166 to 168 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0245] 170. The method of any of paragraphs 166 to 169, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 15 and a light chain with an amino acid sequence of SEQ ID NO:16.

[0246] 171. The method of claim 170, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 115 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 116 encoding the light chain.

[0247] 172. The method of any of paragraphs 166 to 170, wherein the antibody or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0248] 173. The method of any of paragraphs 166 to 172 wherein the mAb or antigen-binding fragment thereof contains an alpha2,6-sialylated glycan.

[0249] 174. The method of any of paragraphs 166 to 173 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.

[0250] 175. The method of any of paragraphs 166 to 174 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0251] 176. The method of any of paragraphs 167 to 175 wherein the recombinant expression vector is AAV8 or AAV9.

[0252] 177. The method of any of paragraphs 167 to 176 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0253] 178. A method of treating HeFH, HoFH, dyslipidemia, cardiovascular disease including atherosclerotic cardiovascular disease (ACD), atherosclerotic plaque formation, abnormally high levels of non-HDL cholesterol and LDL, aortic stenosis, hepatic stenosis, or hypercholesterolemia dyslipidemia in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-PCSK9, anti-ANGPTL3, anti-OxPL mAb or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0254] 179. A method of treating HeFH, HoFH, dyslipidemia, cardiovascular disease including atherosclerotic cardiovascular disease (ACD), atherosclerotic plaque formation, abnormally high levels of non-HDL cholesterol and LDL, aortic stenosis, hepatic stenosis, or hypercholesterolemia in a human subject in need thereof, comprising:

[0255] administering to the liver or muscle of said human subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-PCSK9, anti-OxPL, or anti-ANGPTL3 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells, so that a depot is formed that releases a HuPTM form of the mAb or antigen-binding fragment thereof.

[0256] 180. The method of paragraph 178 or 179 wherein the anti-PCSK9 is alirocumab or evolocumab, or the anti-ANGPTL3 mAb is evinacumab or the anti-OxPL is E06.

[0257] 181. The method of any of paragraphs 178 to 180 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0258] 182. The method of any of paragraphs 178 to 181, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 21 and a light chain with an amino acid sequence of SEQ ID NO:22; or a heavy chain with an amino acid sequence of SEQ ID NO: 23 and a light chain with an amino acid sequence of SEQ ID NO: 24; a heavy chain with an amino acid sequence of SEQ ID NO: 25 and a light chain with an amino acid sequence of SEQ ID NO:26; or a heavy chain with an amino acid sequence of SEQ ID NO: 59 and a light chain with an amino acid sequence of SEQ ID NO: 60.

[0259] 183. The method of claim 182, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 121 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 122 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 123 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 124 encoding the light chain; a nucleotide sequence of SEQ ID NO: 125 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 126 encoding the light chain; or a heavy chain with an amino acid sequence of SEQ ID NO: 159 and a light chain with an amino acid sequence of SEQ ID NO:160.

[0260] 184. The method of any of paragraphs 178 to 182, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0261] 185. The method of any of paragraphs 178 to 184 wherein the mAb or antigen-binding fragment thereof contains an alpha2,6-sialylated glycan.

[0262] 186. The method of any of paragraphs 178 to 185 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.

[0263] 187. The method of any of paragraphs 178 to 186 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0264] 188. The method of any of paragraphs 179 to 187 wherein the recombinant expression vector is AAV8 or AAV9.

[0265] 189. The method of any of paragraphs 179 to 188 in which production of said HuPTM form of the mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or human muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0266] 190. A method of treating osteoporosis, increasing bone mass in breast or prostate cancer patients, or preventing skeletal related events due to bone metastasis in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-RANKL mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0267] 191. A method of treating osteoporosis, increasing bone mass in breast or prostate cancer patients, or preventing skeletal related events due to bone metastasis in a human subject in need thereof, comprising:

[0268] administering to the liver or muscle of said human subject a therapeutically effective amount of a

- recombinant nucleotide expression vector comprising a transgene encoding an anti-RANKL mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells, so that a depot is formed that releases a HuPTM form of the mAb or antigen-binding fragment thereof.
- [0269] 192. The method of paragraph 190 or 191 wherein the anti-RANKL mAb is denosumab.
- [0270] 193. The method of any of paragraphs 190 to 192 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0271] 194. The method of any of paragraphs 190 to 193, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 27 and a light chain with an amino acid sequence of SEQ ID NO: 28.
- [0272] 195. The method of claim 194, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 128 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 127 encoding the light chain.
- [0273] 196. The method of any of paragraphs 190 to 194, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0274] 197. The method of any of paragraphs 190 to 196 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.
- [0275] 198. The method of any of paragraphs 190 to 197 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.
- [0276] 199. The method of any of paragraphs 190 to 198 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.
- [0277] 200. The method of any of paragraphs 191 to 199 wherein the recombinant expression vector is AAV8 or AAV9.
- [0278] 201. The method of any of paragraphs 191 to 200 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.
- [0279] 202. A method of treating metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of a PD-1 blocker mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.
- [0280] 203. A method of treating metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma in a human subject in need thereof, comprising:
- [0281] administering to the liver or muscle of said human subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding a PD-1 blocker mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.
- [0282] 204. The method of paragraph 202 or 203 wherein the PD-1 blocker mAb is nivolumab or pembrolizumab.
- [0283] 205. The method of any of paragraphs 202 to 204 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0284] 206. The method of any of paragraphs 202 to 205, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 29 and a light chain with an amino acid sequence of SEQ ID NO:30; or a heavy chain with an amino acid sequence of SEQ ID NO: 31 and a light chain with an amino acid sequence of SEQ ID NO:32.
- [0285] 207. The method of claim 206, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 129 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 130 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 131 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 132 encoding the light chain.
- [0286] 208. The method of any of paragraphs 202 to 206, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0287] 209. The method of any of paragraphs 202 to 208 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.
- [0288] 210. The method of any of paragraphs 202 to 209 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.
- [0289] 211. The method of any of paragraphs 202 to 210 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.
- [0290] 212. The method of any of paragraphs 203 to 211 wherein the recombinant expression vector is AAV8 or AAV9.
- [0291] 213. The method of any of paragraphs 203 to 212 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.
- [0292] 214. A method of treating systemic lupus erythematosus (SLE) in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-BLyS mAb or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.
- [0293] 215. A method of treating SLE in a human subject in need thereof, comprising:
- [0294] administering to the liver or muscle of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-BLyS mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or in human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

[0295] 216. The method of paragraph 214 or 215 wherein the anti-BLYS mAb is belimumab.

[0296] 217. The method of any of paragraphs 214 to 216 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0297] 218. The method of any of paragraphs 214 to 217, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 41 and a light chain with an amino acid sequence of SEQ ID NO: 42.

[0298] 219. The method of claim 218, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 139 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 142 encoding the light chain.

[0299] 220. The method of any of paragraphs 214 to 218, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0300] 221. The method of any of paragraphs 214 to 220 wherein the mAb or antigen-binding fragment thereof contains an alpha2,6-sialylated glycan.

[0301] 222. The method of any of paragraphs 214 to 221 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0302] 223. The method of any of paragraphs 214 to 222 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0303] 224. The method of any of paragraphs 215 to 223 wherein the recombinant expression vector is AAV8 or AAV9.

[0304] 225. The method of any of paragraphs 215 to 224 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0305] 226. A method of treating an ocular disorder, including neovascular age-related macular degeneration (nAMD), dry AMD, diabetic retinopathy, diabetic macular edema (DME), central retinal vein occlusion (RVO), pathologic myopia, or polypoidal choroidal vasculopathy, in a human subject in need thereof, comprising delivering to the retina of said human subject, a therapeutically effective amount of an anti-MMP9, anti-VEGF or anti-fD mAb, or antigen-binding fragment thereof, produced by human retina cells.

[0306] 227. A method of treating an ocular disorder, including nAMD, dry AMD, diabetic retinopathy, DME, RVO, pathologic myopia, or polypoidal choroidal vasculopathy, in a human subject in need thereof, comprising:

[0307] administering subretinally, intravitreally or suprachoroidally to said human subject, a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-MMP9, anti-VEGF or anti-fD mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human retinal cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

[0308] 228. The method of paragraph 226 or 227 wherein the anti-MMP9, anti-VEGF or anti-fD mAb is anecalciximab, ranibizumab, bevacizumab, brolucizumab, or lampalizumab.

[0309] 229. The method of any of paragraphs 226 to 228 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0310] 230. The method of any of paragraphs 226 to 229, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 33 and a light chain with an amino acid sequence of SEQ ID NO: 34; or a heavy chain with an amino acid sequence of SEQ ID NO: 35 and a light chain with an amino acid sequence of SEQ ID NO: 36; or a heavy chain with an amino acid sequence of SEQ ID NO: 37 and a light chain with an amino acid sequence of SEQ ID NO: 38; or a heavy chain with an amino acid sequence of SEQ ID NO: 39 and a light chain with an amino acid sequence of SEQ ID NO: 40; or a heavy chain with an amino acid sequence of SEQ ID NO: 45 and a light chain with an amino acid sequence of SEQ ID NO: 46.

[0311] 231. The method of claim 230, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 133 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 134 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 135 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 136 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 137 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 138 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 139 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 140 encoding the light chain; or a nucleotide sequence of SEQ ID NO: 145 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 146 encoding the light chain.

[0312] 232. The method of any of paragraphs 226 to 230, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0313] 233. The method of any of paragraphs 226 to 232 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0314] 234. The method of any of paragraphs 226 to 233 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0315] 235. The method of any of paragraphs 226 to 234 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0316] 236. The method of any of paragraphs 227 to 235 wherein the recombinant expression vector is AAV8 or AAV9.

[0317] 237. The method of any of paragraphs 227 to 236 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human retinal cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0318] 238. A method of treating cystic fibrosis (CF), rheumatoid arthritis (RA), UC, CD, solid tumors, pancreatic adenocarcinoma, lung adenocarcinoma, lung squamous cell carcinoma, esophagogastric adenocarcinoma, gastric cancer, colorectal cancer, or breast cancer in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-MMP9 mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.

[0319] 239. A method of treating cystic fibrosis (CF), rheumatoid arthritis (RA), UC, CD, solid tumors, pancreatic

adenocarcinoma, lung adenocarcinoma, lung squamous cell carcinoma, esophagogastric adenocarcinoma, gastric cancer, colorectal cancer, or breast cancer in a human subject in need thereof, comprising:

- [0320] administering to the liver or muscle of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-MMP9 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or in human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.
- [0321] 240. The method of paragraph 238 or 239 wherein the anti-MMP9 mAb is andecaliximab.
- [0322] 241. The method of any of paragraphs 238 to 240 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0323] 242. The method of any of paragraphs 238 to 241, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 45 and a light chain with an amino acid sequence of SEQ ID NO: 46.
- [0324] 243. The method of claim 242, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 145 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 146 encoding the light chain.
- [0325] 244. The method of any of paragraphs 238 to 242, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0326] 245. The method of any of paragraphs 238 to 244 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.
- [0327] 246. The method of any of paragraphs 238 to 245 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.
- [0328] 247. The method of any of paragraphs 238 to 246 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.
- [0329] 248. The method of any of paragraphs 239 to 247 wherein the recombinant expression vector is AAV8 or AAV9.
- [0330] 249. The method of any of paragraphs 239 to 248 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.
- [0331] 250. A method of treating hereditary angioedema in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-kallikrein mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.
- [0332] 251. A method of treating hereditary angioedema in a human subject in need thereof, comprising:

[0333] administering to the liver or muscle of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-kallikrein mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or in human muscle

cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

- [0334] 252. The method of paragraph 250 or 251 wherein the anti-kallikrein mAb is lanadelumab.
- [0335] 253. The method of any of paragraphs 250 to 252 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0336] 254. The method of any of paragraphs 250 to 253, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 47 and a light chain with an amino acid sequence of SEQ ID NO: 48.
- [0337] 255. The method of claim 254, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 147 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 148 encoding the light chain.
- [0338] 256. The method of any of paragraphs 250 to 255, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.
- [0339] 257. The method of any of paragraphs 250 to 256 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.
- [0340] 258. The method of any of paragraphs 250 to 257 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α-Gal.
- [0341] 259. The method of any of paragraphs 250 to 258 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.
- [0342] 260. The method of any of paragraphs 251 to 259 wherein the recombinant expression vector is AAV8 or AAV9.
- [0343] 261. The method of any of paragraphs 251 to 260 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.
- [0344] 262. A method of treating rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, plaque psoriasis, or ulcerative colitis in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-TNF-alpha mAb, or antigen-binding fragment thereof, produced by human liver cells or human muscle cells.
- [0345] 263. A method of treating rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease, plaque psoriasis, or ulcerative colitis in a human subject in need thereof, comprising: administering to the liver or muscle of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-TNF-alpha mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells or in human muscle cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.
- [0346] 264. The method of paragraph 262 or 263 wherein the anti-TNF-alpha mAb is adalimumab or infliximab.
- [0347] 265. The method of any of paragraphs 262 to 264 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.
- [0348] 266. The method of any of paragraphs 262 to 265, wherein the antigen-binding fragment comprises a heavy

chain with an amino acid sequence of SEQ ID NO: 49 and a light chain with an amino acid sequence of SEQ ID NO: 50; or a heavy chain with an amino acid sequence of SEQ ID NO: 51 and a light chain with an amino acid sequence of SEQ ID NO: 52.

[0349] 267. The method of claim **266**, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 149 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 150 encoding the light chain; a nucleotide sequence of SEQ ID NO: 151 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 152 encoding the light chain.

[0350] 268. The method of any of paragraphs 262 to 266, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0351] 269. The method of any of paragraphs 262 to 268 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0352] 270. The method of any of paragraphs 262 to 269 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0353] 271. The method of any of paragraphs 262 to 270 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0354] 272. The method of any of paragraphs 263 to 271 wherein the recombinant expression vector is AAV8 or AAV9.

[0355] 273. The method of any of paragraphs 263 to 272 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

[0356] 274. A method of treating PNH or aHUS in a human subject in need thereof, comprising delivering to the circulation of said human subject, a therapeutically effective amount of an anti-C5 or C5a protein mAb, or antigen-binding fragment thereof, produced by human liver cells.

[0357] 275. A method of treating PNH or aHUS in a human subject in need thereof, comprising: administering to the liver of said subject a therapeutically effective amount of a recombinant nucleotide expression vector comprising a transgene encoding an anti-C5 or C5a protein mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver cells, so that a depot is formed that releases a HuPTM form of said mAb or antigen-binding fragment thereof.

[0358] 276. The method of paragraphs 274 to 275 wherein the anti-C5 or C5a protein mAb, or antigen binding fragment, is eculizumab.

[0359] 277. The method of any of paragraphs 274 to 276 wherein the antigen-binding fragment is a Fab, a F(ab')₂, or an scFv.

[0360] 278. The method of any of paragraphs 274 to 277, wherein the antigen-binding fragment comprises a heavy chain with an amino acid sequence of SEQ ID NO: 43 and a light chain with an amino acid sequence of SEQ ID NO: 44.

[0361] 279. The method of claim **278**, wherein the transgene comprises a nucleotide sequence of SEQ ID NO: 143 encoding the heavy chain and a nucleotide sequence of SEQ ID NO: 144 encoding the light chain.

[0362] 280. The method of any of paragraphs 274 to 279, wherein the mAb or antigen-binding fragment thereof is a hyperglycosylated mutant.

[0363] 281. The method of any of paragraphs 274 to 280 wherein the mAb or antigen-binding fragment thereof contains an alpha 2,6-sialylated glycan.

[0364] 282. The method of any of paragraphs 274 to 281 wherein the mAb or antigen-binding fragment thereof is glycosylated but does not contain detectable NeuGc or α -Gal.

[0365] 283. The method of any of paragraphs 274 to 282 wherein the mAb or antigen-binding fragment thereof contains a tyrosine sulfation.

[0366] 284. The method of any of paragraphs 275 to 283 wherein the recombinant expression vector is AAV8 or AAV9.

[0367] 285. The method of any of paragraphs 275 to 284 in which production of said HuPTM form of said mAb or antigen-binding fragment thereof is confirmed by transducing human liver cells or muscle cells in culture with said recombinant nucleotide expression vector and expressing said mAb or antigen-binding fragment thereof.

Method of Manufacture

[0368] 286. A method of producing recombinant AAVs comprising:

[0369] (a) culturing a host cell containing:

[0370] (i) an artificial genome comprising a cis expression cassette flanked by AAV ITRs, wherein the cis expression cassette comprises a transgene encoding a therapeutic antibody operably linked to expression control elements that will control expression of the transgene in human cells;

[0371] (ii) a trans expression cassette lacking AAV ITRs, wherein the trans expression cassette encodes an AAV rep and capsid protein operably linked to expression control elements that drive expression of the AAV rep and capsid proteins in the host cell in culture and supply the rep and cap proteins in trans;

[0372] (iii) sufficient adenovirus helper functions to permit replication and packaging of the artificial genome by the AAV capsid proteins; and

[0373] (b) recovering recombinant AAV encapsidating the artificial genome from the cell culture.

[0374] 287. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment thereof that comprises the heavy and light chain variable domains of aducanumab, crenezumab, gantenerumab, BAN2401, aTAU, erenumab, eptinezumab, fremanezumab, or galcanezumab.

[0375] 288. The method of claim **286** or **287** in which the AAV capsid protein is an AAV9 or AAVrh10 capsid protein.

[0376] 289. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of ixekizumab, secukinumab or ustekinumab.

[0377] 290. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of natalizumab or vedolizumab.

[0378] 291. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of dupilumab

[0379] 292. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of mepolizumab

[0380] 293. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of alirocumab, evolocumab, evinacumab or E06.

[0381] 294. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of denosumab.

[0382] 295. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of nivolumab or pembrolizumab.

[0383] 296. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of belimumab.

[0384] 297. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of ranibizumab, bevacizumab, or lampalizumab.

[0385] 298. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of andecaliximab.

[0386] 299. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of lanadelumab.

[0387] 300. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of adalimumab or infliximab 301. The method of claim **286** wherein the transgene encodes a mAb or antigen binding fragment that comprises the heavy and light chain variable domains of eculizumab.

[0388] 302. The method of any of claims **286** or **289** to **301** wherein the AAV capsid protein is an AAV8 or AAV9 capsid protein.

4. BRIEF DESCRIPTION OF THE DRAWINGS

[0389] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0390] FIG. 1. A schematic of an rAAV vector genome construct containing an expression cassette encoding the heavy and light chains of the Fab region of a therapeutic mAb controlled by expression elements, flanked by the AAV ITRs.

[0391] FIGS. 2A-F. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies to CNS targets: anti-A β , aducanumab Fab (FIG. 2A); anti-A β , crenezumab Fab (FIG. 2B); anti-A β , gantenerumab Fab (FIG. 2C), ant-tau protein, aTAU Fab (FIG. 2D), and anti-CGRPR, erenumab Fab (FIG. 2E), and anti-A β BAN2401 (FIG. 2F). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green; asparaginal (N) glycosylation sites are highlighted in magenta; and non-consensus asparaginal (N) glycosylation sites are high-

lighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The heavy chain hinge regions are highlighted in grey.

[0392] FIGS. 3A-E. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies to interleukins: anti-IL4R, dupilumab (FIG. 3A); anti-IL-17, ixekizumab (FIG. 3B); secukinumab (FIG. 3C); anti-IL-12/IL-23, ustekinumab (FIG. 3D); and anti-IL5, mepolizumab (FIG. 3E). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The heavy chain hinge regions are highlighted in grey.

[0393] FIGS. 4A-4B. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies to integrin: vedolizumab (FIG. 4A) and natalizumab (FIG. 4B). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The heavy chain hinge regions are highlighted in grey.

[0394] FIGS. 5A-D. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies to PCSK9: alirocumab (FIG. 5A); evolocumab (FIG. 5B); ANGPTL3: evinacumab (FIG. 5C), OxPL: E06-scFv. Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The hinge regions are highlighted in grey.

[0395] FIG. 6. The amino acid sequence of a transgene construct for the Fab region of denosumab, a therapeutic antibody to RANKL. Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The hinge region is highlighted in grey.

[0396] FIGS. 7A and B. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies that are PD-1 blockers: nivolumab (FIG. 7A); and pembrolizumab (FIG. 7B). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The hinge regions are highlighted in grey.

[0397] FIGS. 8A-H. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibodies directed at biological factors: anti-VEGF, ranibizumab (FIG. 8A), bevacizumab (FIG. 8B), and brotuzumab (FIG. 8D); anti- fD , lampalizumab (FIG. 8C); anti-BLyS, belimumab (FIG. 8E); anti-human C5 complement protein, eculizumab (FIG. 8F); anti-MMP 9, andecaliximab (FIG. 8G); and anti-kallikrein, lanadelumab (FIG. 8H). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are highlighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The hinge regions are highlighted in grey.

[0398] FIGS. 9A and B. The amino acid sequence of a transgene construct for the Fab region of therapeutic antibody directed at TNF-alpha: adalimumab (FIG. 9A) and infliximab (FIG. 9B). Glycosylation sites are boldface. Glutamine glycosylation sites are highlighted in green and non-consensus asparaginal (N) glycosylation sites are high-

lighted in blue; tyrosine-O-sulfation sites (*italics*) are highlighted in yellow. The hinge regions are highlighted in grey.

[0399] FIG. 10. Glycans that can be attached to HuGlyFab regions of full length mAbs or the antigen-binding domains. (Adapted from Bondt et al., 2014, Mol & Cell Proteomics 13.1:3029-3039).

[0400] FIGS. 11A and B. Amino acid sequence alignment of the amino acid sequences of the heavy (FIG. 11A) (SEQ ID NOS 283-299, 59, 300-302, 313, 303, 39 and 304-310, respectively, in order of appearance) and light (FIG. 11B) (SEQ ID NOS 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 60, 58, 36, 54, 311, 314, 312, 38, 234, 42, 44, 48, 247 and 235, respectively, in order of appearance) chain Fab portions of the therapeutic antibodies disclosed herein. Positions that may be substituted to produce hyperglycosylated variants of the Fab regions are highlighted in green. Four substitutions (one in the heavy chain and three in the light chain) that should result in hyperglycosylation of the Fab region by human cells are annotated above the amino acid residue positions. (For engineering mAbs or antigen-binding fragments to contain additional glycosylation sites on the Fab domain, see e.g., Courtois et al., 2016, mAbs 8:99-112 for a description of derivatives of antibodies that are hyperglycosylated on the Fab domain of the full-length antibody).

[0401] FIG. 12. Clustal Multiple Sequence Alignment of AAV capsids 1-9. Amino acid substitutions (shown in bold in the bottom rows) can be made to AAV9 and AAV8 capsids by “recruiting” amino acid residues from the corresponding position of other aligned AAV capsids. Sequence shown in Red=hypervariable regions. The amino acid sequences of the AAV capsids are assigned SEQ ID NOS as follows: AAV1 is SEQ ID NO: 71; AAV2 is SEQ ID NO: 72; AAV3-3 is SEQ ID NO: 73; AAV4-4 is SEQ ID NO: 74; AAV5 is SEQ ID NO: 75; AAV6 is SEQ ID NO: 76; AAV7 is SEQ ID NO: 77; AAV8 is SEQ ID NO: 78; AAV9 is SEQ ID NO: 79; hu31 is SEQ ID NO: 81; and hu32 is SEQ ID NO: 82.

5. DETAILED DESCRIPTION OF THE INVENTION

[0402] Compositions and methods are described for the delivery of a fully human post-translationally modified (HuPTM) therapeutic monoclonal antibody (mAb) or a HuPTM antigen-binding fragment of a therapeutic mAb (for example, a fully human-glycosylated Fab (HuGlyFab) of a therapeutic mAb) to a patient (human subject) diagnosed with a disease or condition indicated for treatment with the therapeutic mAb. Delivery may be advantageously accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a therapeutic mAb or its antigen-binding fragment (or a hyperglycosylated derivative of either) to a patient (human subject) diagnosed with a condition indicated for treatment with the therapeutic mAb—to create a permanent depot in a tissue or organ of the patient that continuously supplies the HuPTM mAb or antigen-binding fragment of the therapeutic mAb, e.g., a human-glycosylated transgene product, to a target tissue where the mAb or antigen-binding fragment thereof exerts its therapeutic effect.

[0403] The HuPTM mAb or HuPTM antigen-binding fragment encoded by the transgene can include, but is not limited to, a full-length or an antigen-binding fragment of a therapeutic antibody that binds to:

- [0404]** Nervous system targets, including Amyloid beta (A β or Abeta) peptides, Tau protein, and CGRP receptor;
- [0405]** Interleukins or interleukin receptors, including IL4R, IL17A, IL-5, and IL12/IL23,
- [0406]** Integrins, including integrin-alpha-4,
- [0407]** PCSK9, ANGPTL3, or oxidized phospholipids, such as OxPL,
- [0408]** RANKL,
- [0409]** PD-1, or PD-L1 or PD-L2,
- [0410]** BLyS (B-lymphocyte stimulator, also known as B-cell activating factor (BAFF)),
- [0411]** Ocular Targets including VEGF, fD, and matrix metalloproteinase 9 (MMP9),
- [0412]** TNF-alpha, and
- [0413]** Plasma Protein Targets, such as human complement proteins, including, C5 and C5a complement proteins, and plasma kallikrein,

or such mAbs or antigen-binding fragments engineered to contain additional glycosylation sites on the Fab domain (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety for its description of derivatives of antibodies that are hyperglycosylated on the Fab domain of the full-length antibody). The amino acid sequences of the heavy and light chains of antigen binding fragments of the foregoing are provided in Table 4, *infra*, and codon optimized nucleotide sequences encoding the heavy and light chains of these antigen binding fragments are provided in Table 5.

[0414] The recombinant vector used for delivering the transgene includes non-replicating recombinant adeno-associated virus vectors (“rAAV”). rAAVs are particularly attractive vectors for a number of reasons—they can transduce non-replicating cells, and therefore, can be used to deliver the transgene to tissues where cell division occurs at low levels, such as the CNS; they can be modified to preferentially target a specific organ of choice; and there are hundreds of capsid serotypes to choose from to obtain the desired tissue specificity, and/or to avoid neutralization by pre-existing patient antibodies to some AAVs. Such rAAVs include but are not limited to AAV based vectors comprising capsid components from one or more of AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, AAVrh10 or AAVrh20. In preferred embodiments, AAV based vectors provided herein comprise capsids from one or more of AAV8, AAV9, AAV10, AAV11, AAVrh10 or AAVrh20 serotypes.

[0415] However, other viral vectors may be used, including but not limited to lentiviral vectors; vaccinia viral vectors, or non-viral expression vectors referred to as “naked DNA” constructs. Expression of the transgene can be controlled by constitutive or tissue-specific expression control elements.

[0416] Gene therapy constructs are designed such that both the heavy and light chains are expressed. More specifically, the heavy and light chains should be expressed at about equal amounts, in other words, the heavy and light chains are expressed at approximately a 1:1 ratio of heavy chains to light chains. The coding sequences for the heavy and light chains can be engineered in a single construct in which the heavy and light chains are separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed. In certain embodiments, the coding sequences encode for a Fab or F(ab')₂ or an scFv.

[0417] In certain embodiments, nucleic acids (e.g., polynucleotides) and nucleic acid sequences disclosed herein may be codon-optimized, for example, via any codon-optimization technique known to one of skill in the art (see, e.g., review by Quax et al., 2015, *Mol Cell* 59:149-161). Codon optimized nucleotide sequences of the heavy and light chain variable domains of the therapeutic antibodies are disclosed in Table 5. Each heavy and light chain requires a leader to ensure proper post-translation processing and secretion (unless expressed as an scFv, in which only the N-terminal chain requires a leader sequence). Useful leader sequences for the expression of the heavy and light chains of the therapeutic antibodies in human cells are disclosed herein. An exemplary recombinant expression construct is shown in FIG. 1.

[0418] The production of HuPTMmAb or HuPTM Fab (including an HuPTM scFv) should result in a “biobetter” molecule for the treatment of disease accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a full-length or HuPTM Fab or other antigen binding fragment, such as an scFv, of a therapeutic mAb to a patient (human subject) diagnosed with a disease indication for that mAb, to create a permanent depot in the subject that continuously supplies the human-glycosylated, sulfated transgene product produced by the subject’s transduced cells. The cDNA construct for the HuPTMmAb or HuPTM Fab or HuPTM scFv should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced human cells.

[0419] Pharmaceutical compositions suitable for administration to human subjects comprise a suspension of the recombinant vector in a formulation buffer comprising a physiologically compatible aqueous buffer, a surfactant and optional excipients. Such formulation buffer can comprise one or more of a polysaccharide, a surfactant, polymer, or oil.

[0420] As an alternative, or an additional treatment to gene therapy, the full-length or HuPTM Fab or other antigen binding fragment thereof can be produced in human cell lines by recombinant DNA technology, and the glycoprotein can be administered to patients. Human cell lines that can be used for such recombinant glycoprotein production include but are not limited to human embryonic kidney 293 cells (HEK293), fibrosarcoma HT-1080, HKB-11, CAP, HuH-7, and retinal cell lines, PER.C6, or RPE to name a few (e.g., see Dumont et al., 2015, *Crit. Rev. Biotechnol.* 36 (6): 1110-1122, which is incorporated by reference in its entirety for a review of the human cell lines that could be used for the recombinant production of the HuPTM Fab or HuPTM scFv product, e.g., HuPTM Fab glycoprotein). To ensure complete glycosylation, especially sialylation, and tyrosine-sulfation, the cell line used for production can be enhanced by engineering the host cells to co-express α -2,6-sialyltransferase (or both α -2,3- and α -2,6-sialyltransferases) and/or TPST-1 and TPST-2 enzymes responsible for tyrosine-O-sulfation in human cells.

[0421] It is not essential that every molecule produced either in the gene therapy or protein therapy approach be fully glycosylated and sulfated. Rather, the population of glycoproteins produced should have sufficient glycosylation (including 2,6-sialylation) and sulfation to demonstrate efficacy. The goal of gene therapy treatment of the invention is to slow or arrest the progression of disease.

[0422] Combination therapies involving delivery of the full-length or HuPTM Fab or antigen binding fragment thereof to the patient accompanied by administration of other available treatments are encompassed by the methods of the invention. The additional treatments may be administered before, concurrently or subsequent to the gene therapy treatment. Such additional treatments can include but are not limited to co-therapy with the therapeutic mAb.

[0423] Also provided are methods of manufacturing the viral vectors, particularly the AAV based viral vectors. In specific embodiments, provided are methods of producing recombinant AAVs comprising culturing a host cell containing an artificial genome comprising a cis expression cassette flanked by AAV ITRs, wherein the cis expression cassette comprises a transgene encoding a therapeutic antibody operably linked to expression control elements that will control expression of the transgene in human cells; a trans expression cassette lacking AAV ITRs, wherein the trans expression cassette encodes an AAV rep and capsid protein operably linked to expression control elements that drive expression of the AAV rep and capsid proteins in the host cell in culture and supply the rep and cap proteins in trans; sufficient adenovirus helper functions to permit replication and packaging of the artificial genome by the AAV capsid proteins; and recovering recombinant AAV encapsidating the artificial genome from the cell culture.

5.1 Constructs

[0424] Viral vectors or other DNA expression constructs encoding an HuPTMmAb or antigen-binding fragment thereof, particularly a HuGlyFab, or a hyperglycosylated derivative of a HuPTMmAb antigen-binding fragment are provided herein. The viral vectors and other DNA expression constructs provided herein include any suitable method for delivery of a transgene to a target cell. The means of delivery of a transgene include viral vectors, liposomes, other lipid-containing complexes, other macromolecular complexes, synthetic modified mRNA, unmodified mRNA, small molecules, non-biologically active molecules (e.g., gold particles), polymerized molecules (e.g., dendrimers), naked DNA, plasmids, phages, transposons, cosmids, or episomes. In some embodiments, the vector is a targeted vector, e.g., a vector targeted to retinal pigment epithelial cells, CNS cells, muscle cells, or liver cells.

[0425] In some aspects, the disclosure provides for a nucleic acid for use, wherein the nucleic acid comprises a nucleotide sequence that encodes a HuPTMmAb or HuGlyFab or other antigen-binding fragment thereof, as a transgene described herein, operatively linked to a promoter selected for expression in tissue targeted for expression of the transgene, for example, but not limited to the CB7 promoter (see FIG. 1), cytomegalovirus (CMV) promoter, Rous sarcoma virus (RSV) promoter, GFAP promoter (glial fibrillary acidic protein), MBP promoter (myelin basic protein), MMT promoter, EF-1 alpha promoter, UB6 promoter, chicken beta-actin promoter, CAG promoter, RPE65 promoter and opsin promoter, liver-specific promoters, such as TBG (Thyroxine-binding Globulin) promoter, APOA2 promoter, SERPINA1 (hAAT) promoter, or mIR122 promoter, or muscle-specific promoter, such as a human desmin promoter or Pitx3 promoter, inducible promoters, such as a hypoxia-inducible promoter or a rapamycin-inducible promoter.

[0426] In certain embodiments, provided herein are recombinant vectors that comprise one or more nucleic acids (e.g. polynucleotides). The nucleic acids may comprise DNA, RNA, or a combination of DNA and RNA. In certain embodiments, the DNA comprises one or more of the sequences selected from the group consisting of promoter sequences, the sequence of the gene of interest (the transgene, e.g., the nucleotide sequences encoding the heavy and light chains of the HuPTMmAb or HuGlyFab or other antigen-binding fragment), untranslated regions, and termination sequences. In certain embodiments, viral vectors provided herein comprise a promoter operably linked to the gene of interest.

[0427] In certain embodiments, nucleic acids (e.g., polynucleotides) and nucleic acid sequences disclosed herein may be codon-optimized, for example, via any codon-optimization technique known to one of skill in the art (see, e.g., review by Quax et al., 2015, Mol Cell 59:149-161). Codon optimized nucleotide sequences for expression in human cells are provided herein for the heavy and light chains of the HuGlyFabs in Table 5.

[0428] In a specific embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) one or more control elements, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of anti-VEGF antigen-binding fragment, separated by a self-cleaving furin (F)/F2A linker, ensuring expression of equal amounts of the heavy and the light chain polypeptides. An exemplary construct is shown in FIG. 1.

5.1.1 mRNA Vectors

[0429] In certain embodiments, as an alternative to DNA vectors, the vectors provided herein are modified mRNA encoding for the gene of interest (e.g., the transgene, for example, HuPTMmAb or HuGlyFab or other antigen binding fragment thereof). The synthesis of modified and unmodified mRNA for delivery of a transgene to retinal pigment epithelial cells is taught, for example, in Hansson et al., J. Biol. Chem., 2015, 290 (9): 5661-5672, which is incorporated by reference herein in its entirety. In certain embodiments, provided herein is a modified mRNA encoding for a HuPTMmAb or HuPTM Fab, or HuPTM scFv.

5.1.2 Viral Vectors

[0430] Viral vectors include adenovirus, adeno-associated virus (AAV, e.g., AAV8, AAV9, AAVrh10), lentivirus, helper-dependent adenovirus, herpes simplex virus, poxvirus, hemagglutinin virus of Japan (HVJ), alphavirus, vaccinia virus, and retrovirus vectors. Retroviral vectors include murine leukemia virus (MLV)- and human immunodeficiency virus (HIV)-based vectors. Alphavirus vectors include semliki forest virus (SFV) and sindbis virus (SIN). In certain embodiments, the viral vectors provided herein are recombinant viral vectors. In certain embodiments, the viral vectors provided herein are altered such that they are replication-deficient in humans. In certain embodiments, the viral vectors are hybrid vectors, e.g., an AAV vector placed into a "helpless" adenoviral vector. In certain embodiments, provided herein are viral vectors comprising a viral capsid from a first virus and viral envelope proteins from a second virus. In specific embodiments, the second virus is vesicular stomatitis virus (VSV). In more specific embodiments, the envelope protein is VSV-G protein.

[0431] In certain embodiments, the viral vectors provided herein are HIV based viral vectors. In certain embodiments, HIV-based vectors provided herein comprise at least two polynucleotides, wherein the gag and pol genes are from an HIV genome and the env gene is from another virus.

[0432] In certain embodiments, the viral vectors provided herein are herpes simplex virus-based viral vectors. In certain embodiments, herpes simplex virus-based vectors provided herein are modified such that they do not comprise one or more immediately early (IE) genes, rendering them non-cytotoxic.

[0433] In certain embodiments, the viral vectors provided herein are MLV based viral vectors. In certain embodiments, MLV-based vectors provided herein comprise up to 8 kb of heterologous DNA in place of the viral genes.

[0434] In certain embodiments, the viral vectors provided herein are lentivirus-based viral vectors. In certain embodiments, lentiviral vectors provided herein are derived from human lentiviruses. In certain embodiments, lentiviral vectors provided herein are derived from non-human lentiviruses. In certain embodiments, lentiviral vectors provided herein are packaged into a lentiviral capsid. In certain embodiments, lentiviral vectors provided herein comprise one or more of the following elements: long terminal repeats, a primer binding site, a polypurine tract, att sites, and an encapsidation site.

[0435] In certain embodiments, the viral vectors provided herein are alphavirus-based viral vectors. In certain embodiments, alphavirus vectors provided herein are recombinant, replication-defective alphaviruses. In certain embodiments, alphavirus replicons in the alphavirus vectors provided herein are targeted to specific cell types by displaying a functional heterologous ligand on their virion surface.

[0436] In certain embodiments, the viral vectors provided herein are AAV based viral vectors. In certain embodiments, the AAV-based vectors provided herein do not encode the AAV rep gene (required for replication) and/or the AAV cap gene (required for synthesis of the capsid proteins) (the rep and cap proteins may be provided by the packaging cells in trans). Multiple AAV serotypes have been identified. In certain embodiments, AAV-based vectors provided herein comprise components from one or more serotypes of AAV. In certain embodiments, AAV based vectors provided herein comprise capsid components from one or more of AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, or AAVrh10. In preferred embodiments, AAV based vectors provided herein comprise components from one or more of AAV8, AAV9, AAV10, AAV11, or AAVrh10 serotypes. Provided are viral vectors in which the capsid protein is a variant the AAV8 capsid protein (SEQ ID NO: 78), AAV9 capsid protein (SEQ ID NO: 79), or AAVrh10 capsid protein (SEQ ID NO: 80), more particularly, is at least 95%, 96%, 97%, 98%, 99% or 99.9% identical to the amino acid sequence of the AAV8 capsid protein (SEQ ID NO: 78), AAV9 capsid protein (SEQ ID NO: 79), or AAVrh10 capsid protein (SEQ ID NO: 80), while retaining the biological function of the native capsid. In certain embodiments, the encoded AAV capsid has the sequence of SEQ ID NO: 78, 79 or 80 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions and retaining the biological function of the AAV8 AAV9 or AAVrh10 capsid. FIG. 12 provides a comparative alignment of the amino acid sequences of the capsid proteins of

different AAV serotypes with potential amino acids that may be substituted at certain positions in the aligned sequences based upon the comparison in the row labeled SUBS. Accordingly, in specific embodiments, the AAV vector comprises an AAV8, AAV9 or AAVrh10 capsid variant that has 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions that are not present at that position in the native AAV capsid sequence as identified in the SUBS row of FIG. 12. Sequence for AAVrh10 is provided in Table 4.

[0437] In certain embodiments, the AAV that is used in the compositions and methods described herein is Anc80 or Anc80L65, as described in Zinn et al., 2015, Cell Rep. 12 (6): 1056-1068, which is incorporated by reference in its entirety. In certain embodiments, the AAV that is used in the methods described herein comprises one of the following amino acid insertions: LGETTRP (SEQ ID NO: 162) or LALGETTRP (SEQ ID NO: 163), as described in U.S. Pat. Nos. 9,193,956; 9,458,517; and 9,587,282 and US patent application publication no. 2016/0376323, each of which is incorporated herein by reference in its entirety. In certain embodiments, the AAV that is used in the methods described herein is AAV.7m8 (including variants), as described in U.S. Pat. Nos. 9,193,956; 9,458,517; and 9,587,282, US patent application publication no. 2016/0376323, and International Publication WO 2018/075798, each of which is incorporated herein by reference in its entirety. In certain embodiments, the AAV that is used in the methods described herein is any AAV disclosed in U.S. Pat. No. 9,585,971, such as AAV-PHP.B. In certain embodiments, the AAV used in the compositions and methods described herein is an AAV2/Rec2 or AAV2/Rec3 vector, which have hybrid capsid sequences derived from AAV8 capsids and capsids of serotypes cy5, rh20 or rh39 as described in Charbel Issa et al., 2013, PLOS One 8 (4): e60361, which is incorporated by reference herein for these vectors. In certain embodiments, the AAV that is used in the methods described herein is an AAV disclosed in any of the following patents and patent applications, each of which is incorporated herein by reference in its entirety: U.S. Pat. Nos. 7,906,111; 8,524,446; 8,999,678; 8,628,966; 8,927,514; 8,734,809; 9,284,357; 9,409,953; 9,169,299; 9,193,956; 9,458,517; and 9,587,282 US patent application publication nos. 2015/0374803; 2015/0126588; 2017/0067908; 2013/0224836; 2016/0215024; 2017/0051257; and International Patent Application Nos. PCT/US2015/034799; PCT/EP2015/053335.

[0438] AAV8-based, AAV9-based, and AAVrh10-based viral vectors are used in certain of the methods described herein. Nucleotide sequences of AAV based viral vectors and methods of making recombinant AAV and AAV capsids are taught, for example, in U.S. Pat. No. 7,282,199 B2, U.S. Pat. No. 7,790,449 B2, U.S. Pat. No. 8,318,480 B2, U.S. Pat. No. 8,962,332 B2 and International Patent Application No. PCT/EP2014/076466, each of which is incorporated herein by reference in its entirety. In one aspect, provided herein are AAV (e.g., AAV8, AAV9 or AAVrh10)-based viral vectors encoding a transgene (e.g., an HuPTM Fab). The amino acid sequences of AAV capsids, including AAV8, AAV9 and AAVrh10 are provided in FIG. 12 and Table 4.

[0439] In certain embodiments, a single-stranded AAV (ssAAV) may be used supra. In certain embodiments, a self-complementary vector, e.g., scAAV, may be used (see, e.g., Wu, 2007, Human Gene Therapy, 18 (2): 171-82, McCarty et al, 2001, Gene Therapy, Vol 8, Number 16,

Pages 1248-1254; and U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety).

[0440] In certain embodiments, the viral vectors used in the methods described herein are adenovirus based viral vectors. A recombinant adenovirus vector may be used to transfer in the transgene encoding the HuPTMmAb or HuGlyFab or antigen-binding fragment. The recombinant adenovirus can be a first generation vector, with an E1 deletion, with or without an E3 deletion, and with the expression cassette inserted into either deleted region. The recombinant adenovirus can be a second generation vector, which contains full or partial deletions of the E2 and E4 regions. A helper-dependent adenovirus retains only the adenovirus inverted terminal repeats and the packaging signal (phi). The transgene is inserted between the packaging signal and the 3'ITR, with or without stuffer sequences to keep the genome close to wild-type size of approximately 36 kb. An exemplary protocol for production of adenoviral vectors may be found in Alba et al., 2005, "Gutless adenovirus: last generation adenovirus for gene therapy," Gene Therapy 12: S18-S27, which is incorporated by reference herein in its entirety.

[0441] In certain embodiments, the viral vectors used in the methods described herein are lentivirus based viral vectors. A recombinant lentivirus vector may be used to transfer in the transgene encoding the HuPTM mAb antigen binding fragment. Four plasmids are used to make the construct: Gag/pol sequence containing plasmid, Rev sequence containing plasmids, Envelope protein containing plasmid (i.e. VSV-G), and Cis plasmid with the packaging elements and the anti-VEGF antigen-binding fragment gene.

[0442] For lentiviral vector production, the four plasmids are co-transfected into cells (i.e., HEK293 based cells), whereby polyethylenimine or calcium phosphate can be used as transfection agents, among others. The lentivirus is then harvested in the supernatant (lentiviruses need to bud from the cells to be active, so no cell harvest needs/should be done). The supernatant is filtered (0.45 µm) and then magnesium chloride and benzonase added. Further downstream processes can vary widely, with using TFF and column chromatography being the most GMP compatible ones. Others use ultracentrifugation with/without column chromatography. Exemplary protocols for production of lentiviral vectors may be found in Lesch et al., 2011, "Production and purification of lentiviral vector generated in 293T suspension cells with baculoviral vectors," Gene Therapy 18:531-538, and Ausubel et al., 2012, "Production of CGMP-Grade Lentiviral Vectors," Bioprocess Int. 10 (2): 32-43, both of which are incorporated by reference herein in their entireties.

[0443] In a specific embodiment, a vector for use in the methods described herein is one that encodes an HuPTM mAb antigen binding fragment, such as an HuGlyFab, such that, upon introduction of the vector into a relevant cell, a glycosylated and/or tyrosine sulfated variant of the HuPTM mAb antigen binding fragment or HuGlyFab is expressed by the cell.

5.1.3 Promoters and Modifiers of Gene Expression

[0444] In certain embodiments, the vectors provided herein comprise components that modulate gene delivery or gene expression (e.g., "expression control elements"). In certain embodiments, the vectors provided herein comprise

components that modulate gene expression. In certain embodiments, the vectors provided herein comprise components that influence binding or targeting to cells. In certain embodiments, the vectors provided herein comprise components that influence the localization of the polynucleotide (e.g., the transgene) within the cell after uptake. In certain embodiments, the vectors provided herein comprise components that can be used as detectable or selectable markers, e.g., to detect or select for cells that have taken up the polynucleotide.

[0445] In certain embodiments, the viral vectors provided herein comprise one or more promoters that control expression of the transgene. In certain embodiments, the promoter is a constitutive promoter. In certain embodiments, the promoter is a CB7 promoter (see Dinculescu et al., 2005, *Hum Gene Ther* 16:649-663, incorporated by reference herein in its entirety). In some embodiments, the CB7 promoter includes other expression control elements that enhance expression of the transgene driven by the vector. In certain embodiments, the other expression control elements include chicken β -actin intron and/or rabbit β -globin p₀LA signal. In certain embodiments, the promoter comprises a TATA box. In certain embodiments, the promoter comprises one or more elements. In certain embodiments, the one or more promoter elements may be inverted or moved relative to one another. In certain embodiments, the elements of the promoter are positioned to function cooperatively. In certain embodiments, the elements of the promoter are positioned to function independently. In certain embodiments, the viral vectors provided herein comprise one or more promoters selected from the group consisting of the human CMV immediate early gene promoter, the SV40 early promoter, the Rous sarcoma virus (RS) long terminal repeat, and rat insulin promoter. In certain embodiments, the vectors provided herein comprise one or more long terminal repeat (LTR) promoters selected from the group consisting of AAV, MLV, MMTV, SV40, RSV, HIV-1, and HIV-2 LTRs. In certain embodiments, the vectors provided herein comprise one or more tissue specific promoters (e.g., a retinal pigment epithelial cell-specific promoter, a CNS-specific promoter, a liver-specific promoter or muscle specific). In certain embodiments, the viral vectors provided herein comprise a RPE65 promoter or an opsin promoter (a retinal cell/CNS specific promoter). In certain embodiments, the viral vectors provided herein comprises a liver cell specific promoter, such as, a TBG (Thyroxine-binding Globulin) promoter, an APOA2 promoter, a SERPINA1 (hAAT) promoter, or a MIR122 promoter. In certain embodiments, the viral vector provided herein comprises a muscle specific promoter, such as a human desmin promoter (Jonuschies et al., 2014, *Curr. Gene Ther.* 14:276-288) or a Pitx3 promoter (Coulon et al., 2007, *JBC* 282:33192). In other embodiments, the viral vector comprises a VMD2 promoter.

[0446] In certain embodiments, the promoter is an inducible promoter. In certain embodiments the promoter is a hypoxia-inducible promoter. In certain embodiments, the promoter comprises a hypoxia-inducible factor (HIF) binding site. In certain embodiments, the promoter comprises a HIF-1 α binding site. In certain embodiments, the promoter comprises a HIF-2 α binding site. In certain embodiments, the HIF binding site comprises an RCGTG motif. For details regarding the location and sequence of HIF binding sites, see, e.g., Schödel, et al., *Blood*, 2011, 117 (23): e207-e217, which is incorporated by reference herein in its entirety. In

certain embodiments, the promoter comprises a binding site for a hypoxia induced transcription factor other than a HIF transcription factor. In certain embodiments, the viral vectors provided herein comprise one or more IRES sites that is preferentially translated in hypoxia. For teachings regarding hypoxia-inducible gene expression and the factors involved therein, see, e.g., Kenneth and Rocha, *Biochem J.*, 2008, 414:19-29, which is incorporated by reference herein in its entirety. In specific embodiments, the hypoxia-inducible promoter is the human N-WASP promoter, see, for example, Salvi, 2017, *Biochemistry and Biophysics Reports* 9:13-21 (incorporated by reference for the teaching of the N-WASP promoter) or is the hypoxia-induced promoter of human Epo, see, Tsuchiya et al., 1993, *J. Biochem.* 113:395-400 (incorporated by reference for the disclosure of the Epo hypoxia-inducible promoter). In other embodiments, the promoter is a drug inducible promoter, for example, a promoter that is induced by administration of rapamycin or analogs thereof. See, for example, the disclosure of rapamycin inducible promoters in PCT publications WO94/18317, WO 96/20951, WO 96/41865, WO 99/10508, WO 99/10510, WO 99/36553, and WO 99/41258, and U.S. Pat. No. 7,067,526, which are hereby incorporated by reference in their entireties for the disclosure of drug inducible promoters.

[0447] In certain embodiments, the viral vectors provided herein comprise one or more regulatory elements other than a promoter. In certain embodiments, the viral vectors provided herein comprise an enhancer. In certain embodiments, the viral vectors provided herein comprise a repressor. In certain embodiments, the viral vectors provided herein comprise an intron or a chimeric intron. In certain embodiments, the viral vectors provided herein comprise a polyadenylation sequence.

5.1.4 Signal Peptides

[0448] In certain embodiments, the vectors provided herein comprise components that modulate protein delivery. In certain embodiments, the viral vectors provided herein comprise one or more signal peptides. Signal peptides may also be referred to herein as “leader sequences” or “leader peptides”. In certain embodiments, the signal peptides allow for the transgene product to achieve the proper packaging (e.g. glycosylation) in the cell. In certain embodiments, the signal peptides allow for the transgene product to achieve the proper localization in the cell. In certain embodiments, the signal peptides allow for the transgene product to achieve secretion from the cell.

[0449] There are two general approaches to select a signal sequence for protein production in a gene therapy context or in cell culture. One approach is to use a signal peptide from proteins homologous to the protein being expressed. For example, a human antibody signal peptide may be used to express IgGs in CHO or other cells. Another approach is to identify signal peptides optimized for the particular host cells used for expression. Signal peptides may be interchanged between different proteins or even between proteins of different organisms, but usually the signal sequences of the most abundant secreted proteins of that cell type are used for protein expression. For example, the signal peptide of human albumin, the most abundant protein in plasma, was found to substantially increase protein production yield in CHO cells. However, certain signal peptides may retain

function and exert activity after being cleaved from the expressed protein as “post-targeting functions”. Thus, in specific embodiments, the signal peptide is selected from signal peptides of the most abundant proteins secreted by the cells used for expression to avoid the post-targeting functions. In a preferred embodiment, the signal sequence is

fused to both the heavy and light chain sequences. A preferred sequence is MYRMQLLLIALSLALVINS (SEQ ID NO: 161) (see FIGS. 2-9). Alternatively, signal sequences that are appropriate for expression of the HuPTM mAb or Fab in eye (including CNS), muscle, or liver are provided in Tables 1, 2 and 3, respectively, below.

TABLE 1

Signal peptides for expression in eye/CNS tissue		
Signal Peptide Origin	SEQ ID NO:	Sequence
VEGF-A signal peptide	164	MNFLLSWVHWSLALLLYLHHAKWSQA
Fibulin-1 signal peptide	165	MERAAPSRRVPLPLLLGGLALLAAGVDA
Vitronectin signal peptide	166	MAPLRPLLILALLAWVALA
Complement Factor H signal peptide	167	MRLAKIICMLWAICVA
Opticin signal peptide	168	MRLAFLSLLALVLQETGT
Albumin signal peptide	169	MKWVTFISLLFLFSSAYS
Chymotrypsinogen signal peptide	170	MAFLWLLSCWALLGTTFG
Interleukin-2 signal peptide	171	MYRMQLLSCIALILALVTNS
Trypsinogen-2 signal peptide	172	MNLLLILTFVAAAVA

TABLE 2

Signal peptides for expression in muscle cells.		
Signal Peptide Origin	SEQ ID NO:	Sequence
Human SPARC	173	MRAWIFFLLCLAGRALA
Human Collagen alpha-1(I) chain	174	MFSFVDLRLLLLLAATALLTHG
Human Lactotransferrin	175	MKLVLVLLPLGALGLCLA
Human Complement C3	176	MGPTSGPSLLLLLTHLPLALG
Human Lumican	177	MSLSAFTLFLALIGGTSG
Human Gelsolin isoform 1	178	MAPHRPAPALLCALSLALCALSLPVRA
Human Pro-cathepsin H	179	MWATLPLLCAWLLGVPVCGA
Human SERPINF1	180	MQALVLLLCIGALLGHSSC
Human SERPINE1	181	MQMSPALTCVLGLALVFGEGSA
Human Cathepsin D	182	MQPSSLLPLALCLLAAPASA
Human TIMP1	183	MAPFEPLASGILLLLWLIAPSRA

TABLE 2-continued

Signal peptides for expression in muscle cells.		
Signal Peptide Origin	SEQ ID NO:	Sequence
Human Fibronectin	184	MLRGPGPGLLLLAVQCLGTAVPSTGASKSR
Human Complement Cls subcomponent	185	MWCIVLFSLLAWVYA
Human Cathepsin L1	186	MNPTLILAAFCGLIASA
Human Cathepsin B	187	MWQLWASLCCLLVLANA
Human Salivary acidic proline-rich phosphoprotein 1/2	188	MLLILLSVALLAFSSA
Human Follistatin- related protein 1	189	MWKRWLALALVAVAVVRA

TABLE 3

Signal peptides for expression in liver cells.		
Signal Peptide	SEQ ID NO:	Sequence
Human Serum albumin	169	MKWVTFISLLFLFSSAYS
Human α -1 Antitrypsin (SERPINA1)	190	MPSSVSWGILLLAGLCCCLVPVSLA
Human Apolipoprotein A-1	191	MKAAVLTAVLFLTGSQA
Human Apolipoprotein A-2	192	MKLLAATVLLLTICSLEG
Human Apolipoprotein B- 100	193	MDPPRPALLALLALPALLLLLLLAGARA
Human Coagulation Factor IX	194	MQRVNMIMAESPLITICLLGYLLSAEC
Human Complement C2	195	MGPLMVLFCLLFLYPGLADS
Human Complement Factor H-related Protein 2 (CFHR2)	196	MWLLVSVILISRISSVGG
Human Complement Factor H-related Protein 5 (CFHR5)	197	MLLLFSVILISWVSTVGG
Human Fibrinogen α -chain (FGA)	198	MFSMRIVCLVLSVVGTAWT
Human Fibrinogen β -chain (FGB)	199	MKRMVSWSFHKLKTMKHLLLLLLCVFLVKS
Human Fibrinogen γ -chain (FGG)	200	MSWSLHPRNLILYFYALLFLSSTCVA
Human α -2-HS- Glycoprotein (AHSG)	201	MKSLVLLCLLAQLWGCHS

TABLE 3-continued		
Signal peptides for expression in liver cells.		
Signal Peptide	SEQ ID NO:	Sequence
Human Hemopexin (HPX)	202	MARVLGAPVALGLWSLCWSLAIA
Human Kininogen-1	203	MKLITILFLCSRLLLSLT
Human Mannose-binding protein C (MBL2)	204	MSLFPSLPLLLLSMVAASYS
Human Plasminogen (PLMN)	205	MEHKEVVLLLLFLKSGQG
Human Prothrombin (Coagulation Factor II)	206	MAHVRGLQLPGCLALAALCSLVHS
Human Secreted Phosphoprotein 24	207	MISRMEKMTMMKILIMFALGMNYWSCSG
Human Anti-thrombin-III (SERPINCI)	208	MYSNVIGTVTSGKRKVYLLSLLIGFWDVCVC
Human Serotransferrin (TF)	209	MRLAVGALLVCAVLGLCLA

5.1.5 Polycistronic Messages-IRES and F2A Linkers and scFv Constructs

[0450] Internal ribosome entry sites. A single construct can be engineered to encode both the heavy and light chains separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed by the transduced cells. In certain embodiments, the viral vectors provided herein provide polycistronic (e.g., bicistronic) messages. For example, the viral construct can encode the heavy and light chains separated by an internal ribosome entry site (IRES) elements (for examples of the use of IRES elements to create bicistronic vectors see, e.g., Gurtu et al., 1996, Biochem. Biophys. Res. Comm. 229 (1): 295-8, which is herein incorporated by reference in its entirety). IRES elements bypass the ribosome scanning model and begin translation at internal sites. The use of IRES in AAV is described, for example, in Furling et al., 2001, Gene Ther 8 (11): 854-73, which is herein incorporated by reference in its entirety. In certain embodiments, the bicistronic message is contained within a viral vector with a restraint on the size of the polynucleotide(s) therein. In certain embodiments, the bicistronic message is contained within an AAV virus-based vector (e.g., an AAV8-based, AAV9-based or AAVrh10-based vector).

[0451] Furin-F2A linkers. In other embodiments, the viral vectors provided herein encode the heavy and light chains separated by a cleavable linker such as the self-cleaving furin/F2A (F/F2A) linkers (Fang et al., 2005, Nature Biotechnology 23:584-590, and Fang, 2007, Mol Ther 15:1153-9, each of which is incorporated by reference herein in its entirety). For example, a furin-F2A linker may be incorpo-

rated into an expression cassette to separate the heavy and light chain coding sequences, resulting in a construct with the structure:

[0452] Leader—Heavy chain—Furin site—F2A site—
Leader—Light chain—PolyA.

The F2A site, with the amino acid sequence LLNFDLLKLAGDVESNPGP (SEQ ID NO: 210) is self-processing, resulting in “cleavage” between the final G and P amino acid residues. Additional linkers that could be used include but are not limited to:

T2A: (SEQ ID NO: 211)
(GSG) E G R G S L L T C G D V E E N P G P;
P2A: (SEQ ID NO: 212)
(GSG) A T N F S L L K Q A G D V E E N P G P;
E2A: (SEQ ID NO: 213)
(GSG) Q C T N Y A L L K L A G D V E S N P G P;
F2A: (SEQ ID NO: 214)
(GSG) V K Q T L N F D L L K L A G D V E S N P G P.

[0453] A peptide bond is skipped when the ribosome encounters the F2A sequence in the open reading frame, resulting in the termination of translation, or continued translation of the downstream sequence (the light chain). This self-processing sequence results in a string of additional amino acids at the end of the C-terminus of the heavy chain. However, such additional amino acids are then cleaved by host cell Furin at the furin sites, located imme-

diately prior to the F2A site and after the heavy chain sequence, and further cleaved by carboxypeptidases. The resultant heavy chain may have one, two, three, or more additional amino acids included at the C-terminus, or it may not have such additional amino acids, depending on the sequence of the Furin linker used and the carboxypeptidase that cleaves the linker in vivo (See, e.g., Fang et al., 17 Apr. 2005, *Nature Biotechnol.* Advance Online Publication; Fang et al., 2007, *Molecular Therapy* 15 (6): 1153-1159; Luke, 2012, *Innovations in Biotechnology*, Ch. 8, 161-186). Furin linkers that may be used comprise a series of four basic amino acids, for example, RKRR (SEQ ID NO: 215), RRRR (SEQ ID NO: 216), RRKR (SEQ ID NO: 217), or RKKR (SEQ ID NO: 218). Once this linker is cleaved by a carboxypeptidase, additional amino acids may remain, such that an additional zero, one, two, three or four amino acids may remain on the C-terminus of the heavy chain, for example, R, RR, RK, RKR, RRR, RRK, RKK, RKRR (SEQ ID NO: 215), RRRR (SEQ ID NO: 216), RRKR (SEQ ID NO: 217), or RKKR (SEQ ID NO: 218). In certain embodiments, one the linker is cleaved by a carboxypeptidase, no additional amino acids remain. In certain embodiments, 0.5% to 1%, 1% to 2%, 5%, 10%, 15%, or 20% of the antibody, e.g., antigen-binding fragment, population produced by the constructs for use in the methods described herein has one, two, three, or four amino acids remaining on the C-terminus of the heavy chain after cleavage. In certain embodiments, the furin linker has the sequence R-X-K/R-R, such that the additional amino acids on the C-terminus of the heavy chain are R, RX, RXK, RXR, RXKR, or RXRR, where X is any amino acid, for example, alanine (A). In certain embodiments, no additional amino acids may remain on the C-terminus of the heavy chain.

[0454] Flexible peptide linker. In some embodiments, a single construct can be engineered to encode both the heavy and light chains (preferably the heavy and light chain variable domains) separated by a flexible peptide linker such as those encoding a scFv. A flexible peptide linker can be composed of flexible residues like glycine and serine so that the adjacent heavy chain and light chain domains are free to move relative to one another. The construct may be arranged such that the heavy chain variable domain is at the N-terminus of the scFv, followed by the linker and then the light chain variable domain. Alternatively, the construct may be arranged such that the light chain variable domain is at the N-terminus of the scFv, followed by the linker and then the heavy chain variable domain. That is, the components may be arranged as $\text{NH}_2\text{-V}_L\text{-linker-V}_H\text{-COOH}$ or $\text{NH}_2\text{-V}_H\text{-linker-V}_L\text{-COOH}$.

[0455] In certain embodiments, an expression cassette described herein is contained within a viral vector with a restraint on the size of the polynucleotide(s) therein. In certain embodiments, the expression cassette is contained within an AAV virus-based vector. Due to the size restraints of certain vectors, the vector may or may not accommodate the coding sequences for the full heavy and light chains of the therapeutic antibody but may accommodate the coding sequences of the heavy and light chains of antigen binding fragments, such as the heavy and light chains of a Fab or F(ab')_2 fragment or an scFv. In particular, the AAV vectors described herein may accommodate a transgene of approximately 4.7 kilobases. For constructs such as that in FIG. 1 that contains the CB7 promoter, the chicken β -actin intron, rabbit β -globin polyA signal, and ITRs, the therapeutic

antibody encoded may be approximately 752 amino acids. Substitution of smaller expression elements would permit the expression of larger protein products, such as full length therapeutic antibodies.

5.1.6 Untranslated Regions

[0456] In certain embodiments, the viral vectors provided herein comprise one or more untranslated regions (UTRs), e.g., 3' and/or 5' UTRs. In certain embodiments, the UTRs are optimized for the desired level of protein expression. In certain embodiments, the UTRs are optimized for the mRNA half-life of the transgene. In certain embodiments, the UTRs are optimized for the stability of the mRNA of the transgene. In certain embodiments, the UTRs are optimized for the secondary structure of the mRNA of the transgene.

5.1.7 Inverted Terminal Repeats

[0457] In certain embodiments, the viral vectors provided herein comprise one or more inverted terminal repeat (ITR) sequences. ITR sequences may be used for packaging the recombinant gene expression cassette into the virion of the viral vector. In certain embodiments, the ITR is from an AAV, e.g., AAV8 or AAV2 (see, e.g., Yan et al., 2005, *J. Virol.*, 79 (1): 364-379; U.S. Pat. No. 7,282,199 B2, U.S. Pat. No. 7,790,449 B2, U.S. Pat. No. 8,318,480 B2, U.S. Pat. No. 8,962,332 B2 and International Patent Application No. PCT/EP2014/076466, each of which is incorporated herein by reference in its entirety).

[0458] In certain embodiments, the modified ITRs used to produce self-complementary vector, e.g., scAAV, may be used (see, e.g., Wu, 2007, *Human Gene Therapy*, 18 (2): 171-82, McCarty et al, 2001, *Gene Therapy*, Vol 8, Number 16, Pages 1248-1254; and U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety).

5.1.8 Transgenes

[0459] The transgenes encode a HuPTM mAb, either as a full length antibody or an antigen binding fragment thereof, preferably a Fab fragment (an HuGlyFab) or a F(ab')_2 or an scFv based upon a therapeutic antibody disclosed herein. In specific embodiments, the HuPTM mAb or antigen binding fragment, particularly the HuGlyFab, are engineered to contain additional glycosylation sites on the Fab domain (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety for it description of sites of hyperglycosylation on a Fab domain). FIG. 11 provides alignments of the Fab heavy and light chains of the therapeutic antibodies disclosed herein and highlights in green residues that may be substituted with an asparagine or, in some instances, a serine, resulting in hyperglycosylation.

[0460] In certain embodiments, the transgenes encode either a full-length antibody or an antigen binding fragment thereof with the coding sequence of the heavy and light chains. When using a full-length antibody, a construct encoding a modified mAb may be used. For example, the C-terminal lysines (-K) conserved in the heavy chain genes of all human IgG subclasses are generally absent from antibodies circulating in serum—the C-terminal lysines are cleaved off in circulation, resulting in a heterogeneous population of circulating IgGs. (van den Bremer et al., 2015, mAbs 7:672-680). In the vectored constructs for full length mAbs, the DNA encoding the C-terminal lysine (-K) or

glycine-lysine (-GK) of the Fc terminus can be deleted to produce a more homogeneous antibody product in situ. (See, Hu et al., 2017 *Biotechnol. Prog.* 33:786-794 which is incorporated by reference herein in its entirety).

[0461] Alternatively, antigen binding fragments are advantageously used. FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A, 7B, 8A-8H and 9A-9B provide the amino acid sequences of the heavy and light chains of the Fab fragments and scFv of the therapeutic antibodies (see also Table 4, which provides the amino acid sequences of the heavy and light chains of the therapeutic antibodies). The transgene may comprise the nucleotide sequences encoding the heavy and light chain sequences using nucleotide sequences that encode the Fab portion of the heavy chain plus the constant domain portion of the heavy chain for the appropriate isotype as described further herein and the light chain. Nucleotide sequences that are codon optimized for expression in human cells encoding the Fab fragment portions of the heavy and light chains of the therapeutic antibodies disclosed herein are provided in Table 5. The transgene may encode an Fab fragment using nucleotide sequences encoding the sequences provided in FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5C, 6, 7A, 7B, 8A-8H and 9A-9B but not including the portion of the hinge region on the heavy chain that forms interchain di-sulfide bonds (i.e., the portion containing the sequence CPPCPA (SEQ ID NO: 219)). Heavy chain variable domain sequences that do not contain a CPPCP (SEQ ID NO: 220) sequence of the hinge region at the C-terminus will not form intrachain disulfide bonds and, thus, will form Fab fragments with the corresponding light chain variable domain sequences, whereas those heavy chain variable domain sequences with a portion of the hinge region at the C-terminus containing the sequence CPPCP (SEQ ID NO: 220) will form intrachain disulfide bonds and, thus, will form Fab₂ fragments. For example, in some embodiments, the transgene may encode a scFv comprising a light chain variable domain and a heavy chain variable domain connected by a flexible linker in between (where the heavy chain variable domain may be either at the N-terminal end or the C-terminal end of the scFv), for example, as depicted for brolicizumab in FIG. 8D and E06 in FIG. 5D. Alternatively, in other embodiments, the transgene may encode F(ab')₂ fragments comprising a nucleotide sequence that encodes the light chain and the heavy chain sequence that includes at least the sequence CPPCA (SEQ ID NO: 221) of the hinge region, as depicted in FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5C, 6, 7A, 7B, 8A-8C, 8E-8H and 9A-9B which depict various regions of the hinge region that may be included at the C-terminus of the heavy chain sequence. Pre-existing anti-hinge antibodies (AHA) may cause immunogenicity and reduce efficacy. Thus, in certain embodiments, for the IgG1 isotype, C-terminal ends with D221 or ends with a mutation T225L or with L242 can reduce binding to AHA. (See, e.g., Brezski, 2008, *J Immunol* 181:3183-92 and Kim, 2016, 8:1536-1547). For IgG2, the risk of AHA is lower since the hinge region of IgG2 is not as susceptible to enzymatic cleavage required to generate endogenous AHA. (See, e.g., Brezski, 2011, *MAbs* 3:558-567).

[0462] In certain embodiments, the viral vectors provided herein comprise the following elements in the following order: a) a constitutive or inducible (e.g., hypoxia-inducible or rifamycin-inducible) promoter sequence, and b) a sequence encoding the transgene (e.g., a HuGlyFab). In

certain embodiments, the sequence encoding the transgene comprises multiple ORFs separated by IRES elements. In certain embodiments, the ORFs encode the heavy and light chain domains of the HuGlyFab. In certain embodiments, the sequence encoding the transgene comprises multiple subunits in one ORF separated by F/F2A sequences. In certain embodiments, the sequence comprising the transgene encodes the heavy and light chain domains of the HuGlyFab separated by an F/F2A sequence. In certain embodiments, the sequence comprising the transgene encodes the heavy and light chain variable domains of the HuGlyFab separated by a flexible peptide linker. In certain embodiments, the viral vectors provided herein comprise the following elements in the following order: a) a constitutive or an inducible promoter sequence, and b) a sequence encoding the transgene (e.g., a HuGlyFab), wherein the transgene comprises a nucleotide sequence encoding a signal peptide, a light chain and a heavy chain Fab portion separated by an IRES element. In certain embodiments, the viral vectors provided herein comprise the following elements in the following order: a) a constitutive or a hypoxia-inducible promoter sequence, and b) a sequence encoding the transgene comprising a signal peptide, a light chain and a heavy chain sequence separated by a cleavable F/F2A sequence or a flexible peptide linker.

[0463] In certain embodiments, the viral vectors provided herein comprise the following elements in the following order: a) a first ITR sequence, b) a first linker sequence, c) a constitutive or an inducible promoter sequence, d) a second linker sequence, e) an intron sequence, f) a third linker sequence, g) a first UTR sequence, h) a sequence encoding the transgene (e.g., a HuGlyFab), i) a second UTR sequence, j) a fourth linker sequence, k) a poly A sequence, l) a fifth linker sequence, and m) a second ITR sequence.

[0464] In certain embodiments, the viral vectors provided herein comprise the following elements in the following order: a) a first ITR sequence, b) a first linker sequence, c) a constitutive or an inducible promoter sequence, d) a second linker sequence, e) an intron sequence, f) a third linker sequence, g) a first UTR sequence, h) a sequence encoding the transgene (e.g., HuGlyFab), i) a second UTR sequence, j) a fourth linker sequence, k) a poly A sequence, l) a fifth linker sequence, and m) a second ITR sequence, wherein the transgene comprises a signal, and wherein the transgene encodes a light chain and a heavy chain sequence separated by a cleavable F/F2A sequence.

5.1.9 Manufacture and Testing of Vectors

[0465] The viral vectors provided herein may be manufactured using host cells. The viral vectors provided herein may be manufactured using mammalian host cells, for example, A549, WEHI, 10T1/2, BHK, MDCK, COS1, COS7, BSC 1, BSC 40, BMT 10, VERO, W138, HeLa, 293, Saos, C2C12, L, HT1080, HepG2, primary fibroblast, hepatocyte, and myoblast cells. The viral vectors provided herein may be manufactured using host cells from human, monkey, mouse, rat, rabbit, or hamster.

[0466] The host cells are stably transformed with the sequences encoding the transgene and associated elements (i.e., the vector genome), and the means of producing viruses in the host cells, for example, the replication and capsid genes (e.g., the rep and cap genes of AAV). For a method of producing recombinant AAV vectors with AAV8 capsids, see Section IV of the Detailed Description of U.S.

Pat. No. 7,282,199 B2, which is incorporated herein by reference in its entirety. Genome copy titers of said vectors may be determined, for example, by TAQMAN® analysis. Virions may be recovered, for example, by CsCl₂ sedimentation.

[0467] Alternatively, baculovirus expression systems in insect cells may be used to produce AAV vectors. For a review, see Aponte-Ubillus et al., 2018, Appl. Microbiol. Biotechnol. 102:1045-1054 which is incorporated by reference herein in its entirety for manufacturing techniques.

[0468] In vitro assays, e.g., cell culture assays, can be used to measure transgene expression from a vector described herein, thus indicating, e.g., potency of the vector. For example, the PER. C6® Cell Line (Lonza), a cell line derived from human embryonic retinal cells, or retinal pigment epithelial cells, e.g., the retinal pigment epithelial cell line hTERT RPE-1 (available from ATCC®), can be used to assess transgene expression. Once expressed, characteristics of the expressed product can be determined, including determination of the glycosylation and tyrosine sulfation patterns associated with the HuGlyFab. Glycosylation patterns and methods of determining the same are discussed in Section 5.2.1, while tyrosine sulfation patterns and methods of determining the same are discussed in Section 5.2.2. In addition, benefits resulting from glycosylation/sulfation of the cell-expressed HuGlyFab can be determined using assays known in the art, e.g., the methods described in Sections 5.2.1 and 5.2.2.

5.1.10 Compositions

[0469] Pharmaceutical compositions suitable for administration to human subjects comprise a suspension of the recombinant vector in a formulation buffer comprising a physiologically compatible aqueous buffer, a surfactant and optional excipients. Such formulation buffer can comprise one or more of a polysaccharide, a surfactant, polymer, or oil.

5.2 N-Glycosylation, Tyrosine Sulfation, and O-Glycosylation

[0470] The amino acid sequence (primary sequence) of HuGlyFabs and HuPTM scFvs disclosed herein each comprises at least one site at which N-glycosylation or tyrosine sulfation takes place (see FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H and 9A-9B for glycosylation and/or sulfation positions within the amino acid sequences of the Fab fragments of the therapeutic antibodies).

5.2.1 N-Glycosylation

Reverse Glycosylation Sites

[0471] The canonical N-glycosylation sequence is known in the art to be Asn-X-Ser (or Thr), wherein X can be any amino acid except Pro. However, it recently has been demonstrated that asparagine (Asn) residues of human antibodies can be glycosylated in the context of a reverse consensus motif, Ser (or Thr)-X-Asn, wherein X can be any amino acid except Pro. See Valliere-Douglass et al., 2009, J. Biol. Chem. 284:32493-32506; and Valliere-Douglass et al., 2010, J. Biol. Chem. 285:16012-16022. As disclosed herein, certain HuGlyFabs and HuPTM scFvs disclosed herein comprise such reverse consensus sequences.

Non-Consensus Glycosylation Sites

[0472] In addition to reverse N-glycosylation sites, it recently has been demonstrated that glutamine (Gln) residues of human antibodies can be glycosylated in the context of a non-consensus motif, Gln-Gly-Thr. See Valliere-Douglass et al., 2010, J. Biol. Chem. 285:16012-16022. Surprisingly, certain of the HuGlyFab fragments disclosed herein comprise such non-consensus sequences. In addition, O-glycosylation comprises the addition of N-acetyl-galactosamine to serine or threonine residues by the enzyme. It has been demonstrated that amino acid residues present in the hinge region of antibodies can be O-glycosylated. The possibility of O-glycosylation confers another advantage to the therapeutic antibodies provided herein, as compared to, e.g., antigen-binding fragments produced in *E. coli*, again because the *E. coli* naturally does not contain machinery equivalent to that used in human O-glycosylation. (Instead, O-glycosylation in *E. coli* has been demonstrated only when the bacteria is modified to contain specific O-glycosylation machinery. See, e.g., Farid-Moayer et al., 2007, J. Bacteriol. 189:8088-8098.)

Engineered N-Glycosylation Sites

[0473] In certain embodiments, a nucleic acid encoding a HuGlyFab or HuPTM scFv is modified to include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more N-glycosylation sites (including the canonical N-glycosylation consensus sequence, reverse N-glycosylation site, and non-consensus N-glycosylation sites) than would normally be associated with the HuGlyFab or HuPTM scFv (e.g., relative to the number of N-glycosylation sites associated with the HuGlyFab or HuPTM scFv in its unmodified state). In specific embodiments, introduction of glycosylation sites is accomplished by insertion of N-glycosylation sites (including the canonical N-glycosylation consensus sequence, reverse N-glycosylation site, and non-consensus N-glycosylation sites) anywhere in the primary structure of the antigen-binding fragment, so long as said introduction does not impact binding of the antigen-binding fragment to its antigen. Introduction of glycosylation sites can be accomplished by, e.g., adding new amino acids to the primary structure of the antigen-binding fragment, or the antibody from which the antigen-binding fragment is derived (i.e., the glycosylation sites are added, in full or in part), or by mutating existing amino acids in the antigen-binding fragment, or the antibody from which the antigen-binding fragment is derived, in order to generate the N-glycosylation sites (i.e., amino acids are not added to the antigen-binding fragment/antibody, but selected amino acids of the antigen-binding fragment/antibody are mutated so as to form N-glycosylation sites). Those of skill in the art will recognize that the amino acid sequence of a protein can be readily modified using approaches known in the art, e.g., recombinant approaches that include modification of the nucleic acid sequence encoding the protein.

[0474] In a specific embodiment, a HuGlyMab or antigen-binding fragment is modified such that, when expressed in mammalian cells, such as retina, CNS, liver or muscle cells, it can be hyperglycosylated. See Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety.

N-Glycosylation of HuPTM Antigen-Binding Fragments

[0475] Unlike small molecule drugs, biologics usually comprise a mixture of many variants with different modifi-

cations or forms that could have a different potency, pharmacokinetics, and/or safety profile. It is not essential that every molecule produced either in the gene therapy or protein therapy approach be fully glycosylated and sulfated. Rather, the population of glycoproteins produced should have sufficient glycosylation (including 2,6-sialylation) and sulfation to demonstrate efficacy. The goal of gene therapy treatment provided herein can be, for example, to slow or arrest the progression of a disease or abnormal condition or to reduce the severity of one or more symptoms associated with the disease or abnormal condition.

[0476] When a HuGlyFab or HuPTM scFv is expressed in a human cell, the N-glycosylation sites of the antigen-binding fragment can be glycosylated with various different glycans. N-glycans of antigen-binding fragments have been characterized in the art. For example, Bondt et al., 2014, *Mol. & Cell. Proteomics* 13.11:3029-3039 (incorporated by reference herein in its entirety for its disclosure of Fab-associated N-glycans; see also, FIG. 10) characterizes glycans associated with Fabs, and demonstrates that Fab and Fc portions of antibodies comprise distinct glycosylation patterns, with Fab glycans being high in galactosylation, sialylation, and bisection (e.g., with bisecting GlcNAc) but low in fucosylation with respect to Fc glycans. Like Bondt, Huang et al., 2006, *Anal. Biochem.* 349:197-207 (incorporated by reference herein in its entirety for its disclosure of Fab-associated N-glycans) found that most glycans of Fabs are sialylated. However, in the Fab of the antibody examined by Huang (which was produced in a murine cell background), the identified sialic residues were N-Glycolylneuraminic acid ("Neu5Gc" or "NeuGc") (which is not natural to humans) instead of N-acetylneuraminic acid ("Neu5Ac," the predominant human sialic acid). In addition, Song et al., 2014, *Anal. Chem.* 86:5661-5666 (incorporated by reference herein in its entirety for its disclosure of Fab-associated N-glycans) describes a library of N-glycans associated with commercially available antibodies.

[0477] Importantly, when the HuGlyFab or HuPTM scFv are expressed in human cells, the need for in vitro production in prokaryotic host cells (e.g., *E. coli*) or eukaryotic host cells (e.g., CHO cells or NS0 cells) is circumvented. Instead, as a result of the methods described herein, N-glycosylation sites of the HuGlyFab or HuPTM scFv are advantageously decorated with glycans relevant to and beneficial to treatment of humans. Such an advantage is unattainable when CHO cells, NS0 cells, or *E. coli* are utilized in antibody/antigen-binding fragment production, because e.g., CHO cells (1) do not express 2,6 sialyltransferase and thus cannot add 2,6 sialic acid during N-glycosylation; (2) can add Neu5Gc as sialic acid instead of Neu5Ac; and (3) can also produce an immunogenic glycan, the α -Gal antigen, which reacts with anti- α -Gal antibodies present in most individuals, which at high concentrations can trigger anaphylaxis; and because (4) *E. coli* does not naturally contain components needed for N-glycosylation.

[0478] Assays for determining the glycosylation pattern of antibodies, including antigen-binding fragments are known in the art. For example, hydrazinolysis can be used to analyze glycans. First, polysaccharides are released from their associated protein by incubation with hydrazine (the Ludger Liberate Hydrazinolysis Glycan Release Kit, Oxfordshire, UK can be used). The nucleophile hydrazine attacks the glycosidic bond between the polysaccharide and the carrier protein and allows release of the attached glycans.

N-acetyl groups are lost during this treatment and have to be reconstituted by re-N-acetylation. Glycans may also be released using enzymes such as glycosidases or endoglycosidases, such as PNGase F and Endo H, which cleave cleanly and with fewer side reactions than hydrazines. The free glycans can be purified on carbon columns and subsequently labeled at the reducing end with the fluorophor 2-amino benzamide. The labeled polysaccharides can be separated on a GlycoSep-N column (GL Sciences) according to the HPLC protocol of Royle et al, *Anal Biochem* 2002, 304 (1): 70-90. The resulting fluorescence chromatogram indicates the polysaccharide length and number of repeating units. Structural information can be gathered by collecting individual peaks and subsequently performing MS/MS analysis. Thereby the monosaccharide composition and sequence of the repeating unit can be confirmed and additionally in homogeneity of the polysaccharide composition can be identified. Specific peaks of low or high molecular weight can be analyzed by MALDI-MS/MS and the result used to confirm the glycan sequence. Each peak in the chromatogram corresponds to a polymer, e.g., glycan, consisting of a certain number of repeat units and fragments, e.g., sugar residues, thereof. The chromatogram thus allows measurement of the polymer, e.g., glycan, length distribution. The elution time is an indication for polymer length, while fluorescence intensity correlates with molar abundance for the respective polymer, e.g., glycan. Other methods for assessing glycans associated with antigen-binding fragments include those described by Bondt et al., 2014, *Mol. & Cell. Proteomics* 13.11:3029-3039, Huang et al., 2006, *Anal. Biochem.* 349:197-207, and/or Song et al., 2014, *Anal. Chem.* 86:5661-5666.

[0479] Homogeneity or heterogeneity of the glycan patterns associated with antibodies (including antigen-binding fragments), as it relates to both glycan length or size and numbers glycans present across glycosylation sites, can be assessed using methods known in the art, e.g., methods that measure glycan length or size and hydrodynamic radius. HPLC, such as size exclusion, normal phase, reversed phase, and anion exchange HPLC, as well as capillary electrophoresis, allows the measurement of the hydrodynamic radius. Higher numbers of glycosylation sites in a protein lead to higher variation in hydrodynamic radius compared to a carrier with less glycosylation sites. However, when single glycan chains are analyzed, they may be more homogenous due to the more controlled length. Glycan length can be measured by hydrazinolysis, SDS PAGE, and capillary gel electrophoresis. In addition, homogeneity can also mean that certain glycosylation site usage patterns change to a broader/narrower range. These factors can be measured by Glycopeptide LC-MS/MS.

[0480] In certain embodiments, the HuPTM mAbs, or antigen binding fragments thereof, also do not contain detectable NeuGc and/or α -Gal. By "detectable NeuGc" or "detectable α -Gal" or "does not contain or does not have NeuGc or α -Gal" means herein that the HuPTM mAb or antigen-binding fragment, does not contain NeuGc or α -Gal moieties detectable by standard assay methods known in the art. For example, NeuGc may be detected by HPLC according to Hara et al., 1989, "Highly Sensitive Determination of N-Acetyl- and N-Glycolylneuraminic Acids in Human Serum and Urine and Rat Serum by Reversed-Phase Liquid Chromatography with Fluorescence Detection." *J. Chromatogr., B: Biomed.* 377, 111-119, which is hereby incor-

porated by reference for the method of detecting NeuGc. Alternatively, NeuGc may be detected by mass spectrometry. The α -Gal may be detected using an ELISA, see, for example, Galili et al., 1998, "A sensitive assay for measuring α -Gal epitope expression on cells by a monoclonal anti-Gal antibody." *Transplantation*. 65 (8): 1129-32, or by mass spectrometry, see, for example, Ayoub et al., 2013, "Correct primary structure assessment and extensive glyco-profiling of cetuximab by a combination of intact, middle-up, middle-down and bottom-up ESI and MALDI mass spectrometry techniques." *Landes Bioscience*. 5 (5): 699-710. See also the references cited in Platts-Mills et al., 2015, "Anaphylaxis to the Carbohydrate Side-Chain Alpha-gal" *Immunol Allergy Clin North Am*. 35 (2): 247-260.

Benefits of N-Glycosylation

[0481] N-glycosylation confers numerous benefits on the HuGlyFab or HuPTM scFv described herein. Such benefits are unattainable by production of antigen-binding fragments in *E. coli*, because *E. coli* does not naturally possess components needed for N-glycosylation. Further, some benefits are unattainable through antibody production in, e.g., CHO cells (or murine cells such as NS0 cells), because CHO cells lack components needed for addition of certain glycans (e.g., 2,6 sialic acid and bisecting GlcNAc) and because either CHO or murine cell lines add N—N-Glycolylneuraminic acid ("Neu5Gc" or "NeuGc") which is not natural to humans (and potentially immunogenic), instead of N-Acetylneuraminic acid ("Neu5Ac") the predominant human sialic acid. See, e.g., Dumont et al., 2015, *Crit. Rev. Biotechnol.* 36 (6): 1110-1122; Huang et al., 2006, *Anal. Biochem.* 349:197-207 (NeuGc is the predominant sialic acid in murine cell lines such as SP2/0 and NS0); and Song et al., 2014, *Anal. Chem.* 86:5661-5666, each of which is incorporated by reference herein in its entirety. Moreover, CHO cells can also produce an immunogenic glycan, the α -Gal antigen, which reacts with anti- α -Gal antibodies present in most individuals, which at high concentrations can trigger anaphylaxis. See, e.g., Bosques, 2010, *Nat. Biotech.* 28:1153-1156. The human glycosylation pattern of the HuGlyFab or HuPTM scFv described herein should reduce immunogenicity of the transgene product and improve efficacy.

[0482] While non-canonical glycosylation sites usually result in low level glycosylation (e.g., 1-5%) of the antibody population, the functional benefits may be significant (See, e.g., van de Bovenkamp et al., 2016, *J. Immunol.* 196:1435-1441). For example, Fab glycosylation may affect the stability, half-life, and binding characteristics of an antibody. To determine the effects of Fab glycosylation on the affinity of the antibody for its target, any technique known to one of skill in the art may be used, for example, enzyme linked immunosorbent assay (ELISA), or surface plasmon resonance (SPR). To determine the effects of Fab glycosylation on the half-life of the antibody, any technique known to one of skill in the art may be used, for example, by measurement of the levels of radioactivity in the blood or organs in a subject to whom a radiolabelled antibody has been administered. To determine the effects of Fab glycosylation on the stability, for example, levels of aggregation or protein unfolding, of the antibody, any technique known to one of skill in the art may be used, for example, differential scanning calorimetry (DSC), high performance liquid chromatography (HPLC), e.g., size exclusion high performance

liquid chromatography (SEC-HPLC), capillary electrophoresis, mass spectrometry, or turbidity measurement.

[0483] The presence of sialic acid on HuGlyFab or HuPTM scFv used in the methods described herein can impact clearance rate of the HuGlyFab or HuPTM scFv. Accordingly, sialic acid patterns of a HuGlyFab or HuPTM scFv can be used to generate a therapeutic having an optimized clearance rate. Methods of assessing antigen-binding fragment clearance rate are known in the art. See, e.g., Huang et al., 2006, *Anal. Biochem.* 349:197-207.

[0484] In another specific embodiment, a benefit conferred by N-glycosylation is reduced aggregation. Occupied N-glycosylation sites can mask aggregation prone amino acid residues, resulting in decreased aggregation. Such N-glycosylation sites can be native to an antigen-binding fragment used herein, or engineered into an antigen-binding fragment used herein, resulting in HuGlyFab or HuPTM scFv that is less prone to aggregation when expressed, e.g., expressed in human cells. Methods of assessing aggregation of antibodies are known in the art. See, e.g., Courtois et al., 2016, *mAbs* 8:99-112 which is incorporated by reference herein in its entirety.

[0485] In another specific embodiment, a benefit conferred by N-glycosylation is reduced immunogenicity. Such N-glycosylation sites can be native to an antigen-binding fragment used herein, or engineered into an antigen-binding fragment used herein, resulting in HuGlyFab or HuPTM scFv that is less prone to immunogenicity when expressed, e.g., expressed in human retinal cells, human CNS cells, human liver cells or human muscle cells.

[0486] In another specific embodiment, a benefit conferred by N-glycosylation is protein stability. N-glycosylation of proteins is well-known to confer stability on them, and methods of assessing protein stability resulting from N-glycosylation are known in the art. See, e.g., Sola and Griebelnow, 2009, *J Pharm Sci.*, 98 (4): 1223-1245.

[0487] In another specific embodiment, a benefit conferred by N-glycosylation is altered binding affinity. It is known in the art that the presence of N-glycosylation sites in the variable domains of an antibody can increase the affinity of the antibody for its antigen. See, e.g., Bovenkamp et al., 2016, *J. Immunol.* 196:1435-1441. Assays for measuring antibody binding affinity are known in the art. See, e.g., Wright et al., 1991, *EMBO J.* 10:2717-2723; and Leibiger et al., 1999, *Biochem. J.* 338:529-538.

5.2.2 Tyrosine Sulfation

[0488] Tyrosine sulfation occurs at tyrosine (Y) residues with glutamate (E) or aspartate (D) within +5 to -5 position of Y, and where position -1 of Y is a neutral or acidic charged amino acid, but not a basic amino acid, e.g., arginine (R), lysine (K), or histidine (H) that abolishes sulfation. Surprisingly, the HuGlyFabs and HuPTM scFvs described herein comprise tyrosine sulfation sites (see FIGS. 2A-2F, 3A-3E, 4A-4B, 5A-5D, 6, 7A-7B, 8A-8H, and 9A-9B).

[0489] Importantly, tyrosine-sulfated antigen-binding fragments cannot be produced in *E. coli*, which naturally does not possess the enzymes required for tyrosine-sulfation. Further, CHO cells are deficient for tyrosine sulfation—they are not secretory cells and have a limited capacity for post-translational tyrosine-sulfation. See, e.g., Mikkelsen & Ezban, 1991, *Biochemistry* 30:1533-1537. Advantageously,

the methods provided herein call for expression of HuPTM Fab in human cells that are secretory and have capacity for tyrosine sulfation.

[0490] Tyrosine sulfation is advantageous for several reasons. For example, tyrosine-sulfation of the antigen-binding fragment of therapeutic antibodies against targets has been shown to dramatically increase avidity for antigen and activity. See, e.g., Loos et al., 2015, PNAS 112:12675-12680, and Choe et al., 2003, Cell 114:161-170. Assays for detection tyrosine sulfation are known in the art. See, e.g., Yang et al., 2015, Molecules 20:2138-2164.

5.2.3 O-Glycosylation

[0491] O-glycosylation comprises the addition of N-acetyl-galactosamine to serine or threonine residues by the enzyme. It has been demonstrated that amino acid residues present in the hinge region of antibodies can be O-glycosylated. In certain embodiments, the HuGlyFab comprise all or a portion of their hinge region, and thus are capable of being O-glycosylated when expressed in human cells. The possibility of O-glycosylation confers another advantage to the HuGlyFab provided herein, as compared to, e.g., antigen-binding fragments produced in *E. coli*, again because the *E. coli* naturally does not contain machinery equivalent to that used in human O-glycosylation. (Instead, O-glycosylation in *E. coli* has been demonstrated only when the bacteria is modified to contain specific O-glycosylation machinery. See, e.g., Farid-Moayer et al., 2007, J. Bacteriol. 189:8088-8098.) O-glycosylated HuGlyFab, by virtue of possessing glycans, shares advantageous characteristics with N-glycosylated HuGlyFab (as discussed above).

5.3 Vectored Therapeutic Antibodies

5.3.1 Anti-ABeta HuPTM Constructs and Formulations for Alzheimer's Disease

[0492] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to amyloid beta (A β or Abeta) peptides derived from the amyloid precursor protein that may have benefit in treating Alzheimer's disease (AD) and the like. In particular embodiments, the HuPTM mAb is aducanumab, crenezumab, gantenerumab, or BAN2401, or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of these antibodies are provided in FIGS. 2A-2C and 2F. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an A β -binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of, AD, to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0493] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to A β that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to A β ,

such as aducanumab, crenezumab, gantenerumab, or BAN2401, or variants thereof as detailed herein. The transgene may also encode an anti-A β antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety).

[0494] In certain embodiments, the anti-A β antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of aducanumab (having amino acid sequences of SEQ ID NOs. 1 and 2, respectively, see Table 4 and FIG. 2A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 101 (encoding the aducanumab heavy chain Fab portion) and SEQ ID NO: 102 (encoding the aducanumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161) or the one of the sequences found in Table 1 supra.

[0495] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-A β -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 1 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHL (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 2A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 1 by the hinge region encoding sequences set forth in Table 4 (SEQ ID NO: 101).

[0496] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 2. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 1. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 2 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 1. In specific embodiments, the A β antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 1 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,

14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the A β antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO:2 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0497] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes a hyperglycosylated aducanumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 1 and 2, respectively, with one or more of the following mutations: T119N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0498] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six aducanumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-A β antibody or antigen-binding fragment thereof.

[0499] In certain embodiments, the anti-A β antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of crenezumab (having amino acid sequences of SEQ ID NOs. 3 and 4, respectively, see Table 4 and FIG. 2B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 103 (encoding the crenezumab heavy chain Fab portion) and SEQ ID NO: 104 (encoding the crenezumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVINS (SEQ ID NO: 161) or a signal sequence found in Table 1.

[0500] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-A β -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 3 with additional hinge region sequence starting after the C-terminal tyrosine (Y), contains all or a portion of the amino acid sequence GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPEFLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 2B. These hinge regions may be encoded by nucleotide sequences at the 3' end of

SEQ ID NO: 3 by the hinge region encoding sequences set forth in Table 5 (SEQ ID NO: 103).

[0501] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 4. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 3. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 4 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 3. In specific embodiments, the A β antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 3 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the A β antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 4 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0502] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes a hyperglycosylated crenezumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 3 and 4, respectively, with one or more of the following mutations: T107N (heavy chain), Q165N or Q165S (light chain), and/or E200N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0503] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six crenezumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-A β antibody or antigen-binding fragment thereof.

[0504] In certain embodiments, the anti-A β antigen-binding fragment transgene comprises the nucleotide sequences

encoding the heavy and light chains of the Fab portion of gantenerumab (having amino acid sequences of SEQ ID NOs. 5 and 6, respectively, see Table 4 and FIG. 2C). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 105 (encoding the gantenerumab heavy chain Fab portion) and SEQ ID NO: 106 (encoding the gantenerumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161) or a signal sequence found in Table 1.

[0505] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-A β -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 5 with additional hinge region sequence starting at the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 2C. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 5 by the hinge region encoding sequences set forth in Table 5 (SEQ ID NO: 105).

[0506] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 6. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 5. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 6 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 5. In specific embodiments, the A β antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 5 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2C) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the A β antigen binding fragment comprises a light chain comprising

an amino acid sequence of SEQ ID NO: 6 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2C) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0507] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes a hyperglycosylated gantenerumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 5 and 6, respectively, with one or more of the following mutations: L121N (heavy chain), Q161N or Q161S (light chain), and/or E196N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0508] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six gantenerumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2C which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-A β antibody or antigen-binding fragment thereof.

[0509] In certain embodiments, the anti-A β antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of BAN2401 (having amino acid sequences of SEQ ID NOs. 57 and 58, respectively, see Table 4 and FIG. 2F). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 157 (encoding the BAN2401 heavy chain Fab portion) and SEQ ID NO: 158 (encoding the BAN2401 light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVINS (SEQ ID NO: 161) or the one of the sequences found in Table 1 supra.

[0510] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-A β -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 57 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222) or KTHLCPPCPAPELLGG (SEQ ID NO: 239), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), or KTHLCPPCPA (SEQ ID NO: 226), as set forth in FIG. 2F. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 57 by the hinge region encoding sequences set forth in Table 4 (SEQ ID NO: 157).

[0511] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99%

identical to the sequence set forth in SEQ ID NO: 58. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an A β antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 57. In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 58 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 57. In specific embodiments, the A β antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 57 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2F) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the A β antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 58 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2F) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0512] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes a hyperglycosylated BAN2401Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 57 and 58, respectively, with one or more of the following mutations: T119N (heavy chain), Q165N or Q165S (light chain), and/or E200N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0513] In certain embodiments, the anti-A β antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six BAN2401 CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2F which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-A β antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0514] Provided are methods of treating human subjects for AD by administration of a viral vector containing a transgene encoding an anti-A β antibody, or antigen binding fragment thereof. The antibody may be aducanumab, crenesumab, gantenerumab, or BAN2401 and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In certain embodiments, the patient has been diagnosed with

and/or has symptoms associated with prodromal AD, i.e., a mild cognitive impairment associated with early AD or even pre-AD. Recombinant vectors used for delivering the transgene are described in Section 5.4.1 and shown at FIGS. 2A-C. Such vectors should have a tropism for human CNS cells and can include non-replicating rAAV, particularly those bearing an AAV9, AAVrh10, AAVrh20, AAVrh39, or AAVcy5 capsid. The recombinant vectors can be administered in any manner such that the recombinant vector enters the CNS, preferably by introducing the recombinant vector into the cerebral spinal fluid (CSF). See Section 5.5.1 for details regarding the methods of treatment.

[0515] Subjects to whom such gene therapy is administered can be those responsive to anti-A β therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with AD, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-A β antibody or considered a good candidate for therapy with an anti-A β antibody. In specific embodiments, the patients have previously been treated with aducanumab, crenesumab, gantenerumab, or BAN2401, and have been found to be responsive to one or more of aducanumab, crenesumab, gantenerumab, or BAN2401. To determine responsiveness, the anti-A β antibody or antigen-binding fragment transgene product (e.g., produced in human cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0516] The production of the anti-A β HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of AD accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the antiA β HuPTM Fab, intrathecally, particularly intracisternal or lumbar administration, or intravenous administration to human subjects (patients) diagnosed with or having one or more symptoms of AD, to create a permanent depot in the CNS that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced CNS cells.

[0517] The cDNA construct for the anti-A β HuPTMmAb or anti-A β HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced CNS cells. For example, the signal sequence may be MYRMQLLL-LIALSLALVTNS (SEQ ID NO: 161).

[0518] As an alternative, or an additional treatment to gene therapy, the anti-A β HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with AD, or for whom therapy for AD is considered appropriate.

[0519] In specific embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of aducanumab as set forth in FIG. 2A (with non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N166 of the heavy chain (SEQ ID NO:1) or N158 and/or N210 of the light chain (SEQ ID NO: 2). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy

and light chain variable domain sequences of aducanumab has a sulfation group at Y 94 and/or Y95 of the heavy chain (SEQ ID NO: 1) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 2). In other embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) NeuGc moieties and/or does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) alpha-Gal moieties.

[0520] In specific embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of crenezumab as set forth in FIG. 2B (with non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N52, Q104, N154, and/or N196 of the heavy chain (SEQ ID NO: 3) or Q105, N163 and/or N215 of the light chain (SEQ ID NO: 4). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of crenezumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 3) and/or Y91 and/or Y92 of the light chain (SEQ ID NO: 4). In other embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0521] In specific embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of gantenerumab as set forth in FIG. 2C (with asparagine (N) glycosylation sites highlighted in magenta, non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N52, N77, Q118 and/or N168 of the heavy chain (SEQ ID NO: 5) or Q101, N159 and/or N211 of the light chain (SEQ ID NO: 6). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of gantenerumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 5) and/or Y87 and/or Y88 of the light chain (SEQ ID NO: 6). In other embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0522] In specific embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of BAN2401 as set forth in FIG. 2F (with non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q116 and/or N166 of the heavy chain (SEQ ID NO: 57) or N163 and/or N215 of the light chain (SEQ ID NO: 58). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of

BAN2401 has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 57) and/or Y91 of the light chain (SEQ ID NO: 58). In other embodiments, the anti-A β HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0523] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% 2,6 sialylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated 2,6 sialylation and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of AD, particular cognitive impairment. Efficacy may be monitored by measuring a reduction in plaque formation and/or an improvement in cognitive function or a reduction in the decline in cognitive function.

[0524] Combinations of delivery of the anti-A β HuPTM mAb or antigen-binding fragment thereof, to the CNS accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently or subsequent to the gene therapy treatment. Available treatments for AD that could be combined with the gene therapy provided herein include but are not limited to ARICEPT® (donepezil), RAZADYNE® (galantamine), NAMENDA® (rivastigmine), and NAMZARIC® (donepezil and memantine), to name a few, and administration with anti-A β agents, including but not limited to aducanumab, crenezumab, gantenerumab, or BAN2401, or anti-Tau agents, such as aTAU.

5.3.2. Anti-Tau HuPTM Constructs and Formulations for Tauopathies Like Alzheimer's Disease, Chronic Traumatic Encephalopathy, Progressive Supranuclear Palsy, or Frontotemporal Dementia

[0525] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to Tau protein (Tau), such as monomeric Tau, oligomeric Tau, non-phosphorylated Tau, and phosphorylated Tau, that may have benefit in treating Alzheimer's Disease (AD), Chronic Traumatic Encephalopathy (CTE), Pick's Complex, primary age-related tauopathy, progressive supranuclear palsy (PSP), frontotemporal dementia (FD), and other tauopathies. In particular embodiments, the HuPTM mAb is an antibody having the Fab fragments provided in FIG. 2D (referred to herein as "aTAU") or an antigen binding fragment thereof. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a Tau-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of, AD, CTE, PSP, FD, or other tauopathies, to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0526] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to Tau that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a

nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to Tau, such as aTAU or variants thereof as detailed herein. The transgene may also encode anti-Tau antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety).

[0527] In certain embodiments, the anti-Tau antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of aTAU (having amino acid sequences of SEQ ID NOs. 53 and 54, respectively, see Table 4 and FIG. 2D). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 153 (encoding the aTAU heavy chain Fab portion) and SEQ ID NO: 154 (encoding the aTAU light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161) or the one of the sequences found in Table 1 supra.

[0528] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-Tau-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 53 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence GPPCPPCPA-PEFLGG (SEQ ID NO: 231), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPEFLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 2D. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 53 by the hinge region encoding sequences set forth in Table 4 (SEQ ID NO: 153).

[0529] In certain embodiments, the anti-Tau antigen-binding fragment transgene encodes a Tau antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 54. In certain embodiments, the anti-Tau antigen-binding fragment transgene encodes a Tau antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 53. In certain embodiments, the anti-Tau antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 54 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 53. In specific embodiments, the Tau antigen-binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 53 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions

outside of the CDRs, which CDRs are underlined in FIG. 2D) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the Tau antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 54 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2D) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0530] In certain embodiments, the anti-Tau antigen-binding fragment transgene encodes a hyperglycosylated aTAU Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 53 and 54, respectively, with one or more of the following mutations: T110N (heavy chain), Q164N or Q164S (light chain), and/or E199N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0531] In certain embodiments, the anti-Tau antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six aTAU CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2D which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-Tau antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0532] Provided are methods of treating human subjects for AD, CTE, PSP, FD, or other tauopathies by administration of a viral vector containing a transgene encoding an anti-Tau antibody, or antigen binding fragment thereof. The antibody may be aTAU, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In certain embodiments, the patient has been diagnosed with and/or has symptoms associated with prodromal AD, i.e., a mild cognitive impairment associated with early AD or even pre-AD. A recombinant vector used for delivering the transgene is described in Section 5.4.1 and shown in FIG. 2D. Such vectors should have a tropism for human CNS cells and can include non-replicating rAAV, particularly those bearing an AAV9, AAVrh10, AAVrh20, AAVrh39, or AAVcy5 capsid. The recombinant vectors can be administered in any manner such that the recombinant vector enters the CNS, preferably by introducing the recombinant vector into the cerebral spinal fluid (CSF). See Section 5.5.1 for details regarding the methods of treatment.

[0533] Subjects to whom such gene therapy is administered can be those responsive to anti-Tau therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with AD, PSP, or FD, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-Tau antibody or considered a good candidate for therapy with an anti-Tau antibody. In specific embodiments, the patients have previously been treated with aTAU, and have been found to be responsive to one or more of aTAU. To deter-

mine responsiveness, the anti-Tau antibody or antigen-binding fragment transgene product (e.g., produced in human cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0534] The production of the anti-Tau HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of AD, PSP, or FD accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-Tau HuPTM Fab, intrathecally, particularly intracisternal or lumbar administration, or intravenous administration to human subjects (patients) diagnosed with or having one or more symptoms of AD, PSP, or FD, to create a permanent depot in the CNS that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced CNS cells.

[0535] The cDNA construct for the anti-Tau HuPTMmAb or anti-Tau HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced CNS cells. For example, the signal sequence may be MYRMQLLL-LIALSLALVTNS (SEQ ID NO: 161).

[0536] As an alternative, or an additional treatment to gene therapy, the anti-Tau HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with AD, PSP, or FD, or for whom therapy for AD, PSP, or FD is considered appropriate.

[0537] In specific embodiments, the anti-Tau HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of aTAU as set forth in FIG. 2D (with non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N57 and/or Q107 and/or N157 and/or N199 of the heavy chain (SEQ ID NO:53) or N78 and/or Q104 and/or N162 and/or N214 of the light chain (SEQ ID NO: 54). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of aTAU has a sulfation group at Y96 and/or Y97 and/or Y104 of the heavy chain (SEQ ID NO: 53) and/or Y90 and/or Y91 of the light chain (SEQ ID NO: 54). In other embodiments, the anti-Tau HuPTM mAb or antigen-binding fragment thereof does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) NeuGc moieties and/or does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) alpha-Gal moieties.

[0538] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% 2,6 sialylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated 2,6 sialylation and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of AD, PSP, or FD, particularly cognitive impairment, gross or fine motor skill impairment, or vision impairment. Efficacy may be monitored by measuring a reduction in plaque formation and/or an improvement in cognitive function, with motor skills, or

with vision or a reduction in the decline in cognitive function, motor skills, or vision.

[0539] Combinations of delivery of the anti-Tau HuPTM mAb or antigen-binding fragment thereof, to the CNS accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently or subsequent to the gene therapy treatment. Available treatments for AD, PSP, or FD that could be combined with the gene therapy provided herein include but are not limited to ARICEPT® (donepezil), RAZADYNE® (galantamine), NAMENDA® (rivastigmine), and NAMZARIC® (donepezil and memantine), to name a few, and administration with anti-Tau agents, including but not limited to aTAU and anti-Aβ agents, such as, but not limited to aducanumab, crenezumab, and gantenerumab.

5.3.3. Anti-CGRPR HuPTM Constructs and Formulations for Migraines and Cluster Headaches.

[0540] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to calcitonin gene-related peptide receptor (CGRPR) that may have benefit in treating migraines and cluster headaches (referred to collectively as headache disorders). In particular embodiments, the HuPTM mAb is erenumab, eptinezumab, fremanezumab, galcanezumab or an antigen binding fragment of one of the foregoing. An amino acid sequence for Fab fragments of erenumab is provided in FIG. 2E. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an CGRPR-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of, migraines and cluster headaches, to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0541] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to CGRPR that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to CGRPR, such as erenumab, eptinezumab, fremanezumab, galcanezumab or variants thereof as detailed herein or in accordance with the details herein. The transgene may also encode anti-CGRPR antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al., 2016, mAbs 8:99-112 which is incorporated by reference herein in its entirety).

[0542] In certain embodiments, the anti-CGRPR antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of erenumab (having amino acid sequences of SEQ ID NOs. 55 and 56, respectively, see Table 4 and FIG. 2E). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 155 (encoding the erenumab heavy chain Fab portion) and SEQ ID NO: 156

(encoding the erenumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human CNS cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALLSLALVTNS (SEQ ID NO: 161) or the one of the sequences found in Table 1 supra.

[0543] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-CGRPR-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 55 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence CPPCPAPPVAGG (SEQ ID NO: 232), and specifically, CPPCPA (SEQ ID NO: 219) or CPPCPAPPVAG (SEQ ID NO: 233) as set forth in FIG. 2E. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 55 by the hinge region encoding sequences set forth in Table 4 (SEQ ID NO: 155).

[0544] In certain embodiments, the anti-CGRPR antigen-binding fragment transgene encodes a CGRPR antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 56. In certain embodiments, the anti-CGRPR antigen-binding fragment transgene encodes a CGRPR antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 55. In certain embodiments, the anti-CGRPR antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 56 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 55. In specific embodiments, the CGRPR antigen-binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 55 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2E) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the Tau antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 56 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 2E) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0545] In certain embodiments, the anti-CGRPR antigen-binding fragment transgene encodes a hyperglycosylated erenumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 55 and 56, respectively, with one or more of the following mutations: T125N (heavy chain) and/or Q198N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0546] In certain embodiments, the anti-CGRPR antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six erenumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 2E which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-Tau antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0547] Provided are methods of treating human subjects for migraines and cluster headaches by administration of a viral vector containing a transgene encoding an anti-CGRPR antibody, or antigen binding fragment thereof. The antibody may be erenumab, eptinezumab, fremanezumab, or galcanezumab and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In certain embodiments, the patient has been diagnosed with and/or has symptoms associated with episodic migraines or chronic migraines. In certain embodiments, the patient has been diagnosed with and/or has symptoms associated with episodic cluster headaches or chronic cluster headaches. A recombinant vector used for delivering the transgene is described in Section 5.4.1 and shown in FIG. 2E. Such vectors should have a tropism for human CNS cells and can include non-replicating rAAV, particularly those bearing an AAV9, AAVrh10, AAVrh20, AAVrh39, or AAVcy5 capsid. The recombinant vectors can be administered in any manner such that the recombinant vector enters the CNS, preferably by introducing the recombinant vector into the cerebral spinal fluid (CSF). See Section 5.5.1 for details regarding the methods of treatment.

[0548] Subjects to whom such gene therapy is administered can be those responsive to anti-CGRPR therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with migraines or cluster headaches or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-CGRPR antibody or considered a good candidate for therapy with an anti-CGRPR antibody. In specific embodiments, the patients have previously been treated with erenumab, eptinezumab, fremanezumab, or galcanezumab, and have been found to be responsive to one or more of erenumab, eptinezumab, fremanezumab, and galcanezumab. To determine responsiveness, the anti-CGRPR antibody or antigen-binding fragment transgene product (e.g., produced in human cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0549] The production of the anti-CGRPR HuPTM mAb or HuPTM Fab, should result in a "biobetter" molecule for the treatment of migraines or cluster headaches accom-

plished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-CGRPR HuPTM Fab, intrathecally, particularly intracisternal or lumbar administration, or intravenous administration to human subjects (patients) diagnosed with or having one or more symptoms of migraines or cluster headaches, to create a permanent depot in the CNS that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced CNS cells.

[0550] The cDNA construct for the anti-CGRPR HuPTM mAb or anti-CGRPR HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced CNS cells. For example, the signal sequence may be MYRMQLLLIALSLALVINS (SEQ ID NO: 161).

[0551] As an alternative, or an additional treatment to gene therapy, the anti-CGRPR HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with migraines or cluster headaches, or for whom therapy for migraines or cluster headaches is considered appropriate.

[0552] In specific embodiments, the anti-CGRPR HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of erenumab as set forth in FIG. 2E (with non-consensus asparagine (N) glycosylation sites highlighted in green, glutamine (Q) glycosylation sites highlighted in blue, and Y-sulfation sites highlighted in yellow) has glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77 and/or Q122 and/or N172 and/or N205 and/or N214 of the heavy chain (SEQ ID NO: 55) or N28 and/or N174 of the light chain (SEQ ID NO: 56). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of erenumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 55) and/or Y87 and/or Y88 of the light chain (SEQ ID NO: 56). In other embodiments, the anti-CGRPR HuPTM mAb or antigen-binding fragment thereof does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) NeuGc moieties and/or does not contain any detectable (e.g., as detected by assays known in the art, for example, those described in section 5.2, *infra*) alpha-Gal moieties.

[0553] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% 2,6 sialylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated 2,6 sialylation and/or sulfated. The goal of gene therapy treatment provided herein is to prevent or reduce the intensity or frequency of migraines, cluster headaches, or one or more of the symptoms associated therewith, including nausea, light sensitivity, sound sensitivity, red eye, eyelid edema, forehead and facial sweating, tearing (lacrimation), abnormal small size of the pupil (miosis), nasal congestion, runny nose (rhinorrhea), and drooping eyelid (ptosis). Efficacy may be monitored by measuring a reduction in the intensity or frequency of migraines or cluster headaches, or a reduction in the amount of acute migraine-specific medication used over a defined period of time.

[0554] Combinations of delivery of the anti-CGRPR HuPTM mAb or antigen-binding fragment thereof, to the CNS accompanied by delivery of other available treatments

are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently or subsequent to the gene therapy treatment. Available treatments for cluster headaches or migraines that could be combined with the gene therapy provided herein include but are not limited to triptans, ergotamine derivatives and NSAIDs, to name a few, and administration with anti-CGRPR agents, including but not limited to erenumab, eptinezumab, fremanezumab, and galcanezumab.

5.3.4 Anti-Interleukin and Anti-Interleukin Receptor HuPTM Constructs and Formulations for Autoimmune Disorders

[0555] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to interleukins (IL) or interleukin receptors (ILR) (e.g., IL4R, IL17A, IL12/IL23, or IL-5) derived from anti-ILs or anti-ILRs indicated for treating one or more autoimmune-related disorders, such as atopic dermatitis, psoriasis (e.g., plaque psoriasis, pustular psoriasis, and erythrodermic psoriasis), arthritis (e.g., psoriatic arthritis, and alkylating spondylitis), Crohn's disease, or asthma (collectively referred to hereinafter as "subject AI-Ds"). In particular embodiments, the HuPTM mAb has the amino acid sequence of dupilumab, ixekizumab, secukinumab, ustekinumab, or mepolizumab or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of these antibodies are provided in FIGS. 3A to 3E, respectively. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an IL/ILR-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of atopic dermatitis, psoriasis (e.g., plaque psoriasis), arthritis (e.g., psoriatic arthritis, and alkylating spondylitis), Crohn's disease, or asthma to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0556] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to IL/ILR that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to IL/ILR, such as dupilumab, ixekizumab, secukinumab, ustekinumab, mepolizumab, or variants thereof as detailed herein. The transgene may also encode an anti-IL/ILR antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0557] In certain embodiments, the anti-IL4R antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of dupilumab (having amino acid sequences of SEQ ID NOs. 7 and 8, respectively, see Table 4 and FIG. 3A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 107 (encoding the dupilumab heavy chain Fab portion) and SEQ ID NO: 108 (encoding the dupilumab light chain Fab portion) as set forth

in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or human muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0558] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 7 with additional hinge region sequence starting after the C-terminal tyrosine (Y), contains all or a portion of the amino acid sequence GPPCPPCPAPEFLGG (SEQ ID NO: 231), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPE-FLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 3A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 7 by the hinge region encoding sequences set forth in Table 5.

[0559] In certain embodiments, the anti-IL4R antigen-binding fragment transgene encodes an IL4R antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 8. In certain embodiments, the anti-IL4R antigen-binding fragment transgene encodes an IL4R antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 7. In certain embodiments, the anti-IL4R antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 8 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 7. In specific embodiments, the IL4R antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 7 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the IL4R antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 8 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3A) or are substitutions with an amino acid present at that position in

the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0560] In certain embodiments, the anti-IL4R antigen-binding fragment transgene encodes a hyperglycosylated dupilumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 7 and 8, respectively, with one or more of the following mutations: T120N (heavy chain), Q165N or Q165S (light chain), and/or E200N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0561] In certain embodiments, the anti-IL4R antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six dupilumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 3A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-IL4R antibody or antigen-binding fragment thereof.

[0562] In certain embodiments, the anti-IL17A antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of ixekizumab (having amino acid sequences of SEQ ID NOs: 9 and 10, respectively, see Table 4 and FIG. 3B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 109 (encoding the ixekizumab heavy chain Fab portion) and SEQ ID NO: 110 (encoding the ixekizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0563] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 9 with additional hinge region sequence starting after the C-terminal tyrosine (Y), contains all or a portion of the amino acid sequence GPPCPPCPAPEFLGG (SEQ ID NO: 231), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPE-FLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 3B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 9 by the hinge region encoding sequences set forth in Table 5.

[0564] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an IL17A antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 10. In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an IL17A antigen-binding fragment comprising a heavy chain comprising an amino acid

sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 9. In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 10 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 9. In specific embodiments, the IL17A antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 9 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the IL17A antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 10 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0565] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes a hyperglycosylated ixekizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 9 and 10, respectively, with one or more of the following mutations: L114N (heavy chain), Q165N or Q165S (light chain), and/or E200N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0566] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six ixekizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 3B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-IL17A antibody or antigen-binding fragment thereof.

[0567] In certain embodiments, the anti-IL17A antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of secukinumab (having amino acid sequences of SEQ ID NOs. 11 and 12, respectively, see Table 4 and FIG. 3C). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 111 (encoding the secukinumab heavy chain Fab portion) and SEQ ID NO: 112 (encoding the secukinumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate

for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to myocyte or hepatocyte secreted proteins, respectively.

[0568] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 11 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHT CPPCPAPELLGGPSVFL (SEQ ID NO: 227), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 3C. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 11 by the hinge region encoding sequences set forth in Table 5.

[0569] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an IL17A antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 12. In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an IL17A antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 11. In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 12 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 11. In specific embodiments, the IL17A antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 11 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3C) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the IL17A antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 12 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3C) or are

substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0570] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes a hyperglycosylated secukinumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 11 and 12, respectively, with one or more of the following mutations: L122N (heavy chain), Q161N or Q161S (light chain), and/or E196N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0571] In certain embodiments, the anti-IL17A antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six secukinumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 3C which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-IL/ILR antibody or antigen-binding fragment thereof.

[0572] In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of ustekinumab (having amino acid sequences of SEQ ID NOs. 13 and 14, respectively, see Table 4 and FIG. 3D). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 113 (encoding the ustekinumab heavy chain Fab portion) and SEQ ID NO: 114 (encoding the ustekinumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0573] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 13 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHT CPPCPAPELLGGPSVFL (SEQ ID NO: 227), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 3D. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 13 by the hinge region encoding sequences set forth in Table 5.

[0574] In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene encodes an IL/ILR antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or

99% identical to the sequence set forth in SEQ ID NO: 14. In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene encodes an IL12/IL23 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 13. In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 14 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 13. In specific embodiments, the IL12/IL23 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 13 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3D) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the IL12/IL23 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 14 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3D) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0575] In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene encodes a hyperglycosylated ustekinumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 13 and 14, respectively, with one or more of the following mutations: L114N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0576] In certain embodiments, the anti-IL12/IL23 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six ustekinumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 3D which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-IL12/IL23 antibody or antigen-binding fragment thereof.

[0577] In certain embodiments, the anti-IL-5 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of mepolizumab (having amino acid sequences of SEQ ID NOs. 15 and 16, respectively, see Table 4 and FIG. 3E). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 115 (encoding the

mepolizumab heavy chain Fab portion) and SEQ ID NO: 116 (encoding the mepolizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0578] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 15 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHT CPPCPAPELLGGPSVFL (SEQ ID NO: 227), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 3E. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 15 by the hinge region encoding sequences set forth in Table 5.

[0579] In certain embodiments, the anti-IL-5 antigen-binding fragment transgene encodes an IL-5 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 16. In certain embodiments, the anti-IL-5 antigen-binding fragment transgene encodes an IL-5 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 15. In certain embodiments, the anti-IL-5 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 16 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 15. In specific embodiments, the IL-5 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 15 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3E) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the IL-5 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 16 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more

amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 3E) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0580] In certain embodiments, the anti-IL-5 antigen-binding fragment transgene encodes a hyperglycosylated mepolizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 15 and 16, respectively, with one or more of the following mutations: T114N (heavy chain), Q166N or Q166S (light chain), and/or E201N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0581] In certain embodiments, the anti-IL-5 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six mepolizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 3E which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-IL-5 antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0582] Provided are methods of treating human subjects for one or more of the subject AI-Ds by administration of a viral vector containing a transgene encoding an anti-IL/ILR antibody, or antigen binding fragment thereof. The antibody may be dupilumab, ixekizumab, secukinumab, ustekinumab, or mepolizumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with one or more of the subject AI-Ds. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 3A-3E, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream (or in an alternative embodiment into the hepatic bloodstream, such as through the hepatic artery). See Section 5.5.2 for details regarding the methods of treatment.

[0583] Subjects to whom such gene therapy is administered can be those responsive to anti-IL/ILR therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with one or more of the subject AI-Ds, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-IL/ILR antibody or considered a good candidate for therapy with an anti-IL/ILR antibody. In specific embodiments, the patients have previously been treated with dupilumab, ixekizumab, secukinumab, ustekinumab, or mepolizumab, and have been found to be responsive to dupilumab, ixekizumab, secukinumab, ustekinumab, or mepolizumab. To determine responsiveness, the anti-IL/ILR antibody or antigen-binding fragment transgene product (e.g., produced in human cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0584] The production of the anti-IL/ILR HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of one or more of the subject AI-Ds accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-IL/ILR HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with or having one or more symptoms of one or more of the subject AI-Ds, to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0585] The cDNA construct for the anti-IL/ILR HuPTM-mAb or anti-IL/ILR HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver or muscle cells. For example, the signal sequence may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0586] As an alternative, or an additional treatment to gene therapy, the anti-IL/ILR HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with one or more of the subject AI-Ds, or for whom therapy for one or more of the subject AI-Ds is considered appropriate . . .

[0587] In specific embodiments, the anti-IL4R HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of dupilumab as set forth in FIG. 3A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77, N167, and/or Q117 of the heavy chain (SEQ ID NO:7) or Q105, N163, and/or N215 of the light chain (SEQ ID NO:8). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of dupilumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 7) and/or Y91 and/or Y92 of the light chain (SEQ ID NO: 8). In other embodiments, the anti-IL4R HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0588] In specific embodiments, the anti-IL17A HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of ixekizumab as set forth in FIG. 3B (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q111, N161, and/or N203 of the heavy chain (SEQ ID NO: 9) or Q105, N163 and/or N215 of the light chain (SEQ ID NO: 10). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of ixekizumab has a sulfation group at Y94 and/or

Y95 of the heavy chain (SEQ ID NO: 9) and/or Y91 and/or Y92 of the light chain (SEQ ID NO: 10). In other embodiments, the anti-IL17A HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable α -Gal moieties.

[0589] In specific embodiments, the anti-IL17A HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of secukinumab as set forth in FIG. 3C (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N169 of the heavy chain (SEQ ID NO:11) or Q101, N159, and/or N211 of the light chain (SEQ ID NO:12). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of secukinumab has a sulfation group at Y94 and/or Y95 and/or Y107 and/or Y108 of the heavy chain (SEQ ID NO: 11) and/or Y 87 and/or Y88 of the light chain (SEQ ID NO: 12). In other embodiments, the anti-IL17A HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable α -Gal moieties.

[0590] In specific embodiments, the anti-IL12/IL23 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of ustekinumab as set forth in FIG. 3D (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q111 and/or N161 of the heavy chain (SEQ ID NO: 13) or Q100, N158, and/or N210 of the light chain (SEQ ID NO: 14). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of ustekinumab has a sulfation group at Y86 and/or Y87 of the light chain (SEQ ID NO: 14). In other embodiments, the anti-IL12/IL23 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable α -Gal moieties.

[0591] In specific embodiments, the anti-IL-5 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of mepolizumab as set forth in FIG. 3E (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N76 and/or N161 of the heavy chain (SEQ ID NO: 15) or N22, N34, N164, and/or N216 of the light chain (SEQ ID NO:16). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of mepolizumab has a sulfation group at Y93 and/or Y94 of the heavy chain (SEQ ID NO: 15) and/or Y92 and/or Y93 of the light chain (SEQ ID NO:16). In other embodiments, the anti-IL-5 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0592] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% 2,6-sialylation and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of subject AI-Ds.

[0593] Efficacy may be monitored by scoring the symptoms or degree of inflammation in the affected tissue or area of the body, e.g., such as the skin, colon, or joints. For example, with regard to CD, efficacy can be monitored by assessing Crohn's Disease Activity Index [CDAI] over the course of treatment (e.g., see Best W R et al. (1976) *Gastroenterology* 70 (3): 439-44, "Development of a Crohn's disease activity index. National Cooperative Crohn's Disease Study."). With regard to psoriasis and atopic dermatitis, efficacy can be monitored by assessing changes in the affected skin or in the quality of the patient's life over the course of treatment. One or more standardized assessments can be used to assess the change. (see e.g., Feldman & Krueger, (2005) *Ann. Rheum. Dis.* 64 (Suppl II): ii65-ii68: "Psoriasis assessment tools in clinical trials" describing standardized assessments including the Psoriasis Area and Severity Index (PASI), Physician Global Assessment (PGA), lattice system, NPF Psoriasis Score (NPF-PS), Medical Outcome Survey Short Form 36 (SF-36), the Euro QOL, Dermatology Life Quality Index (DLQI), and the Skindex; Schram et al. (2012) *Allergy*; 67:99-106: "EASI, (objective) SCORAD and POEM for atopic eczema: responsiveness and minimal clinically important difference" describing standardized assessments including Eczema Area and Severity Index (EASI) and the Severity Scoring of Atopic Dermatitis Index (SCORAD)). With regard to arthritis, efficacy can be monitored by assessing one or more of the activity of the disease, the patient's level of function, or the degree of structural damage to patient's joints (e.g., see Zockling & Braun (2005) *Clin. Exp. Rheumatol* 23 (Suppl. 39) S133-S141: "Assessment of ankylosing spondylitis" describing standardized assessment for ankylosing spondylitis; see also Coates et al. (2011) *J. Rheumatol.* 38 (7): 1496-1501: "Development of a disease severity and responder index for psoriatic arthritis (PsA)-report of the OMERACT 10 PsA special interest group" describing standardized assessments for psoriatic arthritis.).

[0594] Combinations of delivery of the anti-IL/ILR HuPTM mAb or antigen-binding fragment thereof, to the liver or muscle accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for subject AI-Ds that could be combined with the gene therapy provided herein include but are not limited to phototherapy for psoriasis, amino-salicylates, immunomodulatory agents (e.g., azathioprine (AZA), 6-mercaptopurine (6-MP), methotrexate (MTX)), oral or topical corticosteroids (e.g., prednisone or budesonide), topical calcineurin inhibitors, inhaled corticosteroids for asthma, and/or antibiotics for Crohn's Disease and administration with anti-IL/ILR agents, including but not limited to dupilumab, ixekizumab, secukinumab, ustekinumab, or mepolizumab.

5.3.5 Anti-Integrin HuPTM Constructs and Formulations for IBD or Multiple Sclerosis

[0595] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments

thereof, such as HuPTM Fabs, that bind to integrin (e.g., $\alpha 4$ or $\alpha 4\beta 7$ integrin) derived from anti- $\alpha 4$ integrin or anti- $\alpha 4\beta 7$ integrin and indicated for treating inflammatory bowel disease (IBD), such as ulcerative colitis (UC) or Crohn's disease (CD), and multiple sclerosis (MS). In particular embodiments, the HuPTM mAb has the amino acid sequence of vedolizumab, natalizumab, or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of vedolizumab and natalizumab are provided in FIGS. 4A and 4B, respectively. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an integrin-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of IBD or MS to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0596] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to integrin that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to integrin, such as vedolizumab, natalizumab, or variants thereof as detailed herein. The transgene may also encode an anti-integrin antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0597] In certain embodiments, the anti-integrin antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of vedolizumab (having amino acid sequences of SEQ ID NOs. 17 and 18, respectively, see Table 4 and FIG. 4A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 117 (encoding the vedolizumab heavy chain Fab portion) and SEQ ID NO: 118 (encoding the vedolizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0598] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 17 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTH-TCPPCPAPELAGA (SEQ ID NO: 236), and specifically, KTHL (SEQ ID NO: 223), KTHL (SEQ ID NO: 224), KTHLTCPPCPA (SEQ ID NO: 225), KTHLTCPPCPA (SEQ ID NO: 226), KTHLTCPPCPAPELAGAPSVFL (SEQ ID

NO: 237) or KTHLCPPCPAPELAGAPSVFL (SEQ ID NO: 238) as set forth in FIG. 4A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 117 by the hinge region encoding sequences set forth in Table 5.

[0599] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an integrin antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 18. In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an integrin antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 17. In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 18 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 17. In specific embodiments, the integrin antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 17 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 4A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the integrin antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 18 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 4A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0600] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes a hyperglycosylated vedolizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 17 and 18, respectively, with one or more of the following mutations: L116N (heavy chain), Q165N or Q165S (light chain), and/or E200N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0601] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six vedolizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 4A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or

light chain variable domain of an anti-integrin antibody or antigen-binding fragment thereof.

[0602] In certain embodiments, the anti-integrin antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of natalizumab (having amino acid sequences of SEQ ID NOs. 19 and 20, respectively, see Table 4 and FIG. 4B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 119 (encoding the natalizumab heavy chain Fab portion) and SEQ ID NO: 120 (encoding the natalizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively. Alternatively, particularly for the treatment of MS, the heavy and light chains have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human CNS cells, for example, any one of the signal sequences set forth in Table 1.

[0603] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 19 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence GPPCPPCPAPEFLGG (SEQ ID NO: 231), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPEFLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 4B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 119 by the hinge region encoding sequences set forth in Table 5.

[0604] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an integrin antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 20. In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an integrin antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 19. In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 20 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 19. In specific embodiments, the integrin antigen binding fragment comprises a heavy chain compris-

ing an amino acid sequence of SEQ ID NO: 19 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 4B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the integrin antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 20 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 4B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0605] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes a hyperglycosylated natalizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 19 and 20, respectively, with one or more of the following mutations: L118N (heavy chain), Q159N or Q159S (light chain), and/or E194N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0606] In certain embodiments, the anti-integrin antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six natalizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 4B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-integrin antibody or antigen-binding fragment thereof.

[0607] In specific embodiments, provided are AAV vectors comprising a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78), AAV9 capsid (SEQ ID NO: 79) or AAVrh10 capsid (SEQ ID NO: 80); and an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-integrin mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver or muscle cells.

Gene Therapy Methods

[0608] Provided are methods of treating human subjects for IBD or MS by administration of a viral vector containing a transgene encoding an anti-integrin antibody, or antigen binding fragment thereof. The antibody may be vedolizumab or natalizumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with IBD, such as UC or CD, or MS. In particular embodiments, IBD can be moderately to severely active. Recombinant vector used for delivering the transgene are described in Section 5.4.1 and 5.4.2. In some embodiments, such vectors should have a tropism for human liver cells and can include non-replicating rAAV, particu-

larly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 4A and 4B, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See 5.5.2 for details regarding the methods of treatment. In other embodiments, such vectors should have a tropism for human CNS cells and can include non-replicating rAAV, particularly those bearing an AAV9, AAVrh10, AAVrh20, AAVrh39, or AAVcy5 capsid. The recombinant vector, such as shown in FIG. 4B, can be administered in any manner such that the recombinant vector enters the CNS, preferably by introducing the recombinant vector into the cerebral spinal fluid (CSF). See Section 5.5.1 for details regarding the methods of treatment.

[0609] Subjects to whom such gene therapy is administered can be those responsive to anti-integrin therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with IBD, MS, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-integrin antibody or considered a good candidate for therapy with an anti-integrin antibody. In specific embodiments, the patients have previously been treated with vedolizumab and/or natalizumab, and have been found to be responsive to vedolizumab or natalizumab. To determine responsiveness, the anti-integrin antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0610] The production of the anti-integrin HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of IBD or MS accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-integrin HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with or having one or more symptoms of IBD or MS, to create a permanent depot in the liver, muscle or CNS tissue that continuously supplies the fully-human post-translationally modified, such as human-glycosylated, sulfated transgene product produced by transduced liver, muscle or CNS cells.

[0611] The cDNA construct for the anti-integrin HuPTM-mAb or anti-integrin HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver or muscle cells. For example, the signal sequence may be MYRMQLLLIALLSLALVINS (SEQ ID NO: 161). Alternatively, in some embodiments, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Tables 1, 2 or 3 that correspond to the proteins secreted by CNS cells, myocytes or hepatocytes, respectively.

[0612] As an alternative, or an additional treatment to gene therapy, the anti-integrin HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with IBD or MS, or for whom therapy for IBD or MS is considered appropriate.

[0613] In specific embodiments, the anti-integrin HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of vedolizumab as set forth in FIG.

4A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q113 and/or N163 of the heavy chain (SEQ ID NO:17) or Q105 and/or N163 and/or N215 of the light chain (SEQ ID NO: 18). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of vedolizumab has a sulfation group at Y94, Y95 and/or Y106 of the heavy chain (SEQ ID NO: 17) and/or Y91 and/or Y92 of the light chain (SEQ ID NO:18). In other embodiments, the anti-integrin HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0614] In specific embodiments, the anti-integrin HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of natalizumab as set forth in FIG. 4B (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q115, N165, and/or N207 of the heavy chain (SEQ ID NO: 19) or Q99, N157 and/or N209 of the light chain (SEQ ID NO:20). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of natalizumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 19) and/or Y86 and/or Y87 of the light chain (SEQ ID NO:20). In other embodiments, the anti-integrin HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0615] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of IBD or MS, particularly a reduction in pain and discomfort for the patient and/or in the case of MS, improvements in mobility. Efficacy may be monitored by scoring the symptoms or degree of inflammation in the affected tissue. For example, with regard to UC, efficacy can be monitored by assessing a Mayo score and an endoscopy subscore over the course of treatment (e.g., see Lobaton et al. (2015) *J. Crohns Colitis*. 2015 October; 9 (10): 846-52, "The Modified Mayo Endoscopic Score (MMES): A New Index for the Assessment of Extension and Severity of Endoscopic Activity in Ulcerative Colitis Patients."). With regard to CD, efficacy can be monitored by assessing Crohn's Disease Activity Index [CDAI] over the course of treatment (e.g., see Best WR et al. (1976) *Gastroenterology*, March; 70 (3): 439-44, "Development of a Crohn's disease activity index. National Cooperative Crohn's Disease Study. "). For example, with regard to MS, efficacy can be monitored by assessing frequency of relapses (e.g., Annualized Relapse Rate), physical disability status (e.g., scoring Kurtzke Expanded Disability Status Scale (EDSS)), and biological markers, including brain scans using MRI (e.g.,

evaluation of T1-weighted gadolinium (Gd)-enhancing lesions and T2-hyperintense lesions through magnetic resonance imaging).

[0616] Combinations of delivery of the anti-integrin HuPTM mAb or antigen-binding fragment thereof, to the CNS, liver, or muscles accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for IBD that could be combined with the gene therapy provided herein include but are not limited to aminosalicylates, corticosteroids, and immunomodulators (e.g. azathioprine, 6-mercaptopurine, and/or methotrexate) and administration with anti-integrin agents, including but not limited to vedolizumab or natalizumab. Available treatments for MS that could be combined with the gene therapy provided herein include but are not limited to interferon beta, interferon beta 1a, glatiramer acetate, cyclophosphamide, corticosteroids, immunomodulators (e.g. azathioprine, 6-mercaptopurine, and/or methotrexate), and mitoxantrone and administration with anti-integrin agents, including but not limited to natalizumab.

5.3.6 Anti-PCSK9 and Anti-ANGPTL3 HuPTM mAbs

[0617] Compositions and methods are described for the delivery of a HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to proprotein convertase subtilisin/kexin type 9 (PCSK9) derived from anti-PCSK9 or angiopoietin-like 3 (ANGPTL3) indicated for treating heterozygous familial hypercholesterolemia (HeFH), homozygous familial hypercholesterolemia (HoFH), or atherosclerotic cardiovascular disease (ACD), lowering low density lipoprotein cholesterol (LDL-C), triglyceride (TG), and/or total cholesterol (TC) levels, and/or reducing or slowing atherosclerotic plaque formation. In particular embodiments, the HuPTM mAb has the amino acid sequence of alirocumab, evolocumab, evinacumab, or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of alirocumab, evolocumab, and evinacumab are provided in FIGS. 5A to 5C, respectively. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an PCSK9-binding or anti-ANGPTL3-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of HeFH, HoFH, or ACD; abnormally high levels of LDL-C, TG, and/or TC; or abnormal atherosclerotic plaque to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0618] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to PCSK9 or ANGPTL3 that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to PCSK9 or ANGPTL3, such as alirocumab, evolocumab, evinacumab, or variants thereof as detailed herein. The transgene may also encode an anti-

PCSK9 or anti-ANGPTL3 antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0619] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of alirocumab (having amino acid sequences of SEQ ID NOs. 21 and 22, respectively, see Table 4 and FIG. 5A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 121 (encoding the alirocumab heavy chain Fab portion) and SEQ ID NO: 122 (encoding the alirocumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or human muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0620] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-PCSK9-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 21 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPPELLGG (SEQ ID NO: 222), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 5A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 21 by the hinge region encoding sequences set forth in Table 5.

[0621] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an PCSK9 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 22. In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an PCSK9 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 21. In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 22 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 21. In specific embodiments, the PCSK9 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 21 with 1, 2, 3,

4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the PCSK9 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 22 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0622] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes a hyperglycosylated alirocumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 21 and 22, respectively, with one or more of the following mutations: L113N (heavy chain), Q166N or Q166S (light chain), and/or E201N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0623] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six alirocumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 5A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-PCSK9 antibody or antigen-binding fragment thereof.

[0624] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of evolocumab (having amino acid sequences of SEQ ID NOs. 23 and 24, respectively, see Table 4 and FIG. 5B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 123 (encoding the evolocumab heavy chain Fab portion) and SEQ ID NO: 124 (encoding the evolocumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0625] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-PCSK9 antigen binding domain has a heavy chain variable domain of SEQ ID NO: 23 with additional hinge region sequence starting after the C-terminal glutamic acid

(E), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 5B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 23 by the hinge region encoding sequences set forth in Table 5.

[0626] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an PCSK9 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 24. In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an PCSK9 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 23. In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 24 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 23. In specific embodiments, the PCSK9 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 23 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the PCSK9 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 24 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0627] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes a hyperglycosylated evolocumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 23 and 24, respectively, with one or more of the following mutations: T110N (heavy chain) and/or Q197N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0628] In certain embodiments, the anti-PCSK9 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six evolocumab CDRs which are underlined in the heavy

and light chain variable domain sequences of FIG. 5B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-PCSK9 antibody or antigen-binding fragment thereof.

[0629] In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains variable domains of evinacumab (having amino acid sequences of SEQ ID NOs. 25 and 26, respectively, see Table 4 and FIG. 5C). These may be fused to heavy chain C1 constant domain and/or the light chain constant domain to form a Fab fragment. The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 25 (encoding the evinacumab heavy chain variable domain) and SEQ ID NO: 26 (encoding the evinacumab light chain variable domain) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0630] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain sequence, all or a portion of the hinge region. In specific embodiments, the anti-ANGPTL3 antigen binding domain has a heavy chain variable domain of SEQ ID NO: 25 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHT CPPCPAPELLGGPSVFL (SEQ ID NO: 227), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 5C. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 25 by the hinge region encoding sequences set forth in Table 5.

[0631] In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene encodes an ANGPTL3 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 26. In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene encodes an ANGPTL3 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 25. In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99%

identical to the sequence set forth in SEQ ID NO: 26 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 25. In specific embodiments, the ANGPTL3 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 25 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5C) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the ANGPTL3 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 26 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5C) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0632] In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene encodes a hyperglycosylated evinacumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 25 and 26, respectively, with one or more of the following mutations: M121N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0633] In certain embodiments, the anti-ANGPTL3 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six evinacumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 5C which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-ANGPTL3 antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0634] Provided are methods of treating human subjects for HeFH, HoFH, or ADC by administration of a viral vector containing a transgene encoding an anti-PCSK9 or anti-ANGPTL3 mAb, or antigen binding fragment thereof. The antibody may be alirocumab, evolocumab, or evinacumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with HeFH, HoFH, or ADC. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 5A-5C, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by

introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0635] Subjects to whom such gene therapy is administered can be those responsive to anti-PCSK9 or anti-ANGPTL3 therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with HeFH, HoFH, or ADC, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-PCSK9 or anti-ANGPTL3 antibody or considered a good candidate for therapy with an anti-PCSK9 or anti-ANGPTL3 antibody. In specific embodiments, the patients have previously been treated with alirocumab, evolocumab, or evinacumab, and have been found to be responsive to alirocumab, evolocumab, or evinacumab. To determine responsiveness, the anti-PCSK9 or anti-ANGPTL3 antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0636] The production of the anti-PCSK9 or anti-ANGPTL3 HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of HeFH, HoFH, or ADC accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-PCSK9 or anti-ANGPTL3 HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with or having one or more symptoms of HeFH, HoFH, or ADC, to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0637] In specific embodiments, the anti-PCSK9 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of alirocumab as set forth in FIG. 5A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N30, N59, and/or N160 of the heavy chain (SEQ ID NO:21) or N22, N35, Q106, N164, and/or N216 of the light chain (SEQ ID NO: 22). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of alirocumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 21) and/or Y92 and/or Y93 of the light chain (SEQ ID NO: 22). In other embodiments, the anti-PCSK9 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0638] In specific embodiments, the anti-PCSK9 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of evolocumab as set forth in FIG. 5B (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q107 and/or N157 and/or N190 and/or N199 of the heavy chain (SEQ ID NO: 23) or

N71 and/or N173 of the light chain (SEQ ID NO: 24). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of evolocumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 23) and/or Y88 and/or Y89 of the light chain (SEQ ID NO: 24). In other embodiments, the anti-PCSK9 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0639] In specific embodiments, the anti-ANGPTL3 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of evinacumab as set forth in FIG. 5C (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77, Q118, and/or N168 of the heavy chain (SEQ ID NO: 25) or Q100, N158 and/or N210 of the light chain (SEQ ID NO: 26). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of evinacumab has a sulfation group at Y95 of the heavy chain (SEQ ID NO: 25) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 26). In other embodiments, the anti-PCSK9 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0640] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of HeFH, HoFH, or ADC and/or to lower the low density lipoprotein cholesterol (LDL-C) levels. Efficacy may be monitored by monitoring LDL-C levels. For example, efficacy can be monitored by assessing mean percent change in LDL-C from baseline.

[0641] Combinations of delivery of the anti-PCSK9 or anti-ANGPTL3 HuPTM mAb or antigen-binding fragment thereof, to the liver or muscle accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for HeFH, HoFH, or ADC that could be combined with the gene therapy provided herein include but are not limited to diet, statins, ezetimibe, and LDL apheresis and administration with anti-PCSK9 or anti-ANGPTL3 agents, including but not limited to alirocumab, evolocumab, or evinacumab.

5.3.7 Anti-OxPL HuPTM mAbs

[0642] Compositions and methods are described for the delivery of a HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to oxidized phospholipids (OxPL) indicated for treating and/or reducing and/or slowing cardiovascular disease including atherosclerotic cardiovascular disease (ACD), atherosclerotic plaque formation, abnormally high levels of non-HDL cholesterol and LDL, aortic stenosis, hepatic stenosis, or hypercholesterolemia. In particular embodiments, the HuPTM mAb has the amino acid sequence of E06-scFv, or an antigen binding fragment thereof. The amino acid sequences of E06-scFv is provided in FIG. 5D. Delivery may be accomplished via

gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an OxPL-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0643] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to OxPL that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to OxPL, such as E06-scFv or variants thereof as detailed herein. The transgene may also encode an anti-OxPL antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0644] In certain embodiments, the anti-OxPL antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chain variable domains of E06-scFv (having amino acid sequences of SEQ ID NOs. 59 and 60, respectively, see Table 4 and FIG. 5D). E06-scFv is a scFv molecule and, thus, contains the heavy and light chain variable domains of an anti-OxPL mAb connected by a flexible linker. The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 159 (encoding the E06-scFv heavy chain variable domain) and SEQ ID NO: 160 (encoding the E06-scFv light chain variable) as set forth in Table 5. E06 is an scFv and, as such, the scFv is expressed as one protein chain, with a linker between the light and heavy chains. The scFv has a leader sequence at the N-terminus for appropriate expression and secretion in human cells, particularly, human liver cells (such as hepatocytes) or human muscle cells. In other embodiments where the heavy and light chains are expressed as separate proteins, both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or human muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0645] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the light chain variable domain sequence, a flexible peptide linker. The flexible peptide linker sequence can comprise flexible residues such as glycine (G) or serine (S). In some embodiments, the flexible peptide linker can comprise 10-30 residues or G, S, or both G and S. Charged residues such as E and K can be used and interspersed to enhance solubility. The flexible peptide linker sequence can have the amino acid sequence of (GGGS)_n, wherein n can be 1, 2, 3, 4, 5, or 6 (SEQ ID NO: 243). In this case, the

signal sequence is fused to the N-terminus of the scFv, either the heavy or light chain variable domain sequence, as the case may be.

[0646] In certain embodiments, the anti-OxPL antigen-binding fragment transgene encodes an OxPL antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 60. In certain embodiments, the anti-OxPL antigen-binding fragment transgene encodes an OxPL antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 59. In certain embodiments, the anti-OxPL antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 60 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 59. In specific embodiments, the OxPL antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 59 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5D) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the OxPL antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 60 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 5D) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0647] In certain embodiments, the anti-OxPL antigen-binding fragment transgene encodes a hyperglycosylated E06-scFv Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 59 and 60, respectively, with optionally the mutation T118N (heavy chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0648] In certain embodiments, the anti-OxPL antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six E06-scFv CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 5D which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-OxPL antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0649] Provided are methods of treating human subjects for cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque by administration of a viral vector containing a transgene encoding an anti-OxPL mAb, or antigen binding fragment thereof. The antibody may be E06-scFv, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as shown in FIG. 5D, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0650] Subjects to whom such gene therapy is administered can be those responsive to anti-OxPL therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-OxPL antibody or considered a good candidate for therapy with an anti-OxPL antibody. In specific embodiments, the patients have previously been treated with E06-scFv or E06, and have been found to be responsive to E06-scFv or E06. To determine responsiveness, the anti-OxPL antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0651] The production of the anti-OxPL HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-OxPL HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with or having one or more symptoms of cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque, to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0652] In specific embodiments, the anti-OxPL HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and

light chain Fab portions of E06-scFv as set forth in FIG. 5D (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N53 of the heavy chain (SEQ ID NO:59). Alternatively or in addition to, the HuPTM scFv or other antigen binding-fragment thereof with the heavy and light chain variable domain sequences of E06-scFv has a sulfation group at Y58 and/or Y62 and/or Y96 and/or Y97 of the heavy chain (SEQ ID NO: 59) and/or Y42 of the light chain (SEQ ID NO: 60). In other embodiments, the anti-OxPL HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0653] In certain embodiments, the HuPTM mAb or Fab of scFv is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to treat, slow and/or arrest the progression of cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque. Efficacy may be monitored by monitoring LDL levels, inflammation markers, or for changes in the degree of aortic stenosis, such as by monitoring for changes in the aortic valve area, peak and mean transvalvular gradients, and/or maximum aortic velocity. For example, efficacy can be monitored by assessing mean percent change in LDL from baseline.

[0654] Combinations of delivery of the anti-OxPL HuPTM mAb or antigen-binding fragment thereof, to the liver or muscle accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for cardiovascular disease, ACD, hypercholesterolemia, abnormally high levels of non-HDL cholesterol and/or LDL, aortic stenosis, hepatic stenosis, and/or abnormal atherosclerotic plaque that could be combined with the gene therapy provided herein include but are not limited to diet, statins, ezetimibe, and LDL apheresis and administration with anti-OxPL agents, including but not limited to E06-scFv.

5.3.8. Anti-RANKL HuPTM Constructs and Formulations for Osteoporosis

[0655] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to receptor activator of nuclear factor kappa-B ligand (RANKL) derived from anti-RANKL antibody, such as denosumab (FIG. 6), and indicated for treating osteoporosis or abnormal bone loss or weakness (e.g., treating giant cell tumor of bone, treating treatment-induced bone loss, slowing the loss of (or increasing) bone mass in breast and prostate cancer patients, preventing skeletal-related events due to bone metastasis or for decreasing bone resorption and turnover. In particular embodiments, the HuPTM mAb has the amino acid sequence of denosumab or an antigen binding fragment thereof. The amino acid sequence of Fab fragment of this antibody is provided in FIG. 6. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an

RANKL-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with osteoporosis or suffering bone loss to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0656] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to RANKL that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to RANKL, such as denosumab or variants thereof as detailed herein. The transgene may also encode an anti-RANKL antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0657] In certain embodiments, the anti-RANKL antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of denosumab (having amino acid sequences of SEQ ID NOs. 27 and 28, respectively, see Table 4 and FIG. 6). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 127 (encoding the denosumab heavy chain Fab portion) and SEQ ID NO: 128 (encoding the denosumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0658] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-RANKL-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 27 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTH-TCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 6. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 27 by the hinge region encoding sequences set forth in Table 5.

[0659] In certain embodiments, the anti-RANKL antigen-binding fragment transgene encodes an RANKL antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 28. In certain embodiments, the anti-RANKL antigen-binding fragment transgene encodes an RANKL antigen-binding

fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 27. In certain embodiments, the anti-RANKL antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 28 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 27. In specific embodiments, the RANKL antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 27 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 6) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the RANKL antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 28 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 6) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0660] In certain embodiments, the anti-RANKL antigen-binding fragment transgene encodes a hyperglycosylated denosumab Fab, comprising a heavy chain and a light chain of SEQ ID NOS: 27 and 28, respectively, with one or more of the following mutations: L117N (heavy chain), Q161N or Q161S (light chain), and/or E196N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0661] In certain embodiments, the anti-RANKL antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six denosumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 6 which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-RANKL antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0662] Provided are methods of treating human subjects for osteoporosis or abnormal bone loss (for example, in breast or prostate cancer patients or due to bone metastases) by administration of a viral vector containing a transgene encoding an anti-RANKL antibody, or antigen binding fragment thereof. The antibody may be denosumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with osteoporosis or abnormal bone loss. Recombinant vectors used

for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vector, such as shown in FIG. 6, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0663] Subjects to whom such gene therapy is administered can be those responsive to anti-RANKL therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with osteoporosis or abnormal bone loss, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-RANKL antibody or considered a good candidate for therapy with an anti-RANKL antibody. In specific embodiments, the patients have previously been treated with denosumab, and have been found to be responsive to denosumab. To determine responsiveness, the anti-RANKL antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0664] The production of the anti-RANKL HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of osteoporosis or bone loss accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-RANKL HuPTM Fab, intravenously to human subjects (patients) diagnosed with or having one or more symptoms of osteoporosis or bone loss, to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0665] The cDNA construct for the anti-RANKL HuPTM-mAb or anti-RANKL HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver or muscle cells. For example, the signal sequence may be MYRMQLLLLIALLSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0666] As an alternative, or an additional treatment to gene therapy, the anti-RANKL HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with osteoporosis or bone loss, or for whom therapy for osteoporosis or bone loss is considered appropriate.

[0667] In specific embodiments, the anti-RANKL HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of denosumab as set forth in FIG. 6 (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77 and/or N164 and/or Q114 of the heavy chain (SEQ ID NO:27) or N159 and/or N211 and/or Q101 of the light chain (SEQ ID NO:28).

Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of denosumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO:27) and/or Y88 of the light chain (SEQ ID NO:28). In other embodiments, the anti-RANKL HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0668] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of osteoporosis or bone loss. Efficacy may be monitored by evaluating bone tissue or skeletal events or the lack of skeletal events. For example, with regard to osteoporosis, efficacy can be monitored by a bone mineral content assessment, assessment of radiographs for vertebral fractures, or diagnostic imaging for clinical fractures confirmation.

[0669] Combinations of delivery of the anti-RANKL HuPTM mAb or antigen-binding fragment thereof, to the liver or muscles accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for osteoporosis or bone loss that could be combined with the gene therapy provided herein include but are not limited to bisphosphonates (e.g., zoledronic acid), parathyroid hormone (e.g., teriparatide [PTH 1-34] and/or full-length PTH 1-84), calcium, vitamin D, and chemotherapy, cryotherapy, or radiotherapy in patients diagnosed with cancer, and administration with anti-RANKL agents, including but not limited to denosumab.

5.3.9 PD Blocker HuPTM Constructs and Formulations for Cancer and Lymphoma

[0670] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to programmed cell death protein 1 (PD-1), programmed death-ligand 1 (PD-L1), or programmed death-ligand 2 (PD-L2) derived from PD-1 blockers (e.g., anti-PD-1, anti-PD-L1, or anti-PD-L2), indicated for treating unresectable/metastatic melanoma, lymphomas (e.g., Hodgkin lymphoma), and carcinomas (e.g., renal cell carcinoma, squamous cell carcinoma, and non-small cell lung carcinomas). In particular embodiments, the HuPTM mAb has the amino acid sequence of nivolumab, pembrolizumab, or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of nivolumab and pembrolizumab are provided in FIGS. 7A and 7B, respectively. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an PD-1/PD-L1/PD-L2 binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of melanoma, carcinomas, or lymphomas to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0671] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to PD-1/PD-L1/PD-L2 that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to PD-1/PD-L1/PD-L2, such as nivolumab, pembrolizumab, or variants thereof as detailed herein. The transgene may also encode an anti-PD-1, anti-PD-L1, or an anti-PD-L2 antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0672] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of nivolumab (having amino acid sequences of SEQ ID NOs. 29 and 30, respectively, see Table 4 and FIG. 7A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 129 (encoding the nivolumab heavy chain Fab portion) and SEQ ID NO: 130 (encoding the nivolumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or human muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0673] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 29 with additional hinge region sequence starting after the C-terminal tyrosine (Y), contains all or a portion of the amino acid sequence GPPCPPCAPEFLG (SEQ ID NO: 240), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCAPEFLGPSVFL (SEQ ID NO: 241) as set forth in FIG. 7A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 29 by the hinge region encoding sequences set forth in Table 5.

[0674] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes an PD-1 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 30. In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes an PD-1 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 29. In certain embodiments, the anti-PD-1, antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%,

93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 30 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 29. In specific embodiments, the PD-1 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 29 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 7A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the PD-1 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 30 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 7A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0675] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes a hyperglycosylated nivolumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 29 and 30, respectively, with one or more of the following mutations: L108N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0676] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six nivolumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 7A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-PD-1 antibody or antigen-binding fragment thereof.

[0677] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of pembrolizumab (having amino acid sequences of SEQ ID NOs. 31 and 32, respectively, see Table 4 and FIG. 7B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 131 (encoding the pembrolizumab heavy chain Fab portion) and SEQ ID NO: 132 (encoding the pembrolizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences

set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0678] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 31 with additional hinge region sequence starting after the C-terminal tyrosine (Y), contains all or a portion of the amino acid sequence GPPCPPCPAPEFLG (SEQ ID NO: 240), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCPAPEFLGPSVFL (SEQ ID NO: 241) as set forth in FIG. 7B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 31 by the hinge region encoding sequences set forth in Table 5.

[0679] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes a PD-1 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 32. In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes a PD-1 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 31. In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 32 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 31. In specific embodiments, the PD-1 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 31 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 7B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the PD-1 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 32 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 7B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0680] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes a hyperglycosylated pembrolizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 31 and 32, respectively, with one or more of the following mutations: T115N (heavy chain)

Q164N or Q164S (light chain), and/or E199N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0681] In certain embodiments, the anti-PD-1 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six pembrolizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 7B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-PD-1 antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0682] Provided are methods of treating human subjects for melanoma, carcinoma, or lymphoma by administration of a viral vector containing a transgene encoding one or more of the anti-PD-1, anti-PD-L1, and anti-PD-L2 antibody, or antigen binding fragment thereof. In particular, methods are provided for treatment of metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma by administration of a viral vector containing a transgene encoding one or more of the anti-PD-1, anti-PD-L1, and anti-PD-L2 antibody, or antigen binding fragment thereof. The antibody may be nivolumab and pembrolizumab, and are preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with melanoma, carcinoma, or lymphoma. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 7A and 7B, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0683] Subjects to whom such gene therapy is administered can be those responsive to anti-PD-1, anti-PD-L1, or anti-PD-L2 therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with melanoma, carcinoma, or lymphoma, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-PD-1, anti-PD-L1, or anti-PD-L2 antibody or considered a good candidate for therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 antibody. In specific embodiments, the patients have previously been treated with nivolumab or pembrolizumab, and have been found to be responsive to nivolumab or pembrolizumab. To determine responsiveness, the anti-PD-1, anti-PD-L1, or anti-PD-L2 antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0684] The production of the anti-PD-1, anti-PD-L1, and/or anti-PD-L2 HuPTM mAb or HuPTM Fab, should result

in a “biobetter” molecule for the treatment of melanoma, carcinomas, or lymphomas accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-PD-1, anti-PD-L1, and/or anti-PD-L2 HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with or having one or more symptoms of melanoma, carcinomas, lymphoma, or other cancers to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0685] The cDNA constructs for the HuPTMmAb or HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver or muscle cells. For example, the signal sequence may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0686] As an alternative, or an additional treatment to gene therapy, the anti-PD-1, anti-PD-L1, or anti-PD-L2 HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma, or for whom therapy for metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma is considered appropriate.

[0687] In specific embodiments, the anti-PD-1 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of nivolumab as set forth in FIG. 7A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77, Q105, and/or N155 of the heavy chain (SEQ ID NO:29) or N93, Q100, N158, and/or N210 of the light chain (SEQ ID NO:30). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of nivolumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 29) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 30). In other embodiments, the anti-PD-1 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0688] In specific embodiments, the anti-PD-1 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of pembrolizumab as set forth in FIG. 7B (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in

yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q112, N162, and/or N204 of the heavy chain (SEQ ID NO: 31) or N162 and/or N214 of the light chain (SEQ ID NO: 32). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of pembrolizumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 31) and/or Y90 and/or Y91 of the light chain (SEQ ID NO: 32). In other embodiments, the anti-PD-1 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0689] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma. Efficacy may be monitored by one or more oncology endpoints including overall survival, progression-free survival, time to progression, time to treatment failure, event-free survival, time to next treatment, objective response rate, or duration of response (see, e.g., U.S. Department of Health and Human Services Food and Drug Administration Center for Drug Evaluation and Research, Center for Biologics Evaluation and Research. Guidance for industry: clinical trial endpoints for the approval of cancer drugs and biologics. <https://www.fda.gov/downloads/Drugs/Guidances/ucm071590.pdf>. Published May 2007. Accessed Oct. 13, 2017; Oncology Endpoints in a Changing Landscape. *Manag. Care.* 2016; 1 (suppl): 1-12).

[0690] Combinations of delivery of the one or more anti-PD-1, anti-PD-L1, and anti-PD-L2 HuPTM mAbs or antigen-binding fragments thereof, to the liver or muscle accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for metastatic melanoma, lymphoma, non-small cell lung carcinoma, head and neck squamous cell cancer, urothelial carcinoma, microsatellite instability-high cancer, gastric cancer, renal cell carcinoma, mismatch repair deficit metastatic colon cancer, or hepatocellular carcinoma that could be combined with the gene therapy provided herein include but are not limited to chemotherapy (e.g., cisplatin, gemcitabine, pemetrexed, carboplatin, and/or paclitaxel), radiotherapy, cryotherapy, targeted small molecule therapies, other antibodies, and vaccine therapy and administration with one or more of the anti-PD-1, anti-PD-L1, and anti-PD-L2 agents, including but not limited to nivolumab and pembrolizumab.

5.3.10 Anti-VEGF or Anti-fD HuPTM Constructs and Formulations for Ocular Disorders

[0691] Compositions and methods are described for the delivery of HuPTM mAb and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to vascular endothelial growth factor (VEGF) or complement (e.g., factor D (fD)) derived from anti-VEGF or anti-complement (e.g.,

anti-fD), respectively, indicated for treating one or more retinal disorders including diabetic retinopathy, myopic choroidal neovascularization (mCNV), macular degeneration (e.g., neovascular (wet) age-related macular degeneration (AMD)), macular edema (e.g., macular edema following a retinal vein occlusion (RVO) or diabetic macular edema (DME)); for suppressing angiogenesis; or, in the of case those derived from anti-VEGF, for treating one or more types of cancer including epithelial ovarian cancer, fallopian tube cancer, peritoneal cancer cervical cancer, metastatic colorectal cancer, metastatic HER2 negative breast cancer, metastatic renal cell carcinoma, glioblastoma, non-small cell lung cancer (NSCLC). In particular embodiments, the HuPTM mAb has the amino acid sequence of ranibizumab, bevacizumab, lampalizumab, brolucizumab, or an antigen binding fragment of one of the foregoing. The amino acid sequences of Fab fragments of ranibizumab, bevacizumab, and lampalizumab, and the scFv of brolucizumab are provided in FIGS. 8A to 8D, respectively. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding a VEGF-binding or Factor D-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof, including an scFv) to patients (human subjects) diagnosed with, or having one or more symptoms of a retinal disorder (e.g. diabetic retinopathy, mCNV, macular degeneration, or macular edema) or cancer (e.g., epithelial ovarian cancer, fallopian tube cancer, peritoneal cancer cervical cancer, metastatic colorectal cancer, metastatic HER2 negative breast cancer, metastatic renal cell carcinoma, glioblastoma, or NSCLC) to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

[0692] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to VEGF or fD that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to VEGF or fD, such as ranibizumab, bevacizumab, lampalizumab, brolucizumab, or variants thereof as detailed herein. The transgene may also encode an anti-VEGF or anti-fD antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0693] In certain embodiments, the anti-VEGF antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of ranibizumab (having amino acid sequences of SEQ ID NOs. 33 and 34, respectively, see Table 4 and FIG. 8A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 133 (encoding the ranibizumab heavy chain Fab portion) and SEQ ID NO: 134 (encoding the ranibizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, one or more cells forming the retina. The signal sequence may have the amino acid sequence of MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 that correspond to the proteins secreted by

one or more cells forming the retina. Alternatively, the signal sequence may be appropriate for expression in muscle or liver cells, such as those listed in Tables 2 and 3 *infra*.

[0694] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-VEGF antigen binding domain has a heavy chain variable domain of SEQ ID NO: 33 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KHTHTCPPCPAPPELLGG (SEQ ID NO: 222), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KHTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KHTHTCPPCPAPPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 8A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 33 by the hinge region encoding sequences set forth in Table 5.

[0695] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a VEGF antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 34. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a VEGF antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 33. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 34 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 33. In specific embodiments, the VEGF antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 33 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the VEGF antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 34 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0696] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a hyperglycosylated ranibizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 33 and 34, respectively, with one or more of the following mutations: L118N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0697] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six ranibizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-VEGF antibody or antigen-binding fragment thereof.

[0698] In certain embodiments, the anti-VEGF antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of bevacizumab (having amino acid sequences of SEQ ID NOs. 35 and 36, respectively, see Table 4 and FIG. 8B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 135 (encoding the bevacizumab heavy chain Fab portion) and SEQ ID NO: 136 (encoding the bevacizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, one or more retina cell or liver cell types. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 or 3 that correspond to the proteins secreted by retina cell or liver cell types, respectively.

[0699] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 35 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KHTHTCPPCPAPPELLGG (SEQ ID NO: 222), and specifically, KTHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KHTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KHTHTCPPCPAPPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 8B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 35 by the hinge region encoding sequences set forth in Table 5.

[0700] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a VEGF antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 36. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a VEGF antigen-binding fragment comprising a heavy chain comprising an amino acid

sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 35. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 36 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 35. In specific embodiments, the VEGF antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 35 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the VEGF antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 36 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0701] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a hyperglycosylated bevacizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 35 and 36, respectively, with one or more of the following mutations: L118N (heavy chain) and/or Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0702] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six bevacizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-VEGF antibody or antigen-binding fragment thereof.

[0703] In certain embodiments, the anti-fD antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of lampalizumab (having amino acid sequences of SEQ ID NOs. 37 and 38, respectively, see Table 4 and FIG. 8C). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 137 (encoding the lampalizumab heavy chain Fab portion) and SEQ ID NO: 138 (encoding the lampalizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus

appropriate for expression and secretion in human cells, in particular, human one or more cells forming the retina. The signal sequence may have the amino acid sequence of MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 that correspond to the proteins secreted by cells forming the retina.

[0704] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-integrin-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 37 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence KTHHT CPPCPAPELLGGPSVFL (SEQ ID NO: 227), and specifically, KTHHT (SEQ ID NO: 224), KTHL (SEQ ID NO: 223), KTHHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227), or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 8C. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 37 by the hinge region encoding sequences set forth in Table 5.

[0705] In certain embodiments, the anti-fD antigen-binding fragment transgene encodes an fD antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 38. In certain embodiments, the anti-fD antigen-binding fragment transgene encodes an fD antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 37. In certain embodiments, the anti-fD antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 38 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 37. In specific embodiments, the fD antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 37 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8C) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the fD antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 38 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8C) or are substitutions with an

amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0706] In certain embodiments, the anti- fD antigen-binding fragment transgene encodes a hyperglycosylated lamlalizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 37 and 38, respectively, with one or more of the following mutations: L110N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0707] In certain embodiments, the anti- fD antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six lamlalizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8C which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti- fD antibody or antigen-binding fragment thereof.

[0708] In certain embodiments, the anti-VEGF antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chain variable domains of brotucizumab (having amino acid sequences of SEQ ID NOs. 39 and 40, respectively, see Table 4 and FIG. 8D). Brotucizumab is a scFv molecule and, thus, contains the heavy and light chain variable domains of an anti-VEGF mAb connected by a flexible linker. The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 139 (encoding the brotucizumab heavy chain variable domain portion) and SEQ ID NO: 142 (encoding the brotucizumab light chain variable domain portion) as set forth in Table 5. In the even the heavy and light chain variable domains are expressed as separate proteins, the heavy and light chain sequences each have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, one or more cells forming the retina. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 that correspond to the proteins secreted by one or more cells forming the retina.

[0709] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the light chain variable domain sequence, a flexible peptide linker. The flexible peptide linker sequence can comprise flexible residues such as glycine (G) or serine (S). In some embodiments, the flexible peptide linker can comprise 10-30 residues or G, S, or both G and S. Charged residues such as E and K can be used and interspersed to enhance solubility. The flexible peptide linker sequence can have the amino acid sequence of (GGGGS)_n, wherein n can be 1, 2, 3, 4, 5, or 6 (SEQ ID NO: 243). In this case, the signal sequence is fused to the N-terminus of the scFv, either the heavy or light chain variable domain sequence, as the case may be.

[0710] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an VEGF antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or

99% identical to the sequence set forth in SEQ ID NO: 40. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an VEGF antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 39. In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 40 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 39. In specific embodiments, the VEGF antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 39 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8D) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the VEGF antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 40 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8D) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0711] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes a hyperglycosylated brotucizumab scFv, comprising a single chain of SEQ ID NOs: 39 and 40, respectively, with the following mutation: L115N (heavy chain) (see FIGS. 11A (heavy chain)).

[0712] In certain embodiments, the anti-VEGF antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six brotucizumab CDRs which are underlined in the single chain variable domain sequences of FIG. 8D which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-VEGF antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0713] Provided are methods of treating human subjects for one or more retinal disorders (such as diabetic retinopathy, mCNV, macular degeneration, or macular edema) or cancer (such as epithelial ovarian cancer, fallopian tube cancer, peritoneal cancer cervical cancer, metastatic colorectal cancer, metastatic HER2 negative breast cancer, metastatic renal cell carcinoma, glioblastoma, or NSCLC) by administration of a viral vector containing a transgene

encoding an anti-VEGF antibody or antigen binding fragment thereof. The antibody or Fab fragment thereof may be ranibizumab bevacizumab, or brolucizumab. In embodiments, the patient has been diagnosed with and/or has symptoms associated with one or more of the various retinal disorders or cancers listed above.

[0714] Also, provided are methods of treating human subjects for one or more retinal disorders (such as diabetic retinopathy, mCNV, macular degeneration, or macular edema) by administration of a viral vector containing a transgene encoding an anti-fd antibody or antigen binding fragment thereof. The antibody may be lampalizumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with one or more of the various retinal disorders listed above.

[0715] Recombinant vector used for delivering the transgene are described in Section 5.4.3. Such vectors should have a tropism for human retina-type cells and can include non-replicating rAAV, particularly those bearing an AAV8 capsid. Alternatively, vectors bearing an AAV7m8 capsid can be used for ocular indications. The recombinant vectors, such as those shown in FIGS. 8A-8D, can be administered in any manner such that the recombinant vector enters the retina, preferably by introducing the recombinant vector into the eye. See Section 5.5.3 for details regarding the methods of treatment. For delivery to the liver, for example, for the treatment of cancer, recombinant vector used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 8A-8C, can be administered in any manner such that the recombinant vector enters the liver, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0716] Subjects to whom such gene therapy is administered can be those responsive to anti-VEGF or anti-fd therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with one or more retinal disorders or types of cancer, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-VEGF antibody or anti-fd antibody, or considered a good candidate for therapy with an anti-VEGF antibody or anti-fd antibody. In specific embodiments, the patients have previously been treated with ranibizumab, bevacizumab, lampalizumab, or brolucizumab, and have been found to be responsive to ranibizumab, bevacizumab, lampalizumab, or brolucizumab. To determine responsiveness, the anti-VEGF or anti-fd antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0717] The production of the anti-VEGF or anti-fd HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of one or more retinal disorders or cancers accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-VEGF or anti-fd HuPTM Fab, subretinally, intravitreally, or suprachoroidally to human subjects (patients) diagnosed with or having one or more symptoms

of one or more retinal disorders, or by administering a viral vector or other DNA expression construct encoding the anti-VEGF HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with a cancer, to create a permanent depot in the retina or liver that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced cells of the retina or liver.

[0718] As an alternative, or an additional treatment to gene therapy, the anti-VEGF or anti-fd HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with a retinal disorder or cancer for whom therapy for a retinal disorder or cancer is considered appropriate.

[0719] In specific embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of ranibizumab as set forth in FIG. 8A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q115 and/or N165 of the heavy chain (SEQ ID NO:33) or Q100, N158, and/or N210 of the light chain (SEQ ID NO:34). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of ranibizumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 33) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 34). In other embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0720] In specific embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of bevacizumab as set forth in FIG. 8B (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q115, and/or N165 of the heavy chain (SEQ ID NO: 35) or Q100, N158, and/or N210 of the light chain (SEQ ID NO: 36). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of bevacizumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO: 35) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 36). In other embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0721] In specific embodiments, the anti-fd HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of lampalizumab as set forth in FIG. 8C (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions Q107 and/or N157 of the

heavy chain (SEQ ID NO: 37) or Q100 and/or N158 and/or N210 of the light chain (SEQ ID NO: 38). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of lampalizumab has a sulfation group at Y60 and/or Y94 and/or Y95 of the heavy chain (SEQ ID NO: 37) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 38). In other embodiments, the anti-fD HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0722] In specific embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain variable domains of brotucizumab as set forth in FIG. 8D (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77 and/or Q112 of the heavy chain (SEQ ID NO: 39) or N97 and/or Q103 of the light chain (SEQ ID NO: 40). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of brotucizumab has a sulfation group at Y32 and/or Y33 and/or Y34 and/or Y59 and/or Y60 and/or Y94 and/or Y95 of the heavy chain (SEQ ID NO: 39) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 40). In other embodiments, the anti-VEGF HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0723] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of a retinal disorder or type of cancer, and/or to suppress angiogenesis. In the case of retinal disorders, efficacy may be monitored by monitoring vision acuity. For example, efficacy can be monitored by assessing change in vision acuity from baseline. In the case of a cancer, efficacy can be monitored by assessing one or more oncology endpoints including overall survival, progression-free survival, time to progression, time to treatment failure, event-free survival, time to next treatment, objective response rate, or duration of response. (see, e.g., U.S. Department of Health and Human Services Food and Drug Administration Center for Drug Evaluation and Research, Center for Biologics Evaluation and Research. Guidance for industry: clinical trial endpoints for the approval of cancer drugs and biologics. <https://www.fda.gov/downloads/Drugs/Guidances/ucm071590.pdf>. Published May 2007. Accessed Oct. 13, 2017; Oncology Endpoints in a Changing Landscape. *Manag. Care.* 2016; 1 (suppl): 1-12).

[0724] Combinations of delivery of the anti-VEGF or anti-fD HuPTM mAb or antigen-binding fragment thereof to the retina or liver accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for diabetic retinopathy, mCNV, macular degeneration, or macular edema that could be combined with the gene therapy provided herein include but are not limited to laser photocoagulation, photodynamic

therapy with verteporfin, aflibercept, and/or intravitreal steroids and administration with anti-VEGF or anti-fD agents, including but not limited to ranibizumab, bevacizumab, lampalizumab, or brotucizumab. Available treatments for epithelial ovarian cancer, fallopian tube cancer, peritoneal cancer cervical cancer, metastatic colorectal cancer, metastatic HER2 negative breast cancer, metastatic renal cell carcinoma, glioblastoma, or NSCLC that could be combined with the gene therapy provided herein include but are not limited to chemotherapy (e.g., cisplatin, gemcitabine, pemetrexed, 5-fluorouracil, carboplatin, irinotecan, interferon alfa, oxaliplatin, paclitaxel pegylated liposomal doxorubicin, and/or topotecan), chemotherapy protective drugs (e.g., leucovorin), radiotherapy, cryotherapy, targeted small molecule therapies, other antibodies, aflibercept, and/or vaccine therapy and administration with anti-VEGF, including but not limited to ranibizumab or bevacizumab.

5.3.11. Anti-BLyS HuPTM Constructs and Formulations for Systemic Lupus Erythematosus

[0725] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to B-lymphocyte stimulator (BLyS) derived from an anti-BLyS antibody, such as belimumab (FIG. 8E), and indicated for treating systemic lupus erythematosus (SLE) and reducing levels of autoreactive B cells and immunoglobulin producing plasma cells. In particular embodiments, the HuPTM mAb has the amino acid sequence of belimumab or an antigen binding fragment thereof. The amino acid sequence of Fab fragment of this antibody is provided in FIG. 8E. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an BLyS-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with SLE to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0726] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to BLyS that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to BLyS, such as belimumab or variants thereof as detailed herein. The transgene may also encode an anti-BLyS antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0727] In certain embodiments, the anti-BLyS antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of belimumab (having amino acid sequences of SEQ ID NOS. 41 and 42, respectively, see Table 4 and FIG. 8E). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 141 (encoding the belimumab heavy chain Fab portion) and SEQ ID NO: 142 (encoding the belimumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both

have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0728] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-BLyS-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 41 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 8E. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 41 by the hinge region encoding sequences set forth in Table 5.

[0729] In certain embodiments, the anti-BLyS antigen-binding fragment transgene encodes an BLyS antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 42. In certain embodiments, the anti-BLyS antigen-binding fragment transgene encodes an BLyS antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 41. In certain embodiments, the anti-BLyS antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 42 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 41. In specific embodiments, the BLyS antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 41 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8E) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the BLyS antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 42 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8E) or are

substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0730] In certain embodiments, the anti-BLyS antigen-binding fragment transgene encodes a hyperglycosylated belimumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 41 and 42, respectively, with one or more of the following mutations: M118N (heavy chain) and/or Q196N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0731] In certain embodiments, the anti-BLyS antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six belimumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8E which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-BLyS antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0732] Provided are methods of treating human subjects for SLE by administration of a viral vector containing a transgene encoding an anti-BLyS antibody, or antigen binding fragment thereof. The antibody may be belimumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with SLE. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIG. 8E, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0733] Subjects to whom such gene therapy is administered can be those responsive to anti-BLyS therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with SLE, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-BLyS antibody or considered a good candidate for therapy with an anti-BLyS antibody. In specific embodiments, the patients have previously been treated with belimumab, and have been found to be responsive to belimumab. To determine responsiveness, the anti-BLyS antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0734] The production of the anti-BLyS HuPTM mAb or HuPTM Fab, should result in a "biobetter" molecule for the treatment of SLE accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-BLyS HuPTM Fab, intravenously to human subjects (patients) diagnosed with or having one or more symptoms of SLE, to create a permanent depot in the

liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0735] The cDNA construct for the anti-BLyS HuPTM-mAb or anti-BLyS HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver or muscle cells. For example, the signal sequence may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0736] As an alternative, or an additional treatment to gene therapy, the anti-BLyS HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with SLE, or for whom therapy for SLE is considered appropriate.

[0737] In specific embodiments, the anti-BLyS HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of belimumab as set forth in FIG. 8E (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N30 and/or N63 and/or N165 of the heavy chain (SEQ ID NO:41) or N68 and/or N95 of the light chain (SEQ ID NO:42). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of belimumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO:41) and/or Y85 and/or Y86 of the light chain (SEQ ID NO:42). In other embodiments, the anti-BLyS HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0738] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of SLE, reduce the levels of pain or discomfort for the patient, or reduce levels of autoreactive B cells and immunoglobulin producing plasma cells. Efficacy may be monitored by scoring the function, symptoms, or degree of inflammation in the affected tissue or area of the body, e.g., such as the skin, joints, kidneys, lungs, blood cells, heart, and brain. For example, efficacy can be monitored by monitoring the presence, extent, or rate of one or more symptoms including seizure, psychosis, organic brain syndrome, visual disturbance, other neurological problems, alopecia, skin rash, muscle weakness, arthritis, blood vessel inflammation, mucosal ulcers, chest pain worse with deep breathing and manifestations of pleurisy and/or pericarditis and fever. Standardized disease indexes can be used, such as Safety of Estrogens in Lupus Erythematosus National Assessment Systemic Lupus Erythematosus Disease Activity Index (SELENA-SLEDAI); British Isles Lupus Assessment Group (BILAG) A, BILAG B, Systemic Lupus Activity Measure (SLAM), or PGA score. (See e.g., Liang M H et al. (1988) "Measurement of systemic lupus erythematosus

activity in clinical research," *Arthritis Rheum.* 31:817-25; Diaz et al. (2011) "Measures of adult systemic lupus erythematosus: updated version of British Isles Lupus Assessment Group (BILAG 2004), European Consensus Lupus Activity Measurements (ECLAM), Systemic Lupus Activity Measure, Revised (SLAM-R), Systemic Lupus Activity Questionnaire for Population Studies (SLAQ), Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2 K), and Systemic Lupus," *International Collaborating Clinics/American College of Rheumatology Damage Index (SDI) Arthritis Care Res.* 63: S37-46).

[0739] Combinations of delivery of the anti-BLyS HuPTM mAb or antigen-binding fragment thereof, to the liver or muscles accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for SLE that could be combined with the gene therapy provided herein include but are not limited to corticosteroids, antimalarials, NSAIDs, and immunosuppressives, and administration with anti-BLyS agents, including but not limited to belimumab.

5.3.12. Anti-CP-C5 HuPTM Constructs and Formulations for Paroxysmal Nocturnal Hemoglobinuria and Atypical Hemolytic Uremic Syndrome

[0740] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to complement protein C5 (or C5a) (CP-C5) derived from an anti-CP-C5 antibody, such as eculizumab (FIG. 8F), and indicated for treating paroxysmal nocturnal hemoglobinuria (PNH), treating atypical hemolytic uremic syndrome (aHUS), reducing the destruction of blood cells, and/or reducing the need for blood transfusions. In particular embodiments, the HuPTM mAb has the amino acid sequence of eculizumab or an antigen binding fragment thereof. The amino acid sequence of the Fab fragment of this antibody is provided in FIG. 8F. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an CP-C5-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with PNH or aHUS to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0741] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to CP-C5 that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to CP-C5, such as eculizumab or variants thereof as detailed herein. The transgene may also encode an anti-CP-C5 antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0742] In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of eculizumab (having amino acid sequences of SEQ

ID NOs. 43 and 44, respectively, see Table 4 and FIG. 8F). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 143 (encoding the eculizumab heavy chain Fab portion) and SEQ ID NO: 144 (encoding the eculizumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes). The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 3 that correspond to the proteins secreted by hepatocytes.

[0743] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-CP-C5-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 43 with additional hinge region sequence starting after the C-terminal glutamic acid (E), contains all or a portion of the amino acid sequence CPPCPAPPVAGG (SEQ ID NO: 232), and specifically, CPPCPA (SEQ ID NO: 219) or CPPCPAPPVAG (SEQ ID NO: 233) as set forth in FIG. 8F. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 43 by the hinge region encoding sequences set forth in Table 5.

[0744] In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene encodes an CP-C5 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 44. In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene encodes an CP-C5 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 43. In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 44 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 43. In specific embodiments, the CP-C5 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 43 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8F) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the CP-C5 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID

NO: 44 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8F) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0745] In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene encodes a hyperglycosylated eculizumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 43 and 44, respectively, with one or more of the following mutations: L117N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0746] In certain embodiments, the anti-CP-C5 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six eculizumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8F which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-CP-C5 antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0747] Provided are methods of treating human subjects for PNH or aHUS by administration of a viral vector containing a transgene encoding an anti-CP-C5 antibody, or antigen binding fragment thereof. The antibody may be eculizumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with PNH or aHUS. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIG. 8F, can be administered in any manner such that the recombinant vector enters the liver, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0748] Subjects to whom such gene therapy is administered can be those responsive to anti-CP-C5 therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with PNH or aHUS, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-CP-C5 antibody or considered a good candidate for therapy with an anti-CP-C5 antibody. In specific embodiments, the patients have previously been treated with eculizumab, and have been found to be responsive to eculizumab. To determine responsiveness, the anti-CP-C5 antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0749] The production of the anti-CP-C5 HuPTM mAb or HuPTM Fab, should result in a "biobetter" molecule for the

treatment of PNH or aHUS accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-CP-C5 HuPTM Fab, intravenously to human subjects (patients) diagnosed with or having one or more symptoms of PNH or aHUS, to create a permanent depot in the liver tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver cells.

[0750] The cDNA construct for the anti-CP-C5 HuPTM-mAb or anti-CP-C5 HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced liver cells. For example, the signal sequence may be MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 3 that correspond to the proteins secreted by hepatocytes.

[0751] As an alternative, or an additional treatment to gene therapy, the anti-CP-C5 HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with PNH or aHUS, or for whom therapy for PNH or aHUS is considered appropriate.

[0752] In specific embodiments, the anti-CP-C5 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of eculizumab as set forth in FIG. 8F (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N63 and/or Q114 and/or N164 and/or N197 and/or N206 of the heavy chain (SEQ ID NO: 43) or N28 and/or Q100 and/or N158 and/or N210 of the light chain (SEQ ID NO:44). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of eculizumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO:43) and/or Y86 and/or Y87 of the light chain (SEQ ID NO:44). In other embodiments, the anti-CP-C5 HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0753] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of PNH or aHUS, reduce the need for a blood transfusion, or reduce the destruction of red blood cells. Efficacy may be monitored by measuring hemoglobin stabilization and/or the number of RBC units transfused or scoring fatigue levels and/or health-related quality of life over the course of treatment.

[0754] Combinations of delivery of the anti-CP-C5 HuPTM mAb or antigen-binding fragment thereof, to the liver accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for PNH or aHUS that could be combined with the gene therapy provided herein include but are not

limited to anticoagulants and steroids/immunosuppressant treatments, and administration with anti-CP-C5 agents, including but not limited to eculizumab.

5.3.13 Anti-MMP9 HuPTM Constructs and Formulations for Ocular Disorders, Cystic Fibrosis, Rheumatoid Arthritis, Inflammatory Bowel Disease, and Cancer

[0755] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to matrix metalloproteinase 9 (MMP9) derived from anti-MMP9 indicated for treating one or more retinal disorders including macular degeneration (e.g., dry age-related macular degeneration (AMD)), cystic fibrosis (CF), rheumatoid arthritis (RA), IBD (e.g., UC and CD), and one or more types of cancer (e.g., solid tumors, pancreatic adenocarcinoma, lung adenocarcinoma, lung squamous cell carcinoma, esophagogastric adenocarcinoma, gastric cancer, colorectal cancer, or breast cancer), or for suppressing extracellular matrix degradation. In particular embodiments, the HuPTM mAb has the amino acid sequence of andecaliximab or an antigen binding fragment thereof. The amino acid sequence of Fab fragments of andecaliximab is provided in FIG. 8G. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an MMP9-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with, or having one or more symptoms of a retinal disorder (e.g. macular degeneration), RA, CF, IBD (e.g., UC or CD), or one or more cancers (such as those listed above) to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

[0756] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to MMP9 that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to MMP9, such as andecaliximab, or variants thereof as detailed herein. The transgene may also encode an anti-MMP9 antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0757] In certain embodiments, the anti-MMP9 antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of andecaliximab (having amino acid sequences of SEQ ID NOs. 45 and 46, respectively, see Table 4 and FIG. 8G). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 145 (encoding the andecaliximab heavy chain Fab portion) and SEQ ID NO: 146 (encoding the andecaliximab light chain Fab portion) as set forth in Table 5. In the case of treating ocular diseases, the heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, one or more cells forming the retina. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 that correspond to the proteins secreted by one or more cells forming the retina. In the case

of treating non-ocular diseases, the heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0758] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-MMP9 antigen binding domain has a heavy chain variable domain of SEQ ID NO: 45 with additional hinge region sequence starting after the C-terminal aspartic acid (D), contains all or a portion of the amino acid sequence GPPCPPCAPEFLGG (SEQ ID NO: 231), and specifically, GPPCPPCPA (SEQ ID NO: 229) or GPPCPPCAPEFLGGPSVFL (SEQ ID NO: 230) as set forth in FIG. 8G. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 45 by the hinge region encoding sequences set forth in Table 5.

[0759] In certain embodiments, the anti-MMP9 antigen-binding fragment transgene encodes an MMP9 antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 46. In certain embodiments, the anti-MMP9 antigen-binding fragment transgene encodes an MMP9 antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 45. In certain embodiments, the anti-MMP9 antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 46 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 45. In specific embodiments, the MMP9 antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 45 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8G) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the MMP9 antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 46 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG.

8G) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0760] In certain embodiments, the anti-MMP9 antigen-binding fragment transgene encodes a hyperglycosylated andecaliximab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 45 and 46, respectively, with one or more of the following mutations: L110N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0761] In certain embodiments, the anti-MMP9 antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six andecaliximab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8G which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-MMP9 antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0762] Provided are methods of treating human subjects for one or more retinal disorders (such as macular degeneration), IBD, CF, RA, or cancers by administration of a viral vector containing a transgene encoding an anti-MMP9 antibody or antigen binding fragment thereof. The antibody may be andecaliximab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with one or more of retinal disorders, IBD, CF, RA, or cancers.

[0763] Recombinant vector used for delivering the transgene are described in Section 5.4.3. Such vectors should have a tropism for human retina-type cells and can include non-replicating rAAV, particularly those bearing an AAV8 capsid. The recombinant vectors, such as those shown in FIG. 8G, can be administered in any manner such that the recombinant vector enters the retina, preferably by introducing the recombinant vector into the eye. See Section 5.5.3 for details regarding the methods of treatment. For delivery to the liver, for example, for the treatment of cancer, recombinant vector used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIG. 8G, can be administered in any manner such that the recombinant vector enters the liver, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment.

[0764] Subjects to whom such gene therapy is administered can be those responsive to anti-MMP9 therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with one or more retinal disorders or types of cancer, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-MMP9 antibody, or considered a good candidate for therapy with an anti-MMP9 antibody. In specific embodiments, the patients have previously been treated with andecaliximab, and have been found to be responsive to andecaliximab. To determine responsiveness,

the anti-MMP9 or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0765] The production of the anti-MMP9 HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of one or more retinal disorders, CF, RA, IBD, or cancers accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-MMP9 HuPTM Fab, subretinally, intravitreally or suprachoroidally to human subjects (patients) diagnosed with or having one or more symptoms of one or more retinal disorders, or by administering a viral vector or other DNA expression construct encoding the anti-MMP9 HuPTM Fab, subcutaneously, intramuscularly, or intravenously to human subjects (patients) diagnosed with RA, CF, IBD, or cancer, to create a permanent depot in the retina or liver that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced cells of the retina or liver.

[0766] The cDNA construct for the anti-MMP9 HuPTM-mAb or anti-MMP9 HuPTM Fab should include a signal peptide that ensures proper co- and post-translational processing (glycosylation and protein sulfation) by the transduced cells of the retina or liver. For example, the signal sequence may be MYRMQLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 1 or 3 that correspond to the proteins secreted by cells of the retina or liver, respectively.

[0767] As an alternative, or an additional treatment to gene therapy, the anti-MMP9 HuPTM mAb or HuPTM Fab can be produced in human cell lines by recombinant DNA technology, and administered to patients diagnosed with a retinal disorder or cancer for whom therapy for a retinal disorder, IBD, CF, RA, or cancer is considered appropriate.

[0768] In specific embodiments, the anti-MMP9 HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of andecaliximab as set forth in FIG. 8G (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N58, N76, Q107, N157, and/or N199 of the heavy chain (SEQ ID NO:45) or N158 and/or N210 of the light chain (SEQ ID NO:46). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of andecaliximab has a sulfation group at Y93 and/or Y94 of the heavy chain (SEQ ID NO: 45) and/or Y86 and/or Y87 of the light chain (SEQ ID NO: 46). In other embodiments, the anti-MMP9 HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0769] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of the disease being treated or alleviate one

or more symptoms thereof. In the case of retinal disorders, efficacy may be monitored by monitoring vision acuity. For example, efficacy can be monitored by assessing change in vision acuity from baseline. In the case of a cancer, efficacy can be monitored by assessing one or more oncology endpoints including overall survival, progression-free survival, time to progression, time to treatment failure, event-free survival, time to next treatment, objective response rate, or duration of response. (see, e.g., U.S. Department of Health and Human Services Food and Drug Administration Center for Drug Evaluation and Research, Center for Biologics Evaluation and Research. Guidance for industry: clinical trial endpoints for the approval of cancer drugs and biologics. <https://www.fda.gov/downloads/Drugs/Guidances/ucm071590.pdf>. Published May 2007. Accessed Oct. 13, 2017; Oncology Endpoints in a Changing Landscape. *Manag. Care.* 2016; 1 (suppl): 1-12). In the case of RA, efficacy can be monitored by assessing one or more of (1) swollen joint count, (2) tender joint count. (3) global physician’s assessment of disease activity, (4) patient self-report of functional status. (5) patient self-report of pain. (6) global patient assessment of disease activity, (7) a laboratory measures of C-reactive protein and erythrocyte sedimentation rate, and (8) radiographic progression. (see, e.g., Smolen JS, Aletaha D. “Assessment of rheumatoid arthritis activity in clinical trials and clinical practice” UptoDate.com Wolters Kluwer Health. Accessed at: www.uptodate.com December 2017) For example, with regard to CD, efficacy can be monitored by assessing Crohn’s Disease Activity Index [CDAI] over the course of treatment (e.g., see Best W R et al. (1976) *Gastroenterology*, March; 70 (3): 439-44, “Development of a Crohn’s disease activity index. National Cooperative Crohn’s Disease Study.”). With regard to UC, efficacy can be monitored by assessing a Mayo score and an endoscopy subscore over the course of treatment (e.g., see Lobaton et al., “The Modified Mayo Endoscopic Score (MMES): A New Index for the Assessment of Extension and Severity of Endoscopic Activity in Ulcerative Colitis Patients,” *J. Crohns Colitis.* 2015 October; 9 (10): 846-52). In the case of CF, efficacy can be monitored by assessing Forced Expiratory Volume in 1s (FEV1), decreased frequency of pulmonary exacerbations, quality of life (QoL) improvement, and, for younger patients, growth improvement. (e.g., see VanDevanter and Konstan, “Outcome measurement for clinical trials assessing treatment of cystic fibrosis lung disease,” *Clin. Investig.* 2 (2): 163-175 (2012)).

[0770] Combinations of delivery of the anti-MMP9 HuPTM mAb or antigen-binding fragment thereof to the retina or liver accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for macular degeneration that could be combined with the gene therapy provided herein include but are not limited to laser photocoagulation, photodynamic therapy with verteporfin, aflibercept, and/or intravitreal steroids and administration with anti-MMP9 agents, including but not limited to andecaliximab. Available treatments for the one or more above listed cancers that could be combined with the gene therapy provided herein include but are not limited to chemotherapy (e.g., cisplatin, gemcitabine, pemetrexed, 5-fluorouracil, carboplatin, irinotecan, interferon alfa, oxaliplatin, paclitaxel pegylated liposomal doxorubicin, and/or topotecan), chemotherapy pro-

fective drugs (e.g., leucovorin), radiotherapy, cryotherapy, targeted small molecule therapies, other antibodies, aflibercept, and/or vaccine therapy and administration with anti-MMP9, including but not limited to andecaliximab. Available treatments for RA that could be combined with the gene therapy provided herein include but are not limited to bisphosphonates, nonsteroidal anti-inflammatory drugs (e.g., celecoxib, naproxen, aspirin, indomethacin, sulfasalazine, and ketoprofen), steroids (e.g., prednisone), disease modifying anti-rheumatic drugs and other immunosuppressants (e.g., leflunomide, methotrexate, tofacitinib, azathioprine, mycophenolate, cyclophosphamide, cyclosporine), hydroxychloroquine, abatacept, anakinra, apremilast, TNF inhibitors, other antibodies (e.g., tocilizumab, secukinimab, rituximab) and administration with anti-MMP9, including but not limited to andecaliximab. Available treatments for IBD that could be combined with the gene therapy provided herein include but are not limited to nonsteroidal anti-inflammatory drugs (e.g., mesalamine, sulfasalazine), steroids (e.g., hydrocortisone, prednisone, budesonide), immunosuppressants (e.g., methotrexate, mercaptopurine, azathioprine), vitamins (e.g., iron, cholecalciferol), antibiotics (e.g., amino salicylic acid, metronidazole), other antibodies (e.g., infliximab, adalimumab) and administration with anti-MMP9, including but not limited to andecaliximab. Available treatments for CF that could be combined with the gene therapy provided herein include but are not limited to antibiotics, vaccines, and cough medicines (e.g., acetylcysteine and dornase alfa) and administration with anti-MMP9, including but not limited to andecaliximab.

5.3.14. Anti-pKal HuPTM Constructs and Formulations for Angioedema

[0771] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to kallikrein (pKal) derived from an anti-pKal antibody and indicated for treating angioedema, such as hereditary angioedema. In particular embodiments, the HuPTM mAb has the amino acid sequence of lanadelumab or an antigen binding fragment thereof. The amino acid sequence of Fab fragment of this antibody is provided in FIG. 8H. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an pKal-binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with angioedema to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0772] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to pKal that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to pKal, such as lanadelumab or variants thereof as detailed herein. The transgene may also encode an anti-pKal antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0773] In certain embodiments, the anti-pKal antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of lanadelumab (having amino acid sequences of SEQ ID NOs. 47 and 48, respectively, see Table 4 and FIG. 8H). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 147 (encoding the lanadelumab heavy chain Fab portion) and SEQ ID NO: 148 (encoding the lanadelumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences both have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0774] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-pKal-antigen binding domain has a heavy chain variable domain of SEQ ID NO: 47 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 8H. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 47 by the hinge region encoding sequences set forth in Table 5.

[0775] In certain embodiments, the anti-pKal antigen-binding fragment transgene encodes an pKal antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 48. In certain embodiments, the anti-pKal antigen-binding fragment transgene encodes an pKal antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 47. In certain embodiments, the anti-pKal antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 48 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 47. In specific embodiments, the pKal antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 47 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which

CDRs are underlined in FIG. 8H) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the pKal antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 48 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 8H) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0776] In certain embodiments, the anti-pKal antigen-binding fragment transgene encodes a hyperglycosylated lanadelumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 47 and 48, respectively, with one or more of the following mutations: M117N (heavy chain) and/or Q159N, Q159S, and/or E194N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0777] In certain embodiments, the anti-pKal antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six lanadelumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 8H which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-pKal antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0778] Provided are methods of treating human subjects for angioedema by administration of a viral vector containing a transgene encoding an anti-pKal antibody, or antigen binding fragment thereof. The antibody may be lanadelumab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptoms associated with angioedema. Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as shown in FIG. 8H, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment

[0779] Subjects to whom such gene therapy is administered can be those responsive to anti-pKal therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with angioedema, or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-pKal antibody or considered a good candidate for therapy with an anti-pKal antibody. In specific embodiments, the patients have previously been treated with lanadelumab, and have been found to be responsive to lanadelumab. To determine responsiveness, the anti-pKal antibody or antigen-binding fragment

transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0780] The production of the anti-pKal HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of angioedema accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-pKal HuPTM Fab, intravenously to human subjects (patients) diagnosed with or having one or more symptoms of angioedema, to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0781] In specific embodiments, the anti-pKal HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of lanadelumab as set forth in FIG. 8H (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N77, Q114 and/or N164 of the heavy chain (SEQ ID NO:47) or Q99, N157, and/or N209 of the light chain (SEQ ID NO:48). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of lanadelumab has a sulfation group at Y94 and/or Y95 of the heavy chain (SEQ ID NO:47) and/or Y86 and/or Y87 of the light chain (SEQ ID NO:48). In other embodiments, the anti-pKal HuPTM mAb or antigen-binding fragment thereof does not contain detectable NeuGc moieties and/or does not contain detectable alpha-Gal moieties.

[0782] In certain embodiments, the HuPTM mAb or Fab (or a hyperglycosylated derivative of either) is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of angioedema, reduce the levels of pain or discomfort for the patient, or reduce levels of autoreactive B cells and immunoglobulin producing plasma cells. Efficacy may be monitored by scoring the function, symptoms, or degree of inflammation in the affected tissue or area of the body, e.g., such as the skin, joints, kidneys, lungs, blood cells, heart, and brain. For example, efficacy can be monitored by assessing changes in attack severity or frequency.

[0783] Combinations of delivery of the anti-pKal HuPTM mAb or antigen-binding fragment thereof, to the liver or muscle accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for angioedema that could be combined with the gene therapy provided herein include but are not limited to danazol, bradykinin receptor antagonist (e.g., icatibant), plasma kallikrein inhibitor (e.g., ecallantide), C1 esterase inhibitor, constant alpha, anti-fibrinolytic agents (e.g., tranexamic acid), omalizumab, and fresh frozen plasma transfusions, antihistamines, and corticosteroids and administration with anti-pKal agents, including but not limited to lanadelumab.

5.3.15. Anti-TNF α HuPTM Constructs and Formulations for Various Auto-Immune Disorders-Adalimumab and Infliximab

[0784] Compositions and methods are described for the delivery of HuPTM mAbs and antigen-binding fragments thereof, such as HuPTM Fabs, that bind to tumor necrosis factor- α (TNF α) derived from an anti-TNF α antibody, such as adalimumab (FIG. 9A) or infliximab (FIG. 9B), and indicated for treating one or more autoimmune-related disorders, such as hidradenitis suppurativa (HS), atopic dermatitis, psoriasis (e.g., plaque psoriasis, pustular psoriasis, and erythrodermic psoriasis), arthritis (e.g., juvenile idiopathic arthritis, rheumatoid arthritis, psoriatic arthritis, and alkylating spondylitis), and/or IBD (e.g., Crohn's disease and ulcerative colitis) (collectively referred to hereinafter as "subject AI-Ds(2)"). In particular embodiments, the HuPTM mAb has the amino acid sequence of adalimumab or an antigen binding fragment thereof. In other embodiments, the HuPTM mAb has the amino acid sequence of infliximab or an antigen binding fragment thereof. Amino acid sequences of Fab fragments of the antibody are provided in FIGS. 9A and 9B. Delivery may be accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding an TNF α -binding HuPTM mAb (or an antigen binding fragment and/or a hyperglycosylated derivative or other derivative, thereof) to patients (human subjects) diagnosed with one or more subject AI-Ds(2) to create a permanent depot that continuously supplies the human PTM, e.g., human-glycosylated, transgene product.

Transgenes

[0785] Provided are recombinant vectors containing a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to TNF α that can be administered to deliver the HuPTM mAb or antigen binding fragment in a patient. The transgene is a nucleic acid comprising the nucleotide sequences encoding an antigen binding fragment of an antibody that binds to TNF α , such as adalimumab or infliximab or variants thereof as detailed herein. The transgene may also encode an anti-TNF α antigen binding fragment that contains additional glycosylation sites (e.g., see Courtois et al.).

[0786] In certain embodiments, the anti-TNF α antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of adalimumab (having amino acid sequences of SEQ ID NOs. 49 and 50, respectively, see Table 4 and FIG. 9A). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 149 (encoding the adalimumab heavy chain Fab portion) and SEQ ID NO: 150 (encoding the adalimumab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences each have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVTNS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0787] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-TNF α -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 49 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHL (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 9A. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 49 by the hinge region encoding sequences set forth in Table 5.

[0788] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an TNF α antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 50. In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an TNF α antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 49. In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 50 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 49. In specific embodiments, the TNF α antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 49 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 9A) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the TNF α antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 50 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 9A) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0789] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes a hyperglycosylated adalimumab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 49 and 50, respectively, with one or more of the following mutations: L116N (heavy chain), Q160N or

Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0790] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six adalimumab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 9A which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-TNF α antibody or antigen-binding fragment thereof.

[0791] In certain embodiments, the anti-TNF α antigen-binding fragment transgene comprises the nucleotide sequences encoding the heavy and light chains of the Fab portion of infliximab (having amino acid sequences of SEQ ID NOs. 51 and 52, respectively, see Table 4 and FIG. 9B). The nucleotide sequences may be codon optimized for expression in human cells and may, for example, comprise the nucleotide sequences of SEQ ID NO: 151 (encoding the infliximab heavy chain Fab portion) and SEQ ID NO: 152 (encoding the infliximab light chain Fab portion) as set forth in Table 5. The heavy and light chain sequences each have a signal or leader sequence at the N-terminus appropriate for expression and secretion in human cells, in particular, human liver cells (e.g., hepatocytes) or muscle cells. The signal sequence may have the amino acid sequence of MYRMQLLLIALSLALVINS (SEQ ID NO: 161). Alternatively, the signal sequence may have an amino acid sequence selected from any one of the signal sequences set forth in Table 2 or 3 that correspond to the proteins secreted by myocytes or hepatocytes, respectively.

[0792] In addition to the heavy and light chain variable domain sequences, the transgenes may comprise, at the C-terminus of the heavy chain variable domain sequence, all or a portion of the hinge region. In specific embodiments, the anti-TNF α -antigen binding domain has a heavy chain variable domain of SEQ ID NO: 51 with additional hinge region sequence starting after the C-terminal aspartate (D), contains all or a portion of the amino acid sequence KTHTCPPCPAPELLGG (SEQ ID NO: 222), and specifically, KTHL (SEQ ID NO: 223), KTHT (SEQ ID NO: 224), KTHTCPPCPA (SEQ ID NO: 225), KTHLCPPCPA (SEQ ID NO: 226), KTHTCPPCPAPELLGGPSVFL (SEQ ID NO: 227) or KTHLCPPCPAPELLGGPSVFL (SEQ ID NO: 228) as set forth in FIG. 9B. These hinge regions may be encoded by nucleotide sequences at the 3' end of SEQ ID NO: 51 by the hinge region encoding sequences set forth in Table 5.

[0793] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an TNF α antigen-binding fragment comprising a light chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 52. In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an TNF α antigen-binding fragment comprising a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 51. In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an antigen-binding fragment comprising a light chain comprising an amino acid sequence that

is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 52 and a heavy chain comprising an amino acid sequence that is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to the sequence set forth in SEQ ID NO: 51. In specific embodiments, the TNF α antigen binding fragment comprises a heavy chain comprising an amino acid sequence of SEQ ID NO: 51 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 9B) or are substitutions with an amino acid present at that position in the heavy chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11A. In specific embodiments, the TNF α antigen binding fragment comprises a light chain comprising an amino acid sequence of SEQ ID NO: 52 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or more amino acid substitutions, insertions or deletions, and the substitutions, insertions or deletions preferably are made in the framework regions (i.e., those regions outside of the CDRs, which CDRs are underlined in FIG. 9B) or are substitutions with an amino acid present at that position in the light chain of one or more of the other therapeutic antibodies, for example, as identified by the alignment in FIG. 11B.

[0794] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes a hyperglycosylated infliximab Fab, comprising a heavy chain and a light chain of SEQ ID NOs: 51 and 52, respectively, with one or more of the following mutations: T115N (heavy chain), Q160N or Q160S (light chain), and/or E195N (light chain) (see FIGS. 11A (heavy chain) and B (light chain)).

[0795] In certain embodiments, the anti-TNF α antigen-binding fragment transgene encodes an antigen-binding fragment and comprises the nucleotide sequences encoding the six infliximab CDRs which are underlined in the heavy and light chain variable domain sequences of FIG. 9B which are spaced between framework regions, generally human framework regions, and associated with constant domains depending upon the form of the antigen-binding molecule, as is known in the art to form the heavy and/or light chain variable domain of an anti-TNF α antibody or antigen-binding fragment thereof.

Gene Therapy Methods

[0796] Provided are methods of treating human subjects for one or more of the subject AI-Ds(2) by administration of a viral vector containing a transgene encoding an anti-TNF α antibody, or antigen binding fragment thereof. The antibody may be adalimumab or infliximab, and is preferably a Fab fragment thereof, or other antigen-binding fragment thereof. In embodiments, the patient has been diagnosed with and/or has symptom(s) associated with one or more of the subject AI-Ds(2). Recombinant vectors used for delivering the transgene are described in Section 5.4.2. Such vectors should have a tropism for human liver or muscle cells and can include non-replicating rAAV, particularly those bearing an AAV8 or AAV9 capsid. The recombinant vectors, such as those shown in FIGS. 9A and 9B, can be administered in any manner such that the recombinant vector enters the liver or muscle tissue, preferably by introducing the recombinant

vector into the bloodstream. See Section 5.5.2 for details regarding the methods of treatment

[0797] Subjects to whom such gene therapy is administered can be those responsive to anti-TNF α therapy. In particular embodiments, the methods encompass treating patients who have been diagnosed with one or more of the subject AI-Ds(2), or have one or more symptoms associated therewith, and identified as responsive to treatment with an anti-TNF α antibody or considered a good candidate for therapy with an anti-TNF α antibody. In specific embodiments, the patients have previously been treated with adalimumab or infliximab, and have been found to be responsive to adalimumab or infliximab. In other embodiments, the patients have been previously treated with an anti-TNF α antibody or fusion protein such as etanercept, golimumab, or certolizumab, or other anti-TNF α agent. To determine responsiveness, the anti-TNF α antibody or antigen-binding fragment transgene product (e.g., produced in cell culture, bioreactors, etc.) may be administered directly to the subject.

Human Post Translationally Modified Antibodies

[0798] The production of the anti-TNF α HuPTM mAb or HuPTM Fab, should result in a “biobetter” molecule for the treatment of one or more subject AI-Ds(2) accomplished via gene therapy—e.g., by administering a viral vector or other DNA expression construct encoding the anti-TNF α HuPTM Fab, intravenously to human subjects (patients) diagnosed with or having one or more symptoms of one or more subject AI-Ds(2), to create a permanent depot in the liver or muscle tissue that continuously supplies the fully-human post-translationally modified, e.g., human-glycosylated, sulfated transgene product produced by transduced liver or muscle cells.

[0799] In specific embodiments, the anti-TNF α HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of adalimumab as set forth in FIG. 9A (with non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N54 and/or N163 and/or Q113 of the heavy chain (SEQ ID NO:49) or Q100 and/or N158 and/or N210 of the light chain (SEQ ID NO:50). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of adalimumab has a sulfation group at Y94 and/or Y95 and/or Y32 of the heavy chain (SEQ ID NO: 49) and/or Y86 and/or Y87 of the light chain (SEQ ID NO:50). In other embodiments, the anti-TNF α HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0800] In specific embodiments, the anti-TNF α HuPTM mAb or antigen-binding fragment thereof has heavy and light chains with the amino acid sequences of the heavy and light chain Fab portions of infliximab as set forth in FIG. 9B (with consensus and non-consensus asparagine (N) glycosylation sites highlighted in aqua, glutamine (Q) glycosylation sites highlighted in green, and Y-sulfation sites highlighted in yellow) has a glycosylation, particularly a 2,6-sialylation, at one or more of the amino acid positions N57 and/or N101 and/or Q112 and/or N162 of the heavy chain

(SEQ ID NO: 51) or N41 and/or N76 and/or N158 and/or N210 of the light chain (SEQ ID NO:52). Alternatively or in addition to, the HuPTM mAb or antigen binding-fragment thereof with the heavy and light chain variable domain sequences of adalimumab has a sulfation group at Y96 and/or Y97 of the heavy chain (SEQ ID NO:51) and/or Y86 and/or Y87 of the light chain (SEQ ID NO:52). In other embodiments, the anti-TNF α HuPTM mAb or antigen-binding fragment thereof does not contain any detectable NeuGc moieties and/or does not contain any detectable alpha-Gal moieties.

[0801] In certain embodiments, the HuPTM mAb or Fab is therapeutically effective and is at least 0.5%, 1% or 2% glycosylated and/or sulfated and may be at least 5%, 10% or even 50% or 100% glycosylated and/or sulfated. The goal of gene therapy treatment provided herein is to slow or arrest the progression of or relieve one or more symptoms of the one or more of the subject AI-Ds(2), such as reduce the levels of pain or discomfort for the patient.

[0802] Efficacy may be monitored by scoring the symptoms or degree of inflammation in the affected tissue or area of the body, e.g., such as the skin, colon, or joints. For example, with regard to CD, efficacy can be monitored by assessing Crohn’s Disease Activity Index [CDAI] over the course of treatment (e.g., see Best W R et al. (1976) *Gastroenterology*, March; 70 (3): 439-44, “Development of a Crohn’s disease activity index. National Cooperative Crohn’s Disease Study.”). With regard to UC, efficacy can be monitored by assessing a Mayo score and an endoscopy subscore over the course of treatment (e.g., see Lobaton et al. (2015) *J. Crohns Colitis*. 2015 October; 9 (10): 846-52, “The Modified Mayo Endoscopic Score (MMES): A New Index for the Assessment of Extension and Severity of Endoscopic Activity in Ulcerative Colitis Patients.”). With regard to psoriasis, HS, and atopic dermatitis, efficacy can be monitored by assessing changes in the affected skin or in the quality of the patient’s life over the course of treatment. One or more standardized assessments can be used to assess the change. (see e.g., Feldman & Krueger, (2005) *Ann. Rheum. Dis.* 64 (Suppl II): ii65-ii68: “Psoriasis assessment tools in clinical trials” describing standardized assessments including the Psoriasis Area and Severity Index (PASI), Physician Global Assessment (PGA), lattice system, NPF Psoriasis Score (NPF-PS), Medical Outcome Survey Short Form 36 (SF-36), the Euro QOL, Dermatology Life Quality Index (DLQI), and the Skindex; Schram et al. (2012) *Allergy*; 67:99-106: “EASI, (objective) SCORAD and POEM for atopic eczema: responsiveness and minimal clinically important difference” describing standardized assessments including Eczema Area and Severity Index (EASI) and the Severity Scoring of Atopic Dermatitis Index (SCORAD); Sisic et al. (2017) *J Cutan Med Surg.* 21 (2): 152-155 “Development of a Quality-of-Life Measure for Hidradenitis Suppurativa.”). With regard to arthritis, efficacy can be monitored by assessing one or more of the activity of the disease, the patient’s level of function, or the degree of structural damage to patient’s joints (e.g., see Zockling & Braun (2005) *Clin. Exp. Rheumatol* 23 (Suppl. 39) S133-S141: “Assessment of ankylosing spondylitis” describing standardized assessment for ankylosing spondylitis; see also Coates et al. (2011) *J. Rheumatol.* 38 (7): 1496-1501: “Development of a disease severity and responder index for

psoriatic arthritis (PsA)-report of the OMERACT 10 PsA special interest group” describing standardized assessments for psoriatic arthritis.).

[0803] Combinations of delivery of the anti-TNF α HuPTM mAb or antigen-binding fragment thereof, to the liver or muscles accompanied by delivery of other available treatments are encompassed by the methods provided herein. The additional treatments may be administered before, concurrently, or subsequent to the gene therapy treatment. Available treatments for subject AI-Ds(2) that could be combined with the gene therapy provided herein include but are not limited to phototherapy for psoriasis, aminosalicylates, immunomodulatory agents (e.g., azathioprine (AZA), 6-mercaptopurine (6-MP), methotrexate (MTX)), oral or topical corticosteroids (e.g., prednisone or budesonide), topical calcineurin inhibitors, antibiotics for IBD, and administration with anti-TNF α agents, including but not limited to adalimumab or infliximab.

5.4 Delivery of Gene Therapy Constructs

5.4.1 Constructs for Delivery to CNS

[0804] Sections 5.3.1, 5.3.2, 5.3.3, and 5.3.4 describe recombinant vectors that contain a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to A β , Tau protein, CGRPR, and integrin, respectively. Such recombinant vector used for delivering the transgene should have a tropism for human CNS cells, such as glial and neuronal cells. Such vectors can include non-replicating recombinant adeno-associated virus vectors (“rAAV”), particularly those bearing an AAV9, AAVrh10, AAVrh20, AAVrh39, or AAVcy5 capsid. However, other viral vectors may be used, including but not limited to lentiviral vectors, vaccinia viral vectors, or non-viral expression vectors referred to as “naked DNA” constructs.

[0805] In specific embodiments, provided are constructs for gene therapy administration to a human subject, comprising an AAV vector, which comprises a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 capsid (SEQ ID NO: 79); and a viral or artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding the heavy and light chains of the therapeutic antibody, operably linked to one or more regulatory sequences that control expression of the transgene in human cells that express and deliver the therapeutic antibody in a therapeutically appropriate manner as disclosed herein, particularly expressed from CNS cells. In certain embodiments, the encoded AAV9 capsid has the sequence of SEQ ID NO: 79 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions, particularly substitutions with amino acid residues found in the corresponding position in other AAV capsids, for example, in the SUBS row of FIG. 12 which provides a comparison of the amino acid sequences of the capsid sequences of various AAVs, highlighting amino acids appropriate for substitution at different positions within the capsid sequence.

[0806] In other specific embodiments, provided are constructs for gene therapy administration to a human subject, comprising an AAV vector, which comprises a viral capsid that is at least 95% identical to the amino acid sequence of an AAVrh10 capsid (SEQ ID NO: 80); and a viral or artificial

genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding the heavy and light chains of the therapeutic antibody, operably linked to one or more regulatory sequences that control expression of the transgene in human cells that express and deliver the therapeutic antibody in a therapeutically appropriate manner as disclosed herein, particularly from CNS cells. In certain embodiments, the encoded AAVrh10 capsid has the sequence of SEQ ID NO: 80 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions, particularly substitutions with amino acid residues found in the corresponding position in other AAV capsids, for example, in the SUBS row of FIG. 12 which provides a comparison of the amino acid sequences of the capsid sequences of various AAVs, highlighting amino acids appropriate for substitution at different positions within the capsid sequence.

[0807] Preferably, the HuPTM mAb or antigen binding fragment thereof, including the HuPTM Fab transgene should be controlled by appropriate expression control elements for expression of the HuPTM Fab in human CNS cells, for example, the CB7 promoter (a chicken β -actin promoter and CMV enhancer), RSV promoter, GFAP promoter (glial fibrillary acidic protein), MBP promoter (myelin basic protein), MMT promoter, EF-1 α , U86 promoter, RPE65 promoter or opsin promoter, an inducible promoter, for example, a hypoxia-inducible promoter or a drug inducible promoter, such as a promoters induced by rapamycin and related agents, and other expression control elements that enhance expression of the transgene driven by the vector (e.g., introns such as the chicken β -actin intron, minute virus of mice (MVM) intron, human factor IX intron (e.g., FIX truncated intron 1), β -globin splice donor/immunoglobulin heavy chain splice acceptor intron, adenovirus splice donor/immunoglobulin splice acceptor intron, SV40 late splice donor/splice acceptor (19S/16S) intron, and hybrid adenovirus splice donor/IgG splice acceptor intron and polyA signals such as the rabbit β -globin polyA signal, human growth hormone (hGH) polyA signal, SV40 late polyA signal, synthetic polyA (SPA) signal, and bovine growth hormone (bGH) polyA signal). See, e.g., Powell and Rivera-Soto, 2015, *Discov. Med.*, 19 (102): 49-57.

[0808] Gene therapy constructs are designed such that both the heavy and light chains are expressed. More specifically, the heavy and light chains should be expressed at about equal amounts, in other words, the heavy and light chains are expressed at approximately a 1:1 ratio of heavy chains to light chains. The coding sequences for the heavy and light chains can be engineered in a single construct in which the heavy and light chains are separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed. The leader sequence for each of the heavy and light chains is preferably MYRMQLLL-LIALSLALVINS (SEQ ID NO: 161). Section 5.1.5, supra, provides specific IRES, 2A, and other linker sequences that can be used with the methods and compositions provided herein. In specific embodiments, the linker is a Furin-F2A linker RKRRAPVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 242). In specific embodiments, the transgene is a nucleotide sequence that encodes the following: Signal sequence-heavy chain Fab portion-Furin-F2A linker-signal sequence-light chain Fab portion. See, e.g., FIGS. 2A-2C and 2F for sequences for aducanumab, crenezumab, gan-

tenerumab, or BAN2401 Fab expression, respectively; FIG. 2D for the sequence for aTAU Fab expression; FIG. 2E for the sequence for erenumab Fab expression; and FIG. 4B for the sequence for natalizumab Fab expression.

[0809] In a specific embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) Control elements, which include a) the CB7 promoter, comprising the CMV enhancer/chicken β -actin promoter, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of the A β -binding, Tau-binding, CGRPR-binding, integrin-binding Fab, separated by a self-cleaving furin (F)/F2A linker, ensuring expression of equal amounts of the heavy and the light chain polypeptides. An exemplary construct is provided in FIG. 1.

[0810] In specific embodiments, provided are AAV vectors comprising a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 capsid (SEQ ID NO: 79) or AAVrh10 (SEQ ID NO: 80); and an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-A β , anti-Tau, anti-CGRPR, or anti-integrin mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human CNS cells.

5.4.2 Constructs for Delivery to Liver or Muscle Cells

[0811] Sections 5.3.4, 5.3.5, 5.3.6, 5.3.7, 5.3.8, 5.3.9, 5.3.10, 5.3.11, 5.3.12, 5.3.13, 5.3.14, 5.3.15 describe recombinant vectors that contain a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to interleukins (IL) or interleukin receptors (ILR), integrin, PCSK9, ANGPTL3, OxPL RANKL, PD-1/PD-L1/PD-L2, VEGF, factor D (fD), BLyS, CP-C5, MMP9, pKal, or TNF α . Such recombinant vector used for delivering the transgene can have a tropism for human liver or muscle cells. Such vectors can include non-replicating recombinant adeno-associated virus vectors ("rAAV"), particularly those bearing an AAV8 or AAV9 capsid are preferred. However, other viral vectors may be used, including but not limited to lentiviral vectors, vaccinia viral vectors, or non-viral expression vectors referred to as "naked DNA" constructs.

[0812] In specific embodiments, provided are constructs for gene therapy administration to a human subject, comprising an AAV vector, which comprises a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78); and a viral or artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding the heavy and light chains of the therapeutic antibody, operably linked to one or more regulatory sequences that control expression of the transgene in human cells (e.g., human muscle or liver cells) that express and deliver the therapeutic antibody in a therapeutically appropriate manner as disclosed herein. In certain embodiments, the encoded AAV8 capsid has the sequence of SEQ ID NO: 78 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions, particularly substitutions with amino acid residues found in the corresponding position in other AAV capsids, for example, in the SUBS row of

FIG. 12 which provides a comparison of the amino acid sequences of the capsid sequences of various AAVs, highlighting amino acids appropriate for substitution at different positions within the capsid sequence.

[0813] In specific embodiments, provided are constructs for gene therapy administration to a human subject, comprising an AAV vector, which comprises a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 capsid (SEQ ID NO: 79); and a viral or artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding the heavy and light chains of the therapeutic antibody, operably linked to one or more regulatory sequences that control expression of the transgene in human cells (e.g., human muscle or liver cells) that express and deliver the therapeutic antibody in a therapeutically appropriate manner as disclosed herein. In certain embodiments, the encoded AAV9 capsid has the sequence of SEQ ID NO: 79 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions, particularly substitutions with amino acid residues found in the corresponding position in other AAV capsids, for example, in the SUBS row of FIG. 12 which provides a comparison of the amino acid sequences of the capsid sequences of various AAVs, highlighting amino acids appropriate for substitution at different positions within the capsid sequence.

[0814] Preferably, the HuPTM mAb or antigen binding fragment thereof, including the HuPTM Fab transgene should be controlled by appropriate expression control elements for expression of the HuPTM Fab in human liver or muscle cells, for example, the CB7 promoter (a chicken β -actin promoter and CMV enhancer), liver specific promoters such as the TBG (Thyroxine-binding Globulin) promoter, the APOA2 promoter, the SERPINA1 (hAAT) promoter or the MIR122 promoter, or muscle specific promoters, such as the human desmin promoter or the human Pitx3 promoter, or inducible promoters, such as hypoxia-inducible promoters or rapamycin-inducible promoter, and can include other expression control elements that enhance expression of the transgene driven by the vector (e.g., introns such as the chicken β -actin intron, minute virus of mice (MVM) intron, human factor IX intron (e.g., FIX truncated intron 1), β -globin splice donor/immunoglobulin heavy chain splice acceptor intron, adenovirus splice donor/immunoglobulin splice acceptor intron, SV40 late splice donor/splice acceptor (19S/16S) intron, and hybrid adenovirus splice donor/IgG splice acceptor intron and polyA signals such as the rabbit β -globin polyA signal, human growth hormone (hGH) polyA signal, SV40 late polyA signal, synthetic polyA (SPA) signal, and bovine growth hormone (bGH) polyA signal). See, e.g., Powell and Rivera-Soto, 2015, *Discov. Med.*, 19 (102): 49-57.

[0815] Gene therapy constructs are designed such that both the heavy and light chains are expressed. More specifically, the heavy and light chains should be expressed at about equal amounts, in other words, the heavy and light chains are expressed at approximately a 1:1 ratio of heavy chains to light chains. The coding sequences for the heavy and light chains can be engineered in a single construct in which the heavy and light chains are separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed. The leader sequence for each of the heavy and light chains is preferably MYRMQLLL-

LIALSLALVINS (SEQ ID NO: 161). Section 5.1.5, supra, provides specific IRES, 2A, and other linker sequences that can be used with the methods and compositions provided herein. In specific embodiments, the linker is a Furin-F2A linker RKRRAPVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 242). In specific embodiments, the transgene is a nucleotide sequence that encodes the following: Signal sequence-heavy chain Fab portion-Furin-F2A linker-signal sequence-light chain Fab portion. See, e.g., FIGS. 3A to 3E for sequences for dupilumab, ixekizumab, secukinumab, ustekinumab, and mepolizumab Fab expression, respectively; FIGS. 4A and 4B for a sequence for vedolizumab and natalizumab Fab expression, respectively; FIGS. 5A to 5D for sequences for alirocumab, evolocumab, evinacumab, and E06-scFv Fab expression, respectively; FIG. 6 for a sequence for denosumab Fab expression; FIGS. 7A and 7B for sequences for nivolumab and pembrolizumab Fab expression, respectively; FIGS. 8A to 8C for sequences for ranibizumab, bevacizumab, and lampalizumab Fab expression, respectively; FIG. 8E for a sequence for belimumab Fab expression; FIG. 8F for a sequence for eculizumab Fab expression; FIG. 8G for a sequence for andecaliximab Fab expression; FIG. 8H for a sequence for lanadelumab Fab expression; and FIGS. 9A and 9B for a sequence for adalimumab Fab expression and infliximab Fab expression, respectively.

[0816] In a specific embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) Control elements, which include a) an inducible promoter, preferably a hypoxia-inducible promoter, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of the IL/ILR-binding, integrin-binding, PCSK9-binding, ANGPTL3-binding, RANKL-binding, OxPL-binding, PD-1/PD-L1/PD-L2 binding, VEGF-binding Fab, α D-binding, BLYS-binding, pKal-binding, or TNF α -binding Fab, separated by a self-cleaving furin (F)/F2A linker, ensuring expression of equal amounts of the heavy and the light chain polypeptides. An exemplary construct is provided in FIG. 1.

[0817] In a specific embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) Control elements, which include a) an inducible promoter, preferably a hypoxia-inducible promoter, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of the IL/ILR-binding, integrin-binding, PCSK9-binding, ANGPTL3-binding, OxPL-binding, RANKL-binding, PD-1/PD-L1/PD-L2 binding, VEGF-binding Fab, α D-binding, BLYS-binding, CP-C5-binding, MMP9-binding, pKal-binding, TNF α -binding Fab, separated by a self-cleaving furin (F)/F2A linker, ensuring expression of equal amounts of the heavy and the light chain polypeptides. An exemplary construct is provided in FIG. 1.

[0818] In specific embodiments, provided are AAV vectors comprising a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78); and an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-IL/ILR, anti-integrin, anti-PCSK9, anti-ANGPTL3, anti-OxPL, anti-RANKL, anti-PD-1, anti-PD-

L1, anti-PD-L2, anti-VEGF, anti- α D, anti-BLYS, anti-CP-C5, anti-MMP9, anti-pKal, or anti-TNF α mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human liver or muscle cells.

[0819] In specific embodiments, provided are AAV vectors comprising a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 (SEQ ID NO: 79); and an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-IL/ILR, anti-integrin, anti-PCSK9, anti-ANGPTL3, anti-RANKL, anti-OxPL, anti-PD-1, anti-PD-L1, anti-PD-L2, anti-VEGF, anti- α D, anti-BLYS, anti-pKal, or anti-TNF α mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human muscle cells.

5.4.3 Constructs for Delivery to Retinal Cell Types

[0820] Sections 5.3.9 and 5.3.12 describe recombinant vectors that contain a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to VEGF, factor D (α D), or MMP9. Such recombinant vectors used for delivering the transgene can have a tropism for one or more human retina cell types. Such vectors can include non-replicating recombinant adeno-associated virus vectors ("rAAV"), particularly those bearing an AAV8 capsid are preferred. Alternatively, an AAV vector bearing an AAV.7m8 capsid can be used. However, other viral vectors may be used, including but not limited to lentiviral vectors, vaccinia viral vectors, or non-viral expression vectors referred to as "naked DNA" constructs.

[0821] In specific embodiments, provided are constructs for gene therapy administration to a human subject, comprising an AAV vector, which comprises a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78); and a viral or artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding the heavy and light chains of the therapeutic antibody, operably linked to one or more regulatory sequences that control expression of the transgene in human cells (e.g., retina cell or liver cell types) that express and deliver the therapeutic antibody in a therapeutically appropriate manner as disclosed herein. In certain embodiments, the encoded AAV8 capsid has the sequence of SEQ ID NO: 78 with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 amino acid substitutions, particularly substitutions with amino acid residues found in the corresponding position in other AAV capsids, for example, in the SUBS row of FIG. 12 which provides a comparison of the amino acid sequences of the capsid sequences of various AAVs, highlighting amino acids appropriate for substitution at different positions within the capsid sequence.

[0822] Preferably, the HuPTM mAb or antigen binding fragment thereof, including the HuPTM Fab transgene should be controlled by appropriate expression control elements for expression of the HuPTM Fab in human retina or liver cell types, for example, CB7 promoter (a chicken β -actin promoter and CMV enhancer), or tissue-specific promoters such as RPE-specific promoters e.g., the RPE65 promoter, or cone-specific promoters, e.g., the opsin pro-

motor, or liver specific promoters, such as, the TBG (Thyroxine-binding Globulin) promoter, the APOA2 promoter, the SERPINA1 (hAAT) promoter or the MIR122 promoter, inducible promoters, for example, hypoxia-induced promoters and drug inducible promoters, such as promoters induced by rapamycin and related agents, and can include other expression control elements that enhance expression of the transgene driven by the vector (e.g., introns such as the chicken β -actin intron, minute virus of mice (MVM) intron, human factor IX intron (e.g., FIX truncated intron 1), β -globin splice donor/immunoglobulin heavy chain splice acceptor intron, adenovirus splice donor/immunoglobulin splice acceptor intron, SV40 late splice donor/splice acceptor (19S/16S) intron, and hybrid adenovirus splice donor/IgG splice acceptor intron and polyA signals such as the rabbit β -globin polyA signal, human growth hormone (hGH) polyA signal, SV40 late polyA signal, synthetic polyA (SPA) signal, and bovine growth hormone (bGH) polyA signal). See, e.g., Powell and Rivera-Soto, 2015, *Discov. Med.*, 19 (102): 49-57.

[0823] Gene therapy constructs are designed such that both the heavy and light chains are expressed. More specifically, the heavy and light chains should be expressed at about equal amounts, in other words, the heavy and light chains are expressed at approximately a 1:1 ratio of heavy chains to light chains. The coding sequences for the heavy and light chains can be engineered in a single construct in which the heavy and light chains are separated by a cleavable linker or IRES so that separate heavy and light chain polypeptides are expressed. The leader sequence for each of the heavy and light chains is preferably MYRMQLLL-LIALSLALVINS (SEQ ID NO: 161). Section 5.1.5, supra, provides specific IRES, 2A, and other linker sequences that can be used with the methods and compositions provided herein. In specific embodiments, the linker is a Furin-F2A linker RKRRAPVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 242). In specific embodiments, the transgene is a nucleotide sequence that encodes the following: Signal sequence-heavy chain Fab portion-Furin-F2A linker-signal sequence-light chain Fab portion. See FIGS. 8A to 8C for sequences for ranibizumab, bevacizumab, and lampalizumab Fab expression, respectively, and FIG. 8G for a sequence for anecalciximab Fab expression.

[0824] In a specific embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) Control elements, which include a) the CB7 promoter, comprising the CMV enhancer/chicken β -actin promoter, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of the VEGF-binding, fD-binding, or MMP9-binding Fab, separated by a self-cleaving furin (F)/F2A linker, ensuring expression of equal amounts of the heavy and the light chain polypeptides. An exemplary construct is provided in FIG. 1.

[0825] In another embodiment, the constructs described herein comprise the following components: (1) AAV2 inverted terminal repeats that flank the expression cassette; (2) Control elements, which include a) the CB7 promoter, comprising the CMV enhancer/chicken β -actin promoter, b) a chicken β -actin intron and c) a rabbit β -globin poly A signal; and (3) nucleic acid sequences coding for the heavy and light chains of the VEGF-binding, fD-binding, or

MMP9-binding Fab, separated by flexible peptide linker, ensuring proper folding and solubility.

[0826] In specific embodiments, provided are AAV vectors comprising a viral capsid that is at least 95% identical to the amino acid sequence of an AAV8 capsid (SEQ ID NO: 78); and an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-VEGF mAb, anti-fD, or anti-MMP9 mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in one or more retina cell types (such as human photoreceptor cells (cone cells, rod cells); horizontal cells; bipolar cells; amacrine cells; retina ganglion cells (midgut cell, parasol cell, bistratified cell, giant retina ganglion cell, photosensitive ganglion cell, and muller glia); and retinal pigment epithelial cells).

5.5 Dose Administration

5.5.1 Administration for Delivering to CNS.

[0827] Sections 5.3.1, 5.3.2, 5.3.3, and 5.3.4 describe recombinant vectors that contain a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to A β , Tau protein, CGRPR, and integrin, respectively. Therapeutically effective doses of any such recombinant vector should be administered in any manner such that the recombinant vector enters the CNS, preferably by introducing the recombinant vector into the cerebral spinal fluid (CSF). In specific, embodiments, the vector is administered intrathecally, specifically intracisternally (such as to the cisterna magna) or, alternatively, lumbar delivery. Alternatively, the recombinant vector may be administered intravenously. In particular, recombinant AAV9 vectors have been shown to cross the blood-brain barrier and, as such, may be useful to deliver the anti-A β , anti-Tau, anti-CGRPR, or anti-integrin antibody transgene product to the CNS. Specifically, an scAAV9 may be particularly useful for intravenous administration. Intrathecal, including intracisternal or lumbar administration, or intravenous administration should result expression of the soluble transgene product in cells of the CNS. The expression of the transgene product (e.g., the encoded anti-A β , anti-Tau, anti-CGRPR, or anti-integrin antibody) results in delivery and maintenance of the transgene product in the CNS. Because the transgene product is continuously produced, maintenance of lower concentrations can be effective. The concentration of the transgene product can be measured in patient samples of the CSF.

[0828] Pharmaceutical compositions suitable for intrathecal, intracisternal, lumbar or intravenous administration comprise a suspension of the recombinant vector comprising the transgene encoding the anti-A β , anti-Tau, anti-CGRPR, or anti-integrin antibody, or antigen-binding fragment thereof, in a formulation buffer comprising a physiologically compatible aqueous buffer. The formulation buffer can comprise one or more of a polysaccharide, a surfactant, polymer, or oil.

5.5.2 Administration for Delivering to Liver or Muscle Tissue

[0829] Sections 5.3.4, 5.3.5, 5.3.6, 5.3.7, 5.3.8, 5.3.9, 5.3.10, 5.3.11, 5.3.12, 5.3.13, and 5.3.14 describe recombi-

nant vectors that contain a transgene encoding a HuPTM mAb or HuPTM Fab (or other antigen binding fragment of the HuPTM mAb) that binds to interleukins (IL) or integrin receptors (ILR), integrin, PCSK9, ANGPTL3, RANKL, PD-1/PD-L1/PD-L2, VEGF, factor D (fD), BLYS, CP-C5, MMP9, pKal, or TNF α . Therapeutically effective doses of any such recombinant vector should be administered in any manner such that the recombinant vector enters the liver or muscle (e.g., skeletal muscle), preferably by introducing the recombinant vector into the bloodstream. Alternatively, the vector may be administered directly to the liver through hepatic blood flow, e.g., via the suprahepatic veins or via the hepatic artery. In specific, embodiments, the vector is administered subcutaneously, intramuscularly or intravenously. Intramuscular, subcutaneous, intravenous or hepatic administration should result in expression of the soluble transgene product in cells of the liver or muscle. Alternatively, the vector may be administered directly to the liver through hepatic blood flow, e.g., via the suprahepatic veins or via the hepatic artery. The expression of the transgene product (e.g., the encoded anti-IL/ILR, anti-integrin, anti-PCSK9, anti-ANGPTL3, anti-RANKL, anti-PD-1, anti-PD-L1, anti-PD-L2, anti-VEGF, anti-fD, anti-BLYS, anti-CP-C5, anti-MMP9, anti-pKal, or anti-TNF α antibody) results in delivery and maintenance of the transgene product in the liver or muscle.

[0830] The concentration of the transgene product can be measured in patient blood serum samples. In specific embodiments, doses that maintain a concentration of the anti-IL/ILR antibody transgene product at a C_{min} of at least 2 $\mu\text{g/mL}$ are desired, such as C_{min} of 5 to 30 $\mu\text{g/mL}$, 5 to 50 $\mu\text{g/mL}$, or 5 to 80 $\mu\text{g/mL}$, or 5 to 100 $\mu\text{g/mL}$, or 5 to 200 $\mu\text{g/mL}$ depending upon the mAb used. For example, to achieve a C_{min} of approximately 60 to 90 $\mu\text{g/mL}$ of dupilumab (comparable to biweekly dosing) or 170 to 200 $\mu\text{g/mL}$ dupilumab (comparable to weekly dosing), or approximately 2 $\mu\text{g/mL}$ to 12 $\mu\text{g/mL}$ of ixekizumab, or approximately 13 $\mu\text{g/mL}$ to 50 $\mu\text{g/mL}$ secukinumab.

[0831] In specific embodiments, doses that maintain a concentration of the anti-TNF α antibody transgene product at a C_{min} of at least 0.5 $\mu\text{g/mL}$ or at least 1 $\mu\text{g/mL}$ (e.g., C_{min} of 1 to 10 $\mu\text{g/mL}$, 3 to 30 $\mu\text{g/mL}$ or 5 to 15 $\mu\text{g/mL}$ or 5 to 30 $\mu\text{g/mL}$) are desired.

[0832] In specific embodiments, doses that maintain a concentration of the anti-integrin antibody transgene product at a C_{min} of at least 10 $\mu\text{g/mL}$ (e.g., C_{min} of 10 to 60 $\mu\text{g/mL}$) are desired.

[0833] In specific embodiments, doses that maintain a concentration of the anti-PCSK9 or anti-ANGPTL3 antibody transgene product at a C_{min} of at least 10 $\mu\text{g/mL}$ are desired, such as C_{min} of 10 to 80 $\mu\text{g/mL}$.

[0834] In specific embodiments, doses that maintain a concentration of the anti-RANKL antibody transgene product at a C_{min} of at least 10 $\mu\text{g/mL}$ (e.g., C_{min} of 10 to 50 $\mu\text{g/mL}$ or 15 to 30 $\mu\text{g/mL}$) are desired.

[0835] In specific embodiments, doses that maintain a concentration of the anti-PD-1, anti-PD-L1, or anti-PD-L2 antibody transgene product at a C_{min} of at least 10 $\mu\text{g/mL}$ are desired, such as C_{min} of 10 to 100 $\mu\text{g/mL}$ or 100 to 300 $\mu\text{g/mL}$ or 300 to 600 $\mu\text{g/mL}$ are desired.

[0836] In specific embodiments, doses that maintain a concentration of the anti-BLYS antibody transgene product at a C_{min} of at least 70 $\mu\text{g/mL}$ (e.g., C_{min} of 70 to 150 $\mu\text{g/mL}$ or 100 to 200 $\mu\text{g/mL}$ or 200 to 350 $\mu\text{g/mL}$) are desired.

[0837] In specific embodiments, doses that maintain a concentration of the anti-pKal antibody transgene product at a C_{min} of at least 70 $\mu\text{g/mL}$ (e.g., C_{min} of 70 to 150 $\mu\text{g/mL}$ or 100 to 200 $\mu\text{g/mL}$ or 200 to 350 $\mu\text{g/mL}$) are desired.

[0838] The expression of the transgene product (e.g., the encoded anti-VEGF antibody) results in delivery and maintenance of the transgene product in the liver. In some embodiments, doses that maintain a concentration of the VEGF transgene product at a C_{min} of at least 90 $\mu\text{g/mL}$ are desired, such as C_{min} of 90 $\mu\text{g/mL}$ to 200 $\mu\text{g/mL}$. In specific embodiments, doses that maintain a concentration of the anti-CP-C5 antibody transgene product at a C_{min} of at least 30 mcg/mL, e.g., C_{min} of 30 to 300 mcg/mL or 100 to 200 mcg/mL, are desired.

[0839] However, in all cases because the transgene product is continuously produced, maintenance of lower concentrations can be effective. Notwithstanding, because the transgene product is continuously produced, maintenance of lower concentrations can be effective. The concentration of the transgene product can be measured in patient blood serum samples.

[0840] Pharmaceutical compositions suitable for intravenous, intramuscular, subcutaneous or hepatic administration comprise a suspension of the recombinant vector comprising the transgene encoding the anti-IL/ILR, anti-integrin, anti-PCSK9, anti-ANGPTL3, anti-RANKL, anti-PD-1, anti-PD-L1, anti-PD-L2, anti-VEGF, anti-fD, anti-BLYS, anti-CP-C5, anti-MMP9, anti-pKal, or anti-TNF α antibody, or antigen-binding fragment thereof, in a formulation buffer comprising a physiologically compatible aqueous buffer. The formulation buffer can comprise one or more of a polysaccharide, a surfactant, polymer, or oil.

5.5.3 Administration for Delivering to Retinal Type Cells

[0841] Therapeutically effective doses of the recombinant vector should be administered in any manner such that the recombinant vector enters the retina, preferably by introducing the recombinant vector directly into the eye. In specific, embodiments, the vector is administered subretinally (a surgical procedure performed by trained retinal surgeons that involves a partial vitrectomy with the subject under local anesthesia, and injection of the gene therapy into the retina; see, e.g., Campochiaro et al., 2016, Hum Gen Ther September 26 epub: doi: 10.1089/hum.2016.117, which is incorporated by reference herein in its entirety), or intravitreally, or suprachoroidally such as by microinjection or microcannulation. (See, e.g., Patel et al., 2012, Invest Ophthalm & Vis Sci 53:4433-4441; Patel et al., 2011, Pharm Res 28:166-176; Olsen, 2006, Am J Ophthalm 142:777-787 each of which is incorporated by reference in its entirety). Subretinal, intravitreal or suprachoroidal administration should result in expression of the soluble transgene product in one or more of the following retinal cell types: human photoreceptor cells (cone cells, rod cells); horizontal cells; bipolar cells; amacrine cells; retina ganglion cells (midget cell, parasol cell, bistratified cell, giant retina ganglion cell, photosensitive ganglion cell, and muller glia); and retinal pigment epithelial cells. The expression of the transgene product (e.g., the encoded anti-VEGF, anti-fD, anti-MMP9 antibody) results in delivery and maintenance of the transgene product in the retina.

[0842] The concentration of the transgene product can be measured in patient samples of the vitreous humour and/or anterior chamber of the treated eye. In specific embodi-

ments, doses that maintain a concentration of the anti-VEGF, anti-fd transgene product at a C_{min} of at least 0.33 $\mu\text{g/mL}$ in the vitreous humour, or 0.11 $\mu\text{g/mL}$ in the aqueous humour (the anterior chamber of the eye) for three months are desired; thereafter, vitreous C_{min} concentrations of the transgene product ranging from 1.70 to 6.60 $\mu\text{g/mL}$, and/or aqueous C_{min} concentrations ranging from 0.567 to 2.20 $\mu\text{g/mL}$ should be maintained. However, because the transgene product is continuously produced, maintenance of lower concentrations can be effective. Alternatively, vitreous humour concentrations can be estimated and/or monitored by measuring the patient's serum concentrations of the transgene product—the ratio of systemic to vitreal exposure to the transgene product is about 1:90,000. (E.g., see, vitreous humor and serum concentrations of ranibizumab reported in Xu L, et al., 2013, Invest. Ophthalm. Vis. Sci. 54:1616-1624, at p. 1621 and Table 5 at p. 1623, which is incorporated by reference herein in its entirety). However, because the transgene product is continuously produced, maintenance of lower concentrations can be effective.

[0843] Pharmaceutical compositions suitable for administration comprise a suspension of the recombinant vector comprising the transgene encoding the anti-VEGF, anti-fd, or anti-MMP9 antibody, or antigen-binding fragment thereof, in a formulation buffer comprising a physiologically compatible aqueous buffer. The formulation buffer can comprise one or more of a polysaccharide, a surfactant, polymer, or oil.

6. EXAMPLES

6.1 Example 1: Aducanumab Fab cDNA-Based Vector

[0844] An aducanumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of aducanumab (amino acid sequences being SEQ ID NOS. 1 and 2, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 101 and 102, respectively. The transgene also comprises nucleotide sequences that encodes a signal peptide, particularly, MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.2 Example 2: Crenezumab Fab cDNA-Based Vector

[0845] A crenezumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of crenezumab (amino acid sequences being SEQ ID NOS. 3 and 4, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 103 and 104, respectively. The transgene also comprises nucleotide sequences that encodes a signal peptide, particularly,

MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2B for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.3 Example 3: Gantenerumab Fab cDNA-Based Vector

[0846] A gantenerumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of gantenerumab (amino acid sequences being SEQ ID NOS. 5 and 6, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 105 and 106, respectively. The transgene also comprises nucleotide sequences that encodes a signal peptide, particularly, MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2C for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.4 Example 4: Dupilumab Fab cDNA-Based Vector

[0847] An dupilumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of dupilumab (amino acid sequences being SEQ ID NOS. 7 and 8, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 107 and 108, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 3A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.5 Example 5: Ixekizumab Fab cDNA-Based Vector

[0848] An ixekizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of ixekizumab (amino acid sequences being SEQ ID NOS. 9 and 10, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human muscle or liver cells and may be the nucleotide sequence of SEQ ID NOS: 109 and 110, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161)

or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 3B for amino acid sequence of a transgene product. Optionally, the vector additionally comprises a hypoxia-inducible promoter.

6.6 Example 6: Secukinumab Fab cDNA-Based Vector

[0849] A secukinumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of secukinumab (amino acid sequences being SEQ ID NOS. 11 and 12, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 111 and 112, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide chosen from the group listed in Table 2 or 3, respectively. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 3C for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.7 Example 7: Ustekinumab Fab cDNA-Based Vector

[0850] An ustekinumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of ustekinumab (amino acid sequences being SEQ ID NOS. 13 and 14, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 113 and 114, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 3D for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.8 Example 8: Mepolizumab Fab cDNA-Based Vector

[0851] A mepolizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of mepolizumab (amino acid sequences

being SEQ ID NOS. 15 and 16, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human muscle or liver cells and may be the nucleotide sequence of SEQ ID NOS: 115 and 116, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 3E for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.9 Example 9: Vedolizumab Fab cDNA-Based Vector

[0852] A vedolizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of vedolizumab (amino acid sequences being SEQ ID NOS. 17 and 18, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 117 and 118, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 4A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.10 Example 10: Natalizumab Fab cDNA-Based Vector

[0853] A natalizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of natalizumab (amino acid sequences being SEQ ID NOS. 19 and 20, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 119 and 120, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 4B for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.11 Example 11: Alirocumab Fab cDNA-Based Vector

[0854] An alirocumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide

sequences encoding the Fab portion of the heavy and light chain sequences of alirocumab (amino acid sequences being SEQ ID NOS. 21 and 22, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 121 and 122, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 5A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.12 Example 12: Evolocumab Fab cDNA-Based Vector

[0855] An evolocumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of evolocumab (amino acid sequences being SEQ ID NOS. 23 and 24, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 123 and 124, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 5B for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.13 Example 13: Evinacumab Fab cDNA-Based Vector

[0856] An evinacumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of evinacumab (amino acid sequences being SEQ ID NOS. 25 and 26, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 125 and 126, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 5C for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.14 Example 14: Denosumab Fab cDNA-Based Vector

[0857] A denosumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of denosumab (amino acid sequences being SEQ ID NOS. 27 and 28, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 127 and 128, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 6 for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.15 Example 15: Nivolumab Fab cDNA-Based Vector

[0858] A nivolumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of nivolumab (amino acid sequences being SEQ ID NOS. 29 and 30 respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 129 and 130, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 7A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.16 Example 16: Pembrolizumab Fab cDNA-Based Vector

[0859] A pembrolizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of pembrolizumab (amino acid sequences being SEQ ID NOS. 31 and 32, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 131 and 132, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 7B for amino acid sequence of a transgene product. The vector additionally includes a

constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.17 Example 17: Ranibizumab Fab cDNA-Based Vector

[0860] A ranibizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of ranibizumab (amino acid sequences being SEQ ID NOS. 33 and 34, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human retinal or liver cells and may be the nucleotide sequence of SEQ ID NOS: 133 and 134, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a retina specific signal sequence from Table 1 or a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8A for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.18 Example 18: Bevacizumab Fab cDNA-Based Vector

[0861] A bevacizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of bevacizumab (amino acid sequences being SEQ ID NOS. 35 and 36, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human retinal or liver cells and may be the nucleotide sequence of SEQ ID NOS: 135 and 136, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a retina specific signal sequence from Table 1 or a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8B for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.19 Example 19: Lampalizumab Fab cDNA-Based Vector

[0862] A lampalizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of lampalizumab (amino acid sequences being SEQ ID NOS. 37 and 38, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human retinal cells and may be the nucleotide sequence of SEQ ID NOS: 137 and 138, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVINS (SEQ ID NO: 161) or is a retina specific signal sequence from Table 1. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a

bicistronic vector. See FIG. 8C for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.20 Example 20: Brolucizumab scFv cDNA-Based Vector

[0863] A brolucizumab scFv cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the variable domain of the heavy and light chain sequences of brolucizumab (amino acid sequences being SEQ ID NOS. 39 and 40, respectively). The nucleotide sequence coding for the variable domains of the heavy and light chain is codon optimized for expression in human retinal or liver cells and may be the nucleotide sequence of SEQ ID NOS: 139 and 140, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a retina specific signal sequence from Table 1 or a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by a flexible peptide linker. See FIG. 8D for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.21 Example 21: Belimumab Fab cDNA-Based Vector

[0864] A belimumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of belimumab (amino acid sequences being SEQ ID NOS. 41 and 42, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver cells and may be the nucleotide sequence of SEQ ID NOS: 141 and 142, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8E for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.22 Example 22: Eculizumab Fab cDNA-Based Vector

[0865] An eculizumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of eculizumab (amino acid sequences being SEQ ID NOS. 43 and 44, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver cells and may be the nucleotide sequence of SEQ ID NOS: 143 and 144, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLLIALSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are

separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8F for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 promoter.

6.23 Example 23: Andecaliximab Fab cDNA-Based Vector

[0866] An andecaliximab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of andecaliximab (amino acid sequences being SEQ ID NOS. 45 and 46, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver cells and may be the nucleotide sequence of SEQ ID NOS: 145 and 146, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8G for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.24 Example 24: Lanadelumab Fab cDNA-Based Vector

[0867] A lanadelumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of lanadelumab (amino acid sequences being SEQ ID NOS. 47 and 48, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver cells and may be the nucleotide sequence of SEQ ID NOS: 147 and 148, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 8H for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.25 Example 25: Adalimumab Fab cDNA-Based Vector

[0868] An adalimumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of adalimumab (amino acid sequences being SEQ ID NOS. 49 and 50, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 149 and 150, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLIALLSLALVINS (SEQ ID NO: 161) or is a liver or muscle specific signal sequence from Table 2 or 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A

cleavage sites to create a bicistronic vector. See FIG. 9A for amino acid sequence of transgene product. The vector additionally includes an inducible promoter, such as a hypoxia-inducible promoter.

6.26 Example 26: Infliximab Fab cDNA-Based Vector

[0869] An infliximab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of infliximab (amino acid sequences being SEQ ID NOS. 51 and 52, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human liver cells and may be the nucleotide sequence of SEQ ID NOS: 151 and 152, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 9B for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 promoter.

6.27 Example 27: ATAU Fab cDNA-Based Vector

[0870] An aTAU Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of aTAU (amino acid sequences being SEQ ID NOS. 53 and 54, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 153 and 154, respectively. The transgene also comprises nucleotide sequences that signal encodes a peptide, particularly, MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2D for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.28 Example 28: Erenumab Fab cDNA-Based Vector

[0871] An erenumab Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of erenumab (amino acid sequences being SEQ ID NOS. 55 and 56, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 155 and 156, respectively. The transgene also comprises nucleotide sequences that encodes a signal peptide, particularly, MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2E for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.29 Example 29: BAN2401 Fab cDNA-Based Vector

[0872] An BAN2401 Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the Fab portion of the heavy and light chain sequences of BAN2401 (amino acid sequences being SEQ ID NOs. 57 and 58, respectively). The nucleotide sequence coding for the Fab portion of the heavy and light chain is codon optimized for expression in human CNS cells and may be the nucleotide sequence of SEQ ID NOS: 157 and 158, respectively. The transgene also comprises nucleotide sequences that encodes a signal peptide, particularly, MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161). The nucleotide sequences encoding the light chain and heavy chain are separated by IRES elements or 2A cleavage sites to create a bicistronic vector. See FIG. 2F for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

6.30 Example 30: E06-scFv Fab cDNA-Based Vector

[0873] An E06-scFv Fab cDNA-based vector is constructed comprising a transgene comprising nucleotide sequences encoding the heavy and light chain variable domain sequences of E06-scFv (amino acid sequences being SEQ ID NOs. 59 and 60, respectively). The nucleotide sequence coding for the heavy and light chain variable domains is codon optimized for expression in human liver or muscle cells and may be the nucleotide sequence of SEQ ID NOS: 159 and 160, respectively. The transgene also comprises nucleotide sequences that encode a signal peptide that may be MYRMQLLLLIALLSLALVTNS (SEQ ID NO: 161) or is a liver specific signal sequence from Table 2 or a muscle specific signal sequence from Table 3. The nucleotide sequences encoding the light chain and heavy chain are separated by a flexible peptide linker. See FIG. 5D for amino acid sequence of a transgene product. The vector additionally includes a constitutive promoter, such as CB7 or an inducible promoter, such as a hypoxia-inducible promoter.

TABLE 4			
Table of Fab Fragment Amino Acid Sequences			
mAb	Chain/ SEQ ID NO.	Sequence	
Aducanumab	Heavy/ SEQ ID NO: 1	XVQLVESGGG VVQPGRSRLR SCAASGFAPF SYGMHWVRQA PGKGLEWVAV IWFDTGKKYY TDSVKGRTFI SRDNSKNTLY LQMNTRLRAED TAVYYCARDR GIGARRGPYY MDVWGKGTTV TVSSASTKGP SVFPLAPSSK STSGGTAAALG CLVKDYFPEP VTWSWNSGAL TSGVHTFPAV LQSSGLYSLS SVVTVPSSSL GTQTYICNVN HKPSNTKVDK RVEPKSCD +/- KTHT (or KTHL) +/- CPPCPA +/- PELLGGPSVFL	
	Light/ SEQ ID NO: 2	DIQMTQSPSS LSASVGRDRT ITCRASQSI SYLNWYQQK GKAPKLLIYA ASSLQSGVPS RFSGSGSGTD FTLTISSLQ EDFATYYCQQ SYSTPLTFGG GTKVEIKRTV AAPSVFIFPP SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK	
Crenezumab	Heavy/ SEQ ID NO: 3	EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYGMSWVRQA PGKGLELVAI INSGGSTYY PDSVKGRTFI SRDNAKNSLY LQMNSLRAED TAVYYCASGD YWQGQTTVTY SSASTKGPSV FPLAPCSRST SESTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV VTPSSSLGT KTYTCNVNTHK PSNTKVDKRV ESKY +/- GPPCPPCPA +/- PEFLGGPSVFL	
	Light/ SEQ ID NO: 4	DIVMTQSPLS LPVTPGEPAS ISCRSSQSLV YSNGDTYLNW YLQKPGQSPQ LLIYKVSNRF SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCSQSTHVP WTFGQGTQVE IKRTVAAPSV FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSYSL SSTLTLSKAD YEKHKVYACE VTHQGLSPV TKSFNRGEC	
Gantenerumab	Heavy/ SEQ ID NO: 5	QVELVESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVAI INASGRTYY ADSVKGRTFI SRDNSKNTLY LQMNSLRAED TAVYYCARGK GNTHKPYGVY RYFDVWGQGT LVTVSSASTK GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS LSSVTVPS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD +/- KTHT (or KTHL) +/- CPPCPA +/- PELLGGPSVFL	
	Light/ SEQ ID NO: 6	DIVLTQSPAT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGVP ARFSGSGSGT DFTLTISSE PEDFATYYCL QIYNMPITFG QGTHKVEIKRT VAAPSVFIFP PSDEQLKSGT ASVCLLNIFY YPREAKVQWK VDNALQSGNS QESVTEQDSK DSTYLSSTLT TSKADYEKH VYACEVTHQ GLSPVTKSF NRGEK	

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences		
mAb	Chain/ SEQ ID NO.	Sequence
Dupilumab	Heavy/ SEQ ID NO: 7	EVQLVESGGG LEQPGGSLRL SCAGSGFTFR DYAMTWVRQA PGKGLEWVSS ISGSGGNTYY ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDR LSITIRPRYY GLDVWGQGT VTVSSASTKG PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL SSVVTVPSSS LGTKTYTCNV DHKPSNTKVD KRVESKY +/- GPPCPPCPA +/- PEFLGGPSVFL
Dupilumab	Light/ SEQ ID NO: 8	DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL YSIGYNYLDW YLQKSGQSPQ LLIYLGSNRA SGVPDRFSGS GSGTDFTLKI SRVEAEDVGF YYCMQALQTP YTFGQGTKLE IKRTVAAPSV FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSL SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC
Ixekizumab	Heavy/ SEQ ID NO: 9	QVQLVQSGAE VKKPGSSVKV SCKASGYSFT DYHIHWVRQA PGQGLEWMGV INPMYGTDDY NQRFKGRVTI TADESTSTAY MELSSLRSED TAVYYCARYD YFTGTGVYWG QGTLVTVSSA STKGPSVFPL APCSRSTSES TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG LYSLSVVTV PSSSLGKTKY TCNVDHKPSN TKVDKRVESK Y +/- GPPCPPCPA +/- PEFLGGPSVFL
Ixekizumab	Light/ SEQ ID NO: 10	DIVMTQTPLS LSVTPGQPAS ISCRSSRSLV HSRGNTYHLW YLQKPGQSPQ LLIYKVSNRD IGVNDRFSGS GSGTDFTLKI SRVEAEDVGF YYCSQSTHLP FTFGQGTKLE IKRTVAAPSV FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDYSL SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNREGC
Secukinumab	Heavy/ SEQ ID NO: 11	EVQLVESGGG LVQPGGSLRL SCAASGFTFS NYWMNWVRQA PGKGLEWVAA INQDGEKYY VGSVKGRFTI SRDNAKNSLY LQMNSLRVED TAVYYCVRDY YDILTDYYIH YWYFDLWGRG TLVTVSSAST KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF PAVLQSSGLY SLSSVVTVPS SSLGTQTYIC NVNHKPSNTK VDKRVEPKSC D +/- KTHT (or KTHL) +/- CPPCPA +/- PELLGGPSVFL
Secukinumab	Light/ SEQ ID NO: 12	EIVLTQSPGT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGIP DRFSGSGSGT DFTLTISRLE PEDFAVYYCQ QYGSSPCTFG QGTRLEIKRT VAAPSVFIFP PSDEQLKSGT ASVVCLLNNF YPREAKVQWK VDNALQSGNS QESVTEQDSK DSTYLSSTL TLKADYEKHK KYACEVTHQ GLSSPVTKSF NRREGC
Ustekinumab	Heavy/ SEQ ID NO: 13	EVQLVQSGAE VKKPGESLKI SCKGSGYSFT TYWLGWVRQM PGKGLDWIGI MSPVDSDIRY SPSFQQQVTM SVDKSITTAY LQWNSLKASD TAMYYCARRR PGQGYFDFWG QGTLVTVSSS STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG LYSLSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCD +/- KTHT (or KTHL) +/- CPPCPA +/- PELLGGPSVFL
Ustekinumab	Light/ SEQ ID NO: 14	DIQMTQSPSS LSASVGDRTV ITCRASQGIS SWLAWYQQKP EKAPKSLIYA ASSLQSGVPS RFGSGSGSGT FTLTISLQ EDFATYYCQQ YNIYPYTFGQ GTKLEIKRTV AAPSVFIFPP SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK DSTYLSSTLT LSKADYEKHK VYACEVTHQ LSSPVTKSFN RREGC
Mepolizumab	Heavy/ SEQ ID NO: 15	QVTLRESGPA LVKPTQTLTL TCTVSGFSLT SYSVHWVRQP PGKGLEWLGW IWASGGTDYN SALMSRLSIS KDTSRNQVVL TMTNMDPVDY ATYYCARDPP SLLRLDYWG RGTPTVTVSSA STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG LYSLSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCD +/- KTHT (or KTHL) +/- CPPCPA +/- PELLGGPSVFL

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences			
mAb	Chain/ SEQ ID NO.	Sequence	
Mepolizumab	Light/ SEQ ID NO: 16	DIVMTQSPDS	LAVSLGERAT INCKSSQSLL NSGNQKNYLA
		WXQQKPGQPP	KLLIYGASTR ESGVPDRFSG SGSGTDFTLT
		ISSLQAEDVA	VYYCQNVHSF PFTFGGGTKL EIKRTVAAPS
		VFIFPPSDEQ	LKSGTASVVC LLNNFYPREA KVQWKVDNAL
		QSGNSQESVT	EQDSKDYSTYS LSSTLTLSKA DYEKKHVVAC
Vedolizumab	Heavy/ SEQ ID NO: 17	EVTHQGLSSP	VTKSFNRGEC
		QVQLVQSGAE	VKKPGASVKV SCKGSGYTFT SYWMHWVRQA
		PGQRLEWIGE	IDPSESNTNY NQKFKGRVTL TVDISASTAY
		MELSSLRSED	TAVYYCARGG YDGDYDAIDY WQGGTLVTVS
		SASTKGPSVF	PLAPSSKSTS GGTAALGCLV KDYPPEPVTV
Vedolizumab	Light/ SEQ ID NO: 18	SWNSGALTSG	VHTFPAVLQS SGLYSLSSVV TVPSSSLGTQ
		TYICNVNHKP	SNTKVDKKVE PKSCD +/- KTHL (or
		KTHL)	+/- CPPCPA +/- PELAGAPSVFL
		DVVMTQSPDS	LPVTPGEPAS ISCRSSQSLA KSYGNTYLSW
		YLQKPGQSPQ	LLIYGISNRF SGVPDRFSGS SGSGTDFTLKI
Natalizumab	Heavy/ SEQ ID NO: 19	SRVEAEDVG	YYCLQGTHTQ YTFGGTKVE IKRTVAAPSV
		FIFPPSDEQL	KSGTASVVCL LLNNFYPREA VQWKVDNALQ
		SGNSQESVTE	QDSKDYSTYS LSSTLTLSKAD YEKHKVYACE
		VTHQGLSSPV	TKSFNRGEC
		QVQLVQSGAE	VKKPGASVKV SCKASGFNIK DTYIHWVRQA
Natalizumab	Light/ SEQ ID NO: 20	PGQRLEWMGR	IDPANGYTKY DPKFQGRVTI TADTSASTAY
		MELSSLRSED	TAVYYCAREG YYGNYGVYAM DYWGQGTLLV
		VSSASTKGPS	VFPLAPCSRS TSESTAALGC LVKDYFPEPV
		TVSWNSGALT	SGVHTFPAVL QSSGLYSLSS VVTVPSSSLG
		TKTYTCNVDH	KPSNTKVDKR VESKY +/- GPPCPPCPA +/-
Alirocumab	Heavy/ SEQ ID NO: 21	PEFLGGPSVFL	
		EVQLVESGGG	LVQPGGSLRL SCAASGFTFN NYAMNWVRQA
		PGKGLDWVST	ISGSGGTNTY ADSVKGRFII SRDSSKHTLY
		LQMNSLRAED	TAVYYCAKDS NWGNFDLWGR GTLVTVSSAS
		TKGPSVFPLA	PSSKSTSGGT AALGCLVKDY FPEPVTVSWN
Alirocumab	Light/ SEQ ID NO: 22	SGALTSGVHT	FPAVLQSSGL YSLSSVVTPV SSSLGTQTYI
		CNVNHKPSNT	KVDKKVEPKS CD +/- KTHL (or KTHL)
		+/- CPPCPA	+/- PELGGPSVFL
		DIVMTQSPDS	LAVSLGERAT INCKSSQSVL YRSNNRNFLG
		WYQQKPGQPP	NLLIYWASTR ESGVPDRFSG SGSGTDFTLT
Evolocumab	Heavy/ SEQ ID NO: 23	ISSLQAEDVA	VYYCQQYYTT PYTFGGGTKL EIKRTVAAPS
		VFIFPPSDEQ	LKSGTASVVC LLNNFYPREA KVQWKVDNAL
		QSGNSQESVT	EQDSKDYSTYS LSSTLTLSKA DYEKKHVVAC
		EVTHQGLSSP	VTKSFNRGEC
		EVQLVQSGAE	VKKPGASVKV SCKASGYTLT SYGISWVRQA
Evolocumab	Light/ SEQ ID NO: 24	PGQGLEWMGW	VSFYNGNTNY AQKLQGRGTM TTDPTSTAY
		MELRSLRSD	TAVYYCARGY GMDVWGQGT VTVSSASTKG
		PSVFPLAPCS	RSTSESTAAL GCLVKDYFPE PVTVSWNSGA
		LTSGVHTFPA	VLQSSGLYSL SSVVTPSSN FGTQTYTCNV
		DHKPSNTKVD	KTVKRCCKVE +/- CPPCPA +/- PPVAG
Evolocumab	Heavy/ SEQ ID NO: 23	ESALTQPASV	SGSPGQSITI SCTGTSSDVG GYNSVSWYQQ
		HPGKAPKLMI	YEVSNRPSGV SNRFSGSKSG NTASLTISGL
		QAEDEADYYC	NSYTSTSMVF GGGTKLTVLG QPKAAPSVTL
		FPPSSEELQA	NKATLVCLIS DFYPGAVTVA WKADSSPVKA
		GVETTTPSKQ	SNNKYAASSY LSLTPQWKS HRSYSCQVTH
Evolocumab	Light/ SEQ ID NO: 24	EGSTVEKTV	PTCCS
		ESALTQPASV	SGSPGQSITI SCTGTSSDVG GYNSVSWYQQ
		HPGKAPKLMI	YEVSNRPSGV SNRFSGSKSG NTASLTISGL
		QAEDEADYYC	NSYTSTSMVF GGGTKLTVLG QPKAAPSVTL
		FPPSSEELQA	NKATLVCLIS DFYPGAVTVA WKADSSPVKA
Evolocumab	Heavy/ SEQ ID NO: 23	GVETTTPSKQ	SNNKYAASSY LSLTPQWKS HRSYSCQVTH
		EGSTVEKTV	PTCCS
		ESALTQPASV	SGSPGQSITI SCTGTSSDVG GYNSVSWYQQ
		HPGKAPKLMI	YEVSNRPSGV SNRFSGSKSG NTASLTISGL
		QAEDEADYYC	NSYTSTSMVF GGGTKLTVLG QPKAAPSVTL
Evolocumab	Light/ SEQ ID NO: 24	FPPSSEELQA	NKATLVCLIS DFYPGAVTVA WKADSSPVKA
		GVETTTPSKQ	SNNKYAASSY LSLTPQWKS HRSYSCQVTH
		EGSTVEKTV	PTCCS
		ESALTQPASV	SGSPGQSITI SCTGTSSDVG GYNSVSWYQQ
		HPGKAPKLMI	YEVSNRPSGV SNRFSGSKSG NTASLTISGL

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences					
mAb	Chain/ SEQ ID NO.	Sequence			
Evinacumab	Heavy/ SEQ ID NO: 25	EVQLVESGGG	VIQPGGSLRL	SCAASGFTFD	DYAMNWVRQG
		PGKGLEWVSA	ISGDGGSTYY	ADSVKGRFTI	SRDNSKNSLY
		LQMNSLRAED	TAFFYCAKDL	RNTIFGVVIP	DAFDIWGQGT
		MVTVSSASTK	GPSVFPLAPC	SRSTSESTAA	LGCLVKDYFP
		EPVTVSWNSG	ALTSQVHTFP	AVLQSSGLYS	LSSVVTVPSS
		SLGTKTYTCN	VDHKPSNTKV	DKRVESKYGP	P +/-
		CPPCPA +/-	PEFLGGPSVFL		
Evinacumab	Light/ SEQ ID NO: 26	DIQMTQSPST	LSASVGRDVT	ITCRASQSIR	SWLAWYQQKP
		GKAPKLLIYK	ASSLESGVPS	RFSGSGSGTE	FTLTISLQP
		DDFATYYCQQ	YNSYSYTFGQ	GTKLEIKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSTLT	LSKADYEKKH	VYACEVTHQG
		LSSPVTKSFN	RGEC		
Denosumab	Heavy/ SEQ ID NO: 27	EVQLVESGGG	LVQPGGSLRL	SCAASGFTFS	SYAMSWVRQA
		PGKGLEWVSG	ITGSGGSTYY	ADSVKGRFTI	SRDNSKNTLY
		LQMNSLRAED	TAVYCAKDP	GTTVIMSWFD	PWGQGTLVTV
		SSASTKGPSV	FPLAPSSKST	SGGTAALGCL	VKDYPPEPVT
		VSWNSGALTS	GVHTFPAVLQ	SSGLYSLSSV	VTVPSSSLGT
		QTYICNVNHK	PSNTKVDKKV	EPKSCD +/-	KTHT (or
		KTHL) +/-	CPPCPA +/-	PELLGGPSVFL	
Denosumab	Light/ SEQ ID NO: 28	EIVLTQSPGT	LSLSPGERAT	LSCRASQSVR	GRYLAWYQQK
		PGQAPRLLIY	GASSRATGIP	DRFSGSGSGT	DFTLTISRLE
		PEDFAVFYCO	QYGSSPRTFG	QGTKVEIKRT	VAAPSVFIFP
		PSDEQLKSGT	ASVVCLLNIF	YPREAKVQWK	VDNALQSGNS
		QESVTEQDSK	DSTYSLSTLT	TLKADYEKH	KVYACEVTHQ
		GLSSPVTKSF	NRGEC		
Nivolumab	Heavy/ SEQ ID NO: 29	QVQLVESGGG	VVQPGRLRL	DKASGITFS	NSGMHWVRQA
		PGKGLEWVAV	IWYDGSKRY	ADSVKGRFTI	SRDNSKNTLF
		LQMNSLRAED	TAVYCATND	DYWGQGTLV	VSSASTKGPS
		VFPLAPCSRS	TSESTAALGC	LVKDYPPEPV	TVSWNSGALT
		SGVHTFPAVL	QSSGLYSLSS	VTVPSSSLG	TKTYTCNVDH
		KPSNTKVDKR	VESKY +/-	GPPCPPCPA +/-	
		PEFLGGPSVFL			
Nivolumab	Light/ SEQ ID NO: 30	EIVLTQSPAT	LSLSPGERAT	LSCRASQSVS	SYLAWYQQKP
		GQAPRLLIYD	ASNRAIGIPA	RFSGSGSGTD	FTLTISLLEP
		EDFAVYCYCO	SSNWPRTFGQ	GTKVEIKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSTLT	LSKADYEKKH	VYACEVTHQG
		LSSPVTKSFN	RGEC		
Pembrolizumab	Heavy/ SEQ ID NO: 31	QVQLVQSGVE	VKKPGASVKV	SCKASGYTFT	NYMYWVRQA
		PGQGLEWMGG	INPSNGGTNF	NEKFKNRVTL	TTDSSTTTAY
		MELKSLQFDD	TAVYICARRD	YRFDMGFDYW	GQGTITVTVSS
		ASTKGPSVFP	LAPCSRSTSE	STAALGCLVK	DYFPEPVTVS
		WNSGALTSGV	HTFPAVLQSS	GLYSLSSVVT	VPSSSLGTKT
		YTCNVDHKPS	NTKVDKRVES	KY +/-	GPPCPPCPA +/-
		PEFLGGPSVFL			
Pembrolizumab	Light/ SEQ ID NO: 32	EIVLTQSPAT	LSLSPGERAT	LSCRASKGVS	TSGYSYLHWY
		QQKPGQAPRL	LIYLYSLES	GVPARFSGSG	SGTDFTLTIS
		SLEPEDFAVY	YQHSRDLPL	TFGGGTKVEI	KRTVAAPSVF
		IFPPSDEQLK	SGTASVVCLL	NNFYPREAKV	QWKVDNALQS
		GNSQESVTEQ	DSKDYSLSL	STLTLSKADY	EKKHKVYACEV
		THQGLSSPVT	KSFNRGEC		
Ranibizumab	Heavy/ SEQ ID NO: 33	EVQLVESGGG	LVQPGGSLRL	SCAASGYDFT	HYGMNWVRQA
		PGKGLEWVGW	INTYTGEPTY	AADFKRRFTF	SLDTSKSTAY
		LQMNSLRAED	TAVYICAKYP	YYGTSHWYF	DVWGQGTLV
		VSSASTKGPS	VFPLAPSSKS	TSGGTAALGC	LVKDYPPEPV
		TVSWNSGALT	SGVHTFPAVL	QSSGLYSLSS	VTVPSSSLG
		TQTYICNVNH	KPSNTKVDKK	VEPKSCD +/-	KTHT (or
		KTHL) +/-	CPPCPA +/-	PELLGGPSVFL	

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences					
mAb	Chain/ SEQ ID NO.	Sequence			
Ranibizumab	Light/ SEQ ID NO: 34	DIQLTQSPSS	LSASVGDRVT	ITCSASQDIS	NYLNWYQQKP
		GKAPKVLIIYF	TSSLHSGVPS	RFSGSGSGTD	FTLTISSLQP
		EDFATYYCQQ	YSTVPWTFGQ	GTKVEIKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSSTLT	LSKADYEKHK	VYACEVTHQG
Bevacizumab	Heavy/ SEQ ID NO: 35	LSSPVTKSFN	RGEC		
		EVQLVESGGG	LVQPGGSLRL	SCAASGYTFT	NYGMNWVRQA
		PGKGLEWVGW	INTYTGETPY	AADFRRRTF	SLDTSKSTAY
		LQMNSLRAED	TAVYYCAKYP	HYYGSSHWYF	DVWGQGLT
		VSSASTKGPS	VFPLAPSSKS	TSGGTAALGC	LVKDYFPEPV
Bevacizumab	Light/ SEQ ID NO: 36	TVSWNSGALT	SGVHTFPAVL	QSSGLYSLSS	VVTVPSSSLG
		TQTYICNVNH	KPSNTKVDKK	VEPKSCD +/-	KTHT
		(KTHL) +/-	CPPCPA +/-	PELLGGPSVFL	
		DIQMTQSPSS	LSASVGDRVT	ITCSASQDIS	NYLNWYQQKP
		GKAPKVLIIYF	TSSLHSGVPS	RFSGSGSGTD	FTLTISSLQP
Lampalizumab	Heavy/ SEQ ID NO: 37	EDFATYYCQQ	YSTVPWTFGQ	GTKVEIKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSSTLT	LSKADYEKHK	VYACEVTHQG
		LSSPVTKSFN	RGEC		
		EVQLVQSGPE	LKKPGASVKV	SCKASGYTFT	NYGMNWVRQA
Lampalizumab	Light/ SEQ ID NO: 38	PGQGLEWMGW	INTYTGETTY	ADDFKGRFVF	SLDTSVSTAY
		LQISSLKAED	TAVYYCEREG	GVNNWGQGLT	VTVSSASTKG
		PSVFPPLAPSS	KSTSGGTAAL	GCLVKDYFPE	PVTVSWNSGA
		LTSGVHTFPA	VLQSSGLYSL	SSVVTVPSSS	LGTQTYICNV
		NHKPSNTKVD	KKVEPKSCD +/-	KTHT (or KTHL) +/-	CPPCPA +/-
Brolucizumab	Heavy/ SEQ ID NO: 39	PELLGGPSVFL			
		EVQLVESGGG	LVQPGGSLRL	SCTASGFSLT	DYYMTWVRQ
		APGKGLEWVG	FIDPDDDPY	ATWAKGRFTI	SRDNSKNTLY
		LQMNSLRAED	TAVYYCAGGD	HNSGWGLDIW	GQGTLLTVSS
		EVQLVQSGAE	VKKPGSSVRV	SCKASGGTFN	NNAINWVRQA
Brolucizumab	Light/ SEQ ID NO: 40	PGQGLEWMGG	IIPMFGTAKY	SQNFGGRVAI	TADESTGTAS
		MELSSLRSER	TAVYYCARSR	DLLLFPFHAL	SPWGRGTMVT
		VSSASTKGPS	VFPLAPSSKS	TSGGTAALGC	LVKDYFPEPV
		TVSWNSGALT	SGVHTFPAVL	QSSGLYSLSS	VVTVPSSSLG
		TQTYICNVNH	KPSNTKVDKK	VEPKSCD +/-	KTHT (or KTHL) +/-
Belimumab	Heavy/ SEQ ID NO: 41	PELLGGPSVFL			
		SSELTQDPAV	SVALGQTVRV	TCQGDSLRSY	YASWYQQKPG
		QAPVLVIYVK	NNRPSGIPDR	FSGSSSGNTA	SLITGAQAE
		DEADYYCSSR	DSSGNHWVFG	GGTELTVLQ	PKAAPSVTLF
		PPSSEELQAN	KATLVCLISD	FYPGAVTVAW	KADSSPVKAG
Belimumab	Light/ SEQ ID NO: 42	VETTTPSKQS	NNKYAASSYL	SLTPEQWKSH	RSYSCQVTHE
		GSTVEKTVP	TECS		
		EVQLVQSGAE	VKKPGASVKV	SCKASGYIFS	
		NYWIQWVRQA	PGQGLEWMGE	ILPGSGSTEY	TENFKDRVTM
		TRDTSTSTVY	MELSSLRSER	TAVYYCARYF	FGSSPNWYFD
Eculizumab	Heavy/ SEQ ID NO: 43	VWGQGTLLTV	SSASTKGPSV	FPLAPCSRST	SESTAALGCL
		VKDYFPEPVT	VSWNSGALTS	GVHTFPAVLQ	SSGLYSLSSV
		VTVPSSNFGT	QTYTCNVDDH	PSNTKVDKTV	ERKCCVE
		+/- CPPCPA	+/- PPVAG		
		EVQLVQSGAE	VKKPGASVKV	SCKASGYIFS	

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences					
mAb	Chain/ SEQ ID NO.	Sequence			
Eculizumab	Light/ SEQ ID NO: 44	DIQMTQSPSS	LSASVGDRTV	ITCGASENIY	GALNHWYQQKP
		GKAPKLLIYG	ATNLADGVPS	RFSGSGSGTD	FTLTISSLQP
		EDFATYYCQN	VLNTPFTFGQ	GTKVEIKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSSTLT	LSKADYEKHK	VYACEVTHQG
Andecaliximab	Heavy/ SEQ ID NO: 45	LSSPVTKSFN	RGEC		
		QVQLQESGPG	LVQPSETLSL	TCTVSGFSL	SYGVHWRQP
		PGKGLEWLG	IWTGGTTN	SALMSRFTIS	KDDSKNTVYL
		KMNSLKTEDT	AIYYCARYYY	GMDYWGQGT	VTVSSASTKG
		PSVFPLAPCS	RSTSESTAAL	GCLVKDYFPE	PVTVSWNSGA
Andecaliximab	Light/ SEQ ID NO: 46	LTSQVHTFPA	VLQSSGLYSL	SSVTVTPSSS	LGTKTYTCNV
		DHKPSNTKVD	KRVESKY +/-	GPPCPPCPA +/-	PEFLGGPSVFL
		DIQMTQSPSS	LSASVGDRTV	ITCKASQDVR	NTVAWYQQKP
		GKAPKLLIYS	SSYRNTGVDP	RFSGSGSGTD	FTLTISSLQA
		EDVAVYYCQ	HYITPYTFGG	GTKVEIKRTV	AAPSVFIFPP
Lanadelumab	Heavy/ SEQ ID NO: 47	SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSSTLT	LSKADYEKHK	VYACEVTHQG
		LSSPVTKSFN	RGEC		
		EVQLLESGGG	LVQPGGSLRL	SCAASGFTFS	HYIMMWVRQA
		PGKGLEWVSG	IYSSGGITVY	ADSVKGRFTI	SRDNSKNTLY
Lanadelumab	Light/ SEQ ID NO: 48	LQMNSLRAED	TAVYYCAYRR	IGVPRRDEFD	IWGQGTMTV
		SSASTKGPSV	FPLAPSSKST	SGGTAALGCL	VKDYFPEPVT
		VSWNSGALTS	GVHTFPAVLQ	SSGLYSLSSV	VTVPSSSLGT
		QTYICNVNHK	PSNTKVDKRV	EPKSCD +/-	KTHT (or
		KTHL) +/-	CPPCPA +/-	PELLGGPSVFL	
Adalimumab	Heavy/ SEQ ID NO: 49	DIQMTQSPST	LSASVGDRTV	ITCRASQSI	SWLAWYQQKP
		GKAPKLLIYK	ASTLESVPS	RFSGSGSGTE	FTLTISSLQP
		DDFATYYCQ	YNTYWTFGQG	TKVEIKRTVA	APSVFIFPPS
		DEQLKSGTAS	VVCLLNNFY	REAKVQWKVD	NALQSGNSQE
		SVTEQDSKDS	TYSLSSTLT	SKADYEKHKV	YACEVTHQGL
Adalimumab	Light/ SEQ ID NO: 50	SSPVTKSFNR	GEC		
		EVQLVESGGG	LVQPGRLRL	SCAASGFTFD	DYAMHWVRQA
		PGKGLEWVSA	ITWNSGHIDY	ADSVGRFTI	SRDNKNSLY
		LQMNSLRAED	TAVYYCAKVS	YLSTASSLDY	WGQGTLTIVTS
		SASTKGPSVF	PLAPSSKSTS	GGTAALGCLV	KDYFPEPVT
Infliximab	Heavy/ SEQ ID NO: 51	SWNSGALTS	VHTFPAVLQ	SGLYSLSSV	TVPSSSLGTQ
		TYICNVNHKP	SNTKVDKKVE	PKSCD +/-	KTHT (KTHL)
		+/- CPPCPA	+/- PELLGGPSVFL		
		RFSGSGSGTD	FTLTISSLQP	EDVATYYCQR	YNRAPYTFGQ
		GTKVEIKRTV	AAPSVFIFPP	SDEQLKSGTA	SVVCLLNNFY
Infliximab	Light/ SEQ ID NO: 52	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYSLSSTLT
		LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC
		EVKLEESGGG	LVQPGGSMKL	SCVASGFIFS	NHWMNWVRQS
		PEKGLEWVAE	IRSKSINSAT	HYAESVKGRF	TISRDDSKSA
		VYLQMTDLRT	EDTGVIYCSR	NYYGSTYD	WGQGTTLTVSS
aTAU	Heavy/ SEQ ID NO: 53	ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS
		WNSGALTS	GVHTFPAVLQ	SSGLYSLSSV	TVPSSSLGTQ
		YICNVNHKPS	NTKVDKKVEP	KSCD +/-	KTHT (KTHL)
		+/- CPPCPA	+/- PELLGGPSVFL		
		DILLTQSPAI	LSVSPGERVS	FSCRASQFVG	SSIHWYQORT
aTAU	Light/ SEQ ID NO: 53	NGSPRLLIYK	ASESMGIP	RFSGSGSGTD	FTLSINTVES
		EDIADYYCQ	SHSWPFTFGS	GTNLEVKRTV	AAPSVFIFPP
		SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ
		ESVTEQDSKD	STYSLSSTLT	LSKADYEKHK	VYACEVTHQG
		LSSPVTKSFN	RGEC		
aTAU	Heavy/ SEQ ID NO: 53	EVKLVESGGG	LVQPGGSMKL	SCVVS GFIFS	NYWVNWVRQA
		PGKGLEWVAQ	IRLKS DNYAT	HYEESVKGRF	TISRDDSKSS
		VYLQMNLR	EDSGIYYCTN	WEDYWGQGT	VTVSSASTKG
		PSVFPLAPCS	RSTSESTAAL	GCLVKDYFPE	PVTVSWNSGA

TABLE 4-continued

Table of Fab Fragment Amino Acid Sequences				
mAb	Chain/ SEQ ID NO.	Sequence		
		LTSGVHTFPA VLQSSGLYSL SSVVTVPSSS LGTKTYTCNV		
		DHKPSNTKVD KRVESKY +/- GPPCPPCPA +/-		
		PEFLGGPSVFL		
aTAU	Light/ SEQ ID NO: 54	DIVLTQSPDS LAVSLGERAT ISCRASQSVS TSRSYIHWY		
		QQKPGQPPKL LIKYASNLES GVPSRFSGSG SGTDFTLNIH		
		PLEPEDFATY YCHHSWEIPL TFGQGTKLEI KRTVAAPSVF		
		IFPPSDEQLK SGTASVVCLL NNFYPREAKV QWKVDNALQS		
		GNSQESVTEQ DSKDSTYSLS STLTLKADY EKHKVYACEV		
		THQGLSSPV KSFNRGEC		
Erenumab	Heavy/ SEQ ID NO: 55	QVQLVESGGG VVQPGRSLRL SCAASGFTFS SFGMHVVRQA		
		PGKGLEWVAV ISFDGSIKYS VDSVKGRFTI SRDNSKNTLF		
		LQMNSLRAED TAVYYCARDL LNYYDSSGYY HYKYYGMAVW		
		GQGTITVTVSS ASTKGPSVFP LAPCSRSTSE STAALGCLVK		
		DYFPEPVTVS WNSGALTSGV HTFPAVLQSS GLYSLSSVVT		
		VPSSNFGTQT YTCNVDHKPS NTKVDKTVR KCCVE +/-		
		CPPCPA +/- PPVAG		
Erenumab	Light/ SEQ ID NO: 56	QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL		
		PGTAPKLLIY DNNKRPSGIP DRFSGSKSGT STTLGITGLQ		
		TGDEADYYCG TWDSRLSAVV FGGGTLKTLV GQPKANPTVT		
		LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADGSPVK		
		AGVETTKPSK QSNNKYAASS YLSLTPEQWK SHRSYSCQVT		
		HEGSTVEKTV APTECS		
BAN2401	Heavy/ SEQ ID NO: 57	EVQLVESGGG LVQPGGSLRL SCSASGFTFS SFGMHVVRQA		
		PGKGLEWVAV ISSGSSSTIYY GDTVKGRTI SRDNAKNSLF		
		LQMSSLRAED TAVYYCAREG GYYGRSYYT MDYWGQGTIV		
		TVSSASTKGP SVFPLAPSSK STSGGTAALG CLVKDYFPEP		
		VTVSWNSGAL TSGVHTFPAV LQSSGLYSL SSVVTVPSSSL		
		GTQTYICNVN HKPSNTKVDK KVEPKSCD +/- KTHT		
		(KTHL) +/- CPPCPA +/- PELLGG		
BAN2401	Light/ SEQ ID NO: 58	DVVMTQSPLS LPVTPGAPAS ISCRSSQSIV HSNNGTYLEW		
		YLQKPGQSPK LLIYKVSNRD SGVPDRFSGS GSGTDFTLRI		
		SRVEAEDVGI YYCFQGSHPV PTFGPGTKLE IKRTVAAPSV		
		FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ		
		SGNSQESVTE QDSKSTYSLS SSTLTLKAD YEKHKVYACE		
		VTHQGLSSPV TKSFNREGC		
E06-scFV	Heavy/ SEQ ID NO: 59	EVKLVESGGG LVQPGGSLRL SCATSGFTFS DFYMEWVRQA		
		PGKRLEWIAA SRNKANDYTT EYADSVKGRF IVSRDTSQSI		
		LYLQMNALRA EDTAIYYCAR DYGSSSYWYF DVWGAGTTVT		
		VSS		
E06-scFV	Light/ SEQ ID NO: 60	DIVMTQSPSS LSVSAGKKVT ISCTASESLY SSKHKVHYLA		
		WYQKKPEQSP KLLIYGASNR YIGVPDRFTG SSGTDFTLT		
		ISSVQVEDLT HYYCAQFYSY PLTFGAGTKL EIK		
AAVrh10	SEQ ID NO: 80	MAADGYLPDW LEDNLSEGIR EWWDLKPGAP KPKANQQKQD		
		DGRGLVLPY KYLGPFNGLD KGEPVNAADA AALEHDKAYD		
		QQLKAGDNPY LRYNHADAEF QERLQEDTSF GGNLGRAVFP		
		AKKRVLEPLG LVEEGAKTAP GKRPVEPSP QRSPDSSTGI		
		GKKGQQPAKK RLNFGQTGDS ESVPDPQPIG EPPAGPSGLG		
		SGTMAAGGGA PMADNNEGAD GVGSSSGNWH CDSTWLGDRV		
		ITTSRTRWAL PTYNNHLYKQ ISNGTSGGST NDNTYFGYST		
		PWGYFDENRF HCHFSRPDWQ RLINNNWGF PKRLNFKLEN		
		IQVKEVTQNE GTKTIANNLT STIQVFTDSE YQLPYVLGSA		
		HQGCLEPPFA DVFMIPQYGY LTLNNGSQAV GRSSFYCLEY		
		FPSQMLRTGN NFEFSYQFED VPFHSSYAHS QSLDRLMNPL		
		IDQYLYLSR TQSTGGTAGT QQLLFSQAGP NNMSAQAKNW		
		LPGPCYRQQR VSTTSLQNNN SNFAWTGATK YHLNGRDSLV		
		NPGVAMATHK DDEERFPSS GVLMFGKQGA GKDNVDYSSV		
		MLTSEEEIKT TNPVATEQYG VVADNLQQQN AAPIVGAVNS		
		QGALPGMVWQ NRDVYLQGP WAKIPHTDGN FHPSPLMGGF		
		GLKHPPPIIL IKNTPVPADP PTTFSQAKLA SFITQYSTGQ		
		VSVEIEWELQ KENSKRWNP IQYTSNYYKS TNVDFAVNTD		
		GTYSEPRPIG TRYLTRNL		

TABLE 5

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
Aducanumab	Heavy/ SEQ ID NO: 101	gtgcagctgg	tggagagcgg	cggcggcggtg	gtgcagcccg
		gcagaagcct	gagactgagc	tgccgcccga	gcggcttcgc
		cttcagcagc	tacggcatgc	actgggtgag	acaggccccc
		ggcaagggcc	tggagtgggt	ggccgtgac	tggttcgacg
		gcaccaagaa	gtactacacc	gacagcgtga	agggcagatt
		caccatcagc	agagacaaca	gcaagaacac	cctgtacctg
		cagatgaaca	ccctgagagc	cgaggacacc	gccgtgtact
		actgcgccag	agacagaggc	atcgccgcca	gaagaggccc
		ctactacatg	gacgtgtggg	gcaagggcac	caccgtgacc
		gtgagcagcg	ccagcaccaa	gggcccagc	gtgttcccc
		tggccccag	cagcaagagc	accagcggcg	gcaccgccc
		cctgggctgc	ctggtgaagg	actacttccc	cgagcccgtg
		accgtgagct	ggaacagcgg	cgccctgacc	agcggcgtgc
		acaccttccc	cgccgtgctg	cagagcagcg	gcctgtacag
		cctgagcagc	gtggtgaccg	tgcccagcag	cagcctgggc
		accagacct	acatctgcaa	cgtgaaccac	aagcccagca
		acaccaagggt	ggacaagaga	gtggagccca	agagctgcga
		c +/- aagaccacacc (or aagaccacctg) +/-			
		tgccccctgccccgcc +/-			
		cccagctgctgggcggccccagcgtgttcctg			
Aducanumab	Light/ SEQ ID NO: 102	gacatccaga	tgaccagag	ccccagcagc	ctgagcgcca
		gcgtgggcga	cagagtgaac	atcacctgca	gagccagcca
		gagcatcagc	agctacctga	actggtacca	gcagaagccc
		ggcaaggccc	ccaagctgct	gatctacgcc	gccagcagcc
		tgacagcggg	cgtgccccagc	agattcagcg	gcagcggcag
		cggcaccgac	ttcacccctga	ccatcagcag	cctgcagccc
		gaggacttcg	ccacctacta	ctgccagcag	agctacagca
		ccccccctgac	cttcggcggc	ggcaccaaagg	tggagatcaa
		gagaaccgtg	gcccgcacca	gcgtgttcat	cttccccccc
		agcgacgagc	agctgaagag	cggcaccgcc	agcgtggtgt
		gcctgctgaa	caacttctac	cccagagagg	ccaaggtgca
		gtggaagggt	gacaacgccc	tgacagcggg	caacagccag
		gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca
		gcctgagcag	caccctgacc	ctgagcaagg	ccgactacga
		gaagcacaag	gtgtacgcct	gcgaggtgac	ccaccagggc
		ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt
		gc			
Crenezumab	Heavy/ SEQ ID NO: 103	gaggtgcagc	tgggtgagag	cggcggcgcc	ctggtgcagc
		ccggcggcag	cctgagactg	agctgcgcgc	ccagcggctt
		caccttcagc	agctacggca	tgagctgggt	gagacaggcc
		cccggcaagg	gcctggagct	ggtgcccgag	atcaacagca
		acggcggcag	cacctactac	cccagacgcg	tgaagggcag
		attcaccatc	agcagagaca	acgccaagaa	cagcctgtac
		ctgcagatga	acagcctgag	agccgaggac	accgccgtgt
		actactgcgc	cagcggcgac	tactggggcc	agggcaccac
		cgtgaccgtg	agcagcgcca	gcaccaaggg	ccccagcgtg
		ttccccctgg	ccccctgcag	cagaagcacc	agcgagagca
		ccgcccctct	gggctgcctg	gtgaaggact	acttccccga
		gcccgtgacc	gtgagctgga	acagcggcgc	cctgaccagc
		ggcgtgcaca	ccttccccgc	cgtgctgcag	agcagcggcc
		tgtacagcct	gagcagcgtg	gtgaccgtgc	ccagcagcag
		cctggggcacc	aagacctaca	cctgcaacgt	ggaccacaag
		cccagcaaca	ccaaggtgga	caagagagtg	gagagcaagt
		ac +/- ggccccctgccccctgccccgcc +/-			
		cccaggttctctgggcggccccagcgtgttcctg			
Crenezumab	Light/ SEQ ID NO: 104	gacatcgtga	tgaccagag	ccccctgagc	ctgcccgtga
		ccccccggcga	gccccccagc	atcagctgca	gaagcagcca
		gagcctgggtg	tacagcaacg	gcgacacctg	cctgcactgg
		tacctgcaga	agccccggcca	gagccccag	ctgctgatct
		acaagggtgag	caacagattc	agcggcgtgc	ccgacagatt
		cagcggcagc	ggcagcggca	ccgacttcac	cctgaagatc
		agcagagtgg	aggccgagga	cgtgggcgtg	tactactgca
		gccagagcac	ccacgtgccc	tggaccttcg	gccagggcac
		caagggtggag	atcaagagaa	ccgtggccgc	ccccagcgtg
		ttcatcttcc	ccccagcga	cgagcagctg	aagagcggca
		ccgcccagcgt	ggtgtgcctg	ctgaacaact	tctaccccag
		agaggccaag	gtgcagtgga	aggtggacaa	cgccctgcag
		agcggcaaca	gccaggagag	cgtgaccgag	caggacagca

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
		aggacagcac	ctacagcctg	agcagcacc	tgaccctgag
		caaggccgac	tacgagaagc	acaaggtgta	cgctgcgag
		gtgaccacc	agggcctgag	cagccccgtg	accaagagct
		tcaacagagg	cgagtgc		
Gantenerumab	Heavy SEQ ID NO: 105	caggtggagc	tggtggagag	cggcggcggc	ctggtgcagc
		ccggcggcag	cctgagactg	agctgcgcgc	ccagcggctt
		caccttcagc	agctacgcca	tgagctgggt	gagacaggcc
		cccggcaagg	gcctggagtg	ggtgagcgcc	atcaacgcca
		gcggcaccag	aacctactac	gccgacagcg	tgaagggcag
		attcaccatc	agcagagaca	acagcaagaa	cacctgttac
		ctgcagatga	acagcctgag	agccgaggac	accgccgtgt
		actactgcgc	cagaggcaag	ggcaacaccc	acaagcccta
		cggtacgtg	agatacttcg	acgtgtgggg	ccagggcacc
		ctggtgaccg	tgagcagcgc	cagcaccgaag	ggccccagcg
		tggtccccct	ggcccccagc	agcaagagca	ccagcggcgg
		caccgccgcc	ctgggctgcc	tggtgaagga	ctacttcccc
		gagccccgtga	ccgtgagctg	gaacagcggc	gccctgacca
		gcggcgtgca	caccttcccc	gccgtgctgc	agagcagcgg
		cctgtacagc	ctgagcagcg	tggtgaccgt	gcccagcagc
		agcctgggca	cccagaccta	catctgcaac	gtgaaccaca
		agcccagcaa	caccaaggtg	gacaagaagg	tggagcccaa
		gagctgcgac +/- aagaccacacc (or			
		aagaccacactg) +/- tgccccctgcgccgcc +/-			
		ccgagctgctgggcggccccagcgtgttctctg			
Gantenerumab	Light/ SEQ ID NO: 106	gacatcgtgc	tgaccagag	ccccgccacc	ctgagcctga
		gccccggcga	gagagccacc	ctgagctgca	gagccagcca
		gagcgtgagc	agcagctacc	tgccctggta	ccagcagaag
		cccggccagg	ccccagact	gctgatctac	ggcgccagca
		gcagagccac	cggcgtgccc	gccagattca	gcggcagcgg
		cagcggcacc	gacttcaccc	tgaccatcag	cagcctggag
		cccaggagct	tcgccaccta	ctactgcctg	cagatctaca
		acatgcccat	caccttcggc	cagggcacca	aggtggagat
		caagagaacc	gtggccgccc	ccagcgtgtt	catcttcccc
		cccagcagc	agcagctgaa	gagcggcacc	gccagcgtgg
		tggtccctgct	gaacaacttc	taccccagag	aggccaaggt
		gcagtggaa	gtggacaacg	ccctgcagag	cgccaacagc
		caggagagcg	tgaccgagca	ggacagcaag	gacagcacct
		acagcctgag	cagcacccctg	accctgagca	aggccgacta
		cgagaagcac	aaggtgtacg	cctgcgaggt	gaccacaccag
		ggcctgagca	gccccgtgac	caagagcttc	aacagaggcg
		agtgc			
Dupilumab	Heavy/ SEQ ID NO: 107	gaggtgcagc	tggtggagag	cggcggcggc	ctggagcagc
		ccggcggcag	cctgagactg	agctgcgcgc	gcagcggctt
		caccttcaga	gactacgcca	tgacctgggt	gagacaggcc
		cccggcaagg	gcctggagtg	ggtgagcagc	atcagcggca
		gcggcggcaa	cacctactac	gccgacagcg	tgaagggcag
		attcaccatc	agcagagaca	acagcaagaa	cacctgttac
		ctgcagatga	acagcctgag	agccgaggac	accgccgtgt
		actactgcgc	caaggacaga	ctgagcatca	ccatcagacc
		cagatactac	ggcctggacg	tggtggggcca	gggcaccacc
		gtgaccgtga	gcagcgccag	caccaagggc	cccagcgtgt
		tccccctggc	cccctgcagc	agaagcacca	gcgagagcac
		cgccgccctg	ggctgcctgg	tgaaggacta	cttccccgag
		cccgtgaccg	tgagctggaa	cagcggcgcc	ctgaccagcg
		gcgtgcacac	cttccccgcc	gtgctgcaga	gcagcggcct
		gtacagcctg	agcagcgtgg	tgaccgtgcc	cagcagcagc
		ctgggcacca	agacctacac	ctgcaacgtg	gaccacaagc
		ccagcaacac	caaggtggac		
		aagagagtgg agagcaagtac +/-			
		ggccccccctgccccctgccccgcc +/-			
		cccagttctctgggcggccccagcgtgttctctg			
Dupilumab	Light/ SEQ ID NO: 108	gacatcgtga	tgaccagag	ccccctgagc	ctgcccgtga
		gccccggcga	gccccgcagc	atcagctgca	gaagcagcca
		gagcctgctg	tacagcatcg	gctacaacta	cctggactgg
		tacctgcaga	agagcggcca	gagccccag	ctgctgatct
		acctgggcag	caacagagcc	agcggcgtgc	ccgacagatt
		cagcggcagc	ggcagcggca	ccgacttcac	cctgaagatc

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences				
mAb	Chain/ SEQ ID NO	Sequence		
Ixekizumab	Heavy/ SEQ ID NO: 109	agcagagtgg	aggccgagga	cgtgggcttc
		tgcaggccct	gcagaccccc	tacaccttcg
		caagctggag	atcaagagaa	ccgtggccgc
		ttcatcttcc	ccccagcga	cgagcagctg
		ccgccagcgt	ggtgtgcctg	ctgaacaact
		agaggccaag	gtgcagtgga	aggtggacaa
		agcggaaca	gccaggagag	cgtgaccgag
		aggacagcac	ctacagcctg	agcagcacc
		caaggccgac	tacgagaagc	acaaggtgta
		gtgaccacc	agggcctgag	cagccccgtg
		tcaacagagg	cgagtgc	
		caggtgcagc	tggtgcagag	cgccgcgag
		ccggcagcag	cgtgaagggtg	agctgcaagg
		cagcttcacc	gactaccaca	tccactgggt
		cccgccagg	gcctggagtg	gatgggctg
		tgtacggcac	caccgactac	aaccagagat
		agtgaaccatc	accgcgcagc	agagcaccag
		atggagctga	gcagcctgag	aagcgaggac
		actactgcgc	cagatacgac	tacttcaccg
		gtactggggc	cagggcacc	tggtgaccgt
Ixekizumab	Light/ SEQ ID NO: 110	agcaccagg	gccccagcgt	gttccccctg
		gcagaagcac	cagcgagagc	accgcgcgcc
		ggtgaaggac	tacttccccg	agcccggtgac
		aacagcggcg	ccctgaccag	cgccgtgcac
		ccgtgctgca	gagcagcggc	ctgtacagcc
		ggtgaccgtg	cccagcagca	gcctgggcac
		acctgcaacg	tggaccacaa	gcccagcaac
		acaagagagt	ggagagcaag	tac +/-
		ggccccccctg	ccccccctg	ccccgcc +/-
		ccgagttcctg	ggcgccccagc	gtgttccctg
		gacatcgtga	tgaccagac	ccccctgagc
		cccccgcca	gccccccagc	atcagctgca
		aagcctgggtg	cacagcagag	gcaacaccta
		tacctgcaga	agccccggca	gagccccag
		acaaggtag	caacagattc	atcgccgtgc
		cagcggcagc	ggcagcggca	ccgacttcac
		agcagagtgg	aggccgagga	cgtgggctgtg
		gccagagcac	ccacctgccc	ttcaccttcg
		caagctggag	atcaagagaa	ccgtggccgc
		ttcatcttcc	ccccagcga	cgagcagctg
Secukinumab	Heavy/ SEQ ID NO: 111	ccgccagcgt	ggtgtgcctg	ctgaacaact
		agaggccaag	gtgcagtgga	aggtggacaa
		agcggaaca	gccaggagag	cgtgaccgag
		aggacagcac	ctacagcctg	agcagcacc
		caaggccgac	tacgagaagc	acaaggtgta
		gtgaccacc	agggcctgag	cagccccgtg
		tcaacagagg	cgagtgc	
		gaggtgcagc	tggtggagag	cgccggcggc
		ccggcggcag	cctgagactg	agctgcgcgc
		caccttcagc	aactactgga	tgaactgggt
		cccggaagg	gcctggagtg	ggtggccgcc
		acggcagcga	gaagtactac	gtgggcagcg
		attcaccatc	agcagagaca	acgccaagaa
		ctgcagatga	acagcctgag	agtggaggac
		actactgcgt	gagagactac	tacgacatcc
		ctacatccac	tactggtact	tcgacctgtg
		accctggtga	ccgtgagcag	cgccagcacc
		gcgtgttccc	cctggccccc	agcagcaaga
		cgccaccgcc	gccctgggct	gcctggtgaa
		cccagagccc	tgaccgtgag	ctggaaacagc
		ccagcggcgt	gcacaccttc	cccgcctgac
		cgccctgtac	agcctgagca	gcgtgggtgac
		agcagcctgg	gcaccagac	ctacatctgc
		acaagcccag	caacaccaag	gtggacaaga
		caagagctgc	gac +/-	aagaccacacc (or
		aagaccacctg)	+/-	tgccccctgccccgcc +/-
		ccgagctgctggg	cgccccagc	gtgttccctg

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences							
mAb	Chain/ SEQ ID NO	Sequence					
Secukinumab	Light/ SEQ ID NO: 112	gagatcgtgc	tgacccagag	ccccggcacc	ctgagcctga		
		gccccggcga	gagagccacc	ctgagctgca	gagccagcca		
		gagcgtgagc	agcagctacc	tggcctggta	ccagcagaag		
		cccggccagg	cccccagact	gctgatctac	ggcgccagca		
		gcagagccac	cggcatcccc	gacagattca	gcggcagcgg		
		cagcggcacc	gacttcaccc	tgaccatcag	cagactggag		
		cccaggagct	tcgcctgtga	ctactgccag	cagtacggca		
		gcagcccctg	cacctctcggc	cagggcacca	gactggagat		
		caagagaacc	gtggccgccc	ccagcgtgtt	catcttcccc		
		cccagcgacg	agcagctgaa	gagcggcacc	gccagcgtgg		
		tgtgcctgct	gaacaacttc	taccccagag	aggccaaggt		
		gcagtggaag	gtggacaacg	ccctgcagag	cggcaacagc		
		caggagagcg	tgaccgagca	ggacagcaag	gacagcacct		
		acagcctgag	cagcacccctg	accctgagca	aggccgacta		
		cgagaagcac	aaggtgtacg	cctgcgaggt	gacccaccag		
		ggcctgagca	gccccctgac	caagagcttc	aacagagggc		
		agtgc					
		Ustekinumab	Heavy/ SEQ ID NO: 113	gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc
				ccggcgagag	cctgaagatc	agctgcaagg	gcagcggcta
				cagcttcacc	acctactggc	tgggctgggt	gagacagatg
cccggcaagg	gctcggactg			gateggcacc	atgagccccc		
tggcacagca	catcagatac			agccccagct	tccagggcca		
ggtgaccatg	agcgtggaca			agagcatcac	caccgcctac		
ctgcagtggc	acagccctgaa			ggccagcgcac	accgccatgt		
actactcgcc	cagaagaaga			cccgccagg	gctacttcga		
cttctggggc	cagggcaccc			tggtgaccgt	gagcagcagc		
agcaccgaag	gccccagcgt			gttccccctg	gccccccagca		
gcaagagcac	cagcgcgggc			accgcgcgcc	tgggctgcct		
ggtgaaggac	tacttccccg			agcccgtgac	cgtgagctgg		
aacagcggcg	ccctgaccag			cggcgtgcac	accttccccg		
ccgtgctgca	gagcagcggc			ctgtacagcc	tgagcagcgt		
ggtgaccctg	cccagcagca			gctcgggcac	ccacaccta		
atctgcaacg	tgaaccacaa			gcccagcaac	accaaggtgg		
acaagagagt	ggagccccaa			agctgcgac	+/-		
aagacccaca	cc (or aagacccacctg)			+/-			
tgccccccctgccccgc	+/-						
cccgagctgtggcgccccagcgtgttctcg							
Ustekinumab	Light/ SEQ ID NO: 114	gacatccaga	tgacccagag	ccccagcagc	ctgagcgcca		
		gcgtggggcga	cagagtgacc	atcacctgca	gagccagcca		
		gggcatcagc	agctgggtgg	cctggtagca	gcagaagccc		
		gagaaggccc	ccaagagcct	gatctacgcc	gccagcagcc		
		tcgacagcgg	cgtgcgccagc	agattcagcg	gcagcggcag		
		cggcaccgac	ttcaccttga	ccatcagcag	cctgcagccc		
		gaggacttcg	ccacctacta	ctgccagcag	tacaacatct		
		accctacac	cttcggccag	ggcaccaagc	tggagatcaa		
		gagaaccctg	ggcgccccca	gcgtgttcat	cttccccccc		
		agcgcagcag	agctgaagag	cggcaccgcc	agcgtgggtg		
		gcctgctgaa	caacttctac	cccagagagg	ccaaggtgca		
		gtggaaggtg	gacaacgccc	tgacagcggg	caacagccag		
		gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca		
		gcctgagcag	caccttgacc	ctgagcaagg	gcgactacga		
		gaagcacaa	gtgtacgcct	gcgaggtgac	ccaccagggc		
		ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt		
gc							
Mepolizumab	Heavy/ SEQ ID NO: 115	caggtgaccc	tgagagagag	cggccccgcc	ctggtagaagc		
		ccaccagac	cctgacctgt	acctgcaccg	tgagcggctt		
		cagcctgacc	agctacagcg	tgcaactggg	gagacagccc		
		cccggcaagg	cctcggagtg	gctggggcgt	atctggggcca		
		gcggcgccag	gcactacaac	agcgccctga	tgagcagact		
		gagcatcagc	aaggacacca	gcagaacca	ggtaggtgtg		
		acctatgacca	acctggaccc	cgtggacacc	gccacctact		
		actgcgcag	agaccctccc	agcagcctgc	tgagactgga		
		ctactggggc	agaggcacc	ccgtgaccgt	gagcagcgcc		
		agcaccgaag	gccccagcgt	gttccccctg	gccccccagca		
		ccaagagcac	cagcggcggc	accgcgcgcc	tgggctgcct		
		ggtgaaggac	tacttccccg	agcccgtgac	cgtgagctgg		
		aacagcggcg	ccctgaccag	cggcgtgcac	accttccccg		
		ccgtgctgca	gagcagcggc	ctgtacagcc	tgagcagcgt		

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences			
mAb	Chain/ SEQ ID NO	Sequence	
		ggtgaccgtg cccagcagca gcctgggcac ccagacctac atctgcaacg tgaaccacaa gcccagcaac accaaggtgg acaagagagt ggagcccaag agctgcgac +/- aagaccacacc (or aagaccacctg) +/- tgccccccctgccccgcc +/- cccgagctgctggcgccccagcgtgttcctg	
Mepolizumab	Light/ SEQ ID NO: 116	gacatcgtga tgaccagag ccccgacagc ctggccgtga gcctgggcga gagagccacc atcaactgca agagcagcca gagcctgctg aacagcggca accagaagaa ctacctggcc tggcagcaga agcccggcca gcccccaag ctgctgatct acggcgccag caccagagag agcggcgtgc ccgacagatt cagcggcagc ggccagcggca ccgacttcac cctgaccatc agcagcctgc agcccgagga cgtggccgtg tactactgcc agaactgtga cagcttcccc ttacacttcg gcggcggcac caagctggag atcaagagaa ccgtggccgc cccagcgtg ttcatcttcc cccccagcga cgagcagctg aagagcggca ccgccagcgt ggtgtgcctg ctgaacaact tctaccccag agaggccaag gtgcagtgga aggtggacaa cgccctgcag agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg agcagcacc tgaccctgag caaggccgac tacgagaagc acaaggtgta cgctgcgag gtgaccaccc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc	
Vedolizumab	Heavy/ SEQ ID NO: 117	caggtgcagc tgggtgcagag cggcgccgag gtgaagaagc ccggcgccag cgtgaagggt agctgcaagg gcagcggcta caccttcacc agctactgga tgcaactgggt gagacaggcc cccggccaga gactggagtg gatcggcgag atcgacccca gcgagagcaa caccactac aaccagaagt tcaagggcag agtgaccctg accgtggaca tcagcgccag caccgcctac atggagctga gcagcctgag aagcgaggac accgccgtgt actactgcgc cagaggcggc tacgacggct gggactacgc catcgactac tggggccagg gcaccctggt gaccgtgagc agcgccagca ccaagggccc cagcgtgttc cccctggccc ccagcagcaa gacaccagc ggcggcaccg ccgccctggg ctgcctgggt aaggactact tccccgagcc cgtgaccgtg agctggaaca gcggcgccct gaccagcggc gtgcacacct tccccgcctg gctgcagagc agcggcctgt acagcctgag cagcgtgggt accgtgcccga gcagcagcct gggcaccag acctacatct gcaacgtgaa ccacaagccc agcaacacca aggtggacaa gaaggtggag cccaagagct gcgac +/- aagaccacacc (or aagaccacctg) +/- tgccccccctgccccgcc +/- cccgagctggcggcgccccagcgtgttcctg	
Vedolizumab	Light/ SEQ ID NO: 118	gacgtggtga tgaccagag cccctgagc ctgcccgtga ccccgccga gcccgcagc atcagctgca gaagcagcca gagcctggcc aagagctacg gcaacaccta cctgagctgg tacctgcaga agcccggcca gagccccag ctgctgatct acggcatcag caacagattc agcggcgtgc ccgacagatt cagcggcagc ggccagcggca ccgacttcac cctgaagatc agcagagtgg agcccgagga cgtgggcgtg tactactgcc tgcagggcac ccaccagccc tacaccttcg gccagggcac caaggtggag atcaagagaa ccgtggccgc cccagcgtg ttcatcttcc cccccagcga cgagcagctg aagagcggca ccgccagcgt ggtgtgcctg ctgaacaact tctaccccag agaggccaag gtgcagtgga aggtggacaa cgccctgcag agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg agcagcacc tgaccctgag caaggccgac tacgagaagc acaaggtgta cgctgcgag gtgaccaccc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc	
Natalizumab	Heavy/ SEQ ID NO: 119	caggtgcagc tgggtgcagag cggcgccgag gtgaagaagc ccggcgccag cgtgaagggt agctgcaagg ccagcggcctt caacatcaag gacacctaca tccactgggt gagacaggcc cccggccaga gactggagtg gatgggcaga atcgacccc ccaacggcta caccagtagc gaccccaagt tccagggcag agtgaccatc accgcggaca ccagcgccag caccgcctac atggagctga gcagcctgag aagcgaggac accgccgtgt	

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences				
mAb	Chain/ SEQ ID NO	Sequence		
Natalizumab	Light/ SEQ ID NO: 120	actactg	cgc	cagagaggc
		tactacg	gca	actacggcg
		gtacgcc	atg	gactactggg
		gccaggg	cac	cctggtgacc
		gtgagcag	cgc	ccagcacc
		gggcccc	cag	gtgttcccc
		tggcccc	ctg	cagcagaagc
		accagcg	aga	gcacgcgcgc
		cctgggctg	c	ctggtgaagg
		actactt	ccc	cgagcccgtg
		accgtgag	ct	ggaacagcgg
		cgccctg	acc	agcggcggtg
		acacctt	ccc	cgccgtgctg
		cagagcag	cgc	gcctgtacag
		cctgagcag	c	gtggtgaccg
		tgcccag	cag	cagcctgggc
		accaagac	ct	acacctgcaa
		cgtggacc	ac	aagcccagca
		acaccaagg	t	ggacaagaga
		gtggagag	ca	agtac +/-
Natalizumab	Light/ SEQ ID NO: 120	ggcccccc	ctgcccccc	ctgccccgc +/-
		cccagagt	tccctggcg	ccccagcgtgttcctg
		gacatcc	aga	ccccagcagc
		ctgagcg	cca	ctgagcgcca
		gcgtggg	cga	agacctgca
		agaccag	cca	agaccagcca
		ggacatc	aac	aagtacatgg
		cctggta	cca	gcagaccccc
		ggcaagg	ccc	ccagactgct
		gatcca	ctac	accagcgcgc
		tgacgcc	cg	catccccagc
		agattca	gcg	gcagcggcag
		cggcagag	ac	tacaccttca
		ccatcag	cag	cctgcagccc
		gaggacat	c	ccacctacta
		ctgctgc	cag	tacgacaacc
		tgtggacc	tt	cgccagggc
		accaaggt	gg	agatcaagag
		aaccgtg	gcc	gccccagcg
		tgttcat	ctt	ccccccagc
Alirocumab	Heavy/ SEQ ID NO: 121	gacgagc	agc	gagcgccagc
		gtggtgt	gcc	gtggtgtgcc
		tgctgaaca	aa	cttctacccc
		agagagg	cca	aggtgacgtg
		gaaggtg	gg	aacgcctgc
		agagcgg	caa	cagccaggag
		agcgtga	ccg	caaggacagc
		acctac	agcc	tgagcagcac
		cctgacc	ctg	agcaaggccg
		actacg	agaa	gcacaagggtg
		tacgcct	gcg	aggtgaccca
		ccaggg	cctg	tgaccaagag
		cttcaac	aga	ggcagtg
		gaggtgc	agc	cgcgcgcg
		ctgagact	g	agctgcgcgc
		ccagcg	gctt	cacettcaac
		aactacg	cca	tgaactgggt
		gagacag	ggc	cccggcaagg
		gcctggac	t	ggtgagcacc
		atcagc	ggca	gcgcggcac
		caccaact	ac	gccgacagcg
		tgaagg	gcag	attcatcatc
		agcagaga	c	agcagagaca
		gcagcaag	cca	cacctgtac
		ctgcagat	ga	acagcctgag
		agccgagg	ac	accgccgtgt
		actactg	cgc	caaggacagc
		aactggg	gca	acttcgacct
		gtggggc	aga	ggcacccctgg
		tgaccgt	gag	cagcgccagc
		accaagg	ggc	ccagcgtgtt
		ccccctg	gcc	cccagcagca
		agagcacc	ag	cgcgccctgg
		gctgcct	ggt	gaaggactac
		ttccccg	agc	ccgtgaccgt
		gagctg	gaac	agcggcgccc
		tgaccag	cgc	cgtgcacacc
		ttccccg	ccg	tgctgcagag
		cagcggc	ctg	tacagcctga
		gcagcgt	ggt	gacctgtccc
		agcagcag	c	tgggcaccca
		gacctac	atc	tgcaacgtga
		accacaag	cc	cagcaacacc
		aaggtg	gaca	agaaaggtgga
		gccccaa	gagc	tgccag +/-
		aagaccac	acc (or aagaccac	ctg) +/-
		tgcccccc	ctgccccgc +/-	
		cccagact	gctggcg	ccccagcgtgttcctg
Alirocumab	Light/ SEQ ID NO: 122	gacatcg	tga	tgaccagag
		ccccgac	agc	ctggccgtga
		gcctggg	cga	gagagccacc
		atcaact	gca	agagcagcca
		gagcgtg	ctg	tacagaagca
		acaacaga	aaa	cttctgggc
		tggtacc	agc	agaagcccgc
		ccagcccc	c	aaactgtctga
		tctactg	ggc	cagcaccaga
		gagagcg	gcg	tgcccagacag
		attcagc	ggc	agcggcagcg
		gcaccgac	tt	cacctgacc
		atcagcag	cc	tgacggccga
		ggacgtg	gcc	gtgtactact
		gccagcag	t	ctacaccacc
		ccctacac	ct	tcggccagg
		caccaag	ctg	gagatcaaga
		gaaccgt	ggc	cgccccagc
		gtgttcac	ct	ccccccag
		cgacgag	cag	ctgaagagcg
		gcaccg	ccag	cgtggtgtgc
		ctgctga	aaca	acttctaccc
		cagagagg	c	aaggtgcagt
		ggaaggt	gga	caacgcctg
		cagagcgg	ca	acagccagga
		gagcgtg	acc	gagcaggaca
		gcaaggac	ag	cacctacagc
		ctgagcag	ca	ccctgacct
		gagcaagg	cc	gactacgaga
		agcacaag	ggt	gtacgcctgc
		gaggtgac	cc	accagggcct
		gagcagcc	c	gtgaccaaga
		gcttcaac	ag	aggcagtg

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences							
mAb	Chain/ SEQ ID NO	Sequence					
Evolocumab	Heavy/ SEQ ID NO: 123	gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc		
		ccggcgccag	cgtgaagggtg	agctgcaag	ccagcggcta		
		caccctgacc	agctacggca	tcagctgggt	gagacaggcc		
		cccgccagg	gcctggagtg	gatgggctgg	gtgagcttct		
		acaacggcaa	caccaactac	gccagaagc	tgcagggcag		
		aggcaccatg	accaccgacc	ccagcaccag	caccgcctac		
		atggagctga	gaagcctgag	aagcgacgac	accgccgtgt		
		actactgcgc	cagaggctac	ggcatggacg	tgtggggcca		
		gggaccacc	gtgaccgtga	gcagcgccag	caccaagggc		
		cccagcgtgt	tccccctggc	cccttcgacg	agaagcacca		
		gcgagagcac	cgccgccttg	ggctgcctgg	tgaaggacta		
		cttccccgag	cccgtgaccg	tgagctggaa	cagcggcgcc		
		ctgaccagcg	gcgtgcacac	cttccccgcc	gtgctgcaga		
		cgagcggcct	gtacagcctg	agcagcgtgg	tgaccgtgcc		
		cagcagcaac	ttcgggaccc	agacctacac	ctgcaacgtg		
		gaccacaagc	ccagcaacac	caaggtggag	aagaccgtgg		
		agagaaagtg	ctcgctggag	+/- tgccccccctgccccgc			
		+/- cccccctggtgccgc					
		Evolocumab	Light/ SEQ ID NO: 124	gagagcgccc	tgaccagacc	cgccagcgtg	agcggcgacc
				ccggccagag	catcaccatc	agctgcaccg	gcaccagcag
cgacgtgggc	ggctacaaca			gcgtgagctg	gtaccagcag		
caccgccgca	aggcccccaa			gctgatgatc	tacgaggtga		
gcaacagacc	cagcggcggtg			agcaacagat	tcagcggcag		
caagagcggc	aacaccgcca			gcctgacctc	cagcggcctg		
caggccgagg	acgaggccga			ctactactgc	aacagctaca		
ccagcaccag	catggtgttc			ggcgccggca	ccaagctgac		
ctgtgtgggc	cagccccagg			ccgccccccag	cgtgaccctg		
ttcccccca	gcagcggagg			gctgcaggcc	aacaaggcca		
ccctgggtgtg	cctgatcagc			gaacttctac	ccggcgccct		
gacctgtggc	tggaaggccg			acagcagccc	cgtgaaggcc		
ggcgtgggag	ccaccacccc			cagcaagcag	agcaacaaca		
agtacgccgc	cagcagctac			ctgagcctga	cccccgagca		
gtggaagagc	cacagaagct			acagctgcc	ggtgaccacc		
gagggcagca	ccgtggagaa			gaccgtggcc	cccaccagat		
gcagc							
Evinacumab	Heavy/ SEQ ID NO: 125			gaggtgcagc	tggtggagag	cggcgccggc	gtgatccagc
				ccggcgccag	cctgagactg	agctgcgccg	ccagcggctt
				caccttcgac	gactacgcc	tgaactgggt	gagacagggc
		cccgccagga	gcctggagtg	ggtagcgcc	atcagcggcg		
		acggcgccag	cacctactac	gccgacagcg	tgaagggcag		
		attcaccatc	agcagagaca	acagcaagaa	cagcctgtac		
		ctgcagatga	acgcctctgag	agccgaggac	accgccttct		
		tctactgcgc	caaggacctg	agaaacacca	tcttcggcgt		
		ggtgatcccc	gcgcctctcg	acatctgggg	ccagggccacc		
		atggttgagg	tgagcagcgc	cagcaccacg	ggccccagcg		
		tgttccccct	ggccccctgc	agcagaagca	ccagcgagag		
		caccgcgcgc	ctgggctgcc	tggtgaagga	ctacttcccc		
		gagcccgctg	ccgtgagctg	gaacagcgcg	gccctgacca		
		gcggcgctga	caccttcccc	gccgtgtctc	agagcagcgg		
		cctgtacagc	ctgagcagcg	tggtgacctg	gcccagcagc		
		agccctggga	ccaagacctc	cacctgcaac	gtggaccaca		
		agcccgagca	caccaagggtg	gacaagagag	tggagagcaa		
		gtacggcccc	ccc +/- tgccccccctgccccgc	+/- cccagtgctc			
		cccgagtcctg	ggcgccccccagcgtgtctg				
		Evinacumab	Light/ SEQ ID NO: 126	gacatccaga	tgaccagagc	ccccagcacc	ctgagcgcca
gcgtgggcga	cagagtgacc			atcacctgca	gagccagcca		
gagcatcaga	agctggctgg			cctggtacca	gcagaagccc		
ggcaaggccc	ccaagctgct			gatctacaag	gccagcagcc		
tggagagcgg	cgtgccagcg			agattcagcg	gcagcggcag		
cggcaccgag	ttcaccctga			ccatcagcag	cctgcagccc		
gacgacttcg	ccacctacta			ctgccagcag	tacaacagct		
acagctacac	cttcggccag			ggcaccacag	tggagatcaa		
gagaaccgtg	gccgccccca			gcgtgttcat	cttccccccc		
agcgacgagc	agctgaagag			cggcaccgcc	agcgtggtgt		
cctgctgtaa	caacttctac			cccagagagg	ccaaggtgca		
gtggaaggtg	gacaacgccc			tgacagcggg	caacagccag		
gagagcgtga	ccgagcagga			cagcaaggac	agcacctaca		
gcctgagcag	cacctgacc			ctgagcaggg	ccgactacga		

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences				
mAb	Chain/ SEQ ID NO	Sequence		
		gaagcacaag	gtgtacgcct	gcgaggtgac
		ctgagcagcc	ccgtgaccaa	gagcttcaac
		gc		agaggcgagt
Denosumab	Heavy/ SEQ ID NO: 127	gaggtgcagc	tgctggagag	cggcggcggc
		ccggcggcag	cctgagactg	agctgcgcgc
		caccttcagc	agctacggca	tgagctgggt
		cccggcaagg	gcctggagtg	ggtgagcggc
		gcggcggcag	cacctactac	gccgacagcg
		attcaccatc	agcagagaca	acagcaagaa
		ctgcagatga	acagcctgag	agccgaggac
		actactgcgc	caaggacccc	ggcaccacgc
		ctggttcgac	ccctggggcc	agggcacccct
		agcagcgcca	gcaccaaggg	ccccagcgtg
		ccccccagcag	caagagcacc	agcggcggca
		gggctgcctg	gtgaaggact	acttccccga
		gtgagctgga	acagcggcgc	cctgaccagc
		ccttccccgc	cgtgctgcag	agcagcggcc
		gagcagcgtg	gtgaccgtgc	ccagcagcag
		cagacctaca	tctgcaacgt	gaaccacaag
		ccaaggtgga	caagaagggt	gagcccaaga
		+/ - aagacccacacc	(aagacccacctg)	+/ -
		tgccccccctgccccgc	+/ -	
		cccagctgctggcgccccagcgtgttctctg		
Denosumab	Light/ SEQ ID NO: 128	gagatcgtgc	tgaccagag	ccccggcacc
		gccccggcga	gagagccacc	ctgagctgca
		gagcgtgaga	ggcagatacc	tggcctggta
		cccggccagg	ccccagact	gctgatctac
		gcagagccac	cggcatcccc	gacagattca
		cagcggcacc	gacttcaccc	tgaccatcag
		cccagaggact	tcgccgtgtt	ctactgccag
		gcagccccag	aaacttcggc	cagggcacca
		caagagaacc	gtggccgccc	ccagcgtgtt
		cccagcgacg	agcagctgaa	gagcggcacc
		tgtgcctgct	gaacaacttc	taccccagag
		gcagtggaa	gtggacaacg	ccctgcagag
		caggagagcg	tgaccgagca	ggacagcaag
		acagcctgag	cagcaccctg	accctgagca
		cgagaagcac	aagggtgtacg	cctgcgaggt
		ggcctgagca	gccccgtgac	caagagcttc
		agtgc		aacagaggcg
Nivolumab	Heavy/ SEQ ID NO: 129	caggtgcagc	tggtggagag	cggcggcggc
		ccggcagaag	cctgagactg	gactgcaagg
		caccttcagc	aacagcggca	tgcaactgggt
		cccggcaagg	gcctggagtg	ggtggccgtg
		acggcagcaa	gagatactac	gccgacagcg
		attcaccatc	agcagagaca	acagcaagaa
		ctgcagatga	acagcctgag	agccgaggac
		actactgcgc	caccaacgcg	gactactggg
		cctggtgacc	gtgagcagcg	ccagcaccaa
		gtgttcccc	tgccccctg	cagcagaagc
		gcaccgcgcg	cctgggctgc	ctggtgaagg
		cgagcccggtg	accgtgagct	ggaacacggg
		agcggcgtgc	acaccttccc	cgccgtgctg
		gcctgtacag	cctgagcagc	gtggtgaccg
		cagcctgggc	accaagacct	acacctgcaa
		aagcccagca	acaccaaggt	ggacaagaga
		agtac	+/ - ggccccccctgccccccctgccccgc	+/ -
		cccagttctctggcgccccagcgtgttctctg		
Nivolumab	Light/ SEQ ID NO: 130	gagatcgtgc	tgaccagag	ccccgccacc
		gccccggcga	gagagccacc	ctgagctgca
		gagcgtgagc	agctacctgg	cctggtacca
		ggccaggccc	ccagactgct	gatctacgac
		gagccaccgg	catccccgcc	agattcagcg
		cggcaccgac	ttcaccctga	ccatcagcag
		gaggacttcg	ccgtgtacta	ctgccagcag
		ggcccagaac	cttcggccag	ggcaccgaag
		gagaaccgtg	gccccccca	gcgtgttcat
				cttccccccc

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
		agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac ccagagagg ccaaggtgca gtggaagggtg gacaacgccc tgcagagcgg caacagccag gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc ctgagcagcc ccgtgaccaa gagcttcaac agaggcagat gc			
Pembrolizumab	Heavy/ SEQ ID NO: 131	caggtgcagc tgggtgcagag cggcgtggag gtgaagaagc ccggcgccag cgtgaagggt agctgcaagg ccagcggcta caccttcacc aactactaca tgtactgggt gagacaggcc cccgccagg gcttgagtg gatggggcgg atcaacccca gcaacggcgg caccacttc aacgagaagt tcaagaacag agtgaacctg accaccgaca gcagcaccac caccgcctac atggagctga agagcctgca gttcgacgac accgccgtgt actactgcgc cagaagagac tacagattcg acatgggctt cgactactgg ggccagggca ccaccgtgac cgtgagcagc gccagacca agggcccccag cgtgttcccc ctggccccc gcagcagaag caccagcag agcaccgcgc ccctgggctg cctggtgaag gactacttcc ccgagcccg gaccgtgagc tggaacagcg gcgcctgac cagcggcgtg cacaccttcc ccgccgtgct gcagagcagc ggccgtgtaca gcctgagcag cgtggtgacc gtgcccagca gcagcctggg caccgaagacc tacacctgca acgtggacca caagcccagc aacaccaagg tggaacaag agtgagagc aagtac +/- ggccccccctgccccccctgccccgc +/- ccgagttcctgggcggccccagcgtgttctg			
Pembrolizumab	Light/ SEQ ID NO: 132	gagatcgtgc tgaccagag ccccgccacc ctgagcctga gccccggcga gagagccacc ctgagctgca gagccagcaa ggcggtgagc accagcggct acagctacct gcactggtac cagcagaagc ccggccaggc cccagactg ctgatctacc tgccagcta cctggagagc ggctgcccgc ccagattcag cggcagcggc agcggcaccg acttcacct gaccatcagc agcctggagc ccgaggact cgccgtgtac tactgccagc acagcagaga cctgcccctg acctcgcg gcggcaccaa ggtggagatc aagagaaccg tggccgcccc cagcgtgttc atcttcccc ccagcgacga gcagctgaag agcggcaccg ccagcgtggt gtgcctgctg aacaacttct accccagaga ggccaagggt cagtggaaagc tggacaacgc cctgcagagc ggcaacagcc aggagagcgt gaccgagcag gacagcaagg acagcaccta cagcctgagc agcaccctga ccctgagcaa ggccgactac gagaagcaca aggtgtacgc ctgcgagggtg accacaccag gctgagcag cccgtgacc aagagcttca acagaggcga gtgc			
Ranibizumab	Heavy/ SEQ ID NO: 133	gaggtgcagc tgggtggagag cggcggcggc ctggtgcagc ccggcgccag cctgagactg agctgcgcgc ccagcggcta cgacttcacc cactacggca tgaactgggt gagacaggcc cccggaagg gcttgagtg ggtgggctgg atcaacacct acaccggcga gcccacctac gccgcccact tcaagagaag attcaccttc agcctggaca ccagcaagag caccgcctac ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc caagtacccc tactactacg gcaccagcca ctggtacttc gacgtgtggg gccagggcac cctggtgacc gtgagcagcg ccagcaccaa gggccccagc gtgttcccc tgccccccag cagcaagagc accagcggcg gcaccgcgc cctgggctgc ctggtgaagg actacttccc cgagcccgtg accgtgagct ggaacagcgg cgccctgacc agcggcgtgc acaccttccc cgccgtgctg cagagcagcg gcctgtacag cctgagcagc gtggtgaccg tgcccagcag cagcctgggc accagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaaggtggagcccaagagctgcgac +/- aagaccacacc (or aagaccacctg) +/- tgccccccctgccccgc +/- ccgagctgctgggcggccccagcgtgttctg			
Ranibizumab	Light/ SEQ ID NO: 134	gacatccagc tgaccagag ccccagcagc ctgagcgcca gcgtgggcga cagagtgacc atcacctgca gcgcagcca ggacatcagc aactacctga actggtacca gcagaagccc			

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
		ggcaaggccc	ccaaggtgct	gatctacttc	accagcagcc
		tgcacagcgg	cgtgcccagc	agattcagcg	gcagcggcag
		cggcaccgac	ttcacctga	ccatcagcag	cctgcagccc
		gaggacttcg	ccacctacta	ctgccagcag	tacagcacccg
		tgccctggac	cttcggccag	ggcaccaagg	tggagatcaa
		gagaaccgtg	gccgccccca	gcgtgttcat	cttccccccc
		agcgacgagc	agctgaagag	cggcaccgcc	agcgtggtgt
		gcctgctgaa	caactttctac	cccagagagg	ccaaggtgca
		gtggaagggtg	gacaacgccc	tgcaagcggg	caacagccag
		gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca
		gcctgagcag	caccctgacc	ctgagcaagg	ccgactacga
		gaagcacaag	gtgtacgcct	gcgaggtgac	ccaccagggc
		ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt
		gc			
Bevacizumab	Heavy/ SEQ ID NO: 135	gaggtgcagc	tggtggagag	cggcgggcggc	ctggtgcagc
		ccggcggcag	cctgagactg	agctgcgcgc	ccagcggcta
		caccttcacc	aactacggca	tgaactgggt	gagacaggcc
		cccggcaagg	gcctggagtg	ggtgggctgg	atcaaacct
		acaccggcga	gcccacctac	gccgcgcgact	tcaagagaag
		attcaccttc	agcctggaca	ccagcaagag	caccgcctac
		ctgcagatga	acagcctgag	agccgaggac	accgccgtgt
		actactgcgc	caagtacccc	cactactacg	gcagcagcca
		ctggtacttc	gacgtgtggg	gccagggcac	cctggtgacc
		gtgagcagcg	ccagcaccaa	gggccccagc	gtgttcccc
		tgccccccag	cagcaagagc	accagcggcg	gcaccgccc
		cctgggctgc	ctggtgaagg	actacttccc	cgagcccgtg
		accgtgagct	ggaacagcgg	cgccctgacc	agcggcgtgc
		acaccttccc	cgccgtgctg	cagagcagcg	gcctgtacag
		cctgagcagc	gtggtgaccg	tgcccagcag	cagcctgggc
		accagacct	acatctgcaa	cgtgaaccac	aagcccagca
		acaccaagggt	ggacaagaag	gtggagccca	agagctgcga
		c +/-	aagaccacacc (aagaccacacc)	+/-	
		tgccccccctgccccgc	+/-		
		cccagctgctggggcgccccagcgtgttcctg			
Bevacizumab	Light/ SEQ ID NO: 136	gacatccaga	tgacccagag	ccccagcagc	ctgagcgcca
		gcgtgggcga	cagagtgacc	atcacctgca	gcgccagcca
		ggacatcagc	aactacctga	actggtacca	gcagaagccc
		ggcaaggccc	ccaaggtgct	gatctacttc	accagcagcc
		tgcacagcgg	cgtgcccagc	agattcagcg	gcagcggcag
		cggcaccgac	ttcacctga	ccatcagcag	cctgcagccc
		gaggacttcg	ccacctacta	ctgccagcag	tacagcacccg
		tgccctggac	cttcggccag	ggcaccaagg	tggagatcaa
		gagaaccgtg	gccgccccca	gcgtgttcat	cttccccccc
		agcgacgagc	agctgaagag	cggcaccgcc	agcgtggtgt
		gcctgctgaa	caactttctac	cccagagagg	ccaaggtgca
		gtggaagggtg	gacaacgccc	tgcaagcggg	caacagccag
		gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca
		gcctgagcag	caccctgacc	ctgagcaagg	ccgactacga
		gaagcacaag	gtgtacgcct	gcgaggtgac	ccaccagggc
		ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt
		gc			
Lampalizumab	Heavy/ SEQ ID NO: 137	gaggtgcagc	tggtgcagag	cggccccgag	ctgaagaagc
		ccggcggccag	cgtgaagggtg	agctgcaagg	ccagcggcta
		caccttcacc	aactacggca	tgaactgggt	gagacaggcc
		cccggccagg	gcctggagtg	gatgggctgg	atcaaacct
		acaccggcga	gaccacctac	gccgacgact	tcaaggcgag
		attcgtgttc	agcctggaca	ccagcgtgag	caccgcctac
		ctgcagatca	gcagcctgaa	ggccgaggac	accgccgtgt
		actactgcga	gagagagggc	ggcgtgaaca	actggggcca
		gggcaccctg	gtgaccgtga	gcagcggccag	caccaagggc
		cccagcgtgt	tccccctggc	ccccagcagc	aagagcacca
		gcggcggcac	cgccgcccctg	ggctgcctgg	tgaaggacta
		cttccccgag	cccgtgaccg	tgagctggaa	cagcggcgcc
		ctgaccagcg	gcgtgcacac	cttccccgcc	gtgctgcaga
		gcagcggcct	gtacagcctg	agcagcgtgg	tgaccgtgcc
		cagcagcagc	ctgggcaccc	agacctacat	ctgcaacgtg
		aaccacaagc	ccagcaaac	caaggtggac	aagaaggtgg
		agcccaagag	ctgcccgc	+/-	aagaccacacc (or

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
		aagaccacctg) +/- tgcacccctgccccgcc +/- cccagactgctggcgccccagcgtgttcctg			
Lampalizumab	Light/ SEQ ID NO: 138	gacatccagg tgacccagag cccagcagc ctgagcgcca gcgtggcgga cagagtgaac atcacctgca tcaccagcac cgacatcgac gacgacatga actggtacca gcagaagccc ggcaaggtgc ccaagctgct gatcagcggc ggcaacaccc tgagaccggg cgtgcccagc agattcagcg gcagcggcag cggcaccgac ttacacctga ccatcagcag cctgcagccc gaggacgtgg ccacctacta ctgcctgcag agcgacagcc tgccctacac cttcgccagc ggaccaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac cccagagagg ccaaggtgca gtggaagggt gacaaacgcc tcagagcgg caacagccag gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc			
Brolucizumab	Light/ SEQ ID NO: 139	gaggtgcagc tggtagagag cggcggcgcc ctggtgcagc ccggcggcag cctgagactg agctgcaccg ccagcggctt cagcctgacc gactactact acatgacctg ggtgagacag gcccccgcca agggcctgga gtgggtgggc ttcatcgacc ccgacgacga cccctactac gccacctggg ccaaggcgag attcaccatc agcagagaca acagcaagaa caccctgtac ctgcagatga acagcctgag agccgaggac acccgggtgt actactgcgc cggcggcgac cacaacagcg gctggggcct ggacatctgg ggccagggca ccctgggtgac cgtgagcagc			
Brolucizumab	Heavy/ SEQ ID NO: 140	gagatcgtga tgacccagag cccagcacc ctgagcgcca gcgtggcgga cagagtgaac atcacctgcc aggccagcga gatcatccac agctggctgg cctggtacca gcagaagccc ggcaaggccc ccaagctgct gatctacctg gccagcacc tggccagcgg cgtgcccagc agattcagcg gcagcggcag cggcggcgag ttacacctga ccatcagcag cctgcagccc gacgacttcg ccacctacta ctgccagaac gtgtacctgg ccagcaccaa cggcgcccaac ttcggccagg gcaccaagct gaccgtgctg ggc			
Belimumab	Heavy/ SEQ ID NO: 141	caggtgcagc tgcagcagag cggcggcgaa gtgaaaaaac cgggcagcag cgtgcgcgtg agctgcaaaag cgagcggcgg cacccttaac aacaacgcga ttaactgggt gcgccaggcg ccggggcagg gcctggaatg gatgggcggc attattccga tgtttggcac cgcgaatat agccagaact ttccgggccc cgtggcgatt accgcggatg aaagcaccgg caccgcgagc atggaactga gcagcctgcg cagcgaagat accgcggtgt attattgcgc gcgcagccgc gatctgctgc tgtttccgca tcatgcgtg agcccgctgg gccgcggcac catggtgacc gtgagcagcg cagcaccacaa agggcccgagc gtgtttccgc tggcggcgag cagcaaaaagc accagcggcg gcaccgcggc gctgggctgc ctggtgaaag attattttcc ggaaccgggtg accgtgagca acagcggcgc gctgaccagc ggcgtgcata cctttccggc ggtgctgcag agcagcggcc tgtatagcct gagcagcgtg gtgaccgtgc cgagcagcag cctgggcacc cagacctata ttgcaacgt gaaccataaa ccgagcaaca ccaaagtga taaaaaagt gaaccgaaaagctcgcat+/- aaaaccataacc (or aaaaccatctg) +/- tgcccgccgtgcccgcg +/- ccggaactgctggcgccccgagcgtgttctg			
Belimumab	Light/ SEQ ID NO: 142	agcagcgaac tgacccagga tccggcggtg agcgtggcgc tgggccagac cgtgcgcgtg acctgccagg gcgatagcct gcgcagctat tatgcgagct ggtatcagca gaaaccgggc caggcggcgg tctggtgat ttatggcaaa aacaaccgcc cgagcggcat tccggatcgc tttagcggca gcagcagcgg caacaccgcg agcctgacca ttaccggcgc gcaggcggaa gatgaagcgg attattattg cagcagccgc gatagcagcg gcaaccattg ggtgtttggc ggccggcacc aactgacgt gctgggcccag ccgaaagcgg gcggcagcgt gacctgttt			

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences				
mAb	Chain/ SEQ ID NO	Sequence		
		ccgccgagca gccaagaact gcaggcgaac aaagcgaccc tggtgtgcct gattagcgat ttttatccgg gcgcggtgac cgtggcggtg aaagcggata gcagcccggt ggccggcggtg gaaaccacca ccccgagcaa acagagcaac aacaaatatg cggcgagcag ctatctgagc ctgaccccggt aacagtggaa aagccatcgc agctatagct gccaggtgac ccatgaaggc agccccgtgg aaaaaaccgt ggccgccgacc gaatgcagc		
Eculizumab	Heavy/ SEQ ID NO: 143	caggtgcagc tggtgcagag cggcgccgag gtgaagaagc ccggcgccag cgtgaagggt agctgcaagg ccagcggcta catcttcagc aactactgga tccagtgggt gagacaggcc cccggccagg gcctggagtg gatggcgag atcctgccc gcagcggcag caccgagtag accgagaact tcaaggacag agtgaccatg accagagaca ccagcaccag caccgtgtac atggagctga gcagcctgag aagcgaggac accgcccgtg actactgcgc cagatacttc ttccgcagca gcccgaactg gtacttcgac gtgtggggcc agggcaccct ggtgaccgtg agcagcgcca gcaccaaggg cccagcgtg tccccctgg ccccctgcag cagaagcacc agcagagaca ccgcgcct gggctgcctg gtgaaggact acttccccga gcccgtgacc gtgagctgga acagcgggc cctgaccagc ggcggtgcaca ccttccccgc cgtgctgcag agcagcgcc tgtacagcct gagcagcgtg gtgaccgtgc ccagcagcaa cttcggcacc cagacctaca cctgcaactg ggaccacaag cccagcaaca ccaaggtgga caagaccgtg gagagaaagt gctgcgtgga g +/- tgcctccccctgccccgc +/- cccccgtggccggc		
Eculizumab	Light/ SEQ ID NO: 144	gacatccaga tgaccagag cccagcagc ctgagcgcca gcgtgggcga cagagtgaac atcacctgcg gcgcagcgca gaacatctac ggccgctga actggtacca gcagaagccc ggcaaggccc ccaagctgct gatctacggc gccaccaacc tgcccgacgg cgtgccacg agattcagcg gcagcggcag cggcaccgac ttcacctga ccatcagcag cctgcagccc gaggacttcg ccacctacta ctgccagAAC gtgctgaaca ccccctgac cttcgccag ggcaccaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactttac cccagagagg ccaaggtgca gtggaagggt gacaacgccc tgcagagcgg caacagccag gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc		
Andecaliximab	Heavy/ SEQ ID NO: 145	caggtgcagc tgcaggagag cggccccggc ctggtgaagc ccagcgagac cctgagcctg acctgcaccg tgagcggtt cagcctgctg agctacggcg tgactgggt gagacagccc cccggaagg gcctggagtg gctggcgctg atctggaccg gcggcaccac caactacaac agcgcctga tgagcagatt caccatcagc aaggacgaca gcaagaacac cgtgtacctg aagatgaaca gcctgaagac cgaggacacc gccatctact actgcgcccag atactactac ggcatggact actggggcca gggcaccctg gtgaccgtga gcagcgccag caccgaaggc cccagcgtgt tccccctggc cccctgcagc agaagcacca gcgagagcag cgcgcctg ggctgcctgg tgaaggacta cttccccgag cccgtgaccg tgagctggaa cagcggcgcc ctgaccagcg gcgtgcacac cttccccgcc gtgctgcaga gcagcgccct gtacagcctg agcagcgtgg tgaccgtgcc cagcagcagc ctgggcacca agacctacac ctgcaactgt gaccacaagc ccagcaacac caaggtggac aagagagtgg agagcaagta c +/- ggccccccctgccccccctgccccgc +/- cccgagttcctggcgccccagcgtgttctg		
Andecaliximab	Light/ SEQ ID NO: 146	gacatccaga tgaccagag cccagcagc ctgagcgcca gcgtgggcga cagagtgaac atcacctgca aggccagcca ggacgtgaga aacaccgtgg cctggtacca gcagaagccc ggcaaggccc ccaagctgct gatctacagc agcagctaca gaaacaccgg cgtgcccagc agattcagcg gcagcggcag cggcaccgac ttcacctga ccatcagcag cctgcaggcc gaggacgtgg ccgtgtacta ctgccagcag cactacatca		

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences				
mAb	Chain/ SEQ ID NO	Sequence		
		ccccctacac cttcggcggc ggcaccaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac ccagagagg ccaagggtgca gtggaagggtg gacaacgccc tgcagagcgg caacagccag gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc		
Lanadelumab	Heavy/ SEQ ID NO: 147	gaggtgcagc tgctggagag cggcggcggc ctggtgcagc ccggcggcag cctgagactg agctgcgcgc ccagcggctt caccttcagc cactacatca tgatgtgggt gagacaggcc cccggcaagg gcctggagtgt ggtgagcggc atctacagca gcggcggcat caccgtgtac gccagacgcg tgaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc ctacagaaga atcggcgtgc ccagaagaga cgagttcgac atctggggcc agggcaccat ggtgaccgtg agcagcgcca gcaccaagg ccagcgcgtg tttccccctg ccccagcag caagagcacc agcggcggca ccgccgcct gggctgcctg gtgaaggact acttccccga gccctgagcc gtgagctgga acagcggcgc cctgaccagc ggcgtgcaca ccttccccgc cgtgctgcag agcagcggcc tgtacagcct gagcagcgtg gtgaccgtgc ccagcagcag cctgggcacc cagacctaca tctgcaactg gaaccacaag ccagcaaca ccaagggtga caagagagtg gagcccaaga gctgcgac +/- aagaccacacc (or aagaccacctg) +/- tgccccccctgccccgc +/- ccgagctgctgggcggccccagcgtgttctctg		
Lanadelumab	Light/ SEQ ID NO: 148	gacatccaga tgaccagag cccagcacc ctgagcgcca gcgtgggcga cagagtgacc atcacctgca gagccagcca gagcatcagc agctggctgg cctggtacca gcagaagccc ggcaaggccc ccaagctgct gatctacaag gccagcacc tggagagcgg cgtgccagc agattcagcg gcagcggcag cggcaccgag ttcaccctga ccatcagcag cctgcagccc gacgacttcg ccacctacta ctgccagcag tacaacacct actggacctt cggccagggc accaagggtg agatcaagag aaccgtggcc gccccagcg tgttcatctt ccccccagc gacgagcagc tgaagagcgg caccgccagc gtggtgtgccc tgctgaacaa cttctacccc agagaggcca aggtgcagtg gaaggtggac aacgcccctg agagcggcaa cagccaggag agcgtgaccg agcaggacag caaggacagc acctacagcc tgagcagcac cctgaccctg agcaaggccg actacgagaa gcacaagggtg tacgcctgcg aggtgaccca ccagggcctg agcagccccg tgaccaagag cttcaacaga ggcgagtgc		
Adalimumab	Heavy/ SEQ ID NO: 149	gaggtgcagc tggtggagag cggcggcggc ctggtgcagc ccggcagaag cctgagactg agctgcgcgc ccagcggctt caccttcgac gactacgcca tgcactgggt gagacaggcc cccggcaagg gcctggagtgt ggtgagcggc atcacctgga acagcggcca catcgactac gccagacgcg tggagggcag attcaccatc agcagagaca acgccaagaa cagcctgtac ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc caaggtgagc tacctgagca ccgccagcag cctggactac tggggccagg gcaccctggt gaccgtgagc agcggcagca ccaagggcc cagcgtgttc cccctggccc ccagcagcaa gacgaccagc ggccgcaccg ccgccctggg ctgcctgggt aaggactact tccccagacc cgtgaccgtg agctggaaca gcggcgccct gaccagcggc gtgcacacct tccccgcctg gctgcagagc agcggcctgt acagcctgag cagcgtgggt accgtgcccc gcagcagcct gggcaccag acctacatct gcaacgtgaa ccacaagccc agcaacacca aggtggacaa gaaggtggag cccaagagct gcgac +/- aagaccacacc (aagaccacctg) +/- tgccccccctgccccgc +/- ccgagctgctgggcggccccagcgtgttctctg		

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
Adalimumab	Light/ SEQ ID NO: 150	agattcagcg	gcagcggcag	cggcaccgac	ttcacccctga
		ccatcagcag	cctgcagccc	gaggacgtgg	ccacctacta
		ctgccagaga	tacaacagag	ccccctacac	cttcgggccag
		ggcaccgaag	tggagatcaa	gagaaccgtg	gccgccccca
		gcgtgttcat	cttccccccc	agcgacgagc	agctgaagag
		cggcaccgcc	agcgtgggtg	gcctgctgaa	caacttctac
		cccagagagg	ccaaggtgca	gtggaagggtg	gacaacgccc
		tgcagagcgg	caacagccag	gagagcgtga	ccgagcagga
		cagcaaggac	agcacctaca	gcctgagcag	cacctgacc
		ctgagcaagg	ccgactacga	gaagcacaag	gtgtacgcct
		gcgaggtgac	ccaccagggc	ctgagcagcc	ccgtgaccaa
		gagcttcaac	agaggcgagt	gc	
Infliximab	Heavy/ SEQ ID NO: 151	gaggtgaagc	tggaggagag	cggcggcgcc	ctggtgcagc
		ccggcggcag	catgaagctg	agctgcgtgg	ccagcggcctt
		catcttcagc	aaccactgga	tgaactgggt	gagacagagc
		cccagagaag	gcctggagtg	ggtggcccgag	atcagaagca
		agagcatcaa	cagcgccacc	cactacgccc	agagcgtgaa
		gggcagattc	accatcagca	gagacgacag	caagagcgcc
		gtgtacctgc	agatgaccga	cctgagaacc	gaggacaccg
		gcgtgtacta	ctgcagcaga	aactactacg	gcagcaccta
		cgactactgg	ggccagggca	ccacctgac	cgtgagcagc
		gccagcacca	agggcccccag	cgtgttcccc	ctggccccca
		gcagcaagag	caccagcgcc	ggcaccgccc	ccctgggctg
		cctgggtgaag	gactacttcc	ccgagcccg	gaccgtgagc
		tggaacagcg	gcgccctgac	cagcggcgtg	cacaccttcc
		ccgcccgtct	gcagagcagc	ggcctgtaca	gcctgagcag
		cgtgggtgacc	gtgcccagca	gcagcctggg	caccagacc
		tacatctgca	acgtgaacca	caagcccagc	aacaccaagg
		tggacaagaa	ggtggagccc	aagagctgcg	ac +/-
		aagaccacacc	(aagaccacctg)	+/-	
		tgccccccctgccccgcc	+/-		
		cccagagctgctggcgccccagcgtgttccctg			
Infliximab	Light/ SEQ ID NO: 152	gacatcctgc	tgaccagag	ccccgccatc	ctgagcgtga
		gccccggcga	gagagtgcag	ttcagctgca	gagccagcca
		gttcgtgggc	agcagcatcc	actggtacca	gcagagaacc
		aacggcagcc	ccagactgct	gatcaagtac	gccagcgaga
		gcagagcgg	catccccagc	agattcagcg	gcagcggcag
		cggcaccgac	ttcacccctga	gcataacac	cgtggagagc
		gaggacatcg	ccgactacta	ctgccagcag	agccacagct
		ggcccttcac	cttcggcagc	ggcaccaccc	tggaggtgaa
		gagaaccgtg	gccgccccca	gcgtgttcat	cttccccccc
		agcgacgagc	agctgaagag	cggcaccgccc	agcgtggtgt
		gcctgctgaa	caacttctac	cccagagagg	ccaaggtgca
		gtggaagggtg	gacaacgccc	tgcaagcgg	caacagccag
		gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca
		gcctgagcag	cacctgacc	ctgagcaagg	ccgactacga
		gaagcacaag	gtgtacgcct	gcgaggtgac	ccaccagggc
		ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt
		gc			
aTAU	Heavy/ SEQ ID NO: 153	gaggtgaagg	tggaggagag	cggcggcgcc	ctggtgcagc
		ccggcggcag	catgaagctg	agctgcgtgg	tgagcggcctt
		caccttcagc	aactactggg	tgaactgggt	gaggcaggcc
		cccggcaagg	gcctggagtg	ggtggccccag	atcaggctga
		agagcgacaa	ctacgccacc	cactacgagg	agagcgtgaa
		gggcaggttc	accatcagca	gggacgacag	caagagcagc
		gtgtacctgc	agatgaacaa	cctgaggggcc	gaggacagcg
		gcactacta	ctgcaccaac	tgggaggact	actggggcca
		gggcaccacc	gtgaccgtga	gcagcgccag	caccaagggc
		cccagcgtgt	tccccctggc	cccctgcagc	aggagcacca
		gcgagagcac	cgccgcccctg	ggctgcctgg	tgaaggacta
		cttccccgag	cccgtgaccg	tgagctggaa	cagcggcgcc
		ctgaccagcg	gcgtgcacac	cttccccgcc	gtgctgcaga
		gcagcggcct	gtacagcctg	agcagcgtgg	tgaccgtgcc
		cagcagcagc	ctgggcacca	agacctacac	ctgcaacgtg
		gaccacaagc	ccagcaacac	caaggtggac	aagaggtgg
		agagcaagta	c +/-	ggccccccctgccccccctgccccgcc	
		+/-	cccaggttccctggcgccccagcgtgttccctg		

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences					
mAb	Chain/ SEQ ID NO	Sequence			
aTAU	Light/ SEQ ID NO: 154	gacatcgtgc gcctgggcga gagcgtgagc cagcagaagc acgccagcaa cgccagcggc cccttgagc acagctggga gctggagatc atcttcccc ccagcgtggt ggccaagggt ggcaacagcc acagcaccta ggccgactac accaccagg acaggggcga	tgacccagag gagggccacc accagcaggt ccggccagcc cctggagagc agcggcaccg ccgaggactt gatccccctg aagaggaccg ccagcgacga gtcctgctg cagtggaaag aggagagcgt cagcctgagc gagaagcaca gcctgagcag gtgc	ccccgacagc atcagctgca acagctacat ccccaaagctg ggcgtgcccc acttcaccct cgccacctac accttcggcc tggccgcccc gcagctgaag aacaacttct tggaacaacg gaccgagcag agcaccctga agggtgtacg ccccgtgacc aagagcttca	ctggccgtga gggccagcca ccactggtac ctgatcaagt gcaggttcag gaacatccac tactgccacc agggcaccaa cagcgtgttc agcggcaccg accccaggga cctgcagagc gacagcaagg ccctgagcaa ctgcgaggtg aagagcttca
Erenumab	Heavy/ SEQ ID NO: 155	caggtgcagc ccggcagaag caccttcagc cccggaagg acggcagcat attcaccatc ctgcagatga actactgcgc cggtacttac ggccagggca agggcccag caccagcgag gactacttec gcgcctgac gcagagcagc gtgcccagca acgtggacca cgtggagaga gcccccccg	tggtggagag cctgagactg agcttcggca gcctggagtg caagtacagc agcagagaca acagcctgag cagagacaga cactacaagt ccaccgtgac cgtgttcccc agcaccgcgc ccgagccgt cagcggcgtg ggcctgtaca gcaacttcgg caagcccagc aagtgtcgtg tgcccggc	cgccggcggc agctgcgcgc tgcaactgggt ggtggccgtg gtggacagcg acagcaagaa agcccaggac ctgaactact actacggcat cgtgagcagc ctggccccct ccctgggctg gaccgtgagc cacaccttec gcctgagcag caccagacc aacaccaagg ggagtgcgcc ggagtgcccc	gtggtgcagc ccagcggctt gagacaggcc atcagcttcg tgaaggcgag caccctgttc accgctgtgt acgacagcag ggcctgtgg gccagcacca gcagcagaag cctggtgaag tggaacagcg ccgcccgtgct cgtggtgacc tacacctgca tggaacaagac ccctgcccc ccttgcccc
Erenumab	Light/ SEQ ID NO: 156	cagagcgtgc ccggccagaa caacatcgcc cccgccaccg agagaccag gagcggcacc accggcgagc gcagactgag gaccgtgctg ctgttcccc ccaccctggt cgtgaccgtg gccggcgtgg acaagtacgc gcagtggagg cacgagggga agtgcagc	tgacccagcc ggtgaccatc aacaactacg cccccaagct cgccatcccc agcaccaccc aggccgacta cgccgtgggt ggccagccca ccagcagcga gtgcctgac agaccaccaa cgccagcagc agccacagaa gcaccgtgga	ccccagcgtg agctgcagcg tgagctggta gctgatctac gacagattca tgggcatcac ctactgcggc ttcgccggcg aggccaaccc ggagctgcag agcgaacttc ccgacggcag gcccagcaag tacctgagcc gctacagctg gaagaccgtg	agcggccccc gcagcagcag ccagcagctg gacaacaaca gcggcagcaa cggcctgcag acctggggaca gcaccaagct caccgtgacc gccaacaagg acccggcgcc ccccgtgaag cagagcaaca tgacccccga ccaggtgacc gccccaccgc
BAN2401	Heavy/ SEQ ID NO: 157	gaggtgcagc ccggcggcag caccttcagc cccggaagg gcagcagcac gttcaccatc ctgcagatga actactgcgc ctactacacc accgtgagca ccctggcccc cgccctgggc gtgaccgtga tgacacacct cagcctgagc	tggtggagag cctgaggctg agcttcggca gcctggagtg catctactac agcagggaca gcagcctgag cagggagggc atggactact gcgccagcac cagcagcaag tgccctgggtga gctggaacag ccccgcctg cagcgtgtga	cgccggcggc agctgcagcg tgcaactgggt ggtggcctac ggcgacaccg acgccaagaa ggccgaggac ggctactact ggggccaggg caaggggccc agcaccagcg aggactactt cggcgccttg ctgcagagca ccgtgcccag	ctggtgcagc ccagcggctt gaggcaggcc atcagcagcg tgaaggcgag cagcctgttc accgcccgtg acggcaggag caccaccgtg agcgtgttcc gcggcaccgc ccccgagccc accagcggcg gcgccctgta cagcagcctg

TABLE 5-continued

Table of Fab Fragment Nucleic Acid Sequences		
mAb	Chain/ SEQ ID NO	Sequence
		ggcaccacaga cctacatctg caacgtgaac cacaagccca gcaacaccaa ggtggacaag aaggtggagcccaagagctgagac +/- aagaccacacac (or aagaccacacctg) +/- tgccccccctgccccgcc +/- ccgagctgctgggcggc
BAN2401	Light/ SEQ ID NO: 158	gacgtggtga tgaccagag cccctgagc ctgcccgtga cccccgccgc ccccgccagc atcagctgca ggagcagcca gagcatcgtg cacagcaacg gcaacaccta cctggagtg tacctgcaga agcccgccga gagcccaag ctgctgatct acaaggtgag caacaggttc agcggcgtgc ccgacaggtt cagcggcagc ggcagcggca ccgacttcac cctgaggatc agcaggggtg aggccgagga cgtgggcac tactactgct tccagggcag ccacgtgccc cccaccttgc gccccggcac caagctggag atcaagagga ccgtggccgc ccccgagctg ttcatcttcc ccccgagcga cgagcagctg aagagcggca ccgcccagct ggtgtgctg ctgaacaact tctaccccag ggagggcaag gtgcagtgga aggtggacaa cgccttcag agcggcaaca gccagagag cgtgaccgag caggacagca aggacagcac ctacagcctg agcagcacc tgaccctgag caagggcgac tacgagaagc acaaggtgta cgcctgcgag gtgaccacac agggcctgag cagccccgtg accaagagct tcaacagggg cgagtgc
E06-scFV	Heavy/ SEQ ID NO: 159	gaggtgaagc tgggtgagag cggcggcggc ctggtgcagc ccggcggcag cctgaggctg agtcgcgcca ccagcgctt caccttcagc gacttttaca tggagtgggt gaggcaggcc cccggaaga ggttgagtg gatcgccgc agcaggaaca aggccaacga ctacaccacc gagtacgccc acagcgtgaa gggcaggctc atcgtgagca gggacaccag ccagagcatc ctgtacctgc agatgaacgc cctgagggcc gaggacaccg ccatctacta ctgcgcagg gactactacg gcagcagcta ctggtacttc gacgtgtggg gcgcggccac caccgtgacc gtgagcagc
E06-scFV	Light/ SEQ ID NO: 160	gacatcgtga tgaccagag ccccgagcgc ctgagcgtga gcgcggcga gaaggtgacc atcagctgca ccgcccagca gagcctgtac agcagcaagc acaaggtgca ctacctggcc tggtaccaga agaagcccca gcagagcccc aagctgctga tctacggcgc cagcaacagg tacatcgccg tgcccagacg gttcaccggc agcggcagcg gcaccgactt caccctgacc atcagcagcg tcaggtgga ggacctgacc cactactact gcgcccagtt ctacagctac cccctgacct tcggcgcggc caccaagctg gagatcaag

EQUIVALENTS

[0874] Although the invention is described in detail with reference to specific embodiments thereof, it will be understood that variations which are functionally equivalent are within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description and accompanying drawings. Such modifications are intended to fall within the scope of the appended claims. Those skilled in the art will recog-

nize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following claims.

[0875] All publications, patents and patent applications mentioned in this specification are herein incorporated by reference into the specification to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference in their entireties.

SEQUENCE LISTING

Sequence total quantity: 314
 SEQ ID NO: 1 moltype = AA length = 249
 FEATURE Location/Qualifiers
 MOD_RES 1
 note = Any amino acid
 SITE 229..249

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VARIANT note = /note="This region may be absent in its entirety"
232

SITE note = L
233..249

SITE note = /note="This region may be absent in its entirety"
239..249

SITE note = /note="This region may be absent in its entirety"
1..249

note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"

REGION 1..249
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

REGION 1..249
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"

source 1..249
mol_type = protein
organism = synthetic construct

SEQUENCE: 1

XVQLVESGGG	VVQPGRLRL	SCAASGFAPS	SYGMHWVRQA	PGKGLEWVAV	IWFDGTKKYY	60
TDSVKGRFTI	SRDNSKNTLY	LQNTLRAED	TAVYYCARDR	GIGARRGPYY	MDVWGKGTTV	120
TVSSASTKGP	SVFPLAPSSK	STSGTAALG	CLVKDYFPEP	VTVSWNSGAL	TSGVHTFPAV	180
LQSSGLYSL	SVVTVPSSSL	GTQTYICNVN	HKPSNTKVDK	RVEPKSCDKT	HTCPPCPAPE	240
LLGGPSVFL						249

SEQ ID NO: 2 moltype = AA length = 214

FEATURE Location/Qualifiers

REGION 1..214
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 2

DIQMTQSPSS	LSASVGRVIT	ITCRASQSI	SYLNWYQQK	GKAPKLLIYA	ASSLQSGVPS	60
RFSGSGSGTD	FTLTISSLQ	EDFATYYCQ	SYSTPLTFG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLNNFY	PREAKVQWK	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQ	LSSPVTKSF	RGEC			214

SEQ ID NO: 3 moltype = AA length = 234

FEATURE Location/Qualifiers

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SITE 224..234
note = /note="This region may be absent in its entirety"

SITE 1..234
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"

REGION 1..234
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

REGION 1..234
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"

source 1..234
mol_type = protein
organism = synthetic construct

SEQUENCE: 3

EVQLVESGGG	LVQPGSLRL	SCAASGFTFS	SYGMSWVRQA	PGKGLELVA	INSNGGSTYY	60
PDSVKGRFTI	SRDNAKSLY	LQNSLRAED	TAVYYCASGD	YWGQGTTVTV	SSASTKGPSV	120
FPLAPCSRST	SESTAALGCL	VKDYFPEPVT	VSWNSGALTS	GVHTFPAVLQ	SSGLYSLSSV	180
VTVPSSSLGT	KTYTCNVDHK	PSNTKVDKRV	ESKYGPPECP	CPAPEFLGGP	SVFL	234

SEQ ID NO: 4 moltype = AA length = 219

FEATURE Location/Qualifiers

REGION 1..219
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..219
mol_type = protein
organism = synthetic construct

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SEQUENCE: 4
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLV YSNGDTYLHW YLQKPGQSPQ LLIYKVS NRF 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVG V YYCSQSTHVP WTFGQGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 5 moltype = AA length = 251
FEATURE Location/Qualifiers
SITE 231..251
 note = /note="This region may be absent in its entirety"
VARIANT 234
 note = L
SITE 235..251
 note = /note="This region may be absent in its entirety"
SITE 241..251
 note = /note="This region may be absent in its entirety"
SITE 1..251
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
REGION 1..251
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
REGION 1..251
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
source 1..251
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 5
QVELVESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA INASGTRTTY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARGK GNTHKPYGYV RYFDVWGQGT 120
LVTSSASTK GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP 180
AVLQSSGLYS LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCPPECPA 240
PELLGGPSVF L 251

SEQ ID NO: 6 moltype = AA length = 215
FEATURE Location/Qualifiers
REGION 1..215
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..215
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 6
DIVLTQSPAT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGVP 60
ARFSGSGSGT DFTLTISLE PEDFATYYCL QIYNMPITFG QGTKVEIKRT VAAPSVFIFP 120
PSDEQLKSGT ASVVCLLNMF YPREAKVQWK VDNALQSGNS QESVTEQDSK DSTYLSSTL 180
TLKADYEKH KVIACEVTHQ GLSSPVTKSF NRGEC 215

SEQ ID NO: 7 moltype = AA length = 247
FEATURE Location/Qualifiers
SITE 228..247
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SITE 237..247
 note = /note="This region may be absent in its entirety"
SITE 1..247
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
REGION 1..247
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
REGION 1..247
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
source 1..247
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 7
EVQLVESGGG LEQPGGSLRL SCAGSGFTFR DYAMTWVRQA PGKGLEWVSS ISGSGGNTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDR LSITIRPRYY GLDVWGQGT 120
VTSSASTKG PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTVPSSS LGTKTYTCNV DHKPSNTKVD KRVESKYGPP CPPCPAPEFL 240

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GGPSVFL 247

SEQ ID NO: 8 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..219
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 8
 DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL YSIGYNYLDW YLQKSGQSPQ LLIYLGSNRA 60
 SGVPDRFSGS GSGTDFTLKI SRVEAEDVGF YYCMQALQTP YTFGQGKLE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVVCL LNNFYPPREK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
 SSTLTLSKAD YEKKHVVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 9 moltype = AA length = 241
 FEATURE Location/Qualifiers
 SITE 222..241
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 SITE 231..241
 note = /note="This region may be absent in its entirety"
 SITE 1..241
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..241
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 Synthetic polypeptide"
 REGION 1..241
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..241
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 9
 QVQLVQSGAE VKKPGSSVKV SCKASGYSFT DYHIHWVRQA PGQGLEWMGV INPMYGTDDY 60
 NQRFKGRVTI TADESTSTAY MELSSLRSED TAVYYCARYD YFTGTGVYWG QGTLVTVSSA 120
 STKGPSVPEL APCSRSTSES TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVQLQSSG 180
 LYSLSVVTVV PSSSLGKTKY TCNVDPKPSN TKVDKRVESK YGPPCPPCPA PEFLGGPSVF 240
 L 241

SEQ ID NO: 10 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..219
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 10
 DIVMTQTPLS LSVTPGQPAS ISCRSSRSRV HSRGNTYLVH YLQKPGQSPQ LLIYKVSNRF 60
 IGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCSQSTHLP FTFGQGKLE IKRTVAAPSV 120
 FIFPPSDEQL KSGTASVVCL LNNFYPPREK VQWKVDNALQ SGNSQESVTE QDSKDYSTSL 180
 SSTLTLSKAD YEKKHVVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 11 moltype = AA length = 252
 FEATURE Location/Qualifiers
 SITE 232..252
 note = /note="This region may be absent in its entirety"
 VARIANT 235
 note = L
 SITE 236..252
 note = /note="This region may be absent in its entirety"
 SITE 242..252
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 SITE 1..252
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..252
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..252
 note = source = /note="See specification as filed for

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detailed description of substitutions and preferred
 embodiments"

source 1..252
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 11
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS NYWMNWVRQA PGKGLEWVAA INQDGSEKYY 60
 VGSVKGRFTI SRDNAKNSLY LQMNSLRVED TAVYYCVRDY YDILTDYYIH YWYFDLWGRG 120
 TLVTVSSAST KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF 180
 PAVLQSSGLY SLSSVTVTPS SSLGTQTYIC NVNHKPSNTK VDKRVEPKSC DKTHTCPPCP 240
 APELLGGPSV FL 252

SEQ ID NO: 12 moltype = AA length = 215
 FEATURE Location/Qualifiers
 REGION 1..215
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"

source 1..215
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 12
 EIVLTQSPGT LSLSPGERAT LSCRASQSVS SSYLAWYQQK PGQAPRLLIY GASSRATGIP 60
 DRFSGSGSGT DFTLTISRLE PEDFAVYCYQ QYGSSPCTFG QGTRLEIKRT VAAPSVFIFP 120
 PSDEQLKSGT ASVVCLLNMF YPREAKVQWK VDNLQSGNS QESVTEQDSK DSTYLSSTL 180
 TISKADYEKH KVIACEVTHQ GLSSPVTKSF NRGEC 215

SEQ ID NO: 13 moltype = AA length = 244
 FEATURE Location/Qualifiers
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 VARIANT 227
 note = L
 SITE 228..244
 note = /note="This region may be absent in its entirety"
 SITE 234..244
 note = /note="This region may be absent in its entirety"
 SITE 1..244
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..244
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..244
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"

source 1..244
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 13
 EVQLVQSGAE VKKPGESLKI SCKGSGYSFT TYWLGWVRQM PGKGLDWIGI MSPVDSDIRY 60
 SPSFQGQVTM SVDKSITTAY LQWNSLKASD TAMYYCARRR PGQGYDFDWFQ QGTLVTVSSS 120
 STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG 180
 LYSLSVTVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPELLGGP 240
 SVFL 244

SEQ ID NO: 14 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 14
 DIQMTQSPSS LSASVGRVIT ITCRASQGIS SWLAWYQQKP EKAPKSLIYA ASSLQSGVPS 60
 RPSGSGSGTD FTLTISLQPD EDFATYCYQQ YNIYPYTFGQ GTKLEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNMFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 15 moltype = AA length = 244
 FEATURE Location/Qualifiers
 SITE 224..244
 note = /note="This region may be absent in its entirety"
 VARIANT 227

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SITE note = L
 228..244
 note = /note="This region may be absent in its entirety"
 SITE 234..244
 note = /note="This region may be absent in its entirety"
 SITE 1..244
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..244
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..244
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..244
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 15
 QVTLRSGPA LVKPTQTLTL TCTVSGFSLT SYSVHWVRQP PGKGLEWLGW IWASGGTDYN 60
 SALMSRLSIS KDTSRNQVVL TMNMDPVDV ATYYCARDPP SSLLRLDYWG RGTPTVTVSSA 120
 STKGPSVFPPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPVAVLQSSG 180
 LYSLSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVPEK SCDKTHTCPP CPAPELLGGP 240
 SVFL 244
 SEQ ID NO: 16 moltype = AA length = 220
 FEATURE Location/Qualifiers
 MOD_RES 42
 note = Any amino acid
 REGION 1..220
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..220
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 16
 DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSGNQKNYLA WXQQKPGQPP KLLIYGASTR 60
 ESGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCQNVHSF PFTFGGGTKL EIKRTVAAPS 120
 VFIFPPSDEQ LKSGTASVVC LLNNFYPREA KVQWKVDNAL QSGNSQESVT EQDSKDSTYS 180
 LSSTLTLSKA DYKHKVYAC EVTHQGLSSP VTKSFNRGEC 220
 SEQ ID NO: 17 moltype = AA length = 246
 FEATURE Location/Qualifiers
 SITE 226..246
 note = /note="This region may be absent in its entirety"
 VARIANT 229
 note = L
 SITE 230..246
 note = /note="This region may be absent in its entirety"
 SITE 236..246
 note = /note="This region may be absent in its entirety"
 SITE 1..246
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..246
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..246
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..246
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 17
 QVQLVQSGAE VKKPGASVKV SCKGSGYTFT SYWMHWVRQA PGQRLEWIGE IDPSESNTNY 60
 NQKFKGRVTL TVDISASTAY MELSSLRSED TAVYYCARGG YDGWDYAIKY WGQGTILVTVS 120
 SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPVAVLQS 180
 SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKKVE PKSCDKTHTC PPCAPELAG 240
 APSVFL 246
 SEQ ID NO: 18 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219

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note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..219
mol_type = protein
organism = synthetic construct

SEQUENCE: 18
DVVMTQSPLS LPVTPGEPAS ISCRSSQSLA KSYGNTYLSW YLQKPGQSPQ LLIYGISNRF 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCLQGTQHP YTFGQGTKVE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYPREAK VQWKVDNALQ SGNSQESVTE QDSKDSYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 19 moltype = AA length = 245
FEATURE Location/Qualifiers
SITE 226..245
note = /note="This region may be absent in its entirety"
SITE 235..245
note = /note="This region may be absent in its entirety"
SITE 1..245
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..245
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
REGION 1..245
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..245
mol_type = protein
organism = synthetic construct

SEQUENCE: 19
QVQLVQSGAE VKKPGASVKV SCKASGFNIK DTYIHVVRQA PGQRLEWMGR IDPANGYTKY 60
DPKFQGRVTI TADTSASTAY MELSSLRSED TAVYYCAREG YYGNYGVYAM DYWGQGTLLV 120
VSSASTKGPS VFPLAPCSRS TSESTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG TKTYTCNVDH KPSNTKVDKR VESKYGPSCP PCPAPEFLGG 240
PSVFL 245

SEQ ID NO: 20 moltype = AA length = 213
FEATURE Location/Qualifiers
REGION 1..213
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..213
mol_type = protein
organism = synthetic construct

SEQUENCE: 20
DIQMTQSPSS LSASVGRVIT ITCKTSQDIN KYMAWYQQTG GKAPRLLIHY TSALQPGIPS 60
RFGSGSGGRD YFTTISLQP EDIATYYCLQ YDNLWTFGQG TKVEIKRTVA APSVFIFPPS 120
DEQLKSGTAS VVCLLNNFYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSSSTLT 180
SKADYEKHKV YACEVTHQGL SSPVTKSFNR GEC 213

SEQ ID NO: 21 moltype = AA length = 243
FEATURE Location/Qualifiers
SITE 223..243
note = /note="This region may be absent in its entirety"
VARIANT 226
note = L
SITE 227..243
note = /note="This region may be absent in its entirety"
SITE 233..243
note = /note="This region may be absent in its entirety"
SITE 1..243
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..243
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
REGION 1..243
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..243
mol_type = protein
organism = synthetic construct

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SEQUENCE: 21
EVQLVESGGG LVQPGGSLRL SCAASGFTFN NYAMNWRQA PGKGLDWVST ISGSGGTTNY 60
ADSVKGRFII SRDSSKHTLY LQMNSLRAED TAVYYCAKDS NWGNFDLWGR GTLVTVSSAS 120
TKGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTVSWN SGALTSGVHT FPAVLQSSGL 180
YSLSSVVTVP SSSLGTQTYI CNVNHKPSNT KVDKKVEPKS CDKTHTCPPC PAPELLGGPS 240
VFL 243

SEQ ID NO: 22 moltype = AA length = 220
FEATURE Location/Qualifiers
REGION 1..220
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..220
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 22
DIVMTQSPDS LAVSLGERAT INCKSSQSVL YRSNNRNFLG WYQQKPGQPP NLLIYWASTR 60
ESGVPRDRFSG SGSGTDFTLT ISSLQAEDVA VYICQQYYTT PYTFGQGTKL EIKRTVAAPS 120
VFIFPPSDEQ LKSGTASVVC LLNMFYPREA KVQWKVDNAL QSGNSQESVT EQDSKDSTYS 180
LSSTLTLSKA DYEKHKVYAC EVTHQGLSSP VTKSFNRGEC 220

SEQ ID NO: 23 moltype = AA length = 231
FEATURE Location/Qualifiers
SITE 221..231
 note = /note="This region may be absent in its entirety"
SITE 227..231
 note = /note="This region may be absent in its entirety"
SITE 1..231
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
REGION 1..231
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
REGION 1..231
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 23
EVQLVQSGAE VKKPGASVKV SCKASGYTLT SYGISWVRQA PGQGLEWMGW VSFYNGNTNY 60
AQKLQGRGTM TTDPTSTAY MELRSLRSD TAVYYCARGY GMDVWGQGT VTVSSASTKG 120
PSVFPPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSN FGTQTYTCNV DHKPSNTKVD KTVRKCCVE CPPCPAPPVA G 231

SEQ ID NO: 24 moltype = AA length = 215
FEATURE Location/Qualifiers
REGION 1..215
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..215
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 24
ESALTQPASV SGSPGQSITI SCTGTSSDVG GYNVSWSYQQ HPGKAPKLM IYEVSNRPSGV 60
SNRFGSKSGS NTASLTISGL QAEDEADYYC NSYTSTSMVF GGGTKLTVLG QPKAAPSVTL 120
FPPSSEELQA NKATLVCLIS DFYPGAVTVA WKADSSPVKA GVETTTPSKQ SNNKYAASSY 180
LSLTPEQWKS HRSYSCQVTH EGSTVEKTVA PTECS 215

SEQ ID NO: 25 moltype = AA length = 248
FEATURE Location/Qualifiers
SITE 232..248
 note = /note="This region may be absent in its entirety"
SITE 238..248
 note = /note="This region may be absent in its entirety"
SITE 1..248
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
REGION 1..248
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
REGION 1..248
 note = source = /note="See specification as filed for

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detailed description of substitutions and preferred
 embodiments"

source 1..248
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 25
 EVQLVESGGG VIQPGGSLRL SCAASGFTFD DYAMNWRQG PGKGLEWVSA ISGDGGSTYY 60
 ADSVKGRFTI SRDNSKNSLY LQMNSLRAED TAFFYCAKDL RNTIFGVVIP DAFDIWGQGT 120
 MVTVSSASTK GPSVFPLAPC SRSTSESTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP 180
 AVLQSSGLYS LSSVVTVPSS SLGKTYTCN VDHKPSNTKV DKRVESKYGP PCPPCPAPEF 240
 LGGPSVFL 248

SEQ ID NO: 26 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 26
 DIQMTQSPST LSASVGRDVT ITCRASQSIR SWLAWYQQKP GKAPKLLIYK ASSLESGVPS 60
 RFGSGSGSTE FTLTISSLQP DDFATYYCQQ YNSYSYTFQG GTKLEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK DSTYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 27 moltype = AA length = 247
 FEATURE Location/Qualifiers
 SITE 227..247
 note = /note="This region may be absent in its entirety"
 VARIANT 230
 note = L
 SITE 231..247
 note = /note="This region may be absent in its entirety"
 SITE 237..247
 note = /note="This region may be absent in its entirety"
 SITE 1..247
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..247
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..247
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"

source 1..247
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 27
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSG ITGSGGSTYY 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDP GTTVIMSWFD PWGQGTLVTV 120
 SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
 SSGLYSLSSV VTVPSSSLGT QTYICNVNHK PSNTKVDKKV EPKSCDKTHT CPPCPAPELL 240
 GGPSVFL 247

SEQ ID NO: 28 moltype = AA length = 215
 FEATURE Location/Qualifiers
 REGION 1..215
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"

source 1..215
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 28
 EIVLTQSPGT LSLSPGERAT LSCRASQSVR GRYLAWYQQK PGQAPRLLIY GASSRATGIP 60
 DRFSGSGSGT DFTLTISRLE PEDFAVFYQC QYGSSPRTFG QGTKVEIKRT VAAPSVFIFP 120
 PSDEQLKSGT ASVVCLLNLF YPREAKVQWK VDNLQSGNS QESVTEQDSK DSTYLSSTLT 180
 TLSKADYEKH KVIACEVTHQ GLSSPVTKSF NRGEK 215

SEQ ID NO: 29 moltype = AA length = 235
 FEATURE Location/Qualifiers
 SITE 216..235
 note = /note="This region may be absent in its entirety"
 SITE 225..235

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SITE          note = /note="This region may be absent in its entirety"
              1..235
              note = /note="Variant residues given in the sequence have
              no preference with respect to those in the annotations for
              variant positions"
REGION        1..235
              note = source = /note="Description of Artificial Sequence:
              Synthetic polypeptide"
REGION        1..235
              note = source = /note="See specification as filed for
              detailed description of substitutions and preferred
              embodiments"
source        1..235
              mol_type = protein
              organism = synthetic construct

SEQUENCE: 29
QVQLVESGGG VVQPGSRSLRL DCKASGITFS NSGMHWVRQA PGKGLEWVAV IWYDGSKRYR 60
ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATND DYWGQGTLLV VSSASTKGPS 120
VPFLAPCSRS TSESTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL QSSGLYSLS 180
VVTVPSSSLG TKTYTCNVDH KPSNTKVDKR VESKYGPPCP PCPAPEPLGG PSVFL 235

SEQ ID NO: 30      moltype = AA length = 214
FEATURE           Location/Qualifiers
REGION           1..214
                 note = source = /note="Description of Artificial Sequence:
                 Synthetic polypeptide"
source          1..214
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 30
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA 60
RFSGSGSGTD FTLTISLEP EDFAVYYCQQ SSNWPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 31      moltype = AA length = 242
FEATURE           Location/Qualifiers
SITE            223..242
                 note = /note="This region may be absent in its entirety"
SITE            232..242
                 note = /note="This region may be absent in its entirety"
SITE            1..242
                 note = /note="Variant residues given in the sequence have
                 no preference with respect to those in the annotations for
                 variant positions"
REGION          1..242
                 note = source = /note="Description of Artificial Sequence:
                 Synthetic polypeptide"
REGION          1..242
                 note = source = /note="See specification as filed for
                 detailed description of substitutions and preferred
                 embodiments"
source          1..242
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 31
QVQLVQSGVE VKKPGASVKV SCKASGYTFT NYMYWVRQA PGQGLEWMGG INPSNGGTNF 60
NEKFKNRVTL TDSSTTTAY MELKSLQFDD TAVYYCARRD YRFDMGFDYW GQGTITVTVSS 120
ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSVVVT VPSSSLGTKT YTCNVDHKPS NTKVDKRVES KYGPPCPPCP APEFLGGPSV 240
FL 242

SEQ ID NO: 32      moltype = AA length = 218
FEATURE           Location/Qualifiers
REGION           1..218
                 note = source = /note="Description of Artificial Sequence:
                 Synthetic polypeptide"
source          1..218
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 32
EIVLTQSPAT LSLSPGERAT LSCRASKGVS TSGYSYLHWY QQKPGQAPRL LIYLASYLES 60
GVPARFSGSG SGTDFTLTIS SLEPEDFAVY YCQHSRDLP LTFGGGTKVEI KRTVAAPSVF 120
IFPPSDEQLK SGTASVCLL NNFYPREKAV QWKVDNALQS GNSQESVTEQ DSKDSTYSL 180
STLTLSKADY EKHKVYACEV THQGLSSPVT KSFNRGEC 218

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SEQ ID NO: 33      moltype = AA  length = 248
FEATURE           Location/Qualifiers
SITE              228..248
                  note = /note="This region may be absent in its entirety"
VARIANT           231
                  note = L
SITE              232..248
                  note = /note="This region may be absent in its entirety"
SITE              238..248
                  note = /note="This region may be absent in its entirety"
SITE              1..248
                  note = /note="Variant residues given in the sequence have
                  no preference with respect to those in the annotations for
                  variant positions"
REGION            1..248
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
REGION            1..248
                  note = source = /note="See specification as filed for
                  detailed description of substitutions and preferred
                  embodiments"
source            1..248
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 33
EVQLVESGGG LVQPGGSLRL SCAASGYDFT HYGMNWVRQA PGKGLEWVGW INTYTGEPTY 60
AADFKRRFTF SLDTSKSTAY LQMNSLRAED TAVYYCAKYP YYYGTSHWYF DVWGQGLT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCDKTH TCPPCPAPEL 240
LGGPSVFL 248

SEQ ID NO: 34      moltype = AA  length = 214
FEATURE           Location/Qualifiers
REGION            1..214
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 34
DIQLTQSPSS LSASVGDVRT ITCSASQDIS NYLWNWQQKP GKAPKVLIIYF TSSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YSTVPWTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 35      moltype = AA  length = 248
FEATURE           Location/Qualifiers
SITE              228..248
                  note = /note="This region may be absent in its entirety"
VARIANT           231
                  note = L
SITE              232..248
                  note = /note="This region may be absent in its entirety"
SITE              238..248
                  note = /note="This region may be absent in its entirety"
SITE              1..248
                  note = /note="Variant residues given in the sequence have
                  no preference with respect to those in the annotations for
                  variant positions"
REGION            1..248
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
REGION            1..248
                  note = source = /note="See specification as filed for
                  detailed description of substitutions and preferred
                  embodiments"
source            1..248
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 35
EVQLVESGGG LVQPGGSLRL SCAASGYTFT NYGMNWVRQA PGKGLEWVGW INTYTGEPTY 60
AADFKRRFTF SLDTSKSTAY LQMNSLRAED TAVYYCAKYP HYYGSSHWYF DVWGQGLT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCDKTH TCPPCPAPEL 240
LGGPSVFL 248

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SEQ ID NO: 36 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 36
 DIQMTQSPSS LSASVGDRVT ITCASQDIS NYLWNWQQKP GKAPKVLIFY TSSLHSGVPS 60
 RFGSGSGGTD FTLTISSLQP EDFATYYCQQ YSTVPWTFGQ GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 37 moltype = AA length = 240
 FEATURE Location/Qualifiers
 SITE 220..240
 note = /note="This region may be absent in its entirety"
 VARIANT 223
 note = L
 SITE 224..240
 note = /note="This region may be absent in its entirety"
 SITE 230..240
 note = /note="This region may be absent in its entirety"
 SITE 1..240
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..240
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..240
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..240
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 37
 EVQLVQSGPE LKKPGASVKV SCKASGYTFT NYGMNWVRQA PGQGLEWMGW INTYTGETTY 60
 ADDPKGRFVF SLDTSVSTAY LQISLLKAED TAVYYCEREG GVNHWGQGTL VTVSSASTKG 120
 PSVFLPLPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTQTYICNV NHPKPSNTKVD KKVEPKSCDK THTCPPCPAP ELLGGPSVFL 240

SEQ ID NO: 38 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 38
 DIQVTQSPSS LSASVGDRVT ITCITSTDID DDMNWYQQKP GKVPKLLISG GNTLRPGVPS 60
 RFGSGSGGTD FTLTISSLQP EDVATYYCLQ SDSLPYTFGQ GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 39 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..120
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 39
 EVQLVESGGG LVQPGGSLRL SCTASGFSLT DYYMTWVRQ APGKGLEWVG FIDPDDDPY 60
 ATWAKGRFTI SRDNSKNTLY LQMNLSRAED TAVYYCAGGD HNSGWGLDIW GQGLTVTVSS 120

SEQ ID NO: 40 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..111

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mol_type = protein
organism = synthetic construct

SEQUENCE: 40
EIVMTQSPST LSASVGDRVI ITCQASEIIH SWLAWYQQKP GKAPKLLIYL ASTLASGVPS 60
RFGSGSGGAE FTLTISSLQP DDFATYYCQN VYLASTNGAN FGQGTKLTVL G 111

SEQ ID NO: 41 moltype = AA length = 248
FEATURE Location/Qualifiers
SITE 228..248
note = /note="This region may be absent in its entirety"
VARIANT 231
note = L
SITE 232..248
note = /note="This region may be absent in its entirety"
SITE 238..248
note = /note="This region may be absent in its entirety"
SITE 1..248
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..248
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
REGION 1..248
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..248
mol_type = protein
organism = synthetic construct

SEQUENCE: 41
QVQLQQSGAE VKKPGSSVRV SCKASGGTFN NNAINWVRQA PGQGLEWMGG IIPMFGTAKY 60
SQNFQGRVAI TADESTGTAS MELSSSLRSED TAVYYCARSR DLLLLPHHAL SPWGRGTMVT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYPPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCDKTH TCPPCPAPEL 240
LGGPSVFL 248

SEQ ID NO: 42 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 42
SSELTQDPAV SVALGQTVRV TCQGDLSLSY YASWYQQKPG QAPVLVIYVK NNRPSGIPDR 60
FSGSSSGNTA SLTITGAQAE DEADYYCSSR DSSGNHWVFG GGTELTVLGQ PKAAPSVTLF 120
PPSSEELQAN KATLVCLISD FYPGAIVTAW KADSSPVKAG VETTPSKQS NNKYAASSYL 180
SLTPQWQKSH RSYSCQVTHE GSTVEKTVAP TECS 214

SEQ ID NO: 43 moltype = AA length = 238
FEATURE Location/Qualifiers
SITE 228..238
note = /note="This region may be absent in its entirety"
SITE 234..238
note = /note="This region may be absent in its entirety"
SITE 1..238
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..238
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
REGION 1..238
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..238
mol_type = protein
organism = synthetic construct

SEQUENCE: 43
QVQLVQSGAE VKKPGASVKV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE ILPGSGSTEY 60
TENFKDRVTM TRDTSTSTVY MELSSSLRSED TAVYYCARYF FGSSPNWYFD VWGQGLTVTV 120
SSASTKGPSV FPLAPCSRST SESTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYSLSSV VTPSSNFGT QTYTCNVNHDH PSNTKVDKTV ERKCCVECPP CPAPPVAG 238

-continued

SEQ ID NO: 44 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 44
 DIQMTQSPSS LSASVGDVRT ITCGASENIY GALNWIYQKP GKAPKLLIYG ATNLADGVPS 60
 RFSGSGSGTD FTLTISSLQP EDFATYYCQN VLNTPLTFGQ GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 45 moltype = AA length = 237
 FEATURE Location/Qualifiers
 SITE 218..237
 note = /note="This region may be absent in its entirety"
 SITE 227..237
 note = /note="This region may be absent in its entirety"
 SITE 1..237
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..237
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..237
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..237
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 45
 QVQLQESGPG LVKPSSETLSL TCTVSGFSLI SYGVHWVRQP PGKGLEWLGIV IWTGGTTNIN 60
 SALMSRFTIS KDDSKNTVYL KMNSLKTEDT AIYYCARYYY GMDYWGQGTLL VVSSASTKG 120
 PSVFPPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTKTYTCNV DHKPSNTKVD KRVEKYGPP CPPCPAPEFL GGPSVFL 237

SEQ ID NO: 46 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 46
 DIQMTQSPSS LSASVGDVRT ITCKASQDVR NTVAWYQKP GKAPKLLIYS SSYRNTGVDP 60
 RFSGSGSGTD FTLTISSLQA EDVAVYYCQV HYITPYTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 47 moltype = AA length = 247
 FEATURE Location/Qualifiers
 SITE 227..247
 note = /note="This region may be absent in its entirety"
 VARIANT 230
 note = L
 SITE 231..247
 note = /note="This region may be absent in its entirety"
 SITE 237..247
 note = /note="This region may be absent in its entirety"
 SITE 1..247
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..247
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..247
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"

-continued

source 1..247
mol_type = protein
organism = synthetic construct

SEQUENCE: 47
EVQLLESGGG LVQPGGSLRL SCAASGFTFS HYIMMWVRQA PGKGLEWVSG IYSSGGITVY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAYRR IGVPRRDEFD IWGQGTMTV 120
SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYSLSV VTPSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCDKTHT CPPCPAPELL 240
GGPSVFL 247

SEQ ID NO: 48 moltype = AA length = 213
FEATURE Location/Qualifiers
REGION 1..213
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..213
mol_type = protein
organism = synthetic construct

SEQUENCE: 48
DIQMTQSPST LSASVGDRVT ITCRASQSI SWLAWYQQK GKAPKLLIYK ASTLESGVPS 60
RFGSGSGTE FTLTISSLQP DDFATYYCQQ YNTYWTFGQG TKVEIKRTVA APSVFIFPPS 120
DEQLKSGTAS VVCLLNHFYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSTLT 180
SKADYEKHKV YACEVTHQGL SSPVTKSFNR GEC 213

SEQ ID NO: 49 moltype = AA length = 246
FEATURE Location/Qualifiers
SITE 226..246
note = /note="This region may be absent in its entirety"
VARIANT 229
note = L
SITE 230..246
note = /note="This region may be absent in its entirety"
SITE 236..246
note = /note="This region may be absent in its entirety"
SITE 1..246
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..246
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
REGION 1..246
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..246
mol_type = protein
organism = synthetic construct

SEQUENCE: 49
EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYAMHWVRQA PGKGLEWVSA ITWNSGHIDY 60
ADSVGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCAKVS YLSTASSLDY WGQGTTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVT SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHK PSNTKVDKVE PKSCDKTHTC PPCPAPELLG 240
GPSVFL 246

SEQ ID NO: 50 moltype = AA length = 154
FEATURE Location/Qualifiers
REGION 1..154
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..154
mol_type = protein
organism = synthetic construct

SEQUENCE: 50
RFGSGSGTD FTLTISSLQP EDVATYYCQR YNRAPYTFGQ GTKVEIKRTV AAPSVFIFPP 60
SDEQLKSGTA SVVCLLNHFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 120
LSKADYEKHK VYACEVTHQ LSSPVTKSFN RGEK 154

SEQ ID NO: 51 moltype = AA length = 245
FEATURE Location/Qualifiers
SITE 225..245
note = /note="This region may be absent in its entirety"
VARIANT 228
note = L
SITE 229..245
note = /note="This region may be absent in its entirety"

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SITE 235..245
 note = /note="This region may be absent in its entirety"
 SITE 1..245
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..245
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..245
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..245
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 51
 EVKLEESGGG LVQPGGSMKL SCVASGFIFS NHWMNWVRQS PEKGLEWVAE IRKSINSAT 60
 HYAESVKGRF TISRDDSKSA VYLQMTDLRT EDTGVYICSR NYYGSTYDYW GQGTTLTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG 240
 PSVFL 245
 SEQ ID NO: 52 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 52
 DILLTQSPAI LSVSPGERVS FSCRASQFVG SSIHWYQRT NGSPRLLIKY ASESMSGIPS 60
 RFGSGSGTD PTLINTVES EDIADYYCQ SHSWPFTFGS GTNLEVKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214
 SEQ ID NO: 53 moltype = AA length = 237
 FEATURE Location/Qualifiers
 SITE 218..237
 note = /note="This region may be absent in its entirety"
 SITE 227..237
 note = /note="This region may be absent in its entirety"
 SITE 1..237
 note = /note="Variant residues given in the sequence have
 no preference with respect to those in the annotations for
 variant positions"
 REGION 1..237
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 REGION 1..237
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..237
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 53
 EVKVVESGGG LVQPGGSMKL SCVVSQFTFS NYWVNWRQA PGKGLEWVAQ IRLKSDNYAT 60
 HYESVKGRF TISRDDSKSS VYLQMNLR AEDSGIYYCTN WEDYWGQGT VTVSSASTKG 120
 PSVFPAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
 SSVVTVPSST LGTKYTCNV DHKPSNTKVD KRVESKYGPP CPPCPAPEFL GGPSVFL 237
 SEQ ID NO: 54 moltype = AA length = 218
 FEATURE Location/Qualifiers
 REGION 1..218
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..218
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 54
 DIVLTQSPDS LAVSLGERAT ISCRASQSVS TSRYSIHWY QQKPGQPPKL LIKYASNLES 60
 GVPSRFGSGS SGTDFTLNIH PLEPEDFATY YCHHSWEIPL TFGQGTKLEI KRTVAAPSVF 120
 IFPPSDEQLK SGTASVVCLL NNFYPREAKV QWKVDNALQS GNSQESVTEQ DSKDSTYSL 180
 STLTLKADY EKKHYACEV THQGLSPVT KSFNRGEC 218

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SEQ ID NO: 55      moltype = AA  length = 246
FEATURE           Location/Qualifiers
SITE              236..246
                  note = /note="This region may be absent in its entirety"
SITE              242..246
                  note = /note="This region may be absent in its entirety"
SITE              1..246
                  note = /note="Variant residues given in the sequence have
                  no preference with respect to those in the annotations for
                  variant positions"
REGION            1..246
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
REGION            1..246
                  note = source = /note="See specification as filed for
                  detailed description of substitutions and preferred
                  embodiments"
source            1..246
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 55
QVQLVESGGG VVQPGKSLRL SCAASGFTFS SFGMHVVRQA PGKGLEWVAV ISFDGSIKYS 60
VDSVKGRFTI SRDNSKNTLF LQMSLRAED TAVYYCARDR LNYDSSGGY HYKYYGMAVW 120
GGTTVTTVSS ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPTVS WNSGALTSGV 180
HTFPAVLQSS GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS NTKVDKTVR KCCVECPPCP 240
APPVAG                                           246

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SEQ ID NO: 56      moltype = AA  length = 216
FEATURE           Location/Qualifiers
REGION            1..216
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..216
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 56
QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT STTLGITGLQ TGDEADYYCG TWDSRLSAVV FGGGTKLTVL GQPKANPTVT 120
LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADGSPVK AGVETTKPSK QSNKYAASS 180
YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS                                           216

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SEQ ID NO: 57      moltype = AA  length = 244
FEATURE           Location/Qualifiers
SITE              229..244
                  note = /note="This region may be absent in its entirety"
VARIANT          232
                  note = L
SITE              233..244
                  note = /note="This region may be absent in its entirety"
SITE              239..244
                  note = /note="This region may be absent in its entirety"
SITE              1..244
                  note = /note="Variant residues given in the sequence have
                  no preference with respect to those in the annotations for
                  variant positions"
REGION            1..244
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
REGION            1..244
                  note = source = /note="See specification as filed for
                  detailed description of substitutions and preferred
                  embodiments"
source            1..244
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 57
EVQLVESGGG LVQPGGSLRL SCSASGFTFS SFGMHVVRQA PGKGLEWVAV ISSGSSTIYY 60
GDTVKGKRFIT SRDNAKNSLF LQMSSLRAED TAVYYCAREG GYVYGRSYYT MDYWGQGTIV 120
TVSSASTKGP SVFPLAPSSK STSGGTAAALG CLVKDYFPEP VTVSWNSGAL TSGVHTFPAV 180
LQSSGLYSLV SVVTVPSSSL GTQTYICNVN HKPSNTKVDK KVEPKSCDKT HTCPCCPAPE 240
LLGG                                           244

```

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SEQ ID NO: 58      moltype = AA  length = 219
FEATURE           Location/Qualifiers
REGION            1..219

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-continued

note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..219
mol_type = protein
organism = synthetic construct

SEQUENCE: 58
DVVMTQSPSL LPVTPGAPAS ISCRSSQSIV HSNNGNTYLEW YLQKPGQSPK LLIYKVSNRF 60
SGVPDRFSGS GSGTDFTLRI SRVEAEDVGI YYCFQGSHVP PTFGPGTKLE IKRTVAAPSV 120
FIFPPSDEQL KSGTASVVCL LNNFYFPREK VQWKVDNALQ SGNSQESVTE QDSKDSTYSL 180
SSTLTLSKAD YEKHKVYACE VTHQGLSSPV TKSFNRGEC 219

SEQ ID NO: 59 moltype = AA length = 123
FEATURE Location/Qualifiers
REGION 1..123
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..123
mol_type = protein
organism = synthetic construct

SEQUENCE: 59
EVKLVESGGG LVQPGGSLRL SCATSGFTFS DFYMEWVRQA PGKRLEWIAA SRNKANDYTT 60
EYADSVKGRF IVSRDTSQSI LYLQMNALRA EDTAIYYCAR DYGSSYWF DVWGAGTTVT 120
VSS 123

SEQ ID NO: 60 moltype = AA length = 113
FEATURE Location/Qualifiers
REGION 1..113
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..113
mol_type = protein
organism = synthetic construct

SEQUENCE: 60
DIVMTQSPSS LSVSAGKKVT ISCTASESLY SSKHKVHYLA WYQKKPEQSP KLLIYGASNR 60
YIGVPDRFTG SSGTDFTLT ISSVQVEDLT HYYCAQFYSY PLTFGAGTKL EIK 113

SEQ ID NO: 61 moltype = length =
SEQUENCE: 61
000

SEQ ID NO: 62 moltype = length =
SEQUENCE: 62
000

SEQ ID NO: 63 moltype = length =
SEQUENCE: 63
000

SEQ ID NO: 64 moltype = length =
SEQUENCE: 64
000

SEQ ID NO: 65 moltype = length =
SEQUENCE: 65
000

SEQ ID NO: 66 moltype = length =
SEQUENCE: 66
000

SEQ ID NO: 67 moltype = length =
SEQUENCE: 67
000

SEQ ID NO: 68 moltype = length =
SEQUENCE: 68
000

SEQ ID NO: 69 moltype = length =
SEQUENCE: 69
000

SEQ ID NO: 70 moltype = length =
SEQUENCE: 70
000

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SEQ ID NO: 71 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus 1

SEQUENCE: 71

MAADGYLPDW	LEDNLSEGIR	EWWDLKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALEHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPVEQSP	QEPDSSSSGIG	KTGQQPAKKR	LNFGQTGDSE	180
SVPDPQPLGE	PPATPAAVGP	TTMASGGGAP	MADNNEGADG	VGNASGNWHC	DSTWLGDRI	240
TTSTRTWALP	TYNHLYKQI	SSASTGASND	NHYFGYSTPW	GYPDFNRFHC	HFSPRDWQRL	300
INNHWGFRPK	RLNFKLFNIQ	VKEVTTNDGV	TTIANNLTST	VQVFSDSSEYQ	LPYVLGSAHQ	360
GCLPPFPADV	FMIPQGYLTL	LNNGSQAVGR	SSFYCLEYFP	SQMLRTGNNF	TFSYTFEEVP	420
FHSSYAHSQS	LDRLMNPLID	QYLYYLNRTQ	NQSGSAQNKD	LLFSRGSPAG	MSVQPKNWLP	480
GPCYRQQRVS	KTKTDNNNSN	FTWTGASKYN	LNGRESIINP	GTAMASHKDD	EDKFFPMSPV	540
MIFGKESAGA	SNTALDNVMI	TDEEEIKATN	PVATERFGTV	AVNPOSSSTD	PATGDVHAMG	600
ALPGMVWQDR	DVYLQGGPIA	KIPHTDGHFH	PSPLMGGFGL	KNPPPQILIK	NTVPANPPA	660
EFSATKFASF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEVQ	YTSNYAKSAN	VDFTVDNNGL	720
YTEPRPIGTR	YLTRPL					736

SEQ ID NO: 72 moltype = AA length = 735
 FEATURE Location/Qualifiers
 source 1..735
 mol_type = protein
 organism = Adeno-associated virus 2

SEQUENCE: 72

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPFNGLD	60
KGEPVNEADA	AALHDKAYD	RQLDSGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEPVKTAP	GKKRPVEHSP	VEPDSSSGTG	KAGQQPARKR	LNFGQTGDAD	180
SVPDPQPLGQ	PPAAPSLGTT	NTMATGSGAP	MADNNEGADG	VGNSSGNWHC	DSTWMGDRVI	240
TTSTRTWALP	TYNHLYKQI	SSQSGASNDN	HYFGYSTPWG	YFDNRFHCH	FSPRDWQRLI	300
NNNWGFRPKR	LNFKLFNIQV	KEVTQNDGTT	TIANNLTSTV	QVFTDSEYQL	PYVLGSAHQG	360
CLPPFPADV	MVPQGYLTL	NNGSAVGRS	SSFYCLEYFSP	QMLRTGNNFT	FSTYTFEDVFP	420
HSSYAHSQSL	DRLMNLIDQ	YLYLSRTNT	PSGTTTQSR	QFSQAGASDI	RDQSRNWLPG	480
PCYRQQRVSK	TSADNNNSEY	SWTGATKYHL	NGRDSLVPNG	PAMASHKDD	EKFPPQSGVL	540
IFGKQSGSEK	NVDIEKVMIT	DEEEIRTTNP	VATEQYGSVS	TNLQRGNRQA	ATADVNTQGV	600
LPGMVWQDRD	VYLQGGPIWAK	IPHTDGHFHP	SPLMGGFGLK	HPPPPQILIK	TPVPANPSTT	660
FSAAKFASFI	TQYSTGQVS	EIEWELQKEN	SKRWNPETQY	TSNYNKSNNV	DFTVDTNGVY	720
SEPRPIGTRY	LTRNL					735

SEQ ID NO: 73 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus 3

SEQUENCE: 73

MAADGYLPDW	LEDNLSEGIR	EWWALKPGVP	QPKANQQHQD	NRRGLVLPGY	KYLGPGNGLD	60
KGEPVNEADA	AALHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRILEPLG	LVEEAAKTAP	GKKGAVDQSP	QEPDSSSGVG	KSGKQPAKKR	LNFGQTGDSE	180
SVPDPQPLGE	PPAAPTSLGS	NTMASGGGAP	MADNNEGADG	VGNSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TYNHLYKQI	SSQSGASNDN	HYFGYSTPWG	YFDNRFHCH	FSPRDWQRLI	300
NNNWGFRPKK	LSFKLFNIQV	RGVTQNDGTT	TIANNLTSTV	QVFTDSEYQL	PYVLGSAHQG	360
CLPPFPADV	MVPQGYLTL	NNGSAVGRS	SSFYCLEYFSP	QMLRTGNNFQ	FSTYTFEDVFP	420
HSSYAHSQSL	DRLMNLIDQ	YLYYLNRTQG	TSGTNTQSR	LLFSQAGPQS	MSLQARNWLP	480
GPCYRQQRLS	KTANDNNNSN	FPWTAASKYH	LNGRDSLVPN	GPAMASHKDD	EEKFFPMHGN	540
LIFGKEGTTA	SNAELDNVMI	TDEEEIRTTN	PVATEQYGTV	ANNLQSSNTA	PTTGTVNHQG	600
ALPGMVWQDR	DVYLQGGPIA	KIPHTDGHFH	PSPLMGGFGL	KHPPPPQIMIK	NTVPANPPPT	660
TFSPAKFASF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEIQ	YTSNYNKSNN	VDFTVDTNGV	720
YSEPRPIGTR	YLTRNL					736

SEQ ID NO: 74 moltype = AA length = 734
 FEATURE Location/Qualifiers
 source 1..734
 mol_type = protein
 organism = Adeno-associated virus 4

SEQUENCE: 74

MTDGYLPDWL	EDNLSEGVR	WWALQPGAPK	PKANQQHQDN	ARGLVLPGYK	YLGPGNGLDK	60
GEPVNAADAA	ALEHDKAYDQ	QLKAGDNPYL	KYNHADADEFQ	QRLQGDTSFG	GNLGRAVFQA	120
KKRVLEPLGL	VEQAGETAPG	KKRPLIESPQ	QPDSSSTGIGK	KGKQPAKKKL	VFEDETGAGD	180
GPPEGSTSGA	MSDDSEMRRA	AGGAAVEGGQ	GADGVGNASG	DWHCDSTWSE	GHVTTTSTRT	240
WVLPYTNHNL	KYTRGSLQS	NTYNGFSTPW	GYPDFNRFHC	HFSPRDWQRL	INNHWGMRPK	300
AMRVKIFNIQ	VKEVTTNNGE	TTVANNLTST	VQIFADSSYE	LPYVMDAGQE	GSLPPFPNDV	360
FMVPQYGYCG	LVTGNTSQQQ	TDNRAPYCLE	YFSPQMLRTG	NNFEITYSFE	KVPFHSMYAH	420
SQSLDRLMNP	LIDQYLWGLQ	STTTGTTLNA	GTATTNFTKL	RPTNFSNFKK	NWLPGPSIKQ	480
QGFSTKATNQ	YKIPATGSDS	LIKYETHSTL	DGRWSALTPG	PPMATAGPAD	SKFSNSQLIF	540
AGPKQNGNTA	TVPGTLIFTS	EEELAATNAT	DTDWNGNLPG	GDQSNNSNLT	VDRLTALGAV	600

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PGMVWQNRDI	YYQGPIWAKI	PHTDGHFHP	PLIGGFGLKH	PPQIFIKNT	PVPANPATT	660
SSTPVNSFIT	QYSTGQVSQ	IDWEIQKERS	KRWNPVQFT	SNYQQNSLL	WAPDAAGKYT	720
EPRAIGTRYL	THHL					734

SEQ ID NO: 75 moltype = AA length = 724
 FEATURE Location/Qualifiers
 source 1..724
 mol_type = protein
 organism = Adeno-associated virus 5

SEQUENCE: 75

MSFVDHPPDW	LEEVEGLRE	FLGLEAGPPK	PKPNQQHQDQ	ARGLVLPGYN	YLGPGNGLDR	60
GEPVNRADVE	AREHDISYNE	QLEAGDNPYL	KYNHADADEFQ	EKLADDTSG	GNLGKAVFQA	120
KKRVLEPFGL	VEEGAKTAPT	GKRIDDHFPK	RKKARTEEDS	KPSTSSDAEA	GPSGSQLQI	180
PAQPASSLGA	DTMSAGGGGP	LGDNNQGADG	VGNASGDWHC	DSTWMGDRV	TKSTRTWVLP	240
SYNNHQYREI	KSGSVDGSNA	NAYFGYSTPW	GYFDFNRFHS	HWSPRDWQL	INNHWGFRPR	300
SLRVKIFNIQ	VKEVTVDQST	TTIANNLST	VQVFTDDDYQ	LPYVVGNGTE	GCLPAFPQV	360
FTLPQYGYAT	LNRDNTENPT	ERSSFFCLEY	FPSKMLRTGN	NFEFTYNFEE	VPFHSSFAPS	420
QNLFKLANPL	VDQYLYRFVS	TNNTGGVQFN	KNLAGRYANT	YKNWFPGPMG	RTQGWNLGSG	480
VNRASVSFA	TTNMELEGA	SYQVPPQPNG	MTNNLQGSNT	YALENTMIFN	SQPANPGTTA	540
TYLEGNMLIT	SESETQPVNR	VAYNVGGQMA	TNNQSSTTAP	ATGTYNLQEI	VPGSVWMERD	600
VYLQGPWAK	IPETGAHFHP	SPAMGGFGLK	HPPPMMLIKN	TPVPGNITSF	SDVPVSSFIT	660
QYSTGQVTVE	MEWELKKENS	KRWNPETQYT	NNYNDPQFVD	FAPDSTGEYR	TTRPIGTRYL	720
TRPL						724

SEQ ID NO: 76 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus 6

SEQUENCE: 76

MAADGYLPDW	LEDNLSEGIR	EWWDLKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LRYNHADADEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPFG	LVEEGAKTAP	GKKRPVEQSP	QEPDSSSGIG	KTGQQPAKKR	LNFGQTGDSE	180
SVPDPQPLGE	PPATPAAVGP	TTMASGGGAP	MADNNEGADG	VGNASGNWHC	DSTWLGDRVI	240
TTSTRTWALP	TYNNHLYKQI	SSASTGASND	NHYFGYSTPW	GYFDFNRFHC	HFSPRDWQL	300
INNHWGFRPK	RLNFKLFNIQ	VKEVTTNDGV	TTIANNLST	VQVFSDEYQ	LPYVLGSAHQ	360
GCLPPFPADV	FMIPQYGYLT	LNNGSQAVGR	SSFYCLEYFP	SQMLRTGNMF	TFSYTFEDVP	420
FHSSYAHSQS	LDRLMNLID	QYLYLNRTO	NQSGSAQNKD	LLFSRGSPAG	MSVQPKNWLP	480
GPCYRQQRVS	KTKTDNNNSN	FTWTGASKYN	LNGRESINP	GTAMASHKDD	KDKFFPMSPV	540
MIFGESAGA	SNTALDNVMI	TDEEEIKATN	PVATERFGTV	AVNLQSSSTD	PATGDVHVMG	600
ALPGMVWQDR	DVYLQGPWA	KIPHTDGHFH	PSPLMGGFGL	KHPPPPQILIK	NTVPVPPANPA	660
EFSATKFASF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEVQ	YTSNYAKSAN	VDFTVDNNGL	720
YTEPRPIGTR	YLTRPL					736

SEQ ID NO: 77 moltype = AA length = 737
 FEATURE Location/Qualifiers
 source 1..737
 mol_type = protein
 organism = Adeno-associated virus 7

SEQUENCE: 77

MAADGYLPDW	LEDNLSEGIR	EWWDLKPGAP	KPKANQQKQD	NGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALHDKAYD	QQLKAGDNPY	LRYNHADADEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	AKKRPVEPSP	QRSPTSSTGI	GKKGQQPARK	RLNFGQTGDS	180
ESVPDPQPLG	EPPAAPSSVG	SGTVAAGGGA	PMADNNEGAD	GVGNASGNWH	CDSTWLGDRV	240
ITSTRTWAL	PTYNNHLYKQ	ISSETAGSTN	DNTYFGYSTP	WGYFDFNRFH	CHFSPRDWQR	300
LINNHWGFRP	KKLRFKLFNI	QVKEVTTNDG	VTTIANNLTS	TIQVFSDEY	QLPYVLGSAH	360
QGCLPPFPAD	VFMIPQYGYL	TLNNGSQSVG	RSSFYCLEYF	PSQMLRTGN	FEFSYSFEDV	420
PFHSSYAHSQ	SLDRLMNLPI	DQYLYYLART	QSNPGGTAGN	RELQFYQGGP	STMAEQAKNW	480
LPGPCFRQQR	VSKTLDQNNN	SNFAWTGATK	YHLNGRNSLV	NPGVAMATHK	DDEDRFFPSS	540
GVLIPGKTGA	TNKTLENLV	MTNEEEIRPT	NPVATEEYGI	VSSNLQAANT	AAQTQVVNNQ	600
GALPGMVWQN	RDVYLQGPW	AKIPHDTGNF	HPSPLMGGFG	LKHPPPPQILI	KNTPVPANPP	660
EVFTPAKFAS	FITQYSTGQV	SVEIEWELQK	ENSKRWNPET	QYTSNFEKQT	GVDFAVDSQG	720
VYSEPRPIGT	RYLTRNL					737

SEQ ID NO: 78 moltype = AA length = 738
 FEATURE Location/Qualifiers
 source 1..738
 mol_type = protein
 organism = Adeno-associated virus 8

SEQUENCE: 78

MAADGYLPDW	LEDNLSEGIR	EWWDLKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALHDKAYD	QQLQAGDNPY	LRYNHADADEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLEPLG	LVEEGAKTAP	GKKRPVEPSP	QRSPTSSTGI	GKKGQQPARK	RLNFGQTGDS	180
ESVPDPQPLG	EPPAAPSSVG	PNTMAAGGGA	PMADNNEGAD	GVGSSSGNWH	CDSTWLGDRV	240
ITSTRTWAL	PTYNNHLYKQ	ISNGTSGGAT	NDNTYFGYST	PWGYFDFNRF	HCHFSPRDWQ	300
RLINNHWGFR	PKRLSFKLFN	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	360

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HQCCLPPPPA	DVFMIPQYGY	LTLNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFOFTYTFED	420
VPFHSSYAHS	QSLDRLMNPL	IDQYLYLSR	TQTTGGTANT	QTLGFSQGGP	NTMANQAKNW	480
LPGPCYRQQR	VSTTTGQNNN	SNFAWTAGTK	YHLNGRNSLA	NPGIAMATHK	DDEERFFPSN	540
GILIFGKQNA	ARDNADYSV	MLTSEEEIKT	TNPVATEEYG	IVADNLQQQN	TAPQIGTVNS	600
QGALPGMVWQ	NRDVYLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPPPQIL	IKNTPVPADP	660
PTTFNQSKLN	SFITQYSTGQ	VSVEIEWELQ	KENSKRWNPE	IQYTSNYYKS	TSVDFAVNTE	720
GVYSEPRPIG	TRYLTRNL					738

SEQ ID NO: 79 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus 9

SEQUENCE: 79

MAADGYLPDW	LEDNLSEGIR	EWWALKPGAP	QPKANQQHQD	NARGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGAQPAKKR	LNFGQGTGDE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDFNRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVI	TNEEEIKTTN	PVATESYGGV	ATNHQSAQAQ	AQTGWVQNQG	600
ILPGMWQDR	DVYLQGGPIA	KIPHTDGNFH	PSPLMGGFGM	KHPPPPQILIK	NTPVPADPPT	660
AFNKDKLNSF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPETQ	YTSNYYKSNN	VEFAVNTEGV	720
YSEPRPIGTR	YLTRNL					736

SEQ ID NO: 80 moltype = AA length = 738
 FEATURE Location/Qualifiers
 source 1..738
 mol_type = protein
 organism = Adeno-associated virus

SEQUENCE: 80

MAADGYLPDW	LEDNLSEGIR	EWWDLPKPGAP	KPKANQQKQD	DGRGLVLPGY	KYLGPFNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LRYNHADAEF	QERLQEDTSF	GGNLGRAVFQ	120
AKKRVLLEPLG	LVEEAAKTAP	GKKRPVEPSP	QRSPTSSTGI	GKKGQPPAKK	RLNFGQTGDS	180
ESVPDPQPIG	EPPAGPSGLG	SGTMAAGGGA	PMADNNEGAD	GVGSSSGNWH	CDSTWLGDRV	240
ITSTRTWAL	PTNNHLYKQ	ISNGTSGGST	NDNTYFGYST	PWGYFDFNRF	HCHFSPRDWQ	300
RLINNNGWFR	KRLNFKLFNI	IQVKEVTQNE	GTKTIANNLT	STIQVFTDSE	YQLPYVLGSA	360
HQCCLPPPPA	DVFMIPQYGY	LTLNNGSQAV	GRSSFYCLEY	FPSQMLRTGN	NFEFSYQFED	420
VPFHSSYAHS	QSLDRLMNPL	IDQYLYLSR	TQSTGGTAGT	QQLLFSQAGP	NNMSAQAKNW	480
LPGPCYRQQR	VSTTLNQNNN	SNFAWTGATK	YHLNGRDSL	NPVAMATHK	DDEERFFPSS	540
GVLMPGKQGA	GKDNVDYSSV	MLTSEEEIKT	TNPVATEQYG	VVADNLQQQN	AAPIVGAVNS	600
QGALPGMVWQ	NRDVYLQGP	WAKIPHTDGN	FHPSPLMGGF	GLKHPPPPQIL	IKNTPVPADP	660
PTTFSQAKLA	SFITQYSTGQ	VSVEIEWELQ	KENSKRWNPE	IQYTSNYYKS	TNVDFAVNTD	720
GTYSERPIGTR	TRYLTRNL					738

SEQ ID NO: 81 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus

SEQUENCE: 81

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120
AKKRLLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQGTGDE	180
SVPDPQPIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDFNRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGGQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVI	TNEEEIKTTN	PVATESYGGV	ATNHQSAQAQ	AQTGWVQNQG	600
ILPGMWQDR	DVYLQGGPIA	KIPHTDGNFH	PSPLMGGFGM	KHPPPPQILIK	NTPVPADPPT	660
AFNKDKLNSF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPETQ	YTSNYYKSNN	VEFAVSTEGV	720
YSEPRPIGTR	YLTRNL					736

SEQ ID NO: 82 moltype = AA length = 736
 FEATURE Location/Qualifiers
 source 1..736
 mol_type = protein
 organism = Adeno-associated virus

SEQUENCE: 82

MAADGYLPDW	LEDTLSEGIR	QWWKLKPGPP	PPKPAERHKD	DSRGLVLPGY	KYLGPGNGLD	60
KGEPVNAADA	AALSHDKAYD	QQLKAGDNPY	LKYNHADAEF	QERLKEDTSF	GGNLGRAVFQ	120

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AKKRLLEPLG	LVEEAAKTAP	GKKRPVEQSP	QEPDSSAGIG	KSGSQPAKKK	LNFGQTGDTE	180
SVDPDPQIGE	PPAAPSGVGS	LTMASGGGAP	VADNNEGADG	VGSSSGNWHC	DSQWLGDRI	240
TTSTRTWALP	TNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDNRFH	CHFSPRDWQR	300
LINNNGWFRP	KRLNFKLFNI	QVKEVTDNNG	VKTIANNLTS	TVQVFTDSY	QLPYVLGSAH	360
EGCLPPFPAD	VFMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	PSQMLRTGNN	FQFSYEFENV	420
PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT	INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	480
GPSYRQQRVS	TTVTQNNNSE	FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG	EDRFFPLSGS	540
LIFGKQGTGR	DNVDADKVM	TNEEEIKTTN	PVATESYGVQ	ATNHQSAQAQ	AQTGWVQNQG	600
ILPGMVWQDR	DVYLQGPWA	KIPHTDGNFH	PSPLMGGFGM	KHPPPPQILIK	NTPVPADPPT	660
AFNKDKLNSF	ITQYSTGQVS	VEIEWELQKE	NSKRWNPEIQ	YTSNYYKSNN	VEFAVNTEGV	720
YSEPRPIGTR	YLTRNL					736

SEQ ID NO: 83 moltype = length =
 SEQUENCE: 83
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SEQ ID NO: 84 moltype = length =
 SEQUENCE: 84
 000

SEQ ID NO: 85 moltype = length =
 SEQUENCE: 85
 000

SEQ ID NO: 86 moltype = length =
 SEQUENCE: 86
 000

SEQ ID NO: 87 moltype = length =
 SEQUENCE: 87
 000

SEQ ID NO: 88 moltype = length =
 SEQUENCE: 88
 000

SEQ ID NO: 89 moltype = length =
 SEQUENCE: 89
 000

SEQ ID NO: 90 moltype = length =
 SEQUENCE: 90
 000

SEQ ID NO: 91 moltype = length =
 SEQUENCE: 91
 000

SEQ ID NO: 92 moltype = length =
 SEQUENCE: 92
 000

SEQ ID NO: 93 moltype = length =
 SEQUENCE: 93
 000

SEQ ID NO: 94 moltype = length =
 SEQUENCE: 94
 000

SEQ ID NO: 95 moltype = length =
 SEQUENCE: 95
 000

SEQ ID NO: 96 moltype = length =
 SEQUENCE: 96
 000

SEQ ID NO: 97 moltype = length =
 SEQUENCE: 97
 000

SEQ ID NO: 98 moltype = length =
 SEQUENCE: 98
 000

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SEQ ID NO: 99      moltype =   length =
SEQUENCE: 99
000

SEQ ID NO: 100     moltype =   length =
SEQUENCE: 100
000

SEQ ID NO: 101     moltype = DNA length = 744
FEATURE
misc_feature       682..744
                    note = /note="This region may be absent in its entirety"
variation          691..693
                    replace = ctg
misc_feature       694..744
                    note = /note="This region may be absent in its entirety"
misc_feature       712..744
                    note = /note="This region may be absent in its entirety"
misc_feature       1..744
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature       1..744
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature       1..744
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source             1..744
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 101
gtgcagctgg tggagagcgg cgggcgcggtg gtgcagcccg gcagaagcct gagactgagc 60
tgcgccgccca gcggcttcgc cttcagcagc tacggcatgc actgggtgag acaggccccc 120
ggcaaggggcc tggagtgggt gggcgtgatc tggttcgacg gcaccaagaa gtactacacc 180
gacagcgtga agggcagatt caccatcagc agagacaaca gcaagaacac cctgtacctg 240
cagatgaaca cctgagagc cgaggacacc gccgtgtact actgcgccag agacagaggc 300
atcgcgccca gaagaggccc ctactacatg gacgtgtggg gcaagggcac caccgtgacc 360
gtgagcagcg ccagcaccaa gggcccccag gtgttcccc tggcccccag cagcaagagc 420
accagcgggc gcaccgcgcg cctgggctgc ctgggtgaagg actacttccc cgagcccgctg 480
accgtgagct ggaacagcgg cgccctgacc agcggcgctg acaccttccc cgccgtgctg 540
cagagcagcg gcctgtacag cctgagcagc gtgggtgaccg tgcccagcag cagcctgggc 600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaga 660
gtggagccca agagctgcga caagacccac acctgcccc cctgccccgc ccccagagctg 720
ctggcgggcc ccagcgtgtt cctg                                     744

SEQ ID NO: 102     moltype = DNA length = 642
FEATURE
misc_feature       1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source             1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 102
gacatccaga tgaccagag cccacagcag ctgagcgcca gcgtgggcga cagagtgacc 60
atcacctgca gagccagcca gagcatcagc agctacctga actggtacca gcagaagccc 120
ggcaaggggcc ccaagctgct gatctacgcc gccagcagcc tgcagagcgg cgtgcccagc 180
agattcagcg gcagcggcag cggcaccgac ttcaccctga ccacacagcag cctgcagccc 240
gaggacttcg ccacctacta ctgccagcag agctacagca ccccccgtgac cttcgcgggc 300
ggcaccgaag tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
cccagagagg ccaaggtgca gtggaagggt gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcgaag ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc                                     642

SEQ ID NO: 103     moltype = DNA length = 702
FEATURE
misc_feature       643..702
                    note = /note="This region may be absent in its entirety"
misc_feature       670..702
                    note = /note="This region may be absent in its entirety"
misc_feature       1..702
                    note = source = /note="Description of Artificial Sequence:

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misc_feature Synthetic polynucleotide"
1..702
note = source = /note="See specification as filed for detailed description of substitutions and preferred embodiments"
source 1..702
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 103
gaggtgcagc tgggtggagag cggcgccggc ctggtgcagc cggcgccgag cctgagactg 60
agctgcgcgc ccagcggtt caccctcagc agctacggca tgagctgggt gagacaggcc 120
cccgcaagg gcctggagct ggtggccagc atcaacagca acggcgccag cactactac 180
ccgacagcg tgaagggcag attcaccatc agcagagaca acgccaagaa cagcctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc cagcgccgac 300
tactggggcc agggcaccac cgtgaccgtg agcagcgcca gcaccaaggg cccagcgtg 360
ttcccccctgg cccctgcagc cagaagcacc agcagagaca ccgcccctt gggtgctctg 420
gtgaaggact acttcccga gccctgacc gtgagctgga acagcgccgc cctgaccagc 480
ggcgtgcaca ccttcccgc cgtgctgcag agcagcgcc tgtacagcct gacgagcgtg 540
gtgaccgtgc ccagcagcag cctgggcacc aagacctaca cctgcaactt ggaccacaag 600
ccagcaaca ccaaggtgga caagagagtg gagagcaagt acggccccc ctgcccccc 660
tgccccccc ccgagttcct gggcgccccc agcgtgttcc tg 702

SEQ ID NO: 104 moltype = DNA length = 657
FEATURE Location/Qualifiers
misc_feature 1..657
note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"
source 1..657
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 104
gacatcgtga tgaccagag cccctcagc ctgccctga ccccgccga gcccgccagc 60
atcagctgca gaagcagcca gagcctggtg tacagcaacg gcgacaccta cctgcactgg 120
tacctgcaga agcccgcca gagccccag ctgctgatct acaaggtgag caacagattc 180
agcgcgctgc ccgacagatt cagcgccagc ggcagcgcca ccgacttcac cctgaagatc 240
agcagagtg aggcagagga cgtggcgctg tactactgca gccagagcac ccacgtgccc 300
tggaccttgc gccagggcac caaggtggag atcaagagaa ccgtggccgc cccagcgtg 360
ttcatcttcc ccccagcga cgagcagctg aagagcgcca ccgcccagct ggtgtgacct 420
ctgaacaact tctaccacc agaggccaag gtgcagtgga aggtggacaa cgccctgcag 480
agcgccaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540
agcagcacc tgaccctgag caaggccgac tacgagaagc acaaggtgta cgccctgcag 600
gtgaccacc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc 657

SEQ ID NO: 105 moltype = DNA length = 752
FEATURE Location/Qualifiers
misc_feature 691..752
note = /note="This region may be absent in its entirety"
misc_feature 700..702
note = Variation - /replace="ctg"
misc_feature 703..752
note = /note="This region may be absent in its entirety"
misc_feature 721..752
note = /note="This region may be absent in its entirety"
misc_feature 1..752
note = /note="Variant nucleotides given in the sequence have no preference with respect to those in the annotations for variant positions"
misc_feature 1..752
note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"
misc_feature 1..752
note = source = /note="See specification as filed for detailed description of substitutions and preferred embodiments"
source 1..752
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 105
caggtggagc tgggtggagag cggcgccggc ctggtgcagc cggcgccgag cctgagactg 60
agctgcgcgc ccagcggtt caccctcagc agctacggca tgagctgggt gagacaggcc 120
cccgcaagg gcctggagtg ggtgagcgcc atcaacggca gcggcaccag aacctactac 180
gccgacagcg tgaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc cagaggcaag 300
ggcaacaccc acaagcccta cggctacgtg agatacttcg acgtgtgggg ccagggcacc 360
ctgggtgacc tgagcagcgc cagcaccaag ggccccagcg tgttccccct ggccccagc 420
agcaagagca ccagcgccg caccgcccgc ctgggctgccc tgggtgaagga ctacttccc 480

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gagcccggtga ccgtgagctg gaacagcggc gccctgacca gcggcggtgca caccctcccc 540
gccgtgctgc agagcagcgg cctgtacagc ctgagcagcg tgggtgacgt gccagcagc 600
agcctgggca cccagaccta catctgcaac gtgaaccaca agcccagcaa caccaggtg 660
gacaagaagg tggagcccaa gagctgcgac aagaccaca cctgcccccc ctgccccgcc 720
ccgagctgct gggcgggccc agcgtgttcc tg 752

SEQ ID NO: 106      moltype = DNA length = 645
FEATURE            Location/Qualifiers
misc_feature       1..645
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..645
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 106
gacatcgtgc tgacccagag ccccgccacc ctgagcctga gcccgcgga gagagccacc 60
ctgagctgca gagccagcca gagcgtgagc agcagctacc tggcctggta ccagcagaag 120
cccgccagg cccccagact gctgatctac ggcgccagca gcagagccac cggcgtgccc 180
gccagattca gcggcagcgg cagcggcacc gacttcacc tgaccatcag cagcctggag 240
ccgaggact tcgccaccta ctactgctg cagatctaca acatgcccat caccctcggc 300
cagggcacca aggtggagat caagagaacc gtggccgccc ccagcgtgtt catcttcccc 360
ccagcgcagc agcagctgaa gagcggcacc gccagcgtgt tgtgctgtgt gaacaacttc 420
taccgccag agggcgaagt gcagtggaa gtggacaacg ccctgcagag cggcaacagc 480
caggagagcg tgaccgagca ggacagcaag gacagcacct acagcctgag cagcaccctg 540
accctgagca aggcgagcta cgagaagcac aaggtgtacg cctgcgaggt gaccaccag 600
ggcctgagca gccccgtgac caagagcttc aacagaggcg agtgc 645

SEQ ID NO: 107      moltype = DNA length = 741
FEATURE            Location/Qualifiers
misc_feature       682..741
                   note = /note="This region may be absent in its entirety"
misc_feature       709..741
                   note = /note="This region may be absent in its entirety"
misc_feature       1..741
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
misc_feature       1..741
                   note = source = /note="See specification as filed for
                   detailed description of substitutions and preferred
                   embodiments"
source            1..741
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 107
gaggtgcagc tgggtggagag cggcgggcggc ctggagcagc ccggcggcag cctgagactg 60
agctgcgcgc gcagcggctt caccctcaga gactacgcca tgacctgggt gagacaggcc 120
cccgcaagg gcctggagtg ggtgagcagc atcagcggca gcggcgga caccctactac 180
gccgacagcg tgaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccagggac accgcgtgt actactgcgc caaggacaga 300
ctgagcatca ccatcagacc cagatactac ggcctggagc tgtggggcca gggcaccacc 360
gtgaccgtga gcagcgccag caccagggc cccagcgtgt tccccctggc cccctgcagc 420
agaagcacca gcagagcac cgccgcccct ggctgccttg tgaaggacta cttccccgag 480
cccgtagcgc tgagctggaa cagcggcgcc ctgaccagcg gcgtgcacac cttccccgcc 540
gtgctgcaga gcagcgccct gtacagcctg agcagcgtgg tgaccgtgcc cagcagcagc 600
ctgggcacca agacctacac ctgcaacgtg gaccacaagc ccagcaacac caaggtggac 660
aagagagtgg agagcaagta cggccccccc tgccccccct gccccgcccc cgagtctctg 720
ggcgcccca gcgtgttct g 741

SEQ ID NO: 108      moltype = DNA length = 657
FEATURE            Location/Qualifiers
misc_feature       1..657
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..657
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 108
gacatcgtga tgacccagag cccctgagc ctgccctga ccccgcgga gcccgccagc 60
atcagctgca gaagcagcca gagcctgctg tacagcatcg gctacaacta cctggactgg 120
tacctgcaga agagcgccca gagccccag ctgctgatct acctgggag caacagagcc 180
agcggcgctgc ccagacgatt cagcggcagc ggcagcgcca ccgacttcac cctgaagatc 240
agcagagtgg aggcagagga cgtgggcttc tactactgca tgcaggccct gcagaccccc 300
tacaccttcg gccaggccac caagctggag atcaagagaa ccgtggccgc cccagcgtg 360
ttcatcttcc cccccagcga cgagcagctg aagagcgcca ccgcccagct ggtgtgcctg 420
ctgaacaact tctaccccc agaggccaag gtgcagtgga aggtggacaa cgccctgcag 480
agcggaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540

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agcagcaccc tgaccctgag caaggccgac tacgagaagc acaaggtgta cgccctgcgag 600
gtgacccacc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc 657

SEQ ID NO: 109      moltype = DNA length = 723
FEATURE            Location/Qualifiers
misc_feature        664..723
                    note = /note="This region may be absent in its entirety"
misc_feature        691..723
                    note = /note="This region may be absent in its entirety"
misc_feature        1..723
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..723
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..723
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 109
caggtgcagc tgggtgcagag cggcgccgag gtgaagaagc ccggcagcag cgtgaagggtg 60
agctgcaagg ccagcgggcta cagcttcacc gactaccaca tccactgggt gagacaggcc 120
cccgccagg ccctggagtg gatgggctg atcaacccca tgtacggcac caccgactac 180
aaccagagat tcaagggcag agtgaccatc accgcccagc agagcaccag caccgcctac 240
atggagctga gcagcctgag aagcagaggac accgcccgtg actactgcgc cagatacgac 300
tacttcaccg gcaccggcgt gtactggggc cagggcaccc tggtgaccgt gagcagcgcc 360
agcaccgaag gccccagcgt gttccccctg gccccctgca gcagaagcac cagcagagagc 420
accgcccggc tgggctgctt ggtgaaggac tacttccccg agcccgtgac cgtgagctgg 480
aacagcggcg ccctgaccag cggcgtgcac accttccccg ccgtgctgca gagcagcgcc 540
ctgtacagcc tgagcagcgt ggtgaccgtg cccagcagca gcctgggac caagacctac 600
acctgcaacg tggaccacaa gccagcaaac accaaggtgg acaagagagt ggagagcaag 660
tacggcccc cctgcccccc ctgccccgcc cccgaggttc tgggcggccc cagcgtgttc 720
ctg 723

SEQ ID NO: 110      moltype = DNA length = 657
FEATURE            Location/Qualifiers
misc_feature        1..657
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..657
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 110
gacatcgtga tgaccagac cccctgagc ctgagcgtga ccccgccca gcccgcagc 60
atcagctgca gaagcagcag aagcctggtg cacagcagag gcaacaccta cctgcactgg 120
tacctgcaga agcccgccca gagccccag ctgctgatct acaaggtgag caacagattc 180
atcggcgtgc ccgacagatt cagcggcagc ggcagcggca ccgacttcac cctgaagatc 240
agcagagtgg aggcgagga cgtgggctg tactactgca gccagagcac ccacctgccc 300
ttcaccttgc cccaggggc cagctggag atcaagagaa ccgtggccgc cccagcgtg 360
ttcatcttcc cccccagcga cgagcagctg aagagcggca ccgcccagct ggtgtgcctg 420
ctgaacaact tctaccccag agaggccaag gtgcagtgga aggtggacaa cgccctgcag 480
agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540
agcagcaccc tgaccctgag caaggccgac tacgagaagc acaaggtgta cgccctgcgag 600
gtgacccacc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc 657

SEQ ID NO: 111      moltype = DNA length = 755
FEATURE            Location/Qualifiers
misc_feature        694..755
                    note = /note="This region may be absent in its entirety"
variation          703..705
                    replace = ctg
misc_feature        706..755
                    note = /note="This region may be absent in its entirety"
misc_feature        724..755
                    note = /note="This region may be absent in its entirety"
misc_feature        1..755
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..755
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..755
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"

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source                1..755
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 111
gaggtgcagc tgggtgagag cggcgccggc ctggtgcagc cggcgccgag cctgagactg 60
agctgcgcgc ccagcggcgt caccctcagc aactactgga tgaactgggt gagacaggcc 120
cccgccaagg gcctggagtg ggtggccggc atcaaccagg acggcagcga gaagtactac 180
gtgggcagcg tgaagggcag attcaccatc agcagagaca acgccaagaa cagcctgtac 240
ctgcagatga acagcctgag agtggaggac accgccgtgt actactgcgt gagagactac 300
tacgacatcc tgaccgacta ctacatccac tactggtact tcgacctgtg gggcagaggc 360
accctggtag ccgtgagcag cggcagcacc aaggggccca gcgtgttccc cctggccccc 420
agcagcaaga gcaccagcgg cggcaccggc gccctgggct gcctggtgaa ggactacttc 480
cccgagcccg tgaccgtgag ctggaacagc ggcgccttga ccagcggcgt gcacaccttc 540
ccgcctgtgc tgcagagcag cggcctgtac agcctgagca gcgtggtgac cgtgcccagc 600
agcagcctgg gcacccagac ctacatctgc aacgtgaacc acaagcccag caacaccaag 660
gtggacaaga gagtggagcc caagagctgc gacaagaccc acacctgccc cccctgcccc 720
gccccgagct gctggggcgg cccagcgtgt tectg                                     755

SEQ ID NO: 112        moltype = DNA length = 645
FEATURE               Location/Qualifiers
misc_feature          1..645
                      note = source = /note="Description of Artificial Sequence:
                      Synthetic polynucleotide"
source                1..645
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 112
gagatcgtgc tgaccagagag ccccggcacc ctgagcctga gcccggcgca gagagccacc 60
ctgagctgca gagccagcca gagcgtgagc agcagctacc tggcctggta ccagcagaag 120
cccgccagag cccccagact gctgatctac ggcgccagca gcagagccac cggcatcccc 180
gacagattca gcggcagcgg cagcggcacc gacttcacc tgaccatcag cagactggag 240
cccgaggact tcgccgtgta ctactgccag cagtacggca gcagccctg caccctcggc 300
cagggcacca gactggagat caagagaacc gtggccgccc ccagcgtgtt catcttcccc 360
cccagcgacg agcagctgaa gagcggcacc gccagcgttg tgtgcctgct gaacaacttc 420
taccgccagag agggccaaggt gcagtggaa gtggacaacg ccctgcagag cggcaacagc 480
caggagagcg tgaccagagca ggacagcaag gacagcacct acagcctgag cagcacccctg 540
accctgagca agggcgacta cgagaagcac aaggtgtacg cctgcgaggt gaccaccag 600
ggcctgagca gcccctgtac caagagcttc aacagaggcg agtgc                                     645

SEQ ID NO: 113        moltype = DNA length = 732
FEATURE               Location/Qualifiers
misc_feature          670..732
                      note = /note="This region may be absent in its entirety"
variation            679..681
                      replace = ctg
misc_feature          682..732
                      note = /note="This region may be absent in its entirety"
misc_feature          700..732
                      note = /note="This region may be absent in its entirety"
misc_feature          1..732
                      note = /note="Variant nucleotides given in the sequence
                      have no preference with respect to those in the
                      annotations for variant positions"
misc_feature          1..732
                      note = source = /note="Description of Artificial Sequence:
                      Synthetic polynucleotide"
misc_feature          1..732
                      note = source = /note="See specification as filed for
                      detailed description of substitutions and preferred
                      embodiments"
source                1..732
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 113
gaggtgcagc tgggtgcagag cggcgccgag gtgaagaagc cggcgagag cctgaagatc 60
agctgcaagg gcagcggcta cagcttcacc acctactggc tgggctgggt gagacagatg 120
cccgccaagg gcctggactg gatcggcatc atgagccccg tggacagcga catcagatac 180
agccccagct tccagggcca ggtgaccatg agcgtggaca agagcatcac caccgcctac 240
ctgcagtgga acagcctgaa ggccagcgac accgccatgt actactgcgc cagaagaaga 300
cccgccagag gctacttcga cttctggggc cagggcaccc tggtgaccgt gaggcagcgc 360
agcaccgaag gcccagcgtg gttcccctg gccccagca gcaagagcac cagcggcggc 420
accgcgcgcc tgggctgctt ggtgaaggac tacttcccgc agcccgtagc cgtgagctgg 480
aacagcggcg ccctgaccag cggcgtgcac accttcccgc ccgtgctgca gaggcagcgc 540
ctgtacagcc tgagcagcgt ggtgaccgtg cccagcagca gcctgggac ccagacctac 600
atctgcaacg tgaaccacaa gccacagcaac accaagggtg acaagagagt ggagcccaag 660
agctgcgaca agaccacac ctgccccccc tgccccggcc cagagctgct gggcgccccc 720

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agcgtgttcc tg                                     732

SEQ ID NO: 114      moltype = DNA  length = 642
FEATURE
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 114
gacatccaga tgacccagag cccccagcagc ctgagcgcca gcgtgggcga cagagtgcacc 60
atcacctgca gagccagcca gggcatcagc agctggctgg cctgggtacca gcagaagccc 120
gagaaggccc ccaagagcct gatctacgcc gccagcagcc tgcagagcgg cgtgcccagc 180
agattcagcg gcagcggcag cggcaccgac ttcaccctga ccatcagcag cctgcagccc 240
gaggacttcg ccacctacta ctgccagcag tacaacatct accctacac cttcggccag 300
ggcaccgaagc tggagatcaa gagaaccgtg gccgcccaca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactcttac 420
cccagagagg ccaaggtgca gtggaagggt gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc                                     642

SEQ ID NO: 115      moltype = DNA  length = 732
FEATURE
misc_feature        670..732
                    note = /note="This region may be absent in its entirety"
variation           679..681
                    replace = ctg
misc_feature        682..732
                    note = /note="This region may be absent in its entirety"
misc_feature        700..732
                    note = /note="This region may be absent in its entirety"
misc_feature        1..732
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..732
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..732
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..732
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 115
caggtgaccc tgagagagag cggccccgcc ctggtgaagc ccaccagac cctgaccctg 60
acctgcaccg tgagcggctt cagcctgacc agctacagcg tgcactgggt gagacagccc 120
cccgcaaggc gcctggagtg gctgggcgtg atctgggcca gccgcggcac cgactacaac 180
agcgccctga tgagcagact gagcatcagc aaggacacca gcagaaacca ggtggtgctg 240
accatgacca acatggaccc cgtggacacc gccacctact actgcgccag agaccccccc 300
agcagcctgc tgagactgga ctactggggc agaggcaccc ccgtgaccgt gagcagcgcc 360
agcaccaaagg gccccagcgt gttccccctg gccccagca gcaagagcac cagcggcggc 420
accgcccccc tgggctgect ggtgaaggac tacttccccg agcccgtgac cgtgagctgg 480
aacagcggcg ccctgaccag cggcgtgcac accttccccg ccgtgctgca gagcagcggc 540
ctgtacagcc tgagcagcgt ggtgaccgtg cccagcagca gcctgggcac ccagacctac 600
atctgcaacg tgaaccacaa gccacgcaac accaaggtgg acaagagagt ggagcccaag 660
agctgcgaca agacccacac ctgccccccc tgccccgccc ccgagctgct gggcgggccc 720
agcgtgttcc tg                                     732

SEQ ID NO: 116      moltype = DNA  length = 657
FEATURE
misc_feature        1..657
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..657
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 116
gacatcgtga tgacccagag ccccgacagc ctggcctgta gcctgggcga gagagccacc 60
atcaactgca agagcagcca gagcctgctg aacagcggca accagaagaa ctacctggcc 120
tggcagcaga agccccggca gccccccaag ctgctgatct acggcgccag caccagagag 180
agcggcgtgc ccgacagatt cagcggcagc ggcagcggca ccgacttcac cctgaccatc 240
agcagcctgc aggcggagga cgtggccgtg tactactgcc agaactgca cagcttcccc 300

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ttcaccttcg cggcgccgac caagctggag atcaagagaa cgtggccgc cccagcgtg 360
ttcatcttcc cccccagcga cgagcagctg aagagcggca cgcacagcgt ggtgtgcctg 420
ctgaacaact tctaccccgag agaggccaag gtgcagtggg aggtggacaa cgccctgcag 480
agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540
agcagcacc tgacctgag caaggccgac tacgagaagc acaagggtga cgctgcgag 600
gtgacccacc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc 657

SEQ ID NO: 117      moltype = DNA length = 738
FEATURE            Location/Qualifiers
misc_feature        676..738
                    note = /note="This region may be absent in its entirety"
variation           685..687
                    replace = ctg
misc_feature        688..738
                    note = /note="This region may be absent in its entirety"
misc_feature        706..738
                    note = /note="This region may be absent in its entirety"
misc_feature        1..738
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..738
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..738
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..738
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 117
cagggtgcagc tgggtgcagag cggcgccgag gtgaagaagc cggcgccag cgtgaagggtg 60
agctgcaagg gcagcggcta caccttcacc agctactgga tgcactgggt gagacaggcc 120
cccgccaga gactggagtg gatcgccgag atcgacccca gcgagagcaa caccaactac 180
aaccagaagt tcaagggcag agtgaccctg accgtggaca tcagcgccag caccgcctac 240
atggagctga gccacctgag aagcgaggac accgcctgt actactgccc cagaggcggc 300
tacgacggct ggcactacgc catcgactac tggggccagg gcacctgggt gaccgtgagc 360
agcggccagc ccaagggccc cagcgtgttc cccctggccc ccagcagcaa gaggcaccagc 420
ggcgccaccg ccgcctctgg ctgcctgtgt aaggactact tcccagagcc cgtgaccgtg 480
agctggaaca gcggcgccct gaccagcggc gtgcacacct tcccgcctg gctgcagagc 540
agcggcctgt acagcctgag cagcgtgtgt accgtgcccc gcagcagcct gggcaccagc 600
acctacatct gcaacgtgaa ccacaagccc agcaacacca aggtggacaa gaagggtggag 660
cccaagagct gccacaagac ccacacctgc ccccctgcc ccgcccgcga gctggccggc 720
gccccagcg tgttctcg 738

SEQ ID NO: 118      moltype = DNA length = 657
FEATURE            Location/Qualifiers
misc_feature        1..657
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..657
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 118
gacgtggtga tgaccagag cccctgagc ctgcccgtga ccccgccga gcccgccagc 60
atcagctgca gaagcagcca gagcctggcc aagagctacg gcaacacct cctgagctgg 120
tacctgcaga agcccgcca gagccccag ctgctgatct acggcatcag caacagattc 180
agcggcgtgc ccgacagatt cagcggcagc ggcagcggca ccgacttcac cctgaagatc 240
agcagagtgg aggcgagga cgtgggcgtg tactactgcc tgcagggcac ccaccagccc 300
tacaccttcg gccaggccac caagggtggag atcaagagaa cgtggccgc cccagcgtg 360
ttcatcttcc cccccagcga cgagcagctg aagagcggca cgcacagcgt ggtgtgcctg 420
ctgaacaact tctaccccgag agaggccaag gtgcagtggg aggtggacaa cgccctgcag 480
agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540
agcagcacc tgacctgag caaggccgac tacgagaagc acaagggtga cgctgcgag 600
gtgacccacc agggcctgag cagccccgtg accaagagct tcaacagagg cgagtgc 657

SEQ ID NO: 119      moltype = DNA length = 735
FEATURE            Location/Qualifiers
misc_feature        676..735
                    note = /note="This region may be absent in its entirety"
misc_feature        703..735
                    note = /note="This region may be absent in its entirety"
misc_feature        1..735
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

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misc_feature 1..735
note = source = /note="See specification as filed for detailed description of substitutions and preferred embodiments"

source 1..735
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 119

cagggtgcagc	tgggtgcagag	cgggcgccgag	gtgaagaagc	cgggcgccag	cgtgaagggtg	60
agctgcaagg	ccagcggctt	caacatcaag	gacacctaca	tccactgggt	gagacaggcc	120
cccgccagag	gactggagtg	gatgggcaga	atcgaccccg	ccaacggcta	caccaagtac	180
gacccaagt	tccagggcag	agtgaccatc	accgcccaga	ccagcgccag	caccgcctac	240
atggagctga	gcagcctgag	aagcagggac	accgccgtgt	actactgcgc	cagagagggc	300
tactacggca	actacggcgt	gtacgccatg	gactactggg	gccagggcac	cctggtgacc	360
gtgagcagcg	ccagcaccaa	gggcccacgc	gtgttccccc	tggcccctcg	cagcagaagc	420
accagcgaga	gcaccgccc	cctgggctgc	ctggtgaagg	actacttccc	cgagcccggtg	480
accgtgagct	ggaacagcgg	cgcctgacc	agcggcgtgc	acaccttccc	cgccgtgctg	540
cagagcagcg	gcctgtacag	cctgagcagc	gtggtgaccg	tgcccagcag	cagcctgggc	600
accaagacct	acacctgcaa	cgtggaccac	aagcccagca	acaccaaggt	ggacaagaga	660
gtggagagca	agtacggccc	cccctgcccc	ccctgccccg	cccccgagtt	cctgggcggc	720
cccagcgtgt	tcctg					735

SEQ ID NO: 120 moltype = DNA length = 639
FEATURE Location/Qualifiers
misc_feature 1..639
note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"

source 1..639
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 120

gacatccaga	tgaccagag	ccccagcagc	ctgagcgcca	gcgtgggcga	cagagtgacc	60
atcacctgca	agaccagcca	ggacatcaac	aagtacatgg	cctggtagca	gcagaccccc	120
ggcaaggccc	ccagactgct	gatccactac	accagcgccc	tgacgcccgg	catccccagc	180
agattcagcg	gcagcggcag	cggcagagac	tacaccttca	ccatcagcag	cctgcagccc	240
gaggacatcg	ccacctacta	ctgcctgcag	tacgacaacc	tgtggacctt	cgccaggggc	300
accaaggtgg	agatcaagag	aaccgtggcc	gccccacgcg	tgttcattct	ccccccagc	360
gacgagcagc	tgaagagcgg	caccgcccgc	gtggtgtgcc	tgctgaacaa	cttctacccc	420
agagaggcca	agggtgcagt	gaagggtggc	aacgcccctg	agagcggcaa	cagccaggag	480
agcgtgaccg	agcaggacag	caaggacagc	acctacagcc	tgagcagcac	cctgaccctg	540
agcaaggccg	actacgagaa	gcacaagggt	tacgcctgcg	aggtgaccga	ccagggcctg	600
agcagccccg	tgaccaagag	cttcaacaga	ggcgagtgc			639

SEQ ID NO: 121 moltype = DNA length = 729
FEATURE Location/Qualifiers
misc_feature 667..729
note = /note="This region may be absent in its entirety"

variation 676..678
replace = ctg

misc_feature 679..729
note = /note="This region may be absent in its entirety"

misc_feature 697..729
note = /note="This region may be absent in its entirety"

misc_feature 1..729
note = /note="Variant nucleotides given in the sequence have no preference with respect to those in the annotations for variant positions"

misc_feature 1..729
note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"

misc_feature 1..729
note = source = /note="See specification as filed for detailed description of substitutions and preferred embodiments"

source 1..729
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 121

gagggtgcagc	tgggtgagag	cgggcgccggc	ctggtgcagc	cgggcgccag	cctgagactg	60
agctgcgcgc	ccagcggctt	caccttcaac	aactacgcca	tgaactgggt	gagacaggcc	120
cccgcaagg	gcttgagctt	ggtgagcacc	atcagcgcca	gcggcgccac	caccaactac	180
gcccagacgc	tgaagggcag	attcatcatc	agcagagaca	gcagcaagca	caccctgtac	240
ctgcagatga	acagcctgag	agccgaggac	accgccgtgt	actactgcgc	caaggacagc	300
aactggggca	acttcgacct	gtggggcaga	ggcaccctgg	tgaccgtgag	cagcgccagc	360
accaagggcc	ccagcgtggt	ccccctggcc	cccagcagca	agagcaccag	cggcggcacc	420
gcccgcctgg	gctgcctggt	gaaggactac	ttccccgagc	ccgtgaccgt	gagctggaac	480

-continued

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agcggcgccc tgaccagcgg cgtgcacacc ttccccgcgg tgctgcagag cagcggcctg 540
tacagcctga gcagcgtggt gaccgtgccc agcagcagcc tgggcaccca gacctacatc 600
tgcaacgtga accacaagcc cagcaacacc aaggtggaca agaaggtgga gcccagagc 660
tgcgacaaga cccacacctg cccccctg cccgcccccg agctgctggg cggccccagc 720
gtgttcctg 729

SEQ ID NO: 122      moltype = DNA length = 660
FEATURE            Location/Qualifiers
misc_feature        1..660
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..660
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 122
gacatcgtga tgaccagcgg ccccgacagc ctggccgtga gcctgggcga gagagccacc 60
atcaactgca agagcagcca gagcgtgctg tacagaagca acaacagaaa ctctctgggc 120
tggtaccagc agaagcccg cccgcccccc aacctgctga tctactgggc cagcaccaga 180
gagagcggcg tgcccgacag attcagcggc agcggcagcg gcaccgactt caccctgacc 240
atcagcagcc tgcagcggcg ggacgtggcc gtgtactact gccagcagta ctacaccacc 300
ccctacacct tcggccaggg caccaagctg gagatcaaga gaaccgtggc cgccccagc 360
gtgttcctct tccccccag cgacgagcag ctgaagagcg gcaccgccc cgtggtgtgc 420
ctgctgaaca acttctaccc cagagaggcc aaggtgcagt ggaaggtgga caacgcccgt 480
cagagcggca acagccagga gagcgtgacc gagcaggaca gcaaggacag cacctacagc 540
ctgagcagca cctgaccct gagcaaggcc gactacgaga agcacaaggt gtacgcctgc 600
gaggtgaccc accagggcct gagcagcccc gtgaccaaga gcttcaacag aggcgagtgc 660

SEQ ID NO: 123      moltype = DNA length = 693
FEATURE            Location/Qualifiers
misc_feature        661..693
                    note = /note="This region may be absent in its entirety"
misc_feature        679..693
                    note = /note="This region may be absent in its entirety"
misc_feature        1..693
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..693
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..693
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 123
gaggtgcagc tggtgcagag cggcgccgag gtgaagaagc ccggcgccag cgtgaagggtg 60
agctgcaagg ccagcggtga caccctgacc agctacggca tcagctgggt gagacaggcc 120
cccgccaggg gcctggagtg gatgggctgg gtgagcttct acaacggcaa caccactac 180
gcccagaagc tgcagggcag aggcaccatg accaccgacc ccagcaccag caccgcctac 240
atggagctga gaagcctgag aagcgacgac accgcggtg actactgcgc cagaggctac 300
ggcatggagc tgtggggcca gggcaccacc gtgaccgtga gcagcgccag caccaagggc 360
cccagcgtgt tccccctggc ccctgcagc agaagcacca gcgagagcac cgccgccctg 420
ggctgcctgg tgaagccta ctccccgag ccctgacggc tgagctggaa cagcgcgccc 480
ctgaccagcg gctgacacac ctccccgcc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcgtgg tgaccgtgcc cagcagcaac ttcggcacc agacctacac ctgcaacgtg 600
gaccacaagc ccagcaacac caaggtggac aagaccgtgg agagaaagtg ctgcgtggag 660
tgccccccct gccccgcccc ccccgctggc ggc 693

SEQ ID NO: 124      moltype = DNA length = 645
FEATURE            Location/Qualifiers
misc_feature        1..645
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..645
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 124
gagagcgccc tgaccagcgg cgccagcgtg agcggcagcc ccggccagag catcaccatc 60
agctgcaccg gcaccagcag cgacgtgggc ggctacaaca gcctgagctg gtaccagcag 120
caccgccgga agggccccaa gctgatgatc tacgaggtga gcaacagacc cagcggcgtg 180
agcaacagat tcagcggcag caagagcggc aacaccgcca gcctgacct cagcggcctg 240
caggccgagg accagggcca ctactactgc aacagctaca ccagcaccag catggtgttc 300
ggcgggcgga ccaagctgac cgtgctgggc cagcccaagg ccgccccag cgtgaccctg 360
tccccccca gcagcgagga gctgcaggcc aacaaggcca ccctgggtgt cctgatcagc 420
gacttctacc ccggcgccgt gaccgtggcc tggaaggccc acagcagccc cgtgaaggcc 480
ggcgtggaga ccaccacccc cagcaagcag agcaacaaca agtacgcgc cagcagctac 540
ctgagcctga cccccgagca gtggaagagc cacagaagct acagctgcca ggtgaccac 600

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gagggcgagca ccgtggagaa gaccgtggcc cccaccgagt gcage 645

SEQ ID NO: 125 moltype = DNA length = 744
 FEATURE Location/Qualifiers
 misc_feature 694..744
 note = /note="This region may be absent in its entirety"
 misc_feature 712..744
 note = /note="This region may be absent in its entirety"
 misc_feature 1..744
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 misc_feature 1..744
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..744
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 125

gaggtgcagc	tgggtggagag	cggcgggcggc	gtgatccagc	ccggcggcag	cctgagactg	60
agctgcgccc	ccagcggcgt	caccttcgac	gactacgcca	tgaactgggt	gagacagggc	120
cccgcaagg	gcctggagtg	ggtgagcgcc	atcagcgggc	acggcggcag	cacctactac	180
gccgacagcg	tgaagggcag	attcaccatc	agcagagaca	acagcaagaa	cagcctgtac	240
ctgcagatga	acagcctgag	agccgaggac	accgccttct	tctactgcgc	caaggacctg	300
agaaacacca	tcttcggcgt	ggtgatcccc	gacgccttcg	acatctgggg	ccagggcacc	360
atgggtgaccg	tgagcagcgc	cagcaccaag	ggccccagcg	tggtcccccct	ggccccctgc	420
agcagaagca	ccagcgagag	caccgcccgc	ctgggctgcc	tggtgaagga	ctacttcccc	480
gagcccgta	ccgtgagctg	gaacagcggc	gccctgacca	gcggcggtgca	caccttcccc	540
gccgtgctgc	agagcagcgg	cctgtacagc	ctgagcagcg	tggtgacgct	gcccagcagc	600
agcctgggca	ccaagaccta	cacctgcaac	gtggaccaca	agcccagcaa	caccaaggtg	660
gacaagagag	tggagagcaa	gtacggcccc	ccttgccccc	cctgcccgcg	ccccgagttc	720
ctggcgcgcc	ccagcgtggt	ctcg				744

SEQ ID NO: 126 moltype = DNA length = 642
 FEATURE Location/Qualifiers
 misc_feature 1..642
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..642
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 126

gacatccaga	tgaccagag	ccccagcacc	ctgagcgcca	gcgtgggcga	cagagtgacc	60
atcacctgca	gagccagcca	gagcatcaga	agctggctgg	cctgggtacca	gcagaagccc	120
ggcaaggccc	ccaagctgct	gatctacaag	gccagcagcc	tgagagcgcg	cgtgcccagc	180
agattcagcg	gcagcgccag	cggcaccgag	ttcaccctga	ccatcagcag	cctgcagccc	240
gacgacttcg	ccacctacta	ctgccagcag	tacaacagct	acagctacac	cttcggccag	300
ggcaccaaag	tggagatcaa	gagaaccgtg	gccgccccca	gcgtgttcat	cttccccccc	360
agcgacgagc	agctgaagag	cggcaccgcc	agcgtggtgt	gcctgctgaa	caactttcac	420
cccagagagg	ccaaggtgca	gtggaaggtg	gacaacgccc	tgacagcgcg	caacagccag	480
gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca	gcctgagcag	caccctgacc	540
ctgagcaagg	ccgactacga	gaagcacaa	gtgtacgcct	gcgaggtgac	ccaccagggc	600
ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt	gc		642

SEQ ID NO: 127 moltype = DNA length = 741
 FEATURE Location/Qualifiers
 misc_feature 679..741
 note = /note="This region may be absent in its entirety"
 variation 688..690
 replace = ctg
 misc_feature 691..741
 note = /note="This region may be absent in its entirety"
 misc_feature 709..741
 note = /note="This region may be absent in its entirety"
 misc_feature 1..741
 note = /note="Variant nucleotides given in the sequence
 have no preference with respect to those in the
 annotations for variant positions"
 misc_feature 1..741
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 misc_feature 1..741
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
 source 1..741

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 127

gaggtgcagc	tgctggagag	cggcgccggc	ctggtgcagc	cggcgccgag	cctgagactg	60
agctgcgcgc	ccagcggett	caccttcagc	agctacgcca	tgagctgggt	gagacaggcc	120
cccgccaagg	gcctggagtg	ggtgagcggc	atcacccgca	gcgccggcag	cacctactac	180
gcccagacgc	tgaagggcag	attcaccatc	agcagagaca	acagcaagaa	cacctgttac	240
ctgcagatga	acagcctgag	agccgaggac	accgcccgtg	actactgcgc	caaggacccc	300
ggcaccaccg	tgatcatgag	ctggttcgac	ccctggggcc	agggcaccct	ggtgaccgtg	360
agcagcgcca	gcaccaaggg	ccccagcgtg	ttccccctgg	ccccagcag	caagagcacc	420
agcgccggca	ccgcccgcct	gggctgcctg	gtgaaggact	acttcccga	gcccgtgacc	480
gtgagctgga	acagcggcgc	cctgaccagc	ggcgtgcaca	ccttcccgcg	cgtgctgcag	540
agcagcgccc	tgtacagcct	gagcagcgtg	gtgaccgtgc	ccagcagcag	cctggggcacc	600
cagacctaca	tctgcaactg	gaaccacaag	cccagcaaca	ccaaggtgga	caagaagggtg	660
gagcccaaga	gctgcgacaa	gacccacacc	tgccccccct	gccccgcccc	cgagctgctg	720
ggcgcccca	gcgtgttctc	g				741

SEQ ID NO: 128

FEATURE

misc_feature

1..645

note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"

source

1..645

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 128

gagatcgtgc	tgacccagag	ccccggcacc	ctgagcctga	gccccggcga	gagagccacc	60
ctgagctgca	gagccagcca	gagcgtgaga	ggcagatacc	tggcctggta	ccagcagaag	120
cccgccagag	ccccagact	gctgatctac	ggcgccagca	gcagagccac	cgccatcccc	180
gacagattca	gcgccagcgg	cagcggcacc	gacttcaccc	tgaccatcag	cagactggag	240
cccgaggact	tcgcccgtgt	ctactgccag	cagtagcgca	gcagcccccag	aaccttcggc	300
cagggcacca	aggtggagat	caagagaacc	gtggccgccc	ccagcgtgtt	catcttcccc	360
cccagcgacg	agcagctgaa	gagcggcacc	gccagcgtgg	tgtgcctgct	gaacaacttc	420
taccccagag	agggccaagg	gcagtggaag	gtggacaacg	ccctgcagag	cgccaacagc	480
caggagagct	tgaccgagca	ggacagcaag	gacagcacct	acagcctgag	cagcaccctg	540
accctgagca	agggcgacta	cgagaagcac	aaggtgtacg	cctgcgaggt	gacccaccag	600
ggcctgagca	gccccgtgac	caagagcttc	aacagaggcg	agtgcc		645

SEQ ID NO: 129

FEATURE

misc_feature

646..705

note = /note="This region may be absent in its entirety"

misc_feature

673..705

note = /note="This region may be absent in its entirety"

misc_feature

1..705

note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"

misc_feature

1..705

note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"

source

1..705

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 129

caggtgcagc	tggtggagag	cggcgccggc	gtggtgcagc	cggcgagaag	cctgagactg	60
gactgcaagg	ccagcggcat	caccttcagc	aacagcgcca	tgactgggtg	gagacaggcc	120
cccgccaagg	gcctggagtg	ggtggccgtg	atctggtagc	acggcagcaa	gagatactac	180
gcccagacgc	tgaagggcag	attcaccatc	agcagagaca	acagcaagaa	cacctgttcc	240
ctgcagatga	acagcctgag	agccgaggac	accgcccgtg	actactgcgc	caccaacgac	300
gactactggg	gccagggcac	ctgggtgacc	gtgagcagcg	ccagcaccaa	gggccccagc	360
gtgttcccc	tggccccctg	cagcagaagc	accagcgaga	gcaccgcccg	cctgggctgc	420
ctggtgaaag	actacttccc	cgagcccgtg	accgtgagct	ggaacagcgg	cgccctgacc	480
agcgccgtgc	acaccttccc	cgccgtgctg	cagagcagcg	gcctgtacag	cctgagcagc	540
gtggtgaccg	tgcccagcag	cagcctgggg	accaagacct	acacctgcaa	cgtggaccac	600
aagcccagca	acaccaaggt	ggacaagaga	gtggagagca	agtacggccc	ccctgcccc	660
cctgccccg	cccccgagtt	cctgggcggc	cccagcgtgt	tcctg		705

SEQ ID NO: 130

FEATURE

misc_feature

1..642

note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"

source

1..642

mol_type = other DNA

organism = synthetic construct

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SEQUENCE: 130

gagatcgtgc	tgaccagag	ccccgccacc	ctgagcctga	gccccggcga	gagagccacc	60
ctgagctgca	gagccagcca	gagcgtgagc	agctacctgg	cctggtagca	gcagaagccc	120
ggccaggccc	ccagactgct	gatctacgac	gccagcaaca	gagccaccgg	catccccgcc	180
agattcagcg	gcagcggcag	cggcacccgac	ttcacccctga	ccatcagcag	cctggagccc	240
gaggacttcg	ccgtgtacta	ctgccagcag	agcagcaact	ggcccagaac	cttcggccag	300
ggcaccaagg	tggagatcaa	gagaaccgtg	gccgccccca	gcgtgttcat	cttccccccc	360
agcgacgagc	agctgaagag	cggcacccgcc	agcgtgggtg	gcctgctgaa	caactttctac	420
cccagagagg	ccaaggtgca	gtggaagggtg	gacaacgccc	tgacagcggg	caacagccag	480
gagagcgtga	ccgagcagga	cagcaaggac	agcacctaca	gcctgagcag	caccctgacc	540
ctgagcaagg	ccgactacga	gaagcacaag	gtgtacgcct	gcgagggtgac	ccaccagggc	600
ctgagcagcc	ccgtgaccaa	gagcttcaac	agaggcgagt	gc		642

SEQ ID NO: 131 moltype = DNA length = 726

FEATURE Location/Qualifiers

misc_feature 667..726

 note = /note="This region may be absent in its entirety"

misc_feature 694..726

 note = /note="This region may be absent in its entirety"

misc_feature 1..726

 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"

misc_feature 1..726

 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"

source 1..726

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 131

cagggtgcagc	tggtgcagag	cggcgtggag	gtgaagaagc	ccggcgccag	cgtgaagggtg	60
agctgcaagg	ccagcggcta	caccttcacc	aactactaca	tgtactgggtg	gagacaggcc	120
cccggccagg	gcctggagtg	gatggggcggc	atcaacccca	gcaacggcgg	caccaacttc	180
aacgagaagt	tcaagaacag	agtgaccctg	accaccgaca	gcagcaccac	caccgcctac	240
atggagctga	agagcctgca	gttcgacgac	accgccgtgt	actactgcgc	cagaagagac	300
tacagattcg	acatgggctt	cgactactgg	ggccagggca	ccaccgtgac	cgtgagcagc	360
gcagcaccca	agggccccag	cgtgttcctc	ctggccccct	gcagcagaag	caccagcgag	420
agcacccgcc	ccctgggctg	cctgggtgaag	gactacttcc	ccgagcccg	gaccgtgagc	480
tggacacagc	gcgcctgac	cagcggcggtg	cacaccttcc	ccgccgtgct	gcagagcagc	540
ggcctgtaca	gcctgagcag	cgtgggtgacc	gtgcccagca	gcagcctggg	caccaagacc	600
tacacctgca	acgtggacca	caagcccagc	aacaccaagg	tggacaagag	agtggaagagc	660
aagtacggcc	ccccctgccc	cccctgcccc	gcccccgagt	tcctgggcgg	ccccagcgtg	720
ttcctg						726

SEQ ID NO: 132 moltype = DNA length = 654

FEATURE Location/Qualifiers

misc_feature 1..654

 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"

source 1..654

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 132

gagatcgtgc	tgaccagag	ccccgccacc	ctgagcctga	gccccggcga	gagagccacc	60
ctgagctgca	gagccagcaa	gggcgtgagc	accagcggct	acagctacct	gcactggtac	120
cagcagaagc	ccggccaggc	ccccagactg	ctgatctacc	tggccagcta	cctggagagc	180
ggcgtgcccc	ccagattcag	cggcagcggc	agcggcaccg	acttcacct	gaccatcagc	240
agcctggagc	ccgaggactt	cgccgtgtac	tactgccagc	acagcagaga	cctgccccctg	300
accttcggcg	gcggcaccaa	ggtggagatc	aagagaaccg	tggccgcccc	cagcgtgttc	360
atcttcccc	ccagcgacga	gcagctgaag	agcggcaccg	ccagcgtggt	gtgcctgctg	420
aacaacttct	accccagaga	ggccaagggt	cagtggaagg	tggacaacgc	cctgcagagc	480
ggcaacagcc	aggagagcgt	gaccgagcag	gacagcaagg	acagcaccta	cagcctgagc	540
agcaccctga	ccctgagcaa	ggccgactac	gagaagcaca	aggtgtacgc	ctgcgagggtg	600
acccaccagg	gcctgagcag	ccccgtgacc	aagagcttca	acagaggcga	gtgc	654

SEQ ID NO: 133 moltype = DNA length = 744

FEATURE Location/Qualifiers

misc_feature 682..744

 note = /note="This region may be absent in its entirety"

variation 691..693

 replace = ctg

misc_feature 694..744

 note = /note="This region may be absent in its entirety"

misc_feature 712..744

 note = /note="This region may be absent in its entirety"

misc_feature 1..744

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note = /note="Variant nucleotides given in the sequence
have no preference with respect to those in the
annotations for variant positions"
misc_feature 1..744
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
misc_feature 1..744
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..744
mol_type = other DNA
organism = synthetic construct

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SEQUENCE: 133
gaggtgcagc tgggtggagag cggcgcgccg ctggtgcagc cggcgcgccg cctgagactg 60
agctgcgccc ccagcgggcta cgacttcacc cactacggca tgaactgggt gagacaggcc 120
cccggaagg gcttgagtg ggtgggctgg atcaacacct acacggcgga gcccacctac 180
gcccgcgact tcaagagaag attcaccttc agcctggaca ccagcaagag caccgcctac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc caagtacccc 300
tactactacg gcaccagcca ctggtacttc gacgtgtggg gccagggcac cctggtgacc 360
gtgagcagcg ccagcaccaa gggccccagc gtgttcccc tggccccag cagcaagagc 420
accagcggcg gcaccgccc cctgggctgc ctggtgaagg actacttccc cgagccgtg 480
accgtgagct ggaacacggg cgccctgacc agcggcgtgc acaccttccc cgccgtgctg 540
cagagcagcg gcctgtacag cctgagcagc gtggtgaccg tgcccagcag cagcctgggc 600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaagg ggacaagaag 660
gtggagccca agagctgcga caagacccac acctgcccc cctgccccg ccccgagctg 720
ctggcgcgcc ccagcgtgtt ctg 744

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SEQ ID NO: 134 moltype = DNA length = 642
FEATURE Location/Qualifiers
misc_feature 1..642
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
source 1..642
mol_type = other DNA
organism = synthetic construct

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SEQUENCE: 134
gacatccagc tgaccagag cccagcagc ctgagcgcca gcgtggcgga cagagtgacc 60
atcacctgca gcgcccagcca ggacatcagc aactacctga actggtacca gcagaagccc 120
ggcaaggccc ccaagggtgt gatctacttc accagcagcc tgcacagcgg cgtgcccagc 180
agattcagcg gcagcggcag cggcaccgac ttcaccttga ccatcagcag cctgcagccc 240
gaggacttcg ccacctacta ctgccagcag tacagcaccg tgccctggag cttcggccag 300
ggcaccgaag tgagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgccc agcgtggtgt gcctgctgaa caacttctac 420
cccagagagg ccaagggtgca gtggaagggt gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc 642

```

```

SEQ ID NO: 135 moltype = DNA length = 744
FEATURE Location/Qualifiers
misc_feature 682..744
note = /note="This region may be absent in its entirety"
variation 691..693
replace = ctg
misc_feature 694..744
note = /note="This region may be absent in its entirety"
misc_feature 712..744
note = /note="This region may be absent in its entirety"
misc_feature 1..744
note = /note="Variant nucleotides given in the sequence
have no preference with respect to those in the
annotations for variant positions"
misc_feature 1..744
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
misc_feature 1..744
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source 1..744
mol_type = other DNA
organism = synthetic construct

```

```

SEQUENCE: 135
gaggtgcagc tgggtggagag cggcgcgccg ctggtgcagc cggcgcgccg cctgagactg 60
agctgcgccc ccagcgggcta caccttcacc aactacggca tgaactgggt gagacaggcc 120

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ccccgcaagg gacctggagt ggtgggctgg atcaaacacct acaccggcga gcccacctac 180
gccgccgact tcaagagaag attcaccttc agcctggaca ccagcaagag caccgcctac 240
ctgcagatga acagcctgag agccgaggac accgcccgtg actactgcgc caagtacccc 300
cactactacg gcagcagcca ctggctacttc gacgtgtggg gccagggcac cctggtgacc 360
gtgagcagcg ccagcaccaa gggcccccag gtgttcccc tggcccccag cagcaagagc 420
accagcggcg gcaccgccc cctgggctgc ctggtgaagg actactccc cgagcccgtg 480
accgtgagct ggaacacggg cgccctgacc agcggcgtgc acacctccc cgccgtgctg 540
cagagcagcg gcctgtacag cctgagcagc gtggtgaccg tgcccagcag cagcctgggc 600
acccagacct acatctgcaa cgtgaaccac aagcccagca acaccaaggt ggacaagaag 660
gtggagccca agagctgcga caagaccac acctgcccc cctgcccgcg ccccgagctg 720
ctgggcgggc ccagcgtgtt cctg          744

```

```

SEQ ID NO: 136      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 136
gacatccaga tgaccagag cccagcagc ctgagcgcca gcgtggcgga cagagtgacc 60
atcacctgca gcgccagcca ggacatcagc aactacctga actggtacca gcagaagccc 120
ggcaaggccc ccaagggtgt gatctacttc accagcagcc tgcacagcgg cgtgcccagc 180
agattcagcg gcagcggcag cggcaccgac ttcacctga ccacagcag cctgcagccc 240
gaggacttcg ccacctacta ctgccagcag tacagcacgg tgccctggac cttcggccag 300
ggcaccaggg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactttac 420
cccagagagg ccaagggtga gtggaagggt gacaacgccc tgcaagcggc caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagcttaaac agaggcgagt gc          642

```

```

SEQ ID NO: 137      moltype = DNA length = 720
FEATURE            Location/Qualifiers
misc_feature        658..720
                    note = /note="This region may be absent in its entirety"
variation           667..669
                    replace = ctg
misc_feature        670..720
                    note = /note="This region may be absent in its entirety"
misc_feature        688..720
                    note = /note="This region may be absent in its entirety"
misc_feature        1..720
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..720
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..720
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..720
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 137
gagggtgcagc tgggtcagag cggcccccag ctgaagaagc ccggcgccag cgtgaagggt 60
agctgcaagg ccagcggcta caccttcacc aactacggca tgaactgggt gagacaggcc 120
cccgccaggg gcctggagtgt gatgggctgg atcaaacacct acaccggcga gaccacctac 180
gccgacgact tcaagggcag attcgtgttc agcctggaca ccagcgtgag caccgcctac 240
ctgcagatca gcagcctgaa ggcgaggac accgcccgtg actactgcga gagagagggc 300
ggcgtgaaca actggggcca gggcaccctg gtgaccgtga gcagcgccag caccaagggc 360
cccagcgtgt tcccctggc cccagcagc aagagcacca gcggcggcac cgccgcccgt 420
ggctgcctgg tgaaggacta cttccccgag cccgtgaccg tgagctggaa cagcgcgccc 480
ctgaccagcg gcgtgcacac cttccccgcc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcgtgg tgaccgtgcc cagcagcagc ctgggcaccc agacctacat ctgcaacgtg 600
aaccacaagc ccagcaacac caaggtggac aagaaggtgg agcccaagag ctgcgacaag 660
acccacacct gcccccctg ccccgcccc gagctgctgg gcggccccag cgtgttcctg 720

```

```

SEQ ID NO: 138      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

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source                1..642
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 138
gacatccagg tgaccagag cccagcagc ctgagcgcca gcgtgggcca cagagtgacc 60
atcacctgca tcaccagcac cgacatcgac gacgacatga actggtacca gcagaagccc 120
ggcaaggtgc ccaagctgct gatcagcggc ggcaaacacc tgagaccggc cgtgcccagc 180
agattcagcg gcagcggcag cggcacgcgac ttcacctga ccatagcag cctgcagccc 240
gaggacgtgg ccactacta ctgcctgcag agcgacagcc tgcctacac cttcggccag 300
ggcaccaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat ctccccccc 360
agcgacgagc agctgaagag cggcacgcgc agcgtggtgt gcctgctgaa caactttac 420
cccagagagg ccaaggtgca gtggaaggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagcttcaac agaggcagat gc 642

SEQ ID NO: 139        moltype = DNA length = 360
FEATURE               Location/Qualifiers
misc_feature          1..360
                      note = source = /note="Description of Artificial Sequence:
                      Synthetic polynucleotide"
source                1..360
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 139
gaggtgcagc tgggtgagag cggcggcggc ctggtgcagc ccggcggcag cctgagactg 60
agctgcaccg ccagcggcgt cagcctgacc gactactact acatgacctg ggtgagacag 120
gcccccgcca agggcctgga gtgggtgggc ttcacgacc ccgacgacga cccctactac 180
gccacctggg ccaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgccgtgt actactgcgc cggcggcgac 300
cacaacagcg gctggggcct ggacatctgg ggccagggca cctggtgac cgtgagcagc 360

SEQ ID NO: 140        moltype = DNA length = 333
FEATURE               Location/Qualifiers
misc_feature          1..333
                      note = source = /note="Description of Artificial Sequence:
                      Synthetic polynucleotide"
source                1..333
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 140
gagatcgtga tgaccagag cccagcacc ctgagcgcca gcgtgggcca cagagtgatc 60
atcacctgcc aggcagcga gatcatccac agctggctgg cctggtacca gcagaagccc 120
ggcaaggccc ccaagctgct gatctacctg gccagcaccc tggccagcgg cgtgcccagc 180
agattcagcg gcagcggcag cggcgccgag ttcacctga ccatagcag cctgcagccc 240
gacgacttcg ccactacta ctgccagaac gtgtacctgg ccagcaccaa cggcgccaac 300
ttcggccagg gcaccaagct gaccgtgctg ggc 333

SEQ ID NO: 141        moltype = DNA length = 741
FEATURE               Location/Qualifiers
misc_feature          679..741
                      note = /note="This region may be absent in its entirety"
variation            688..690
                      replace = ctg
misc_feature          691..741
                      note = /note="This region may be absent in its entirety"
misc_feature          709..741
                      note = /note="This region may be absent in its entirety"
misc_feature          1..741
                      note = /note="Variant nucleotides given in the sequence
                      have no preference with respect to those in the
                      annotations for variant positions"
misc_feature          1..741
                      note = source = /note="Description of Artificial Sequence:
                      Synthetic polynucleotide"
misc_feature          1..741
                      note = source = /note="See specification as filed for
                      detailed description of substitutions and preferred
                      embodiments"
source                1..741
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 141
caggtgcagc tgcagcagag cggcgccgaa gtgaaaaaac cgggcagcag cgtgcgcgtg 60
agctgcaaaag cgagcggcgg cacctttaac aacaacgcga ttaactgggt gcgccaggcg 120
ccgggccagg gcctggaatg gatggcgggc attattccga tgtttggcac cgcgaaatat 180

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agccagaact ttcaggggcgc cgtggcgatt accgcggatg aaagcaccgg caccgcgagc 240
atgggaactga gcagcctgcg cagcgaagat accgcgggtg attattgcgc gcgcagccgc 300
gatctgctgc tgtttccgca tcatgcgctg agcccggtgg gccgcggcac catggtgacc 360
gtgagcagcg cgagcaccaa aggcgccgagc gtgtttccgc tggcgccgag cagcaaaagc 420
accagcggcg gcacgcggcg gctgggctgc ctggtgaaag attattttcc ggaaccggtg 480
accgtgagca acagcggcgc gctgaccagc ggctgacata cctttccggc ggtgctgcag 540
agcagcggcc tgtatagctg gagcagcgtg gtgaccgtgc cgagcagcag cctgggcacc 600
cagacctata tttgcaacgt gaaccataaa ccgagcaaca ccaaagtgga taaaaaagtg 660
gaaccgaaaa gctgcgataa aaccataacc tgcccgcgct gcccgcgcc ggaactgctg 720
ggcgccccga gcgtgtttct g 741

```

SEQ ID NO: 142 moltype = DNA length = 639
FEATURE
misc_feature 1..639
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..639
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 142

```

agcagcgaac tgaccaggga tccggcggtg agcgtggcgc tgggcccagc cgtgcgcgtg 60
acctgccagg gcgtagcctt gcgcagctat tatgcgagct ggtatcagca gaaaccgggc 120
caggcgccgg tgetgggtgat ttatggcaaa aacaaccgcc cgagcggcat tccggatcgc 180
tttagcggca gcagcagcgg caacaccgcg agcctgacca ttaccggcgc gcaggcggaa 240
gatgaagcgg attattattg cagcagccgc gatagcagcg gcaaccattg ggtgtttggc 300
ggcggcaccg aactgaccgt gctggggcag ccgaaagcgg cgcgcgagct gacctgttt 360
ccgcgcagca gcgaagaact gcaggcgaac aaagcgaccc tgggtgtgct gattagcgat 420
tttatccggc gcgcggtgac cgtggcggtg aaagcggata gcagcccggg gccggggcgtg 480
gaaaccacca ccccgagcaa acagagcaac aacaatatg ccgcgagcag ctatctgagc 540
ctgaccccgg aacagtggaa aagccatcgc agctatagct gccaggtgac ccatgaaggc 600
agcaccgtgg aaaaaaccgt ggcgccgacc gaatgcagc 639

```

SEQ ID NO: 143 moltype = DNA length = 714
FEATURE
misc_feature 682..714
 note = /note="This region may be absent in its entirety"
misc_feature 700..714
 note = /note="This region may be absent in its entirety"
misc_feature 1..714
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
misc_feature 1..714
 note = source = /note="See specification as filed for
 detailed description of substitutions and preferred
 embodiments"
source 1..714
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 143

```

cagggtgcagc tgggtgcagag cggcgccgag gtgaagaagc ccggcgccag cgtgaagggtg 60
agctgcaagg ccagcggcta catcttcagc aactactgga tccagtggtg gagacaggcc 120
cccgccaggg ccttgagtgat gatggcgagc atcctgcccgc gcagcgccag caccaggtac 180
accgagaact tcaaggacag agtgaccatg accagagaca ccagcaccag caccgtgtac 240
atggagctga gcagcctgag aagcgaggac accgcgctgt actactgcgc cagatacttc 300
ttcggcagca gccccaaactg gtacttcgac gtgtggggcc agggcacccct ggtgaccgtg 360
agcagcgcca gccaccaaggg cccagcgtg ttcctccctg cccctgcag cagaagcacc 420
agcgagagca ccgcgcctct gggctgcctg gtgaaggact acttcccga gcccgtagc 480
gtgagctgga acagcggcgc cctgaccagc ggctgcgaca ccttcccgc cgtgctgcag 540
agcagcggcc tgtacagcct gagcagcgtg gtgaccgtgc ccagcagcaa cttcggcacc 600
cagacctaca cctgcaacgt ggaccacaag cccagcaaca ccaaggtgga caagaccgtg 660
gagagaaagt gctgcgtgga gtgccccccc tgccccgccc ccccggtggc cggc 714

```

SEQ ID NO: 144 moltype = DNA length = 642
FEATURE
misc_feature 1..642
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..642
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 144

```

gacatccaga tgaccagag cccagcagc ctgagcgcca gcgtgggcca cagagtgacc 60
atcacctgcg gcgcccagca gaacatctac ggccgacctg actggtacca gcagaagccc 120
ggcaaggccc ccaagctgct gatctacggc gccaccaacc tggccgacgg cgtgcccagc 180
agattcagcg gcagcggcga cggcacccag ttcacctga ccatcagcag cctgcagccc 240
gaggacttcg ccacctacta ctgccagaac gtgctgaaca cccctcgac cttcgccagc 300

```


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```

ggcaccaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat ctccccccc 360
agcgacgagc agctgaagag cggcacccgc agcgtggtgt gcctgctgaa caacttttac 420
cccagagagc ccaaggtgca gtggaaggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc cctgaccaa gagcttcaac agaggcgagt gc 642

SEQ ID NO: 145      moltype = DNA length = 711
FEATURE            Location/Qualifiers
misc_feature        652..711
                    note = /note="This region may be absent in its entirety"
misc_feature        679..711
                    note = /note="This region may be absent in its entirety"
misc_feature        1..711
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..711
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..711
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 145
caggtgcagc tgcaggagag cggccccggc ctggtgaagc ccagcgagac cctgagcctg 60
acctgcaccg tgagcggctt cagcctgctg agctacggcg tgcactgggt gagacagccc 120
cccgcaaggg gcctgggagt gctggggcgt atctgggacc gcggcaccac caactacaac 180
agcgccctga tgagcagatt caccatcagc aaggacgaca gcaagaacac cgtgtacctg 240
aagatgaaca gctgaagac cgaggacacc gccatctact actgcgccag atactactac 300
ggcatggact actggggcca gggcaccctg gtgaccgtga gcagcgccag caccaagggc 360
cccagcgtgt tccccctggc cccctgcagc agaagcacca gcgagagcac cgccgcctg 420
ggctgcctgg tgaaggacta ctcccccgag cccgtgacgc tgagctggaa cagcgcgccc 480
ctgaccagcg gcgtgcacac cttccccgcc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcgtgg tgacgtgccc cagcagcagc ctgggcacca agacctacac ctgcaacgtg 600
gaccacaagc ccagcaacac caaggtggac aagagagtgg agagcaagta cggccccccc 660
tgccccccct gccccgcccc cgagttcctg gccgccccca gcgtgttct g 711

SEQ ID NO: 146      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 146
gacatccaga tgaccagag cccagcagc ctgagcgcca gcgtggcgga cagagtgacc 60
atcacctgca aggccagcca ggacgtgaga aacaccgtgg cctggtagca gcagaagccc 120
ggcaaggccc ccaagctgct gatctacagc agcagctaca gaaacaccgg cgtgcccagc 180
agattcagcg gcagcgccag cggcaccgac ttccaccctga ccatcagcag cctgcaggcc 240
gaggacgtgg ccgtgtacta ctgccagcag cactacatca cccctacac cttcgcgccg 300
ggcaccgaag tggagatcaa ggaaccgtg gccgccccca gcgtgttcat ctccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttttac 420
cccagagagg ccaaggtgca gtggaaggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc cctgaccaa gagcttcaac agaggcgagt gc 642

SEQ ID NO: 147      moltype = DNA length = 741
FEATURE            Location/Qualifiers
misc_feature        679..741
                    note = /note="This region may be absent in its entirety"
variation          688..690
                    replace = ctg
misc_feature        691..741
                    note = /note="This region may be absent in its entirety"
misc_feature        709..741
                    note = /note="This region may be absent in its entirety"
misc_feature        1..741
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..741
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..741

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note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source      1..741
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 147
gaggtgcagc tgctggagag cggtggcgcc ctggtgcagc ccggcgccag cctgagactg 60
agctgcgcgc ccagcggcct cacttcagc cactacatca tgatgtgggt gagacaggcc 120
cccggcaagg gcctggagtg ggtgagcgcc atctacagca gcggcgccat caccgtgtac 180
gccgacagcg tgaagggcag attcaccatc agcagagaca acagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgcccgtg actactgcgc ctacagaaga 300
atcggcgtgc ccagaagaga cgagttcgac atctggggcc agggcaccat ggtgaccgtg 360
agcagcgcca gcaccaaggg ccccgagcgt tccccctgg ccccgagcag caagagcacc 420
agcggcgcca ccgcccctt gggctgcctg gtgaaggact acttccccga gcccgtagcc 480
gtgagctgga acagcggcgc cctgaccagc ggctgtgcaca ccttccccgc cgtgctgcag 540
agcagcgccc tgtacagcct gagcagcgtg gtgacgcgtg ccagcagcag cctgggcacc 600
cagacctaca tctgcaacgt gaaccacaag cccagcaaca ccaaggtgga caagagagtg 660
gagcccaaga gctgcgacaa gacccacacc tgccccccct gccccgcccc cgagctgctg 720
ggcgccccca gcgtgttct g 741

SEQ ID NO: 148      moltype = DNA length = 639
FEATURE            Location/Qualifiers
misc_feature       1..639
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
source            1..639
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 148
gacatccaga tgacccagag ccccgagcacc ctgagcgcca gcgtggcgga cagagtgacc 60
atcacctgca gagccagcca gagcatcagc agctggctgg cctggtagca gcagaagccc 120
ggcaaggccc ccaagctgct gatctacaag gccagcaccg tggagagcgg cgtgcccagc 180
agattcagcg gcagcggcag cgccaccgag ttcaccctga ccacagcag cctgcagccc 240
gacgacttgc ccacctacta ctgccagcag tacaacacct actggacctt cggccagggc 300
accaaggttg agatcaagag aaccgtggcc gcccccagcg tgttcattt cccccccagc 360
gacgagcagc tgaagagcgg caccgcccag gtggtgtgccc tgcgtaacaa cttctacccc 420
agagagggcca aggtgcagtg gaaggtggac aacgcccctc agagcggcaa cagccaggag 480
agcgtgaccg agcaggacag caaggacagc acctacagcc tgagcagcac cctgaccctg 540
agcaaggccg actacgagaa gcaacaagtg tacgcctgcg aggtgaccca ccagggcctg 600
agcagccccg tgaccaagag cttcaacaga ggcgagtg 639

SEQ ID NO: 149      moltype = DNA length = 737
FEATURE            Location/Qualifiers
misc_feature       676..737
note = /note="This region may be absent in its entirety"
variation         685..687
replace = ctg
misc_feature       688..737
note = /note="This region may be absent in its entirety"
misc_feature       706..737
note = /note="This region may be absent in its entirety"
misc_feature       1..737
note = /note="Variant nucleotides given in the sequence
have no preference with respect to those in the
annotations for variant positions"
misc_feature       1..737
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
misc_feature       1..737
note = source = /note="See specification as filed for
detailed description of substitutions and preferred
embodiments"
source            1..737
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 149
gaggtgcagc tggtggagag cggtggcgcc ctggtgcagc ccggcagaag cctgagactg 60
agctgcgcgc ccagcggcct cacttcgac gactacgcca tgactgggt gagacaggcc 120
cccggcaagg gcctggagtg ggtgagcgcc atcacctgga acagcgccca catcgactac 180
gccgacagcg tggagggcag attcaccatc agcagagaca acgccaagaa cagcctgtac 240
ctgcagatga acagcctgag agccgaggac accgcccgtg actactgcgc caaggtgagc 300
tacctgagca ccgcccagcag cctggactac tggggccagg gcaccctggt gaccgtgagc 360
agcggccagc ccaagggccc cagcgtgttc cccctggccc ccagcagcaa gagcaccagc 420
ggcggcaccg ccgcccctgg ctgcctgggt aaggactact tccccgagcc cgtgaccgtg 480
agctggaaca gcggcgccct gaccagcgcc gtgcacacct tccccgcgtg gctgcagagc 540

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agcggcctgt acagcctgag cagcgtggtg accgtgccc a gcagcagcct gggcaccag 600
acctacatct gcaacgtgaa ccacaagccc agcaacacca aggtggacaa gaaggtggag 660
cccaagagct gcgacaagac ccacacctgc cccccctgcc cggccccgag ctgctgggag 720
gccccagcgt gttcctg 737

SEQ ID NO: 150      moltype = DNA length = 462
FEATURE            Location/Qualifiers
misc_feature        1..462
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..462
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 150
agattcagcg gcagcggcag cggcaccgac ttcaccctga ccatcagcag cctgcagccc 60
gaggacgtgg ccacctacta ctgccagaga tacaacagag cccctacac cttcgccag 120
ggcaccaaagg tggagatcaa gagaaccgtg gccgccccca gcgtgttcat cttccccccc 180
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactctac 240
ccagagaggg ccaaggtgca gtggaagggt gacaacgccc tgcagagcgg caacagccag 300
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 360
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 420
ctgagcagcc ccgtgaccaa gagcttcaac agaggcgagt gc 462

SEQ ID NO: 151      moltype = DNA length = 735
FEATURE            Location/Qualifiers
misc_feature        673..735
                    note = /note="This region may be absent in its entirety"
variation           682..684
                    replace = ctg
misc_feature        685..735
                    note = /note="This region may be absent in its entirety"
misc_feature        703..735
                    note = /note="This region may be absent in its entirety"
misc_feature        1..735
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..735
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..735
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..735
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 151
gaggtgaagc tggaggagag cggcggcgcc ctggtgcagc ccggcgcgag catgaagctg 60
agctgcgtgg ccagcggcct catcttcagc aaccactgga tgaactgggt gagacagagc 120
cccgagaagg gcctggagtg ggtggccgag atcagaagca agagcatcaa cagcgccacc 180
cactacgccc agagcgtgaa gggcagattc accatcagca gagacgacag caagagcgcc 240
gtgtacctgc agatgaccca cctgagaacc gaggacaccg gcgtgtacta ctgcagcaga 300
aactactacg gcagcaccta cgactactgg ggccagggca ccacctgac cgtgagcagc 360
gccagcacca agggccccc cgtgttcccc ctggccccca gcagcaagag caccagcgcc 420
ggcaccgccc ccctgggctg cctggtgaag gactacttcc ccgagcccgt gaccgtgagc 480
tggaaacagc gcgcccctgac cagcggcgctg cacaccttcc ccgccgtgct gcagagcagc 540
ggcctgtaca gcctgagcag cgtggtgacc gtgcccagca gcagcctggg caccagagc 600
tacatctgca acgtgaacca caagcccagc aacaccaagg tggacaagaa ggtggagccc 660
aagagctgcg acaagaccca cacctgcccc ccctgccccg ccccagagct gctgggagcg 720
cccagcgtgt tcctg 735

SEQ ID NO: 152      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 152
gacatctctg tgaccagag ccccgccatc ctgagcgtga gcccggcgga gagagtgagc 60
ttcagctgca gagccagcca gttcgtgggc agcagcatcc actggtacca gcagagaacc 120
aacggcagcc ccagactgct gatcaagtac gccagcgaga gcatgagcgg catccccagc 180
agattcagcg gcagcggcag cggcaccgac ttcaccctga gcataaacac cgtggagagc 240
gaggacatcg ccgactacta ctgccagcag agccacagct ggcccttcac cttcggcagc 300

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ggcaccaacc tggaggtgaa gagaaccgtg gccgccccca gcgtgttcat ctccccccc 360
agcgacgagc agctgaagag cggcacccgc agcgtggtgt gcctgctgaa caactttac 420
cccagagagc ccaaggtgca gtggaaggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgagcagcc ccgtgaccaa gagttcaac agaggcgagt gc 642

SEQ ID NO: 153      moltype = DNA length = 711
FEATURE            Location/Qualifiers
misc_feature        652..711
                    note = /note="This region may be absent in its entirety"
misc_feature        679..711
                    note = /note="This region may be absent in its entirety"
misc_feature        1..711
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..711
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..711
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 153
gaggtgaagg tgggtggagag cggcgccggc ctggtgcagc ccggcgccag catgaagctg 60
agctgcgtgg tgagcggctt caccctcagc aactactggg tgaactgggt gaggcaggcc 120
cccggcaagg gcctggagtg ggtggcccag atcaggctga agagcgacaa ctacgccacc 180
cactacgagg agagcgtgaa gggcaggttc accatcagca gggacgacag caagagcagc 240
gtgtacctgc agatgaacaa cctgagggcc gaggcagcg gcattacta ctgcaccaac 300
tgaggaggact tctggggcca gggcaccacc gtgaccgtga gcagcgccag caccaggggc 360
cccagcgtgt tccccctggc cccctgcagc aggcagcaca gcgagagcac cgccgcctg 420
ggctgcctgg tgaaggacta ctcccccgag cccgtgaccg tgagctggaa cagcgcgcc 480
ctgaccagcg gcgtgcacac ctcccccgcc gtgtgcgaga gcagcgccct gtacagcctg 540
agcagcgtgg tgacgtgccc cagcagcagc ctgggcacca agacctacac ctgcaacgtg 600
gaccacaagc ccagcaacac caaggtggac aagaggggtg agagcaagta cggccccccc 660
tgccccccct gccccgcccc cgagttcctg ggcggcccca gcgtgttct g 711

SEQ ID NO: 154      moltype = DNA length = 654
FEATURE            Location/Qualifiers
misc_feature        1..654
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..654
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 154
gacatcgtgc tgaccagag ccccgacagc ctggccgtga gcctgggcca gagggccacc 60
atcagctgca gggccagcca gagcgtgagc accagcaggt acagctacat ccactggtag 120
cagcagaagc ccggccagcc ccccaagctg ctgatcaagt acgccaagca cctggagagc 180
ggcgtgcccc gcaggttcag cggcagcggc agcggcaccg acttcaccct gaacatccac 240
cccctggagg ccgaggactt cgccacctac tactgccacc acagctggga gatccccctg 300
accttcggcc agggcaccaa gctggagatc aagaggaccg tggccgcccc cagcgtgttc 360
atcttcccc cccgcgacga gcagctgaag agcggcaccg ccagcgtggt gtgcctgctg 420
aacaacttct accccaggga ggccaagggt cagtgggaag tggacaacgc cctgcagagc 480
ggcaacagcc aggcagcgtg gaccgagcag gacagcaagg acagcaccta cagcctgagc 540
agcaccctga ccctgagcaa ggcgcgactac gagaagcaca aggtgtacgc ctgcgaggtg 600
accaccagcg gcctgagcag ccccgtagcc aagagcttca acaggggcca gtgc 654

SEQ ID NO: 155      moltype = DNA length = 738
FEATURE            Location/Qualifiers
misc_feature        1..738
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..738
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 155
caggtgcagc tgggtggagag cggcgccggc gtggtgcagc ccggcagaag cctgagactg 60
agctgcgccg ccagcggctt caccctcagc agcttcggca tgcaactgggt gagacaggcc 120
cccggcaagg gcctggagtg ggtggccgtg atcagcttcg accgcagcat caagtacagc 180
gtggacagcg tgaaggccag attcaccatc agcagagaca acagcaagaa caccctgttc 240
ctgcagatga acagcctgag agccgaggac accgcctgtg actactgcgc cagagacaga 300
ctgaactact acgacagcag cggctactac cactacaagt actacggcat ggccgtgtgg 360
ggccagggca ccaccgtgac cgtgagcagc gccagcacca agggccccag cgtgttcccc 420
ctggccccct ccagcagaag caccagcgag agcaccggcg ccctgggctg cctggtgaag 480
gactacttcc ccgagcccggt gaccgtgagc tggaacagcg gcgcctgac cagcggcgtg 540

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cacaccttcc cgcgcgtgct gcagagcagc ggccctgtaca gcctgagcag cgtggtgacc 600
gtgcccagca gcaacttcgg caccagacc tacacctgca acgtggacca caagcccagc 660
aacaccaagg tggacaagac cgtggagaga aagtgtctgcg tggagtgcc cccctgcccc 720
gcccccccg tggccggc 738

SEQ ID NO: 156      moltype = DNA length = 648
FEATURE            Location/Qualifiers
misc_feature        1..648
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..648
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 156
cagagcgtgc tgaccagccc cccagcgtg agcgccgccc ccggccagaa ggtgaccatc 60
agctgcagcg gcagcagcag caacatcggc aacaactacg tgagctggta ccagcagctg 120
cccgccaccg cccccaagct gctgatctac gacaacaaca agagaccag ccgcatcccc 180
gacagattca gcggcagcaa gagcggcacc agcaccaccc tgggcatcac cgccctgcag 240
accggcgacg aggcgcgacta ctactgcggc acctgggaca gcagactgag cgccgtggtg 300
ttcgggggcg gcaccaagct gaccgtgctg ggccagccca aggccaaacc caccgtgacc 360
ctgttcccc ccagcagcga ggagctgcag gccacaagg ccaccctggt gtgcctgatc 420
agcgacttct acccgggcgc cgtgaccgtg gcctggaagg ccgacggcag cccgtgaag 480
gcccggcgtg agaccaccaa gcccagcaag cagagcaaca acaagtacgc cgccagcagc 540
tacctgagcc tgacccccga gcagtgaag agccacagaa gctacagctg ccaggtgacc 600
cacgagggca gcaccgtgga gaagaccgtg gccccaccg agtgacgc 648

SEQ ID NO: 157      moltype = DNA length = 731
FEATURE            Location/Qualifiers
misc_feature        685..731
                    note = /note="This region may be absent in its entirety"
variation           694..696
                    replace = ctg
misc_feature        697..731
                    note = /note="This region may be absent in its entirety"
misc_feature        715..731
                    note = /note="This region may be absent in its entirety"
misc_feature        1..731
                    note = /note="Variant nucleotides given in the sequence
                    have no preference with respect to those in the
                    annotations for variant positions"
misc_feature        1..731
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
misc_feature        1..731
                    note = source = /note="See specification as filed for
                    detailed description of substitutions and preferred
                    embodiments"
source              1..731
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 157
gaggtgcagc tgggtggagag cggcgccggc ctggtgcagc ccggcgccag cctgaggctg 60
agctgcagcg ccagcggctt caccctcagc agcttcggca tgcactgggt gaggcaggcc 120
cccgccaagg gcctggagtg ggtggcctac atcagcagcg gcagcagcac catctactac 180
ggcgacaccg tgaagggcag gttcaccatc agcagggaca acgccaagaa cagcctgttc 240
ctgcagatga gcagcctgag ggcgaggac accgcccgtg actactgcgc caggggagggc 300
ggctactact acggcaggag ctactacacc atggactact ggggcagggg caccaccgtg 360
accgtgagca gcgccagcac caaggggcccc agcgtgttcc ccctggcccc cagcagcaag 420
agcaccagcg gcggcaccgc cgccctgggc tgcctggtga aggactactt ccccgagccc 480
gtgaccgtga gctggaacag cggcgccctg accagcgcg tgcacacctt ccccgccgtg 540
ctgcagagca gcgcctgta cagcctgagc agcgtggtga ccgtgccca cagcagcctg 600
ggcaccacga cctacatctg caacgtgaac cacaagccca gcaacaccaa ggtggacaag 660
aaggtggagc ccaagagctg cgacaagacc cacacctgcc cccctgccc cgcccagc 720
tgctggcgcg c 731

SEQ ID NO: 158      moltype = DNA length = 657
FEATURE            Location/Qualifiers
misc_feature        1..657
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..657
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 158
gacgtggtga tgaccagag cccctgagc ctgccctga ccccgcgcg ccccgccagc 60
atcagctgca ggagcagcca gagcatcgtg cacagcaacg gcaacaccta cctggagtgg 120

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tacctgcaga agcccgccca gagccccaag ctgctgatct acaaggtgag caacaggttc 180
agcggcgctgc ccgacagggt cagcggcagc ggcagcggca ccgacttcac cctgaggatc 240
agcagggttg aggccgagga cgtgggcac tactactgct tccagggcag ccacgtgccc 300
cccaccttcg gcccggcac caagctggag atcaagagga ccgtggccgc ccccagcgtg 360
ttcatcttcc ccccagcga cgagcagctg aagagcggca ccgccagcgt ggtgtgctg 420
ctgaacaact tctacccag ggaggccaag gtgcagtggg aggtggacaa cgccctgcag 480
agcggcaaca gccaggagag cgtgaccgag caggacagca aggacagcac ctacagcctg 540
agcagcacc tgacctgag caaggccgac tacgagaagc acaaggtgta cgccctgcag 600
gtgacccacc agggcctgag cagccccgtg accaagagct tcaacagggg cgagtgc 657

SEQ ID NO: 159      moltype = DNA length = 369
FEATURE            Location/Qualifiers
misc_feature        1..369
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..369
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 159
gaggtgaagc tgggtgagag cggcggcggc ctggtgcagc ccggcggcag cctgaggctg 60
agctgcgccca ccagcggctt cacttcagc gacttctaca tggagtgggt gaggcaggcc 120
cccgcaaga ggctggagt gctgcggcc agcaggaaca aggccaaaga ctacaccacc 180
gagtacgcgc acagcgtgaa gggcagggtc atcgtgagca gggacaccag ccagagcacc 240
ctgtacctgc agatgaacgc cctgaggggc gaggcaccgc ccatctacta ctgcgccagg 300
gactactacg gcagcagcta ctggtacttc gacgtgtggg gcgccggcac caccgtgacc 360
gtgagcagc
369

SEQ ID NO: 160      moltype = DNA length = 339
FEATURE            Location/Qualifiers
misc_feature        1..339
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..339
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 160
gacatcgtga tgaccagag cccagcagc ctgagcgtga gcgccggcaa gaaggtgacc 60
atcagctgca ccgccagcga gagcctgtac agcagcaagc acaaggtgca ctacctggcc 120
tggtaccaga agaagcccg gcagagcccc aagctgctga tctacggcgc cagcaacagg 180
tacatcggcg tgcccgacag gttcaccggc agcggcagcg gcaccgactt caccctgacc 240
atcagcagcg tgcaggtgga ggacctgacc cactactact gcgccagtt ctacagctac 300
cccctgacct tcggcgccg caccaagctg gagatcaag 339

SEQ ID NO: 161      moltype = AA length = 20
FEATURE            Location/Qualifiers
REGION              1..20
                    note = source = /note="Description of Unknown: signal
                    sequence"
source              1..20
                    mol_type = protein
                    organism = unidentified

SEQUENCE: 161
MYRMQLLLLI ALSLALVTNS 20

SEQ ID NO: 162      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION              1..7
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 162
LGETTRP 7

SEQ ID NO: 163      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION              1..9
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 163
LALGETTRP 9

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SEQ ID NO: 164	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
REGION	1..26	
	note = source = /note="Description of Unknown: VEGF-A signal peptide"	
source	1..26	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 164		
MNFLLSWVHW SLALLLYLHH AKWSQA		26
SEQ ID NO: 165	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
REGION	1..29	
	note = source = /note="Description of Unknown: Fibulin-1 signal peptide"	
source	1..29	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 165		
MERAAPSRRV PLPLLLGGL ALLAAGVDA		29
SEQ ID NO: 166	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Unknown: Vitronectin signal peptide"	
source	1..19	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 166		
MAPLRPLLIL ALLAWVALA		19
SEQ ID NO: 167	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = source = /note="Description of Unknown: Complement Factor H signal peptide"	
source	1..18	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 167		
MRLLAKIICL MLWAICVA		18
SEQ ID NO: 168	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Unknown: Opticin signal peptide"	
source	1..19	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 168		
MRLLAFLSLL ALVLQETGT		19
SEQ ID NO: 169	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = source = /note="Description of Unknown: Albumin signal peptide"	
source	1..18	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 169		
MKWVTFISLL FLFSSAYS		18
SEQ ID NO: 170	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = source = /note="Description of Unknown: Chymotrypsinogen signal peptide"	
source	1..18	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 170		
MAFLWLLSCW ALLGTTFG		18

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SEQ ID NO: 171	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = source = /note="Description of Unknown: Interleukin-2 signal peptide"	
source	1..20	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 171		
MYRMQLLSCL ALILALVTNS		20
SEQ ID NO: 172	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = source = /note="Description of Unknown: Trypsinogen-2 signal peptide"	
source	1..15	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 172		
MNLLLLILTFV AAVA		15
SEQ ID NO: 173	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 173		
MRAWIFFLLC LAGRALA		17
SEQ ID NO: 174	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 174		
MFSFVDLRLL LLLAATALT HG		22
SEQ ID NO: 175	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 175		
MKLVLVLVLLF LGALGLCLA		19
SEQ ID NO: 176	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 176		
MGPTSGPSLL LLLLTHLPLA LG		22
SEQ ID NO: 177	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 177		
MSLSAFTLFL ALIGGTSG		18
SEQ ID NO: 178	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 178		
MAPHRPAPAL LCALSLALCA LSLPVRA		27
SEQ ID NO: 179	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = Homo sapiens	

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SEQUENCE: 179		
MWATLPLLCA GAWLLGVPVC GA		22
SEQ ID NO: 180	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 180		
MQALVLLLCI GALLGHSSC		19
SEQ ID NO: 181	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 181		
MQMSPALTCL VLGLALVFGE GSA		23
SEQ ID NO: 182	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 182		
MQPSSLLPLA LCLLAAPASA		20
SEQ ID NO: 183	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 183		
MAPFEPLASG ILLLLWLIAP SRA		23
SEQ ID NO: 184	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
source	1..31	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 184		
MLRGPGPGLL LLAVQCLGTA VPSTGASKSK R		31
SEQ ID NO: 185	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 185		
MWCIVLFSLL AWVYA		15
SEQ ID NO: 186	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 186		
MNPTLILAAF CLGIASA		17
SEQ ID NO: 187	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 187		
MWQLWASLCC LLVLANA		17
SEQ ID NO: 188	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 188		
MLLILLSVAL LAFSSA		16
SEQ ID NO: 189	moltype = AA length = 20	

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FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 189		
MWKRWLALAL ALVAVAVRA		20
SEQ ID NO: 190	moltype = AA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 190		
MPSSVSWGIL LLAGLCCLVP VSLA		24
SEQ ID NO: 191	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 191		
MKAAVLTAV LFLTGSQA		18
SEQ ID NO: 192	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 192		
MKLLAATVLL LTICSLEG		18
SEQ ID NO: 193	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 193		
MDPPRPALLA LLALPALLLL LLAGARA		27
SEQ ID NO: 194	moltype = AA length = 28	
FEATURE	Location/Qualifiers	
source	1..28	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 194		
MQRVNMIMAE SPGLITICLL GYLLSAEC		28
SEQ ID NO: 195	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 195		
MGPLMVLFL LFLYPGLADS		20
SEQ ID NO: 196	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 196		
MWLLVSVILI SRISSVGG		18
SEQ ID NO: 197	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 197		
MLLLFSVILI SWVSTVGG		18
SEQ ID NO: 198	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Homo sapiens	

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SEQUENCE: 198 MFSMRIVCLV LSVVGTAWT	19
SEQ ID NO: 199 FEATURE source	moltype = AA length = 30 Location/Qualifiers 1..30 mol_type = protein organism = Homo sapiens
SEQUENCE: 199 MKRMVSWSFH KLKTMKHL LLLCVFLVKS	30
SEQ ID NO: 200 FEATURE source	moltype = AA length = 26 Location/Qualifiers 1..26 mol_type = protein organism = Homo sapiens
SEQUENCE: 200 MSWSLHPRNL ILYFYALLFL SSTCVA	26
SEQ ID NO: 201 FEATURE source	moltype = AA length = 18 Location/Qualifiers 1..18 mol_type = protein organism = Homo sapiens
SEQUENCE: 201 MKSLVLLCL AQLWGCHS	18
SEQ ID NO: 202 FEATURE source	moltype = AA length = 23 Location/Qualifiers 1..23 mol_type = protein organism = Homo sapiens
SEQUENCE: 202 MARVLGAPVA LGLWSLCWSL AIA	23
SEQ ID NO: 203 FEATURE source	moltype = AA length = 18 Location/Qualifiers 1..18 mol_type = protein organism = Homo sapiens
SEQUENCE: 203 MKLITILFLC SRLLSLT	18
SEQ ID NO: 204 FEATURE source	moltype = AA length = 20 Location/Qualifiers 1..20 mol_type = protein organism = Homo sapiens
SEQUENCE: 204 MSLFPSLPLL LLSMVAASYS	20
SEQ ID NO: 205 FEATURE source	moltype = AA length = 19 Location/Qualifiers 1..19 mol_type = protein organism = Homo sapiens
SEQUENCE: 205 MEHKEVVLLL LFLKSGQG	19
SEQ ID NO: 206 FEATURE source	moltype = AA length = 24 Location/Qualifiers 1..24 mol_type = protein organism = Homo sapiens
SEQUENCE: 206 MAHVRGLQLP GCLALAALCS LVHS	24
SEQ ID NO: 207 FEATURE source	moltype = AA length = 29 Location/Qualifiers 1..29 mol_type = protein organism = Homo sapiens
SEQUENCE: 207 MISRMKMTM MMKILIMFAL GMNYWSCSG	29
SEQ ID NO: 208	moltype = AA length = 32

-continued

FEATURE	Location/Qualifiers	
source	1..32	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 208		
MYSNVIGTVT SGKRKVYLLS LLLIGFWDCV TC		32
SEQ ID NO: 209	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 209		
MRLAVGALLV CAVLGLCLA		19
SEQ ID NO: 210	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 210		
LLNFDLLKLA GDVESNPGP		19
SEQ ID NO: 211	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
VARIANT	1..3	
	note = /replace=" "	
SITE	1..21	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..21	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 211		
GSGEGRGSLL TCGDVEENPG P		21
SEQ ID NO: 212	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
VARIANT	1..3	
	note = /replace=" "	
SITE	1..22	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..22	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 212		
GS GATNFSLL KQAGDVEENP GP		22
SEQ ID NO: 213	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
VARIANT	1..3	
	note = /replace=" "	
SITE	1..23	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..23	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..23	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 213		
GS GQCTNYAL LKLAGDVESN PGP		23

-continued

SEQ ID NO: 214	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
VARIANT	1..3	
	note = /replace=" "	
SITE	1..25	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..25	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 214		
GSQVKTQTLNF DLLKLAGDVE SNPGP		25
SEQ ID NO: 215	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 215		
RKRR		4
SEQ ID NO: 216	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 216		
RRRR		4
SEQ ID NO: 217	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 217		
RRKR		4
SEQ ID NO: 218	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 218		
RKKR		4
SEQ ID NO: 219	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 219		
CPPCPA		6
SEQ ID NO: 220	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	

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	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 220		
CPPCP		5
SEQ ID NO: 221	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 221		
CPPCA		5
SEQ ID NO: 222	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 222		
KTHTCPPCPA PELLGG		16
SEQ ID NO: 223	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
KTHL		4
SEQ ID NO: 224	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 224		
KTHT		4
SEQ ID NO: 225	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
KTHTCPPCPA		10
SEQ ID NO: 226	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
KTHLCPPCPA		10
SEQ ID NO: 227	moltype = AA length = 21	
FEATURE	Location/Qualifiers	

-continued

REGION	1..21	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
KHTTCCPPCPA PELLGGPSVF L		21
SEQ ID NO: 228	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
REGION	1..21	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
KTHLCPPCPA PELLGGPSVF L		21
SEQ ID NO: 229	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
GPPCPCCPA		9
SEQ ID NO: 230	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 230		
GPPCPCCPAP EFLGGPSVFL		20
SEQ ID NO: 231	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 231		
GPPCPCCPAP EFLGG		15
SEQ ID NO: 232	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 232		
CPPCPAPPVA GG		12
SEQ ID NO: 233	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 233		
CPPCPAPPVA G		11
SEQ ID NO: 234	moltype = AA length = 213	

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FEATURE Location/Qualifiers
 REGION 1..213
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..213
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 234
 IQMTQSPSSL SASVGDRVTI TCKASQDVRN TVAWYQQKPG KAPKLLIYSS SYRNTGVPDR 60
 FSGSGSGTDF TLTISSLQAE DVAVYYCQQH YITPYTFGGG TKVEIKRTVA APSVFIFPPS 120
 DEQLKSGTAS VVCLLNNFYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSTLTLL 180
 SKADYEKHKV YACEVTHQGL SSPVTKSFNR GEC 213

SEQ ID NO: 235 moltype = AA length = 213
 FEATURE Location/Qualifiers
 REGION 1..213
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..213
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 235
 DILLTQSPAI LSVSPGERVS FSCRASQFVS SIHWYQQRTN GSPRLLIKYA SESMSGIPSR 60
 FSGSGSGTDF TLSINTVESE DIADYYCQQS HSWPFTFGSG TNLEVKRTVA APSVFIFPPS 120
 DEQLKSGTAS VVCLLNNFYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSTLTLL 180
 SKADYEKHKV YACEVTHQGL SSPVTKSFNR GEC 213

SEQ ID NO: 236 moltype = AA length = 16
 FEATURE Location/Qualifiers
 REGION 1..16
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..16
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 236
 KTHTCPPCPA PELAGA 16

SEQ ID NO: 237 moltype = AA length = 21
 FEATURE Location/Qualifiers
 REGION 1..21
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..21
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 237
 KTHTCPPCPA PELAGAPSVF L 21

SEQ ID NO: 238 moltype = AA length = 21
 FEATURE Location/Qualifiers
 REGION 1..21
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..21
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 238
 KTHLCPPCPA PELAGAPSVF L 21

SEQ ID NO: 239 moltype = AA length = 16
 FEATURE Location/Qualifiers
 REGION 1..16
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..16
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 239
 KTHLCPPCPA PELLGG 16

SEQ ID NO: 240 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"

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source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 240		
GPPCPPCPAP EFLG		14
SEQ ID NO: 241	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 241		
GPPCPPCPAP EFLGPSVFL		19
SEQ ID NO: 242	moltype = AA length = 28	
FEATURE	Location/Qualifiers	
REGION	1..28	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..28	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 242		
RKRRAPVKQT LNFDLLKLAG DVESNPGP		28
SEQ ID NO: 243	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
SITE	1..30	
	note = /note="This sequence may encompass 1-6 'Gly Gly Gly Gly Ser' repeating units"	
REGION	1..30	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 243		
GGGSGGGGS GGGSGGGGS GGGSGGGGS		30
SEQ ID NO: 244	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Unknown: signal sequence"	
source	1..19	
	mol_type = protein	
	organism = unidentified	
SEQUENCE: 244		
MYRMQLLLIA LSLALVTNS		19
SEQ ID NO: 245	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 245		
GGGSGGGGS GGGGS		15
SEQ ID NO: 246	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 246		
GGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 247	moltype = AA length = 214	

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FEATURE	Location/Qualifiers	
REGION	1..214	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..214	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 247		
DIQMTQSPSS LSASVGDRVIT	ITCRASQGIR NYLAWYQQKP	GKAPKLLIYA ASTLQSGVPS 60
RFSGSGSGTD PTLTISSLQP	EDVATYYCQR YNRAPYTFGQ	GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY	PREAKVQWKV DNALQSGNSQ	ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG	LSSPVTKSFN RGEC	214
SEQ ID NO: 248	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 248		
EDTAVYY		7
SEQ ID NO: 249	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 249		
DKTHTCPPCP APELLGG		17
SEQ ID NO: 250	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 250		
EDFATYY		7
SEQ ID NO: 251	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 251		
EDVGVYY		7
SEQ ID NO: 252	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 252		
NWEDYW		6
SEQ ID NO: 253	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	

-continued

SEQUENCE: 253 VECPPCPAPP VAG	13
SEQ ID NO: 254 FEATURE REGION	moltype = AA length = 17 Location/Qualifiers 1..17 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..17 mol_type = protein organism = synthetic construct
SEQUENCE: 254 DKTHLCPPCP APELLGG	17
SEQ ID NO: 255 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..6 mol_type = protein organism = synthetic construct
SEQUENCE: 255 EDVGIY	6
SEQ ID NO: 256 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..7 mol_type = protein organism = synthetic construct
SEQUENCE: 256 GDEADYY	7
SEQ ID NO: 257 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..7 mol_type = protein organism = synthetic construct
SEQUENCE: 257 EDVGFYY	7
SEQ ID NO: 258 FEATURE REGION	moltype = AA length = 10 Location/Qualifiers 1..10 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..10 mol_type = protein organism = synthetic construct
SEQUENCE: 258 DILTDYIHY	10
SEQ ID NO: 259 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..7 mol_type = protein organism = synthetic construct
SEQUENCE: 259 EDFAVYY	7
SEQ ID NO: 260 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..6 mol_type = protein

-continued

SEQUENCE: 260	organism = synthetic construct	
DTATYY		6
SEQ ID NO: 261	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 261		
EDVAVYY		7
SEQ ID NO: 262	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 262		
DGWDYADY		9
SEQ ID NO: 263	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 263		
DKHTCPPCP APELAGA		17
SEQ ID NO: 264	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 264		
EDIATYY		7
SEQ ID NO: 265	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 265		
DDTAVYY		7
SEQ ID NO: 266	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 266		
EDEADYY		7
SEQ ID NO: 267	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	

-continued

	mol_type = protein organism = synthetic construct	
SEQUENCE: 267 EDTAFYY		7
SEQ ID NO: 268 FEATURE REGION	moltype = AA length = 12 Location/Qualifiers 1..12 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..12 mol_type = protein organism = synthetic construct	
SEQUENCE: 268 CPPCPAPEFL GG		12
SEQ ID NO: 269 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 269 DDFATYY		7
SEQ ID NO: 270 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 270 YQKKPE		6
SEQ ID NO: 271 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 271 DYTTY		6
SEQ ID NO: 272 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 272 EDTAIYY		7
SEQ ID NO: 273 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 273 EDFAVYF		7
SEQ ID NO: 274 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	

-continued

source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 274 EDVATYY		7
SEQ ID NO: 275 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 275 TFGQGT		6
SEQ ID NO: 276 FEATURE REGION	moltype = AA length = 8 Location/Qualifiers 1..8 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 276 ETTYADDF		8
SEQ ID NO: 277 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 277 SKTDYYY		7
SEQ ID NO: 278 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 278 DDDPYY		6
SEQ ID NO: 279 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 279 DDTAIYY		7
SEQ ID NO: 280 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 280 FTFDDYA		7
SEQ ID NO: 281 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = source = /note="Description of Artificial Sequence:	

-continued

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source          Synthetic peptide"
                1..7
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 281
EDTGVVY                                               7

SEQ ID NO: 282          moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                1..7
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source                1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 282
EDIADYY                                               7

SEQ ID NO: 283          moltype = AA  length = 232
FEATURE                Location/Qualifiers
MOD_RES              1
                    note = Any amino acid
REGION                1..232
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..232
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 283
XVQLVESGGG VVQPGSLRL SCAASGFADS SYGMHWVRQA PGKGLEWVAV IWFDTGKKYY 60
TDSVKGRFTI SRDNSKNTLY LQMNTLRAED TAVYYCARDR GIGARRGPYY MDVWGKGTTV 120
TVSSASTKGP SVFPLAPSSK STSGGTAALG CLVKDYFPEP VTVSWNSGAL TSGVHTFPAV 180
LQSSGLYSL SVVTPVSSSL GTQTYICNVN HKPSNTKVDK RVEPKSCDKT HT 232

SEQ ID NO: 284          moltype = AA  length = 217
FEATURE                Location/Qualifiers
REGION                1..217
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..217
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 284
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYGMSWVRQA PGKGLELVA INSNGGSTYY 60
PDSVKGRFTI SRDNAKNTLY LQMNSLRAED TAVYYCASGD YWQGTTVTV SSASTKGPSV 120
FPLAPCSRST SESTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ SSGLYSLSSV 180
VTVPSSSLGT KTYTCNVDHK PSNTKVDKRV ESKYGGP 217

SEQ ID NO: 285          moltype = AA  length = 234
FEATURE                Location/Qualifiers
REGION                1..234
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..234
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 285
QVELVESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA INASGTRTTY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARGK GNTHKPYGYV RYFDVWGQGT 120
LVTVSASTK GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP 180
AVLQSSGLYS LSSVTVPS SSGLTQTYICN VNHKPSNTKV DKKVEPKSCD KTHT 234

SEQ ID NO: 286          moltype = AA  length = 230
FEATURE                Location/Qualifiers
REGION                1..230
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..230
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 286
EVQLVESGGG LEQPGGSLRL SCAGSGFTFR DYAMTWVRQA PGKGLEWVSS ISGSGGNTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDR LSITIRPRYY GLDVWGQGT 120
VTVSSASTKG PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTVPS SS LGTKTYTCNV DHKPSNTKVD KRVESKYGGP 230

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SEQ ID NO: 287 moltype = AA length = 224
FEATURE Location/Qualifiers
REGION 1..224
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..224
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 287
QVQLVQSGAE VKKPGSSVKV SCKASGYSTF DYHIHWVRQA PGQGLEWMGV INPMYGTDDY 60
NQRFKGRVTI TADESTSTAY MELSSLRSED TAVYYCARYD YFTGTGVYWG QGTLVTVSSA 120
STKGPSVFPL APCSRSTSES TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVTV PSSSLGTKTY TCNVNHNKPSN TKVDKRVESK YGPP 224

SEQ ID NO: 288 moltype = AA length = 235
FEATURE Location/Qualifiers
REGION 1..235
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..235
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 288
EVQLVESGGG LVQPGGSLRL SCAASGFTFS NYWMNWVRQA PGKGLEWVAA INQDGSEKYY 60
VGSVKGRFTI SRDNAKNSLY LQMNLSLRVED TAVYYCVRDY YDILTDYYIH YWYFDLWGRG 120
TLVTVSSAST KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF 180
PAVLQSSGLY SLSSVVTVP SSGLTQTYIC NVNHNKPSNTK VDKRVEPKSC DKTHT 235

SEQ ID NO: 289 moltype = AA length = 227
FEATURE Location/Qualifiers
REGION 1..227
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..227
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 289
EVQLVQSGAE VKKPGESLKI SCKGSGYSTF TYWLGWVRQM PGKGLDWIGI MSPVDSDIRY 60
SPSPQGQVTM SVDKSITTAY LQMNLSLKASD TAMYYCARRR PGQGYFDFWG QGTLVTVSSS 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVTV PSSSLGTQTY ICNVNHNKPSN TKVDKRVEPK SCDKTHT 227

SEQ ID NO: 290 moltype = AA length = 227
FEATURE Location/Qualifiers
REGION 1..227
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..227
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 290
QVTLRSGPA LVKPTQTTLT TCTVSGFSLT SYSVHWVRQP PGKGLEWLGW IWASGGTDYN 60
SALMSRLSIS KDTSRNQVVL TMTNMDPVDI ATYYCARDPP SLLRLDYWG RGTPTVTVSSA 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TFPVAVLQSSG 180
LYSLSSVTV PSSSLGTQTY ICNVNHNKPSN TKVDKRVEPK SCDKTHT 227

SEQ ID NO: 291 moltype = AA length = 229
FEATURE Location/Qualifiers
REGION 1..229
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..229
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 291
QVQLVQSGAE VKKPGASVKV SCKGSGYTFT SYWMHWVRQA PGQRLIEWIGE IDPSESNTNY 60
NQKFKGRVTI TVDISASTAY MELSSLRSED TAVYYCARGG YDGWDYDAIDY WGQGLTVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTW SWNSGALTSG VHTFPVAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHNK SNTKVDKKVE PKSCDKTHT 229

SEQ ID NO: 292 moltype = AA length = 228
FEATURE Location/Qualifiers
REGION 1..228
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..228

-continued

mol_type = protein
organism = synthetic construct

SEQUENCE: 292
QVQLVQSGAE VKKPGASVKV SCKASGFNIK DTYIHWRQA PGQRLEWMGR IDPANGYTKY 60
DPKFQGRVTI TADTSASTAY MELSSLRSED TAVYYCAREG YYGNYGVYAM DYWGQGLVT 120
VSSASTKGPS VFPLAPCSRS TSESTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYLSLS VVTVFSSSLG TKYTCNVNDH KPSNTKVDKR VESKYGPP 228

SEQ ID NO: 293 moltype = AA length = 226
FEATURE Location/Qualifiers
REGION 1..226
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..226
mol_type = protein
organism = synthetic construct

SEQUENCE: 293
EVQLVESGGG LVQPGGSLRL SCAASGFTFN NYAMNWRQA PGKGLDWVST ISGSGGTTNY 60
ADSVKGRFTI SRDSSKHTLY LQMNSLRAED TAVYYCAKDS NWGNFDLWGR GTLVTVSSAS 120
TKGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTVSWN SGALTSGVHT FPAVLQSSGL 180
YLSVSVVTP SSSLGTQTYI CNVNHKPSNT KVDKKVEPKS CDKTHT 226

SEQ ID NO: 294 moltype = AA length = 220
FEATURE Location/Qualifiers
REGION 1..220
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..220
mol_type = protein
organism = synthetic construct

SEQUENCE: 294
EVQLVQSGAE VKKPGASVKV SCKASGYTLT SYGISWVRQA PGQGLEWMGW VSFYNGNTNY 60
AQKLQGRGTM TTDPTSTAY MELRSLRSD TAVYYCARGY GMDVWQGQT VTVSSASTKG 120
PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVTVTPSSN FGTQTYTCNV DHKPSNTKVD KTVKCKCCE 220

SEQ ID NO: 295 moltype = AA length = 231
FEATURE Location/Qualifiers
REGION 1..231
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..231
mol_type = protein
organism = synthetic construct

SEQUENCE: 295
EVQLVESGGG VIQPGGSLRL SCAASGFTFD DYAMNWRQG PGKGLEWVSA ISGDGGSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAFFYCAKDL RNTIFGVVIP DAFDIWGQGT 120
MVTVSSASTK GPSVFPLAPC SRSTSESTA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP 180
AVLQSSGLYS LSSVTVTPSS SLGKTYTCN VDHKPSNTKV DKRVESKYGP P 231

SEQ ID NO: 296 moltype = AA length = 230
FEATURE Location/Qualifiers
REGION 1..230
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..230
mol_type = protein
organism = synthetic construct

SEQUENCE: 296
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSG ITGSGGSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDP GTTVIMSWFD PWGQGLVTV 120
SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYLSLV VTVPSSSLGT QTYICNVNHK PSNTKVDKIV EPKSCDKTHT 230

SEQ ID NO: 297 moltype = AA length = 218
FEATURE Location/Qualifiers
REGION 1..218
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..218
mol_type = protein
organism = synthetic construct

SEQUENCE: 297
QVQLVESGGG VVQPGSLRL DCKASGITFS NSGMHWVRQA PGKGLEWVAV IWYDGSKRY 60
ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATND DYWGQGLVT VSSASTKGPS 120
VFPLAPCSRS TSESTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL QSSGLYLSLS 180

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VVTVPSSSLG TKTYTCNVDH KPSNTKVDKR VESKYGPP 218

SEQ ID NO: 298 moltype = AA length = 225
 FEATURE Location/Qualifiers
 REGION 1..225
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..225
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 298
 QVQLVQSGVE VKKPGASVKV SCKASGYTFT NYMYWVRQA PGQGLEWMGG INPSNGGTNF 60
 NEKFKNRVTL TDSSTTAY MELKSLQFDD TAVYYCARRD YRFDMGFDYW GQGTTVTVSS 120
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSKV HTFPAVLQSS 180
 GLYSLSVVT VPSSSLGTKT YTCNVDHKPS NTKVDKRVES KYGPP 225

SEQ ID NO: 299 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 299
 EVQLVESGGG LVQPGGSLRL SCAASGYDFT HYGMNWRQA PGKGLEWVGW INTYTGEPTY 60
 AADFKRRFTF SLDTSKSTAY LQMNSLRAED TAVYYCAKYP YYYGTSHWYF DVWGQGLT 120
 VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
 QSSGLYSLS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCDKTH L 231

SEQ ID NO: 300 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..228
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 300
 EVQLVESGGG LVQPGGSLRL SCSASGFTFS SFGMHWRQA PGKGLEWVAV ISSGSSTIYY 60
 GDTVKGRTFI SRDPAKNSLF LQMNSLRAED TAVYYCAREG GYYGTSHWYF DVWGQGLT 120
 TVSSASTKGP VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
 LQSSGLYSLS SVTVPSSSLG GTQTYICNVN HKPSNTKVDK VEPKSCDKT 228

SEQ ID NO: 301 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 301
 EVQLVESGGG LVQPGGSLRL SCAASGYTFT NYGMNWRQA PGKGLEWVGW INTYTGEPTY 60
 AADFKRRFTF SLDTSKSTAY LQMNSLRAED TAVYYCAKYP HYYGSSHWYF DVWGQGLT 120
 VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
 QSSGLYSLS VVTVPSSSLG TQTYICNVNH KPSNTKVDK VEPKSCDKT T 231

SEQ ID NO: 302 moltype = AA length = 106
 FEATURE Location/Qualifiers
 REGION 1..106
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..106
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 302
 QVQLVESGGG VVQPGSLRL SCAASGFTFS SFGMHWRQA PGKGLEWVAV ISFDGSIKYS 60
 VDSVKGRFTI SRDPAKNSLF LQMNSLRAED TAVYYCARDR LNYDYS 106

SEQ ID NO: 303 moltype = AA length = 220
 FEATURE Location/Qualifiers
 REGION 1..220
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..220

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mol_type = protein
organism = synthetic construct

SEQUENCE: 303
EVKVVESGGG LVQPGGSMKL SCVVSGETFS NYWVNWVRQA PGKGLEWVAQ IRLKSDNYAT 60
HYEESVKGRF TISRDDSKSS VYLQMNLRRA EDSGIYYCTN WEDYWGQGT VTVSSASTKG 120
PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTKTYTCNV DHKPSNTKVD KRVESKYGPP 220

SEQ ID NO: 304 moltype = AA length = 223
FEATURE Location/Qualifiers
REGION 1..223
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..223
mol_type = protein
organism = synthetic construct

SEQUENCE: 304
EVQLVQSGPE LKKPGASVKV SCKASGYTFT NYGMNWVRQA PGQGLEWMGW INTYTGETTY 60
ADDFKGRFVF SLDTSVSTAY LQSSSLKAED TAVYYCEREG GVNNWGQGT VTVSSASTKG 120
PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTQTYTCNV NHKPSNTKVD KKEPKSCDK THT 223

SEQ ID NO: 305 moltype = AA length = 220
FEATURE Location/Qualifiers
REGION 1..220
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..220
mol_type = protein
organism = synthetic construct

SEQUENCE: 305
QVQLQESGPG LVKPSSETLSL TCTVSGFSL SYGVHWVRQP PGKGLEWLVG IWTGGTTNIN 60
SALMSRFTIS KDDSKNTVYL KMNSLKTED AIYYCARYYY GMDYWGQGT VTVSSASTKG 120
PSVFPLAPCS RSTSESTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL 180
SSVVTVPSSS LGTKTYTCNV DHKPSNTKVD KRVESKYGPP 220

SEQ ID NO: 306 moltype = AA length = 231
FEATURE Location/Qualifiers
REGION 1..231
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..231
mol_type = protein
organism = synthetic construct

SEQUENCE: 306
QVQLQESGAE VKKPGSSVRV SCKASGGTFN NNAINWVRQA PGQGLEWMGG IIPMFGTAKY 60
SQNFQGRVAI TADESTGTAS MELSSLRSED TAVYYCARSR DLLLLPHHAL SPWGRGTMVT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYLSLS VVTVPSSSLG TQTYICNVNH KPSNTKVDKK VEPKSCDKTH T 231

SEQ ID NO: 307 moltype = AA length = 227
FEATURE Location/Qualifiers
REGION 1..227
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..227
mol_type = protein
organism = synthetic construct

SEQUENCE: 307
QVQLVQSGAE VKKPGASVKV SCKASGYIFS NYWIQWVRQA PGQGLEWMGE ILPGSGSTEY 60
TENFKDRVTM TRDTSTSTVY MELSSLRSED TAVYYCARYF FGSSPNWYFD VWGQGLVTV 120
SSASTKGPSV FPLAPCSRST SESTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYLSVSV VTPSSNFGT QTYTCNVNHDH PSNTKVDKTV ERKCCVE 227

SEQ ID NO: 308 moltype = AA length = 230
FEATURE Location/Qualifiers
REGION 1..230
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..230
mol_type = protein
organism = synthetic construct

SEQUENCE: 308
EVQLLESGGG LVQPGGSLRL SCAASGFTFS HYIMMWVRQA PGKGLEWVSG IYSSGGITVY 60
ADSVKGRFTI SRDNSKNTLY LQMNLSRAED TAVYYCAYRR IGVPRRDEFD IWGQGTMTV 120
SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180

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SSGLYSLSSV VTPSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCDKTHT 230

SEQ ID NO: 309 moltype = AA length = 229
 FEATURE Location/Qualifiers
 REGION 1..229
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..229
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 309
 EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYAMHWVRQA PGKGLEWVSA ITWNSGHIDY 60
 ADSVEGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCAKVS YLSTASSLDY WQGTLVTVS 120
 SASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
 SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKKEV PKSCDKTHT 229

SEQ ID NO: 310 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..228
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 310
 EVKLEESGGG LVQPGGSMKL SCVASGFIFS NHWMNWRQS PEKGLEWVAE IRKSINSAT 60
 HYAESVKGRI TISRDDSKSA VYLQMTDLRT EDTGVYYCSR NYYGSTYDYW GQGTTTLTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHT 228

SEQ ID NO: 311 moltype = AA length = 55
 FEATURE Location/Qualifiers
 REGION 1..55
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..55
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 311
 QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL PGTAPKLLIY DNNKR 55

SEQ ID NO: 312 moltype = AA length = 110
 FEATURE Location/Qualifiers
 REGION 1..110
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..110
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 312
 EIVMTQSPST LSASVGRVI ITCQASEIIS WLAWYQQKPG KAPKLLIYLA STLASGVPSR 60
 FSGSGSGAEF TLTISLQPD DFATYYCQNV YLASTNGANF GQGTKLTVLG 110

SEQ ID NO: 313 moltype = AA length = 123
 FEATURE Location/Qualifiers
 REGION 1..123
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..123
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 313
 KYYGMAVWGQ GTTIVVSSAS TKGPSVFPLA PCSRSTSEST AALGCLVKDY FPEPVTVSWN 60
 SGALTSVHT PPAVLQSSGL YSLSSVVTVP SSNFGTQYTT CNVDHKPSNT KVDKTVKRC 120
 CVE 123

SEQ ID NO: 314 moltype = AA length = 158
 FEATURE Location/Qualifiers
 REGION 1..158
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..158
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 314
 PDRFSGSKS GTSTTLGITG LQTGDEADYY CGTWDSRLSA VVFGGGTKLT VLGQPKANPT 60

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VTLFPPSSEE LQANKATLVC LISDFYPGAV TVAWKADGSP VKAGVETTKP SKQSNKNYAA	120
SSYLSLTPEQ WKSHRSYSCQ VTHEGSTVEK TVAPTECS	158

1. A pharmaceutical composition for treating Alzheimer's disease, migraines, cluster headaches, or tauopathies including chronic traumatic encephalopathy, progressive supranuclear palsy, and frontotemporal dementia in a human subject in need thereof, comprising an adeno-associated virus (AAV) vector having:

- (a) a viral capsid that is at least 95% identical to the amino acid sequence of an AAV9 capsid (SEQ ID NO: 78) or AAVrh10 (SEQ ID NO: 80); and
- (b) an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the expression cassette comprises a transgene encoding an anti-amyloid beta, anti-Tau, or anti-CGRPR mAb, or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human CNS cells;

wherein said AAV vector is formulated for intrathecal administration to the CNS of said subject.

2.-18. (canceled)

19. A pharmaceutical composition for treating an ocular disorder in a human subject in need thereof, wherein the pharmaceutical composition comprises an adeno-associated virus (AAV) vector comprising:

- (a) a viral capsid that is at least 95% identical to the amino acid sequence of SEQ ID NO: 78; and
- (b) an artificial genome comprising an expression cassette flanked by AAV inverted terminal repeats (ITRs) wherein the expression cassette comprises a transgene encoding an anti-VEGF monoclonal antibody (mAb), or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human retinal cells;

wherein said AAV vector is formulated for suprachoroidal administration to the eye of said subject.

20. The pharmaceutical composition of claim 19, wherein the anti-VEGF mAb or antigen-binding fragment thereof is ranibizumab, bevacizumab, or brolucizumab.

21.-26. (canceled)

27. The pharmaceutical composition of claim 19, wherein the anti-VEGF mAb or antigen-binding fragment thereof is a Fab, a F(ab')₂, an scFv, or a full-length antibody.

28. The pharmaceutical composition of claim 19, wherein the anti-VEGF mAb or antigen-binding fragment thereof comprises (i) a heavy chain having the amino acid sequence of SEQ ID NO: 33 and a light chain having the amino acid sequence of SEQ ID NO: 34, or (ii) a heavy chain having the amino acid sequence of SEQ ID NO: 35 and a light chain having the amino acid sequence of SEQ ID NO: 36; or (iii) a heavy chain having the amino acid sequence of SEQ ID NO: 39 and a light chain having the amino acid sequence of SEQ ID NO: 40.

29. The pharmaceutical composition of claim 28, wherein the transgene comprises (i) the nucleotide sequence of SEQ ID NO: 133 encoding the heavy chain and the nucleotide sequence of SEQ ID NO: 134 encoding the light chain; or (ii) the nucleotide sequence of SEQ ID NO: 135 encoding the heavy chain and the nucleotide sequence of SEQ ID NO: 136 encoding the light chain; or (iii) the nucleotide sequence

of SEQ ID NO: 139 encoding the heavy chain and the nucleotide sequence of SEQ ID NO: 140 encoding the light chain.

30. The pharmaceutical composition of claim 19, wherein the transgene further comprises a cleavable linker comprising a furin site encoded by the nucleotide sequence of SEQ ID NO: 215, SEQ ID NO: 216, SEQ ID NO: 217 or SEQ ID NO: 218.

31. The pharmaceutical composition of claim 30, wherein the cleavable linker has the amino acid sequence of SEQ ID NO: 211, SEQ ID NO: 212, SEQ ID NO: 213 or SEQ ID NO: 214.

32. The pharmaceutical composition of any one of claim 19-20 or 27-31, wherein the heavy chain and the light chain of said antigen-binding fragment each have an N-terminus, and said transgene further comprises a signal sequence encoded at the N-termini of the heavy chain and the light chain, wherein said signal sequence directs secretion and post translational modification of the heavy chain and the light chain in said human retinal cells.

33. The pharmaceutical composition of claim 32, wherein said signal sequence has the amino acid sequence of SEQ ID NO: 164, SEQ ID NO: 165, SEQ ID NO: 166, SEQ ID NO: 167, SEQ ID NO: 168, SEQ ID NO: 169, SEQ ID NO: 170, SEQ ID NO: 171, SEQ ID NO: 161, or SEQ ID NO: 172.

34. The pharmaceutical composition of any one of claim 19-20 or 27-33, wherein the ocular disorder is age-related macular degeneration.

35. A method of producing recombinant adeno-associated virus (AAV) vectors, the method comprising:

- (a) culturing a host cell containing:
 - i. an artificial genome comprising a cis expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the cis expression cassette comprises a transgene encoding an anti-VEGF monoclonal antibody (mAb), or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human retinal cells;
 - ii. a trans expression cassette lacking AAV ITRs, wherein the trans expression cassette encodes an AAV rep protein and an AAV cap protein operably linked to expression control elements that drive expression of the rep and cap proteins in the host cell in culture and supply the rep and cap proteins in trans, wherein the AAV cap protein is at least 95% identical to the amino acid sequence of SEQ ID NO: 78;
 - iii. sufficient adenovirus helper functions to permit replication and packaging of the artificial genome by the AAV rep and cap proteins; and
- (b) recovering recombinant AAV encapsidating the artificial genome from the cell culture.

36. A host cell comprising a plasmid comprising a cis expression cassette flanked by AAV inverted terminal repeats (ITRs), wherein the cis expression cassette comprises a transgene encoding an anti-VEGF monoclonal antibody (mAb), or an antigen-binding fragment thereof, operably linked to one or more regulatory sequences that control expression of the transgene in human retinal cells.

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